

Project Data Page:





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DESIGN CALCULATION

In Accordance with ASME Section VIII Division 1

ASME Code Version : 2025

Analysis Performed by : SL User (Modify in Config->Save Options)

Job File : ...\\401-V-32001 - 7%.pvdb

Date of Analysis : Jul 20,2026 10:59am

PV Elite 28, January 2026



Vessel Design Summary:

ASME Code, Section VIII Division 1, 2025

Diameter Spec : 2344.000 x 2300.000 mm. ID
 Vessel Design Length, Tangent to Tangent 11000.00 mm.
 Specified Datum Line Distance 0.00 mm.
 Internal Design Temperature 150 °C
 Internal Design Pressure 23.000 bars
 External Design Temperature 0 °C
 Maximum Allowable Working Pressure 23 bars
 Shop Test Pressure 30.201 bars
 Required Minimum Design Metal Temperature -33.0 °C
 Warmest Computed Minimum Design Metal Temperature -29.0 °C

Warning: Computed overall MDMT was higher than the required value !

Wind Design Code IS 875 (PART 3) Amd. 1&2 (2020)
 Earthquake Design Code IS 1893 RSM Amd. 1&2 (2018)

Materials of Construction:

Component Type	Material	Class	Thickness	UNS #	Material Treatment Options
Shell	SA-516 70	...	...	K02700	
Head	SA-516 70	...	...	K02700	+IT
Nozzle	SA-516 70	...	...	K02700	+IT
Nozzle	SA-333 6	...	...	K03006	+IT
Nozzle	SA-350 LF2	1	...	K03011	+IT
Re-Pad	SA-516 70	...	...	K02700	+IT
Nozzle Flg	SA-350 LF2	2	...	K03011	+IT
Nozzle Flg	SA-350 LF2	1	...	K03011	+IT
Nozzle Flg	SA-105	...	...	K03504	
Hrz Bolting	SA-285 C	...	...	K02801	
Saddles	SA-516 70	...	...	K02700	

+FG - Fine Grained, +R - rolled, +T - Tempered, +N - Normalized
 +Q - quenched, +CBV - Cb & V added, +IT - Impact tested

Element Pressures and MAWP (bars & mm.):

Element Description or Type	Design Pressure + Stat. head	Ext. Press.	Element M.A.W.P	Total Corrosion Allowance	Str. Flg. Gov.	In Creep Range
Left D end	23.137	0.00	23.000	1.5000	No	No
Shell 1	23.137	0.00	24.000	1.5000	N/A	No
Shell 2	23.137	0.00	24.000	1.5000	N/A	No
Shell 3	23.137	0.00	24.000	1.5000	N/A	No
Shell 4	23.137	0.00	24.000	1.5000	N/A	No
Shell 5	23.137	0.00	24.000	1.5000	N/A	No
Right D end	23.137	0.00	23.000	1.5000	No	No

Liquid Level: 2300.00 mm. Dens.: 0.001 kg./cm3 Sp. Gr.: 0.608

Element Types and Properties:

Element Type	"To" Elev mm.	Element Length mm.	Nominal Thickness mm.	Finished Thickness mm.	Reqd Thk Internal mm.	Reqd Thk External mm.	Long Eff	Circ Eff
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Ellipse	50.0	50.0	25.0	21.0	20.8	...	1.00	1.00
Cylinder	2050.0	2000.0	22.0	22.0	21.0	...	1.00	1.00
Cylinder	4050.0	2000.0	22.0	22.0	21.0	...	1.00	1.00
Cylinder	7050.0	3000.0	22.0	22.0	21.0	...	1.00	1.00
Cylinder	9050.0	2000.0	22.0	22.0	21.0	...	1.00	1.00
Cylinder	10950.0	1900.0	22.0	22.0	21.0	...	1.00	1.00
Ellipse	11000.0	50.0	25.0	21.0	20.8	...	1.00	1.00

Local Stress Analysis Results:

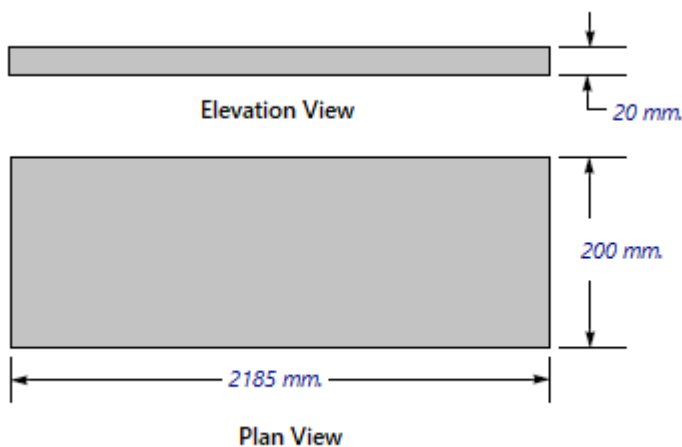
Description	Analysis Type	Max Stress Ratio	High Stress Location	Pass Fail
N1_8"	WRC-297	0.942	Shell	Passed
N2_2"	WRC-297	0.942	Shell	Passed
N6_4"	WRC-297	0.949	Shell	Passed
K9_12"	WRC-297	0.942	Shell	Passed
N3_6"	WRC-297	0.942	Shell	Passed
N8_8"	WRC-297	0.949	Shell	Passed

Foundation/Support Summary

Saddle Parameters:

Saddle Width	180.000 mm.
Saddle Bearing Angle	137.000 deg.
Centerline Dimension	1430.000 mm.
Wear Pad Width	230.000 mm.
Wear Pad Thickness	12.000 mm.
Wear Pad Bearing Angle	150.000 deg.
Distance from Saddle to Tangent	1450.000 mm.
Baseplate Length	2185.000 mm.
Baseplate Thickness	20.000 mm.
Baseplate Width	200.000 mm.
Number of Ribs (including outside ribs)	3
Rib Thickness	16.000 mm.
Web Thickness	16.000 mm.
Height of Center Web	306.000 mm.
Number of Bolts in Baseplate	4

Baseplate Sketch



Summary of Maximum Saddle Loads, Operating Case, Un-Factored:

Maximum Vertical Saddle Load	41043.99 Kg
Maximum Transverse Saddle Shear Load	4411.27 Kg
Maximum Longitudinal Saddle Shear Load	5578.53 Kg

Summary of Maximum Saddle Loads, Pressure Test Case:



Maximum Vertical Saddle Load	42319.79	Kgf
Maximum Transverse Saddle Shear Load	1455.72	Kgf
Maximum Longitudinal Saddle Shear Load	669.15	Kgf

Weights:

Fabricated - Bare w/o removable internals	25658.7	kg.
Shop Test - Fabricated + water (full)	74725.3	kg.
Shipping - Fab. + rem. intls.+ shipping app.	30733.4	kg.
Erected - Fab. + rem. intls.+ insul. (etc)	30933.4	kg.
Empty - Fab. + intls. + details + wghts.	30933.4	kg.
Operating - Empty + operating liquid (no ca)	61358.0	kg.

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Nozzle Calculation Summary (bars & mm.):

Description	MAWP	Ext	MAPNC	UG-45	[tr]	Weld Path	Areas or Stresses
MH1_24"	23	...	...	...		OK	Passed
N1_8"	24	...	...	OK	8.66	OK	Passed
N2_2"	24	...	...	OK	4.92	OK	No Calc[*]
N6_4"	24	...	...	OK	6.76	OK	Passed
K8_1"	24	...	...	OK	4.92	OK	No Calc[*]
N10_1"	24	...	...	OK	4.92	OK	No Calc[*]
N7_20"	24	...	...	OK	9.83	OK	Passed
K3A_2"	24	...	...	OK	6.30	OK	No Calc[*]
K3B_2"	24	...	...	OK	6.30	OK	No Calc[*]
K1A_2"	24	...	...	OK	6.30	OK	Passed
K1A_2"	24	...	...	OK	6.30	OK	No Calc[*]
K1B_2"	24	...	...	OK	6.30	OK	Passed
K1B_2"	24	...	...	OK	6.30	OK	No Calc[*]
K2A_2"	24	...	...	OK	6.30	OK	Passed
K2A_2"	24	...	...	OK	6.30	OK	No Calc[*]
K2B_2"	24	...	...	OK	6.30	OK	Passed
K2B_2"	24	...	...	OK	6.30	OK	No Calc[*]
N4_3"	24	...	...	OK	6.30	OK	Passed
K9_12"	24	...	...	OK	9.83	OK	Passed
K7_1"	24	...	...	OK	4.92	OK	No Calc[*]
N3_6"	24	...	...	OK	7.72	OK	Passed
N8_8"	24	...	...	OK	8.66	OK	Passed
MH2_24"	23	...	...	...		OK	Passed

Nozzle MAWP Summary:

Minimum MAWP Nozzles : 23 Nozzle : MH124"
 Minimum MAWP Shells/Flanges : 23 Element : Left D end
 Minimum MAPnc Shells/Flanges : 25 Element : Left D end

Computed Vessel M.A.W.P. : 23 bars

[*] - This was a small opening and the areas were not computed.

Note: MAWPs (Internal Case) shown above are at the High Point.

Multiple output lines for the same nozzle indicates required Code calculations in both the longitudinal and circumferential planes of reinforcement where applicable.

Check the spatial relationship between the nozzles:

From Node	Nozzle Description	X Coordinate mm.	Layout Angle deg	Dia. Limit mm.
10	MH1_24"	0.000	0.000	1225.200
20	N1_8"	450.000	90.000	399.700
20	N2_2"	850.000	90.000	101.325
20	N6_4"	800.000	270.000	204.641
30	K8_1"	3250.000	90.000	94.848
30	N10_1"	3550.000	90.000	94.848
40	N7_20"	5300.000	90.000	1022.000
40	K3A_2"	6550.000	90.000	125.328
40	K3B_2"	6550.000	270.000	125.328
40	K1A_2"	6050.000	54.153	125.328
40	K1B_2"	6050.000	-54.153	125.328
40	K2A_2"	6550.000	54.153	125.328
40	K2B_2"	6550.000	-54.153	125.328
50	N4_3"	8450.000	90.000	157.130
60	K9_12"	10200.001	90.000	615.600
60	K7_1"	10550.000	46.490	94.848
60	N3_6"	10750.002	90.000	304.145
60	N8_8"	10420.001	270.000	399.700
70	MH2_24"	0.000	0.000	1225.200



The nozzle spacing is computed by the following:

$$= \sqrt{ll^2 + lc^2} \text{ where}$$

ll - Arc length along the inside vessel surface in the long. direction.

lc - Arc length along the inside vessel surface in the circ. direction

If any interferences/violations are found, they will be noted below.

No interference violations have been detected!

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Nozzle Schedule:

Description	Nominal or Actual Size	Schd or FVC Type	Flg Type	Nozzle O/Dia in	Wall Thk mm	Reinforcing Pad Diameter	Thk mm	Cut Length mm	Flg Class
K8_1"	1.000 in	LWN	LWN	2.120	14.224	...	...	535.23	300
N10_1"	1.000 in	LWN	LWN	2.120	14.224	...	...	535.23	300
K7_1"	1.000 in	LWN	LWN	2.120	14.224	...	...	298.92	300
N2_2"	2.000 in	160	WNF	2.375	8.738	...	...	477.24	600
K3A_2"	2.000 in	Actual	LWN	3.320	16.764	...	...	530.20	300
K3B_2"	2.000 in	Actual	LWN	3.320	16.764	...	...	229.15	300
K1A_2"	2.000 in	Actual	LWN	3.320	16.764	...	...	311.26	300
K1B_2"	2.000 in	Actual	LWN	3.320	16.764	...	...	311.26	300
K2A_2"	2.000 in	Actual	LWN	3.320	16.764	...	...	311.26	300
K2B_2"	2.000 in	Actual	LWN	3.320	16.764	...	...	311.26	300
N4_3"	3.000 in	80	WNF	3.500	7.620	152.40	22.00	422.34	300
N6_4"	4.000 in	80	WNF	4.500	8.560	203.20	22.00	467.09	300
N3_6"	6.000 in	80	WNF	6.625	10.973	280.00	22.00	456.05	300
N1_8"	8.000 in	80	WNF	8.625	12.700	360.00	22.00	445.50	300
N8_8"	8.000 in	80	WNF	8.625	12.700	360.00	22.00	445.50	300
K9_12"	12.000 in	Actual	WNF	12.787	10.000	580.00	22.00	433.00	300
N7_20"	20.000 in	Actual	WNF	20.787	10.000	980.00	22.00	420.18	300
MH1_24"	24.000 in	Actual	WNF	24.630	8.000	1060.00	22.00	202.95	300
MH2_24"	24.000 in	Actual	WNF	24.630	8.000	1060.00	22.00	202.95	300

General Notes for the above table:

The cut length is the outside projection + inside projection + drop + in-plane shell thickness. This value does not include weld gaps, nor does it account for shrinkage.

In the case of oblique nozzles, the outside diameter must be increased. The re-pad width around the nozzle is calculated as follows:

Width of Pad = (Pad Outside Dia. (per above) - Nozzle Outside Dia.)/2

For hub nozzles, the thickness and diameter shown are those of the smaller and thinner section.

Nozzle Material and Weld Fillet Leg Size Details (mm.):

Description	Material	Shl Grve Weld	Noz Shl/Pad Weld	Pad OD Weld	Pad Grve Weld	Inside Weld
K8_1"	SA-350 LF2	22.000	10.000	...	...	...
N10_1"	SA-350 LF2	22.000	10.000	...	...	...
K7_1"	SA-350 LF2	22.000	10.000	...	...	...
N2_2"	SA-333 6	22.000	10.000	...	...	...
K3A_2"	SA-350 LF2	22.000	10.000	...	...	...
K3B_2"	SA-350 LF2	22.000	10.000	...	...	...
K1A_2"	SA-350 LF2	22.000	10.000	...	...	...
K1B_2"	SA-350 LF2	22.000	10.000	...	...	...
K2A_2"	SA-350 LF2	22.000	10.000	...	...	...
K2B_2"	SA-350 LF2	22.000	10.000	...	...	...
N4_3"	SA-333 6	22.000	10.000	18.000	22.000	...
N6_4"	SA-333 6	22.000	10.000	18.000	22.000	...
N3_6"	SA-333 6	22.000	10.000	18.000	22.000	...
N1_8"	SA-333 6	22.000	10.000	18.000	22.000	...
N8_8"	SA-333 6	22.000	10.000	18.000	22.000	...
K9_12"	SA-516 70	22.000	10.000	18.000	22.000	...
N7_20"	SA-516 70	22.000	10.000	18.000	22.000	...
MH1_24"	SA-516 70	22.000	10.000	18.000	22.000	...
MH2_24"	SA-516 70	22.000	10.000	18.000	22.000	...

Note: The Outside projections below do not include the flange thickness.

Nozzle Miscellaneous Data:

Description	Elev/Distance From Datum mm.	Layout Angle deg	Proj Outside mm.	Proj Inside mm.	Installed in Component
-------------	------------------------------	------------------	------------------	-----------------	------------------------



K8_1"	3250.000	90.0	512.25	0.00	Shell 2
N10_1"	3550.000	90.0	512.25	0.00	Shell 2
K7_1"	10550.000	46.5	211.14	0.00	Shell 5
N2_2"	850.000	90.0	454.85	0.00	Shell 1
K3A_2"	6550.000	90.0	507.43	0.00	Shell 3
K3B_2"	6550.000	270.0	206.38	0.00	Shell 3
K1A_2"	6050.000	54.2	206.38	0.00	Shell 3
K1B_2"	6050.000	-54.2	206.38	0.00	Shell 3
K2A_2"	6550.000	54.2	206.38	0.00	Shell 3
K2B_2"	6550.000	-54.2	206.38	0.00	Shell 3
N4_3"	8450.000	90.0	399.48	0.00	Shell 4
N6_4"	800.000	270.0	443.67	0.00	Shell 1
N3_6"	10750.002	90.0	430.97	0.00	Shell 5
N1_8"	450.000	90.0	418.27	0.00	Shell 1
N8_8"	10420.001	270.0	418.27	0.00	Shell 5
K9_12"	10200.000	90.0	399.48	0.00	Shell 5
N7_20"	5300.000	90.0	367.47	0.00	Shell 3
MH1_24"	...	0.0	158.38	0.00	Left D end
MH2_24"	...	0.0	158.38	0.00	Right D end



Bill of Materials:

QTY	DESCRIPTION	MATERIAL
2	ELLIPTICAL HEAD: 2.0 X 1, 25.0mm. THK X 2300.0mm. ID X 50.0mm.	SA-516 70
1	PLATFORM: 1200mm X 2344mm WIDE	
1	LADDER: 2602.0mm	...
2	INSULATION: 625mm X 70mm THK	Cold
2	SADDLE: 180.0mm X 137 DEG	SA-516 70
3	INSULATION: 2000mm X 70mm THK	Cold
1	INSULATION: 3000mm X 70mm THK	Cold
1	INSULATION: 1899mm X 70mm THK	Cold
2	CLASS 300 GR 1.1, 24.0" BLIND FLANGE(S)	SA-350 LF2
1	CLASS 300 GR 1.1, 20.0" BLIND FLANGE(S)	SA-350 LF2
4	CYLINDRICAL SEGMENT 22.0mm. THK X 3612.8mm. X 2000.0mm.	SA-516 70
1	CYLINDRICAL SEGMENT 22.0mm. THK X 3412.1mm. X 2000.0mm.	SA-516 70
1	CYLINDRICAL SEGMENT 22.0mm. THK X 3813.5mm. X 2000.0mm.	SA-516 70
2	CYLINDRICAL SEGMENT 22.0mm. THK X 3612.8mm. X 3000.0mm.	SA-516 70
2	CYLINDRICAL SEGMENT 22.0mm. THK X 3612.8mm. X 1900.0mm.	SA-516 70
1	NAMEPLATE	...



Minimum Design Metal Temperature Results Summary (°C):

Description	Curve	Basic MDMT	Reduced MDMT	UG-20(f) MDMT	Thickness ratio	Gov Thk mm.	E*	PWHT reqd
Notes								
Left D end	[10]	!	-33	-33	1.000	21.000	1.00	No
Left D end	[7]	!	-33	-42	0.838	25.000	1.00	No
Shell 1	[8]	B	-21	-23	-29	0.962	22.000	1.00
Shell 2	[8]	B	-21	-23	-29	0.961	22.000	1.00
Shell 3	[8]	B	-21	-23	-29	0.961	22.000	1.00
Shell 4	[8]	B	-21	-23	-29	0.961	22.000	1.00
Shell 5	[8]	B	-21	-23	-29	0.961	22.000	1.00
Right D end	[10]	B	-6	-6	-29	1.000	21.000	1.00
Right D end	[7]	B	-1	-10	-29	0.838	25.000	1.00
Fixed Saddle	[24]	B	-40	-42		0.962	12.000	1.00
New Saddle	[24]	B	-40	-42		0.961	12.000	1.00
MH124"	[1]	!	-33	-39		0.889	21.000	1.00
Nozzle Flg	[5]	!	-18	-57				
N18"	[1]	B	-21	-29	-29	0.946	22.000	1.00
Nozzle Flg	[5]	!	-46	-85				
N22"	[1]	B	-46	-104		0.077	7.645	1.00
Nozzle Flg	[5]	!	-46	-104				
N64"	[1]	B	-21	-29	-29	0.951	22.000	1.00
Nozzle Flg	[5]	!	-46	-85				
K81"	[1]	B	-21	-29	-29	0.946	22.000	1.00
Nozzle Flg	[5]	!	-46	-46				
N101"	[1]	B	-21	-29	-29	0.946	22.000	1.00
Nozzle Flg	[5]	!	-46	-46				
N720"	[1]	B	-21	-29	-29	0.946	22.000	1.00
Nozzle Flg	[5]	!	-46	-85				
K3A2"	[1]	B	-29	-104		0.030	16.764	1.00
Nozzle Flg	[5]	A	-18	-48				
K3B2"	[1]	B	-29	-104		0.030	16.764	1.00
Nozzle Flg	[5]	A	-18	-48				
K1A2"	[1]	B	-29	-104		0.030	16.764	1.00
Nozzle Flg	[5]	A	-18	-48				
K1B2"	[1]	B	-29	-104		0.030	16.764	1.00
Nozzle Flg	[5]	A	-18	-48				
K2A2"	[1]	B	-29	-104		0.030	16.764	1.00
Nozzle Flg	[5]	A	-18	-48				
K2B2"	[1]	B	-29	-104		0.030	16.764	1.00
Nozzle Flg	[5]	A	-18	-48				
N43"	[1]	B	-21	-29	-29	0.946	22.000	1.00
Nozzle Flg	[5]	!	-46	-85				
K912"	[1]	B	-21	-29	-29	0.946	22.000	1.00
Nozzle Flg	[5]	!	-46	-85				
K71"	[1]	B	-21	-29	-29	0.946	22.000	1.00
Nozzle Flg	[5]	!	-46	-46				
N36"	[1]	B	-21	-29	-29	0.946	22.000	1.00
Nozzle Flg	[5]	!	-46	-85				
N88"	[1]	B	-21	-29	-29	0.951	22.000	1.00
Nozzle Flg	[5]	!	-46	-85				
MH224"	[1]	B	-6	-29	-29	0.889	21.000	1.00
Nozzle Flg	[5]	!	-46	-85				
Noz1. Bolting[22]			-48		1.000	64.000	1.00	No

Warmest MDMT: -1 -6

Required Minimum Design Metal Temperature -33.0 °C
 Warmest Computed Minimum Design Metal Temperature -29.0 °C

Warning: Computed overall MDMT was higher than the required value !

Notes:

- [!] - This was an impact tested material.
- [1] - Governing Nozzle Weld.
- [4] - ASME Flange MDMT Calcs; Thickness ratio per UCS-66(b)(1)(-c).
- [5] - ASME Flange MDMT Calcs; Thickness ratio per UCS-66(b)(1)(-b).



- [6] - MDMT Calculations at the Shell/Head Joint.
- [7] - MDMT Calculations for the Straight Flange.
- [8] - Cylinder/Cone/Flange Junction MDMT.
- [9] - Calculations in the Spherical Portion of the Head.
- [10] - Calculations in the Knuckle Portion of the Head.
- [11] - Calculated (Body Flange) Flange MDMT.
- [12] - Calculated Flat Head MDMT per UCS-66.3
- [13] - Tubesheet MDMT, shell side, if applicable
- [14] - Tubesheet MDMT, tube side, if applicable
- [15] - Nozzle Material
- [16] - Shell or Head Material
- [17] - Impact Testing required
- [18] - Impact Testing not required, see UCS-66(b)(3)
- [20] - Cylinder/Cone Junction MDMT based on Longitudinal Stress considerations
- [21] - Body Flange Bolting Material
- [22] - Nozzle Flange Bolting Material
- [23] - Stiffening Ring to Shell Weld
- [24] - Saddle to Shell Weld

UG-84(b)(2) was not considered.

UCS-66(g) was not considered.

UCS-66(i) was not considered.

Notes:

Impact test temps were not entered in and not considered in the analysis.

UCS-66(i) applies to impact tested materials not by specification and

UCS-66(g) applies to materials impact tested per UG-84.1 General Note (c).

The Basic MDMT includes the (30F) PWHT credit if applicable.

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Class From To : Basic Element Checks.

Class From To: Check of Additional Element Data

There were no geometry errors or warnings.

PV Elite performs all calculations internally in Imperial Units to remain compliant with the ASME Code and any built in assumptions in the ASME Code formulas. The finalized results are reflected to show the set of selected units for this analysis.

Information Regarding Nozzle Loads

Please note that nozzle loadings, if included, are assumed to be local in nature and will not contribute to or create a net section bending moment. Therefore, the addition of nozzle loads will not affect the support load calculation. If you wish to create a load on the support(s) from nozzle loads, you can enable that feature in tools->configuration->Job Specific Settings.



One or more nozzles in this model was specified using the actual thickness basis and a standard flange is also specified. In many cases a matching nominal diameter cannot be determined and the weight of the flange and optional blind flange cannot be determined. When actual thickness basis is specified, you must calculate the weight of the assembly (by hand) and enter it into the 'Overriding Weight' field in the nozzle dialog when there is a standard flange specified. However, for FVC nozzles that have their nominal diameter specified in the FVC dialog, entering the overriding weight for that nozzle is not required.



The level of precision is set to a value of less than 3. At this level you may not obtain proper resolution on some results. This value can be changed in Tools->Set Configuration Parameters->Default Values->Level of Precision.



The ASME Code Section VIII Division 1 does not explicitly require stresses on elements to be computed under test conditions. Please see interpretation VIII-1-10-12 for more information. PV Elite also computes weight due to test fluid and uses the weight in various calculations such as saddle and leg loads. PV Elite also computes static pressure due to operating and test liquids. The static pressure is used in conjunction with the design and test pressures to determine a total pressure in the determination of required thickness, MAWP etc. This ensures compliance with UG-22 requirements.

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Units used in this Analysis (barsUniCode):

Name	System Unit	Constant	User Unit
Length	Feet	304.8000	mm.
Force	Pounds	0.4536	Kgf
Mass	Pounds	0.4536	kg.
Area	sq. inches	6.4516	cm2
Moment	ft. lbs.	0.1383	Kg-m.
Stress	lbs./sq.in.	0.0069	N./mm2
Temperature	Degrees F	0.5556	°C
Pressure	psig	0.0689	bars
Elast. Modulus	lbs./sq.in.	6.8946	KPa.
Pipe Density	lbs./cu.in.	0.0277	kg./cm3
Ins. Density	lbs./cu.ft.	0.1600E-04	kg./cm3
Fluid Density	lbs./cu.ft.	0.1600E-04	kg./cm3
Wind Speed	miles/hr	1.6093	Km/hr
Tray Weight	lbs./sq.ft.	0.0005	Kgs/cm2
Inertia	in.4	41.6231	cm4
G Load	G's	1.0000	g's
Wind Load	lbs./sq.ft.	4.8825	Kgs/m2
Elevation	Feet	304.8000	mm.
Volume	in.3	16.3900	cm3
Diameter	inches	25.4000	mm.
Thickness	inches	25.4000	mm.

PV Elite Vessel Analysis Program: Input Data

Design Internal Pressure (for Hydrotest)	23 bars
Design Internal Temperature	150.0 °C
Projection of Nozzle from Vessel Top	0 mm.
Projection of Nozzle from Vessel Bottom	0 mm.
Minimum Design Metal Temperature	-33.0 °C
Type of Construction	Welded
Special Service	None
Degree of Radiography	RT-1
Use Higher Longitudinal Stresses (Flag)	Y
Select t for Internal Pressure (Flag)	N
Select t for External Pressure (Flag)	N
Select t for Axial Stress (Flag)	N
Select Location for Stiff. Rings (Flag)	N

User Spectrum Data points specified for increments of Time Period

Time Period	Sa/g
0.010	0.070
0.050	0.117
0.100	0.175
0.130	0.175
0.200	0.175
0.270	0.175
0.300	0.175
0.400	0.175
0.500	0.140
0.550	0.127
0.600	0.117
0.670	0.105
0.800	0.087
1.000	0.070
1.200	0.058
1.400	0.050
1.600	0.044
1.800	0.039
2.000	0.035
2.400	0.029
0.000	0.000



Consider Vortex Shedding N

Shop Pressure Test:

Type of Pressure Test UG-99(b)
 Pressure Test Position Horizontal
 Test Performed in Corroded Condition No

Load Case 1	NP+EW+WI+FW+BW
Load Case 2	NP+EW+EE+FS+BS
Load Case 3	NP+OW+WI+FW+BW
Load Case 4	NP+OW+EQ+FS+BS
Load Case 5	NP+HW+HI
Load Case 6	NP+HW+HE
Load Case 7	IP+OW+WI+FW+BW
Load Case 8	IP+OW+EQ+FS+BS
Load Case 9	EP+OW+WI+FW+BW
Load Case 10	EP+OW+EQ+FS+BS
Load Case 11	HP+HW+HI
Load Case 12	HP+HW+HE
Load Case 13	IP+WE+EW
Load Case 14	IP+WF+CW
Load Case 15	IP+VO+OW
Load Case 16	IP+VE+EW
Load Case 17	NP+VO+OW
Load Case 18	FS+BS+IP+OW
Load Case 19	FS+BS+EP+OW
Load Case 20	BL+IP+OW

Wind Design Code	IS 875 (PART 3):2015-2020	
Basic Wind Speed for IS-875	180	Km/hr
Wind Zone Number	5	
Base Elevation	2600	mm.
Percent Wind for Hydrotest	33.0	
Risk Factor	[k1] 1.0	
Terrain Category	2	
Equipment Class	Class A	
Topography Factor	[k3] 1.0	
Importance Factor	[k4] 1.15	
Damping Factor (Beta) for Wind (Ope)	0.0100	
Damping Factor (Beta) for Wind (Empty)	0.0000	
Damping Factor (Beta) for Wind (Filled)	0.0000	

Seismic Design Code	IS 1893-RSM
Importance Factor for IS-1893	2.000
Soil Factor	1.000
Zone Number	2.000
Percent Seismic for Hydrotest	0.000
Period of Vibration (User defined) Sec	0.000
Damping Factor	2.000

User Spectrum Data points specified for increments of Time Period

Time Period	Sa/g
0.010	0.070
0.050	0.117
0.100	0.175
0.130	0.175
0.200	0.175
0.270	0.175
0.300	0.175
0.400	0.175
0.500	0.140
0.550	0.127
0.600	0.117
0.670	0.105
0.800	0.087
1.000	0.070
1.200	0.058
1.400	0.050



1.600 0.044
 1.800 0.039
 2.000 0.035
 2.400 0.029
 0.000 0.000

Consider Vertical Acceleration Yes
 Design Pressure + Static Head Y
 Consider MAP New and Cold in Noz. Design N
 Consider External Loads for Nozzle Des. Y
 Use ASME VIII-1 Appendix 1-9 N
 Perform Blast Load Analysis No
 Material Database Year Current w/Addenda or Code Year

Configuration Directives:

g0 is taken as min(g0, attached shell thk) Yes
 Do not use Nozzle MDMT Interpretation VIII-1 01-37 No
 Use Table G instead of exact equation for "A" Yes
 Shell Head Joints are Tapered Yes
 Compute "K" in corroded condition Yes
 Use Code Case 2286 No
 Use the MAWP to compute the MDMT Yes
 For thickness ratios <= 0.35, MDMT will be -155F (-104C) Yes
 For PWHT & P1 Materials the MDMT can be < -55F (-48C) No
 Using Metric Material Databases, ASME II D Yes
 Calculate B31.3 type stress for Nozzles with Loads Yes
 Reduce the MDMT due to lower membrane stress Yes
 Consider Longitudinal Stress in MDMT Calculations Yes

Complete Listing of Vessel Elements and Details:

Element# 1/7
 Element From Node 10
 Element To Node 20
 Element Type Elliptical
 Description Left D end
 Distance "FROM" to "TO" 50 mm.
 Inside Diameter 2300 mm.
 Element Thickness 21 mm.
 Internal Corrosion Allowance 1.5 mm.
 Nominal Thickness 25 mm.
 External Corrosion Allowance 0 mm.
 Design Internal Pressure 23 bars
 Design Temperature Internal Pressure 150 °C
 Design External Pressure 0 bars
 Design Temperature External Pressure 0 °C
 Effective Diameter Multiplier 1.2
 Material Name SA-516 70 +IT
 Allowable Stress, Ambient 138 N./mm2
 Allowable Stress, Operating 138 N./mm2
 Allowable Stress, Hydrotest 234 N./mm2
 Material Density 0.00775 kg./cm3
 P Number Thickness 32.004 mm.
 Yield Stress, Operating 232 N./mm2
 UCS-66 Chart Curve Designation Impact Tested
 External Pressure Chart Name CS-2
 UNS Number K02700
 Product Form Plate
 Efficiency, Longitudinal Seam 1.0
 Efficiency, Circumferential Seam 1.0
 Elliptical Head Factor 2.0
 Weld is pre-Heated No
 Element From Node 10
 Detail Type Platform
 Detail ID PLAT:[1 OF 1]



Dist. from "FROM" Node / Offset dist	0	mm.
Platform Start Angle (degrees)	0.0	
Platform End Angle (degrees)	0.0	
Platform Wind Area	171089	cm2
Platform Weight	4481.8	Kgf
Platform Railing Weight	0	Kgf/mm.
Platform Grating Weight	0.017	Kgs/cm2
Platform Width	2344	mm.
Platform Height	1200	mm.
Platform Clearance or End Offset	230	mm.
Platform Force Coefficient	1.2	
Ladder Layout Angle	0.0	
Ladder Start Elevation	0	mm.
Ladder End Elevation	2602	mm.
Unit Weight of Ladder	0.037	Kgf/mm.
Platform Length (top head platform)	11000	mm.

Element From Node	10	
Detail Type	Liquid	
Detail ID	Liquid: 10	
Dist. from "FROM" Node / Offset dist	0	mm.
Height/Length of Liquid	2300	mm.
Liquid Density	0.0006069	kg./cm3

Element From Node	10	
Detail Type	Insulation	
Detail ID	Ins: 10	
Dist. from "FROM" Node / Offset dist	-575	mm.
Height/Length of Insulation	625	mm.
Thickness of Insulation	70	mm.
Density	0.40000E-04	kg./cm3

Element From Node	10	
Detail Type	Nozzle	
Detail ID	MH124"	
Dist. from "FROM" Node / Offset dist	0	mm.
Nozzle Diameter	24	in.
Nozzle Schedule	None	
Nozzle Class	300	
Layout Angle	0.0	
Blind Flange (Y/N)	Y	
Weight of Nozzle (Used if > 0)	705.06	Kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-516 70 +IT	

Element From Node	10	
Detail Type	Weight	
Detail ID	manway davit	
Dist. from "FROM" Node / Offset dist	-575	mm.
Miscellaneous Weight	360	Kgf
Offset from Element Centerline	0	mm.

Element#	2/7	
Element From Node	20	
Element To Node	30	
Element Type	Cylinder	
Description	Shell 1	
Distance "FROM" to "TO"	2000	mm.
Element Outside Diameter	2344	mm.
Element Thickness	22	mm.
Internal Corrosion Allowance	1.5	mm.
Nominal Thickness	22	mm.
External Corrosion Allowance	0	mm.
Design Internal Pressure	23	bars
Design Temperature Internal Pressure	150	°C
Design External Pressure	0	bars
Design Temperature External Pressure	0	°C
Effective Diameter Multiplier	1.2	



Material Name	SA-516 70
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Weld is pre-Heated	No
Element From Node	20
Detail Type	Saddle
Detail ID	Fixed Saddle
Dist. from "FROM" Node / Offset dist	1400 mm.
Width of Saddle	180 mm.
Height of Saddle at Bottom	1430 mm.
Saddle Contact Angle	137.0
Height of Composite Ring Stiffener	0 mm.
Width of Wear Plate	230 mm.
Thickness of Wear Plate	12 mm.
Contact Angle, Wear Plate (degrees)	150.0
Friction coefficient	0.0
Moment Factor	3.0
Dimension E at base (optional)	0 mm.
Circumferential Eff. over Saddle	1.0
Circumferential Eff. at Midspan	1.0
Tangent to Tangent dist. (optional)	0 mm.
Element From Node	20
Detail Type	Liquid
Detail ID	Liquid 20
Dist. from "FROM" Node / Offset dist	0 mm.
Height/Length of Liquid	2300 mm.
Liquid Density	0.0006069 kg./cm3
Element From Node	20
Detail Type	Insulation
Detail ID	Ins: 10
Dist. from "FROM" Node / Offset dist	0 mm.
Height/Length of Insulation	2000 mm.
Thickness of Insulation	70 mm.
Density	0.40000E-04 kg./cm3
Element From Node	20
Detail Type	Nozzle
Detail ID	N18"
Dist. from "FROM" Node / Offset dist	400 mm.
Nozzle Diameter	8 in.
Nozzle Schedule	80
Nozzle Class	300
Layout Angle	90.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	68.695 Kg
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-333 6 +IT
Element From Node	20
Detail Type	Nozzle
Detail ID	N22"
Dist. from "FROM" Node / Offset dist	800 mm.
Nozzle Diameter	2 in.
Nozzle Schedule	160
Nozzle Class	600
Layout Angle	90.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	9.4586 Kg
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-333 6 +IT
Element From Node	20
Detail Type	Nozzle
Detail ID	N64"
Dist. from "FROM" Node / Offset dist	750 mm.
Nozzle Diameter	4 in.
Nozzle Schedule	80



Nozzle Class 300
 Layout Angle 270.0
 Blind Flange (Y/N) N
 Weight of Nozzle (Used if > 0) 19.243 Kgf
 Grade of Attached Flange GR 1.1
 Nozzle Matl SA-333 6 +IT

Element From Node 20
 Detail Type Weight
 Detail ID N1Dip pipe
 Dist. from "FROM" Node / Offset dist 400 mm.
 Miscellaneous Weight 140 Kgf
 Offset from Element Centerline 0 mm.

Element From Node 20
 Detail Type Weight
 Detail ID N2Dip pipe
 Dist. from "FROM" Node / Offset dist 800 mm.
 Miscellaneous Weight 25 Kgf
 Offset from Element Centerline 0 mm.

Element From Node 20
 Detail Type Weight
 Detail ID vortex N6
 Dist. from "FROM" Node / Offset dist 750 mm.
 Miscellaneous Weight 2 Kgf
 Offset from Element Centerline 1150 mm.

Element# 3/7
 Element From Node 30
 Element To Node 40
 Element Type Cylinder
 Description Shell 2
 Distance "FROM" to "TO" 2000 mm.
 Inside Diameter 2300 mm.
 Element Thickness 22 mm.
 Internal Corrosion Allowance 1.5 mm.
 Nominal Thickness 22 mm.
 External Corrosion Allowance 0 mm.
 Design Internal Pressure 23 bars
 Design Temperature Internal Pressure 150 °C
 Design External Pressure 0 bars
 Design Temperature External Pressure 0 °C
 Effective Diameter Multiplier 1.2
 Material Name SA-516 70
 Efficiency, Longitudinal Seam 1.0
 Efficiency, Circumferential Seam 1.0
 Weld is pre-Heated No

Element From Node 30
 Detail Type Liquid
 Detail ID Liquid 30
 Dist. from "FROM" Node / Offset dist 0 mm.
 Height/Length of Liquid 2300 mm.
 Liquid Density 0.0006069 kg./cm3

Element From Node 30
 Detail Type Insulation
 Detail ID Ins: 10
 Dist. from "FROM" Node / Offset dist 0 mm.
 Height/Length of Insulation 2000 mm.
 Thickness of Insulation 70 mm.
 Density 0.40000E-04 kg./cm3

Element From Node 30
 Detail Type Nozzle
 Detail ID K81"
 Dist. from "FROM" Node / Offset dist 1200 mm.



Nozzle Diameter 1 in.
 Nozzle Schedule None
 Nozzle Class 300
 Layout Angle 90.0
 Blind Flange (Y/N) N
 Weight of Nozzle (Used if > 0) 5.2464 Kgf
 Grade of Attached Flange GR 1.1
 Nozzle Matl SA-350 LF2 +IT

Element From Node 30
 Detail Type Nozzle
 Detail ID N101"
 Dist. from "FROM" Node / Offset dist 1500 mm.
 Nozzle Diameter 1 in.
 Nozzle Schedule None
 Nozzle Class 300
 Layout Angle 90.0
 Blind Flange (Y/N) N
 Weight of Nozzle (Used if > 0) 5.2464 Kgf
 Grade of Attached Flange GR 1.1
 Nozzle Matl SA-350 LF2 +IT

Element From Node 30
 Detail Type Weight
 Detail ID Baffle type A
 Dist. from "FROM" Node / Offset dist 1000 mm.
 Miscellaneous Weight 325 Kgf
 Offset from Element Centerline 0 mm.

Element# 4/7
 Element From Node 40
 Element To Node 50
 Element Type Cylinder
 Description Shell 3
 Distance "FROM" to "TO" 3000 mm.
 Inside Diameter 2300 mm.
 Element Thickness 22 mm.
 Internal Corrosion Allowance 1.5 mm.
 Nominal Thickness 22 mm.
 External Corrosion Allowance 0 mm.
 Design Internal Pressure 23 bars
 Design Temperature Internal Pressure 150 °C
 Design External Pressure 0 bars
 Design Temperature External Pressure 0 °C
 Effective Diameter Multiplier 1.2
 Material Name SA-516 70
 Efficiency, Longitudinal Seam 1.0
 Efficiency, Circumferential Seam 1.0
 Weld is pre-Heated No

Element From Node 40
 Detail Type Liquid
 Detail ID Liquid 40
 Dist. from "FROM" Node / Offset dist 0 mm.
 Height/Length of Liquid 2300 mm.
 Liquid Density 0.0006069 kg./cm3

Element From Node 40
 Detail Type Insulation
 Detail ID Ins: 10
 Dist. from "FROM" Node / Offset dist 0 mm.
 Height/Length of Insulation 3000 mm.
 Thickness of Insulation 70 mm.
 Density 0.40000E-04 kg./cm3

Element From Node 40
 Detail Type Nozzle
 Detail ID N720"



Dist. from "FROM" Node / Offset dist	1250 mm.
Nozzle Diameter	20 in.
Nozzle Schedule	None
Nozzle Class	300
Layout Angle	90.0
Blind Flange (Y/N)	Y
Weight of Nozzle (Used if > 0)	551.97 Kg
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-516 70 +IT

Element From Node	40
Detail Type	Nozzle
Detail ID	K3A2"
Dist. from "FROM" Node / Offset dist	2500 mm.
Nozzle Diameter	2 in.
Nozzle Schedule	None
Nozzle Class	300
Layout Angle	90.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	10.577 Kg
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-350 LF2 +IT

Element From Node	40
Detail Type	Nozzle
Detail ID	K3B2"
Dist. from "FROM" Node / Offset dist	2500 mm.
Nozzle Diameter	2 in.
Nozzle Schedule	None
Nozzle Class	300
Layout Angle	270.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	10.577 Kg
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-350 LF2 +IT

Element From Node	40
Detail Type	Nozzle
Detail ID	K1A2"
Dist. from "FROM" Node / Offset dist	2000 mm.
Nozzle Diameter	2 in.
Nozzle Schedule	None
Nozzle Class	300
Layout Angle	54.152699
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	10.577 Kg
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-350 LF2 +IT

Element From Node	40
Detail Type	Nozzle
Detail ID	K1B2"
Dist. from "FROM" Node / Offset dist	2000 mm.
Nozzle Diameter	2 in.
Nozzle Schedule	None
Nozzle Class	300
Layout Angle	-54.152699
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	10.577 Kg
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-350 LF2 +IT

Element From Node	40
Detail Type	Nozzle
Detail ID	K2A2"
Dist. from "FROM" Node / Offset dist	2500 mm.
Nozzle Diameter	2 in.
Nozzle Schedule	None
Nozzle Class	300
Layout Angle	54.152699



Blind Flange (Y/N) N
 Weight of Nozzle (Used if > 0) 10.577 Kg
 Grade of Attached Flange GR 1.1
 Nozzle Matl SA-350 LF2 +IT

Element From Node 40
 Detail Type Nozzle
 Detail ID K2B2"
 Dist. from "FROM" Node / Offset dist 2500 mm.
 Nozzle Diameter 2 in.
 Nozzle Schedule None
 Nozzle Class 300
 Layout Angle -54.152699
 Blind Flange (Y/N) N
 Weight of Nozzle (Used if > 0) 10.577 Kg
 Grade of Attached Flange GR 1.1
 Nozzle Matl SA-350 LF2 +IT

Element From Node 40
 Detail Type Weight
 Detail ID Clip Wt
 Dist. from "FROM" Node / Offset dist 1500 mm.
 Miscellaneous Weight 200 Kg
 Offset from Element Centerline 0 mm.

Element From Node 40
 Detail Type Weight
 Detail ID Lifting Wt
 Dist. from "FROM" Node / Offset dist 1700 mm.
 Miscellaneous Weight 630 Kg
 Offset from Element Centerline 0 mm.

Element From Node 40
 Detail Type Weight
 Detail ID Exchanger weigh
 Dist. from "FROM" Node / Offset dist 1250 mm.
 Miscellaneous Weight 2500 Kg
 Offset from Element Centerline 3000 mm.

Element From Node 40
 Detail Type Weight
 Detail ID baffle B
 Dist. from "FROM" Node / Offset dist 2000 mm.
 Miscellaneous Weight 325 Kg
 Offset from Element Centerline 0 mm.

Element From Node 40
 Detail Type Weight
 Detail ID Vessel internal
 Dist. from "FROM" Node / Offset dist 1250 mm.
 Miscellaneous Weight 700 Kg
 Offset from Element Centerline 2000 mm.

Element From Node 40
 Detail Type Weight
 Detail ID N7 ext int
 Dist. from "FROM" Node / Offset dist 1300 mm.
 Miscellaneous Weight 200 Kg
 Offset from Element Centerline 2000 mm.

Element# 5/7
 Element From Node 50
 Element To Node 60
 Element Type Cylinder
 Description Shell 4
 Distance "FROM" to "TO" 2000 mm.
 Inside Diameter 2300 mm.
 Element Thickness 22 mm.



Internal Corrosion Allowance	1.5	mm.
Nominal Thickness	22	mm.
External Corrosion Allowance	0	mm.
Design Internal Pressure	23	bars
Design Temperature Internal Pressure	150	°C
Design External Pressure	0	bars
Design Temperature External Pressure	0	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-516 70	
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Weld is pre-Heated	No	

Element From Node	50	
Detail Type	Liquid	
Detail ID	Liquid 50	
Dist. from "FROM" Node / Offset dist	0	mm.
Height/Length of Liquid	2300	mm.
Liquid Density	0.0006069	kg./cm ³

Element From Node	50	
Detail Type	Insulation	
Detail ID	Ins: 10	
Dist. from "FROM" Node / Offset dist	0	mm.
Height/Length of Insulation	2000	mm.
Thickness of Insulation	70	mm.
Density	0.40000E-04	kg./cm ³

Element From Node	50	
Detail Type	Nozzle	
Detail ID	N43"	
Dist. from "FROM" Node / Offset dist	1400	mm.
Nozzle Diameter	3	in.
Nozzle Schedule	80	
Nozzle Class	300	
Layout Angle	90.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	17.759	Kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-333 6 +IT	

Element#	6/7	
Element From Node	60	
Element To Node	70	
Element Type	Cylinder	
Description	Shell 5	
Distance "FROM" to "TO"	1900	mm.
Inside Diameter	2300	mm.
Element Thickness	22	mm.
Internal Corrosion Allowance	1.5	mm.
Nominal Thickness	22	mm.
External Corrosion Allowance	0	mm.
Design Internal Pressure	23	bars
Design Temperature Internal Pressure	150	°C
Design External Pressure	0	bars
Design Temperature External Pressure	0	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-516 70	
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Weld is pre-Heated	No	

Element From Node	60	
Detail Type	Saddle	
Detail ID	New Saddle	
Dist. from "FROM" Node / Offset dist	600	mm.
Width of Saddle	180	mm.
Height of Saddle at Bottom	1430	mm.



Saddle Contact Angle	137.0	
Height of Composite Ring Stiffener	0	mm.
Width of Wear Plate	230	mm.
Thickness of Wear Plate	12	mm.
Contact Angle, Wear Plate (degrees)	150.0	
Friction coefficient	0.1	
Moment Factor	3.0	
Dimension E at base (optional)	0	mm.
Circumferential Eff. over Saddle	1.0	
Circumferential Eff. at Midspan	1.0	
Tangent to Tangent dist. (optional)	0	mm.
Element From Node	60	
Detail Type	Liquid	
Detail ID	Liquid 60	
Dist. from "FROM" Node / Offset dist	0	mm.
Height/Length of Liquid	2300	mm.
Liquid Density	0.0006069	kg./cm3
Element From Node	60	
Detail Type	Insulation	
Detail ID	Ins: 10	
Dist. from "FROM" Node / Offset dist	0	mm.
Height/Length of Insulation	1900	mm.
Thickness of Insulation	70	mm.
Density	0.40000E-04	kg./cm3
Element From Node	60	
Detail Type	Nozzle	
Detail ID	K912"	
Dist. from "FROM" Node / Offset dist	1150	mm.
Nozzle Diameter	12	in.
Nozzle Schedule	None	
Nozzle Class	300	
Layout Angle	90.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	130.71	Kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-516 70	+IT
Element From Node	60	
Detail Type	Nozzle	
Detail ID	K71"	
Dist. from "FROM" Node / Offset dist	1500	mm.
Nozzle Diameter	1	in.
Nozzle Schedule	None	
Nozzle Class	300	
Layout Angle	46.4902	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	6.2779	Kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-350 LF2	+IT
Element From Node	60	
Detail Type	Nozzle	
Detail ID	N36"	
Dist. from "FROM" Node / Offset dist	1700	mm.
Nozzle Diameter	6	in.
Nozzle Schedule	80	
Nozzle Class	300	
Layout Angle	90.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	35.863	Kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-333 6	+IT
Element From Node	60	
Detail Type	Nozzle	
Detail ID	N88"	
Dist. from "FROM" Node / Offset dist	1370	mm.



Nozzle Diameter 8 in.
 Nozzle Schedule 80
 Nozzle Class 300
 Layout Angle 270.0
 Blind Flange (Y/N) N
 Weight of Nozzle (Used if > 0) 68.695 Kgf
 Grade of Attached Flange GR 1.1
 Nozzle Matl SA-333 6 +IT

Element From Node 60
 Detail Type Weight
 Detail ID N3Dip pipe
 Dist. from "FROM" Node / Offset dist 1700 mm.
 Miscellaneous Weight 90 Kgf
 Offset from Element Centerline 0 mm.

Element From Node 60
 Detail Type Weight
 Detail ID vortex N8
 Dist. from "FROM" Node / Offset dist 1370 mm.
 Miscellaneous Weight 5 Kgf
 Offset from Element Centerline 1150 mm.

Element# 7/7
 Element From Node 70
 Element To Node 80
 Element Type Elliptical
 Description Right D end
 Distance "FROM" to "TO" 50 mm.
 Inside Diameter 2300 mm.
 Element Thickness 21 mm.
 Internal Corrosion Allowance 1.5 mm.
 Nominal Thickness 25 mm.
 External Corrosion Allowance 0 mm.
 Design Internal Pressure 23 bars
 Design Temperature Internal Pressure 150 °C
 Design External Pressure 0 bars
 Design Temperature External Pressure 0 °C
 Effective Diameter Multiplier 1.2
 Material Name SA-516 70
 Efficiency, Longitudinal Seam 1.0
 Efficiency, Circumferential Seam 1.0
 Elliptical Head Factor 2.0
 Weld is pre-Heated No

Element From Node 70
 Detail Type Liquid
 Detail ID Liquid 70
 Dist. from "FROM" Node / Offset dist 0 mm.
 Height/Length of Liquid 2300 mm.
 Liquid Density 0.0006069 kg./cm3

Element From Node 70
 Detail Type Insulation
 Detail ID Ins: 10
 Dist. from "FROM" Node / Offset dist 0 mm.
 Height/Length of Insulation 625 mm.
 Thickness of Insulation 70 mm.
 Density 0.40000E-04 kg./cm3

Element From Node 70
 Detail Type Nozzle
 Detail ID MH224"
 Dist. from "FROM" Node / Offset dist 0 mm.
 Nozzle Diameter 24 in.
 Nozzle Schedule None
 Nozzle Class 300
 Layout Angle 0.0



Blind Flange (Y/N)	Y
Weight of Nozzle (Used if > 0)	747.03 Kgf
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-516 70 +IT

Element From Node	70
Detail Type	Weight
Detail ID	Manway davit
Dist. from "FROM" Node / Offset dist	625 mm.
Miscellaneous Weight	360 Kgf
Offset from Element Centerline	0 mm.

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XY Coordinate Calculations:

From	To	X (Horiz.) mm.	Y (Vert.) mm.	DX (Horiz.) mm.	DY (Vert.) mm.
Left D end		50	...	50	...
Shell 1		2050	...	2000	...
Shell 2		4050	...	2000	...
Shell 3		7050	...	3000	...
Shell 4		9050	...	2000	...
Shell 5		10950	...	1900	...
Right D end		11000	...	50	...

The calculated dimensions are based on the given element lengths. Due to variability in manufacturing (weld gaps etc.), the Tangent to Tangent length may not be exact. The length includes gasket thicknesses and raised face heights.

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Internal Pressure Results Summary:

Element Thickness, Pressure, Diameter and Allowable Stress :

From	To	Int. Press + Liq. Hd bars	Nominal Thickness mm.	Total Corr Allowance mm.	Element Diameter mm.	Allowable Stress(SE) N./mm ²	Element#
Left D end		23.137	25	1.5	2300	138	1
Shell 1		23.137	22	1.5	2344	138	2
Shell 2		23.137	22	1.5	2300	138	3
Shell 3		23.137	22	1.5	2300	138	4
Shell 4		23.137	22	1.5	2300	138	5
Shell 5		23.137	22	1.5	2300	138	6
Right D end		23.137	25	1.5	2300	138	7

Element Required Thickness and MAWP :

From	To	Design Pressure bars	M.A.W.P. Corroded bars	M.A.P. New & Cold bars	Minimum Thickness mm.	Required Thickness mm.	Element#
Left D end		23	23.2315	25.1521	21	20.8065	1
Shell 1		23	24.1693	26.0984	22	21.0204	2
Shell 2		23	24.1693	26.0984	22	21.0038	3
Shell 3		23	24.1693	26.0984	22	21.0038	4
Shell 4		23	24.1693	26.0984	22	21.0038	5
Shell 5		23	24.1693	26.0984	22	21.0038	6
Right D end		23	23.2315	25.1522	21	20.8065	7

Minimum 23 25

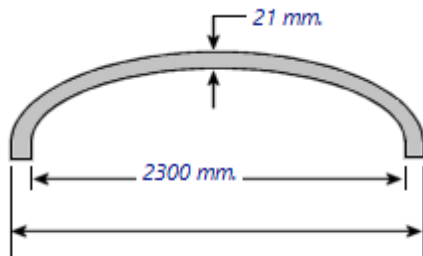
MAWP: 23 bars, limited by: Left D end.

Elements Suitable for Design Internal Pressure.

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Internal Pressure Calculation Results:

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Internal Pressure Results for: Left D end

Elliptical Head Design Information:							
Design Pressure	P	23.137 bars		Int. Design Temperature		150 °C	
Material		SA-516 70		Ext. Design Temperature		0 °C	
Length		50.000 mm.		External Pressure Chart		CS-2	
				UNS Number		K02700	
Int. Corr. All.	c	1.5000 mm.		Allowable Stress (ope)	S	138.0 N./mm^2	
Ext. Corr. All.	ce	0.0000 mm.		Allowable Stress (amb)	Sa	138.0 N./mm^2	
Inside Diameter	Di	2300.000 mm.		Actual Stress		136.6 N./mm^2	
Spec. Min. Thk t _e , or sa		21.0000 mm.		Spec. Nominal Thk.		25.000 mm.	
Surface Area		63554.5 cm^2		SG of Contents		0.608	
Element Volume		1800717.2 cm3		Weight of Contents		1093.9 Kg	
Empty Weight		1324.4 Kg		Operating Weight		2418.3 Kg	
				Comp. All. Ope. Stress		125.3 N./mm^2	
				Aspect Ratio		2.00	

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 28.750 mm.
 Head inner surface maximum deviation inside the specified shape, 0.625% of D: 14.375 mm.

Material treatment options: +IT

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot D \cdot K_{cor}) / (2 \cdot S \cdot E - 0.2 \cdot P) \text{ Appendix 1-4(c)}$$

$$= (23.137 \cdot 2303.0 \cdot 0.998) / (2 \cdot 138.0 \cdot 1.0 - 0.2 \cdot 23.137)$$

$$= 19.3065 + 1.5000 = 20.8065 \text{ mm.}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.137 bars

$$= (2 \cdot S \cdot E \cdot t) / (K_{cor} \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 138.0 \cdot 1.0 \cdot 19.5) / (0.998 \cdot 2303.0 + 0.2 \cdot 19.5)$$

$$= 23 - 0 = 23 \text{ bars}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (2 \cdot S \cdot E \cdot t) / (K \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 138.0 \cdot 1.0 \cdot 21.0) / (1.0 \cdot 2300.0 + 0.2 \cdot 21.0)$$

$$= 25 \text{ bars}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (K_{cor} \cdot D + 0.2 \cdot t)) / (2 \cdot E \cdot t)$$

$$= (23.137 \cdot (0.998 \cdot 2303.0 + 0.2 \cdot 19.5)) / (2 \cdot 1.0 \cdot 19.5)$$



= 136.630 N./mm²

Straight Flange Required Thickness:

$$\begin{aligned} &= (P \cdot R) / (S \cdot E - 0.6 \cdot P) + c \quad \text{per UG-27 (c)(1)} \\ &= (23.137 \cdot 1151.5) / (138.0 \cdot 1.0 - 0.6 \cdot 23.137) + 1.5 \\ &= 21.004 \text{ mm.} \end{aligned}$$

Straight Flange Maximum Allowable Working Pressure:

Less Operating Hydrostatic Head Pressure of 0.137 bars

$$\begin{aligned} &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \quad \text{per UG-27 (c)(1)} \\ &= (138.0 \cdot 1.0 \cdot 23.5) / (1151.5 + 0.6 \cdot 23.5) \\ &= 28 - 0 = 28 \text{ bars} \end{aligned}$$

Factor K, corroded condition [Kcor]:

$$\begin{aligned} &= (2 + (\text{Inside Diameter} / (2 \cdot \text{Inside Head Depth}))^2) / 6 \\ &= (2 + (2303.0 / (2 \cdot 576.5))^2) / 6 \\ &= 0.998267 \end{aligned}$$

Percent Elong. per UCS-79, VIII-1-01-57 $(75 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o)$ 4.647 %

MDMT Calculations in the Knuckle Portion:

Note: This Element/Detail was specified as being Impact Tested.

MDMT Calculations in the Head Straight Flange:

Note: This Element/Detail was specified as being Impact Tested.

Tolerance for Formed Heads per UG-81(a):

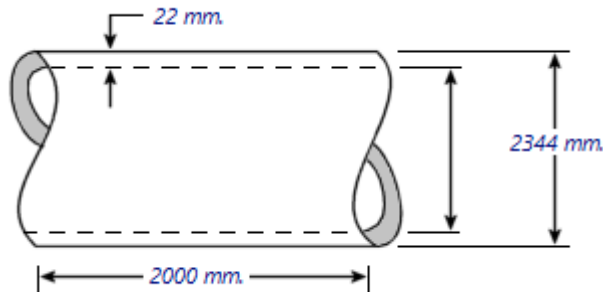
Head inner surface maximum deviation outside the specified shape, 1.25% of D: 28.750 mm.
Head inner surface maximum deviation inside the specified shape, 0.625% of D: 14.375 mm.

Material treatment options: +IT

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Internal Pressure Calculation Results:

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Internal Pressure Results for: Shell 1

Cylindrical Shell Design Information:					
Design Pressure	P	23.137 bars	Int. Design Temperature		150 °C
Material		SA-516 70	Ext. Design Temperature		0 °C
Length		2000.000 mm.	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	1.5000 mm.	Allowable Stress (ope)	S	138.0 N./mm^2
Ext. Corr. All.	ce	0.0000 mm.	Allowable Stress (amb)	Sa	138.0 N./mm^2
Outside Diameter	Do	2344.000 mm.	Actual Stress		131.4 N./mm^2
Spec. Min. Thk t _e , or s _a		22.0000 mm.	Spec. Nominal Thk.		22.000 mm.
Surface Area		147277.9 cm^2	SG of Contents		0.608
Element Volume		8311000.5 cm3	Weight of Contents		5048.9 Kg
Empty Weight		2487.7 Kg	Operating Weight		7536.6 Kg
			Comp. All. Ope. Stress		126.6 N./mm^2

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R_o) / (S \cdot E + 0.4 \cdot P) \text{ per Appendix 1-1 (a)(1)}$$

$$= (23.137 \cdot 1172.0) / (138.0 \cdot 1.0 + 0.4 \cdot 23.137)$$

$$= 19.5204 + 1.5000 = 21.0204 \text{ mm.}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.137 bars

$$= (S \cdot E \cdot t) / (R_o - 0.4 \cdot t) \text{ per Appendix 1-1 (a)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 20.5) / (1172.0 - 0.4 \cdot 20.5)$$

$$= 24 - 0 = 24 \text{ bars}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (R_o - 0.4 \cdot t) \text{ per Appendix 1-1 (a)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 22.0) / (1172.0 - 0.4 \cdot 22.0)$$

$$= 26 \text{ bars}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R_o - 0.4 \cdot t)) / (E \cdot t)$$

$$= (23.137 \cdot (1172.0 - 0.4 \cdot 20.5)) / (1.0 \cdot 20.5)$$

$$= 131.359 \text{ N./mm}^2$$

% Elongation per Table UG-79-1 (50 * t_{nom} / R_f * (1 - R_f / R_o)) 0.947 %

Minimum Design Metal Temperature Results:

Govrn. thk, t_g = 22.0, t_r = 19.714, c = 1.5 mm., E* = 1.0



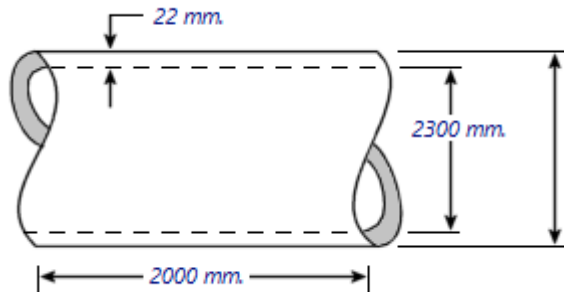
Thickness Ratio = $tr * E^* / (tg - c) = 0.962$, Temp. Reduction = 2 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -5 °C
 Min Metal Temp per UCS-66 and UCS-68(c),PWHT credit -21 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -23 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	23.369 bars
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-33 °C
Governing Thk.	22.000 mm.	Basic MDMT	-21 °C
Stress/Thk. Ratio	0.962	Temperature Difference	2 °C
UCS-66(c) red.	Yes	Computed Min. Temp.	-23 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E*	100 %

Internal Pressure Calculation Results:

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Internal Pressure Results for: Shell 2

Cylindrical Shell Design Information:					
Design Pressure	P	23.137 bars	Int. Design Temperature		150 °C
Material		SA-516 70	Ext. Design Temperature		0 °C
Length		2000.000 mm.	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	1.5000 mm.	Allowable Stress (ope)	S	138.0 N./mm^2
Ext. Corr. All.	ce	0.0000 mm.	Allowable Stress (amb)	Sa	138.0 N./mm^2
Inside Diameter	Di	2300.000 mm.	Actual Stress		131.4 N./mm^2
Spec. Min. Thk t _e , or sa		22.0000 mm.	Spec. Nominal Thk.		22.000 mm.
Surface Area		147277.9 cm^2	SG of Contents		0.608
Element Volume		8311002.5 cm3	Weight of Contents		5048.9 Kg
Empty Weight		2487.7 Kg	Operating Weight		7536.6 Kg
			Comp. All. Ope. Stress		126.6 N./mm^2

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (23.137 \cdot 1151.5) / (138.0 \cdot 1.0 - 0.6 \cdot 23.137)$$

$$= 19.5038 + 1.5000 = 21.0038 \text{ mm.}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.137 bars

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 20.5) / (1151.5 + 0.6 \cdot 20.5)$$

$$= 24 - 0 = 24 \text{ bars}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 22.0) / (1150.0 + 0.6 \cdot 22.0)$$

$$= 26 \text{ bars}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t)$$

$$= (23.137 \cdot (1151.5 + 0.6 \cdot 20.5)) / (1.0 \cdot 20.5)$$

$$= 131.359 \text{ N./mm}^2$$

% Elongation per Table UG-79-1 (50 * t_{nom} / R_f * (1 - R_f / R_o)) 0.947 %

Minimum Design Metal Temperature Results:

Govrn. thk, tg = 22.0, tr = 19.701, c = 1.5 mm., E* = 1.0



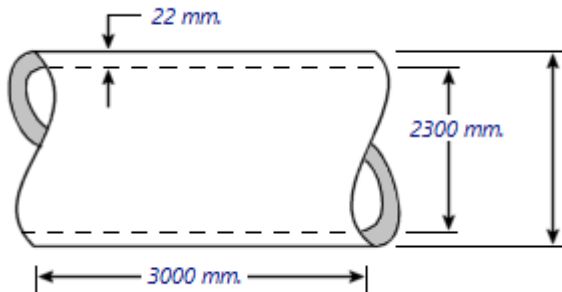
Thickness Ratio = $tr * E^* / (tg - c) = 0.961$, Temp. Reduction = 2 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -5 °C
 Min Metal Temp per UCS-66 and UCS-68(c),PWHT credit -21 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -23 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	23.369 bars
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-33 °C
Governing Thk.	22.000 mm.	Basic MDMT	-21 °C
Stress/Thk. Ratio	0.961	Temperature Difference	2 °C
UCS-66(c) red.	Yes	Computed Min. Temp.	-23 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E*	100 %

Internal Pressure Calculation Results:

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Internal Pressure Results for: Shell 3

Cylindrical Shell Design Information:					
Design Pressure	P	23.137 bars	Int. Design Temperature		150 °C
Material		SA-516 70	Ext. Design Temperature		0 °C
Length		3000.000 mm.	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	1.5000 mm.	Allowable Stress (ope)	S	138.0 N./mm^2
Ext. Corr. All.	ce	0.0000 mm.	Allowable Stress (amb)	Sa	138.0 N./mm^2
Inside Diameter	Di	2300.000 mm.	Actual Stress		131.4 N./mm^2
Spec. Min. Thk t _e , or sa		22.0000 mm.	Spec. Nominal Thk.		22.000 mm.
Surface Area		220916.8 cm^2	SG of Contents		0.608
Element Volume		12466504.0 cm3	Weight of Contents		7573.4 Kg
Empty Weight		3731.5 Kg	Operating Weight		11304.9 Kg
			Comp. All. Ope. Stress		126.6 N./mm^2

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (23.137 \cdot 1151.5) / (138.0 \cdot 1.0 - 0.6 \cdot 23.137)$$

$$= 19.5038 + 1.5000 = 21.0038 \text{ mm.}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.137 bars

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 20.5) / (1151.5 + 0.6 \cdot 20.5)$$

$$= 24 - 0 = 24 \text{ bars}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 22.0) / (1150.0 + 0.6 \cdot 22.0)$$

$$= 26 \text{ bars}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t)$$

$$= (23.137 \cdot (1151.5 + 0.6 \cdot 20.5)) / (1.0 \cdot 20.5)$$

$$= 131.359 \text{ N./mm}^2$$

% Elongation per Table UG-79-1 (50 * t_{nom} / R_f * (1 - R_f / R_o)) 0.947 %

Minimum Design Metal Temperature Results:

Govrn. thk, tg = 22.0, tr = 19.701, c = 1.5 mm., E* = 1.0



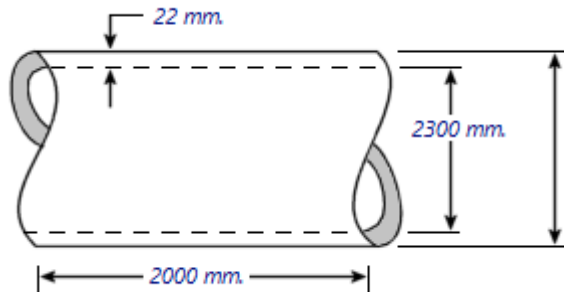
Thickness Ratio = $tr * E^* / (tg - c) = 0.961$, Temp. Reduction = 2 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -5 °C
 Min Metal Temp per UCS-66 and UCS-68(c),PWHT credit -21 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -23 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	23.369 bars
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-33 °C
Governing Thk.	22.000 mm.	Basic MDMT	-21 °C
Stress/Thk. Ratio	0.961	Temperature Difference	2 °C
UCS-66(c) red.	Yes	Computed Min. Temp.	-23 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E*	100 %

Internal Pressure Calculation Results:

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Internal Pressure Results for: Shell 4

Cylindrical Shell Design Information:					
Design Pressure	P	23.137 bars	Int. Design Temperature		150 °C
Material		SA-516 70	Ext. Design Temperature		0 °C
Length		2000.000 mm.	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	1.5000 mm.	Allowable Stress (ope)	S	138.0 N./mm^2
Ext. Corr. All.	ce	0.0000 mm.	Allowable Stress (amb)	Sa	138.0 N./mm^2
Inside Diameter	Di	2300.000 mm.	Actual Stress		131.4 N./mm^2
Spec. Min. Thk t _e , or s _a		22.0000 mm.	Spec. Nominal Thk.		22.000 mm.
Surface Area		147277.9 cm^2	SG of Contents		0.608
Element Volume		8311002.5 cm3	Weight of Contents		5048.9 Kg
Empty Weight		2487.7 Kg	Operating Weight		7536.6 Kg
			Comp. All. Ope. Stress		126.6 N./mm^2

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (23.137 \cdot 1151.5) / (138.0 \cdot 1.0 - 0.6 \cdot 23.137)$$

$$= 19.5038 + 1.5000 = 21.0038 \text{ mm.}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.137 bars

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 20.5) / (1151.5 + 0.6 \cdot 20.5)$$

$$= 24 - 0 = 24 \text{ bars}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 22.0) / (1150.0 + 0.6 \cdot 22.0)$$

$$= 26 \text{ bars}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t)$$

$$= (23.137 \cdot (1151.5 + 0.6 \cdot 20.5)) / (1.0 \cdot 20.5)$$

$$= 131.359 \text{ N./mm}^2$$

% Elongation per Table UG-79-1 (50 * t_{nom} / R_f * (1 - R_f / R_o)) 0.947 %

Minimum Design Metal Temperature Results:

Govrn. thk, tg = 22.0, tr = 19.701, c = 1.5 mm., E* = 1.0



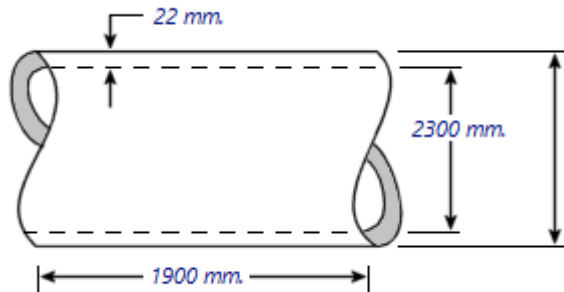
Thickness Ratio = $tr * E^* / (tg - c) = 0.961$, Temp. Reduction = 2 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -5 °C
 Min Metal Temp per UCS-66 and UCS-68(c),PWHT credit -21 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -23 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	23.369 bars
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-33 °C
Governing Thk.	22.000 mm.	Basic MDMT	-21 °C
Stress/Thk. Ratio	0.961	Temperature Difference	2 °C
UCS-66(c) red.	Yes	Computed Min. Temp.	-23 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E*	100 %

Internal Pressure Calculation Results:

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Internal Pressure Results for: Shell 5

Cylindrical Shell Design Information:					
Design Pressure	P	23.137 bars	Int. Design Temperature		150 °C
Material		SA-516 70	Ext. Design Temperature		0 °C
Length		1900.000 mm.	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	1.5000 mm.	Allowable Stress (ope)	S	138.0 N./mm^2
Ext. Corr. All.	ce	0.0000 mm.	Allowable Stress (amb)	Sa	138.0 N./mm^2
Inside Diameter	Di	2300.000 mm.	Actual Stress		131.4 N./mm^2
Spec. Min. Thk t _e , or s _a		22.0000 mm.	Spec. Nominal Thk.		22.000 mm.
Surface Area		139914.0 cm^2	SG of Contents		0.608
Element Volume		7895452.0 cm3	Weight of Contents		4796.5 Kg
Empty Weight		2363.3 Kg	Operating Weight		7159.8 Kg
			Comp. All. Ope. Stress		126.6 N./mm^2

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (23.137 \cdot 1151.5) / (138.0 \cdot 1.0 - 0.6 \cdot 23.137)$$

$$= 19.5038 + 1.5000 = 21.0038 \text{ mm.}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.137 bars

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 20.5) / (1151.5 + 0.6 \cdot 20.5)$$

$$= 24 - 0 = 24 \text{ bars}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 22.0) / (1150.0 + 0.6 \cdot 22.0)$$

$$= 26 \text{ bars}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t)$$

$$= (23.137 \cdot (1151.5 + 0.6 \cdot 20.5)) / (1.0 \cdot 20.5)$$

$$= 131.359 \text{ N./mm}^2$$

% Elongation per Table UG-79-1 (50 * t_{nom} / R_f * (1 - R_f / R_o)) 0.947 %

Minimum Design Metal Temperature Results:

Govrn. thk, tg = 22.0, tr = 19.701, c = 1.5 mm., E* = 1.0



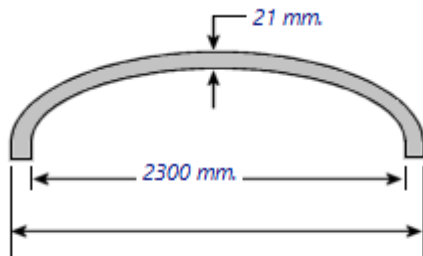
Thickness Ratio = $tr * E^* / (tg - c) = 0.961$, Temp. Reduction = 2 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -5 °C
 Min Metal Temp per UCS-66 and UCS-68(c),PWHT credit -21 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -23 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	23.369 bars
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-33 °C
Governing Thk.	22.000 mm.	Basic MDMT	-21 °C
Stress/Thk. Ratio	0.961	Temperature Difference	2 °C
UCS-66(c) red.	Yes	Computed Min. Temp.	-23 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E*	100 %

Internal Pressure Calculation Results:

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Internal Pressure Results for: Right D end

Elliptical Head Design Information:							
Design Pressure	P	23.137 bars		Int. Design Temperature		150 °C	
Material		SA-516 70		Ext. Design Temperature		0 °C	
Length		50.000 mm.		External Pressure Chart		CS-2	
				UNS Number		K02700	
Int. Corr. All.	c	1.5000 mm.		Allowable Stress (ope)	S	138.0 N./mm^2	
Ext. Corr. All.	ce	0.0000 mm.		Allowable Stress (amb)	Sa	138.0 N./mm^2	
Inside Diameter	Di	2300.000 mm.		Actual Stress		136.6 N./mm^2	
Spec. Min. Thk t,e, or sa		21.0000 mm.		Spec. Nominal Thk.		25.000 mm.	
Surface Area		63554.5 cm^2		SG of Contents		0.608	
Element Volume		1800717.2 cm3		Weight of Contents		1093.9 Kg	
Empty Weight		1324.4 Kg		Operating Weight		2418.3 Kg	
				Comp. All. Ope. Stress		126.6 N./mm^2	
				Aspect Ratio		2.00	

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 28.750 mm.
 Head inner surface maximum deviation inside the specified shape, 0.625% of D: 14.375 mm.

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot D \cdot K_{cor}) / (2 \cdot S \cdot E - 0.2 \cdot P) \text{ Appendix 1-4(c)}$$

$$= (23.137 \cdot 2303.0 \cdot 0.998) / (2 \cdot 138.0 \cdot 1.0 - 0.2 \cdot 23.137)$$

$$= 19.3065 + 1.5000 = 20.8065 \text{ mm.}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.137 bars

$$= (2 \cdot S \cdot E \cdot t) / (K_{cor} \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 138.0 \cdot 1.0 \cdot 19.5) / (0.998 \cdot 2303.0 + 0.2 \cdot 19.5)$$

$$= 23 - 0 = 23 \text{ bars}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (2 \cdot S \cdot E \cdot t) / (K \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 138.0 \cdot 1.0 \cdot 21.0) / (1.0 \cdot 2300.0 + 0.2 \cdot 21.0)$$

$$= 25 \text{ bars}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (K_{cor} \cdot D + 0.2 \cdot t)) / (2 \cdot E \cdot t)$$

$$= (23.137 \cdot (0.998 \cdot 2303.0 + 0.2 \cdot 19.5)) / (2 \cdot 1.0 \cdot 19.5)$$

$$= 136.630 \text{ N./mm}^2$$

Straight Flange Required Thickness:
 $= (P \cdot R) / (S \cdot E - 0.6 \cdot P) + c$ per UG-27 (c)(1)
 $= (23.137 \cdot 1151.5) / (138.0 \cdot 1.0 - 0.6 \cdot 23.137) + 1.5$
 $= 21.004$ mm.

Straight Flange Maximum Allowable Working Pressure:
 Less Operating Hydrostatic Head Pressure of 0.137 bars
 $= (S \cdot E \cdot t) / (R + 0.6 \cdot t)$ per UG-27 (c)(1)
 $= (138.0 \cdot 1.0 \cdot 23.5) / (1151.5 + 0.6 \cdot 23.5)$
 $= 28 - 0 = 28$ bars

Factor K, corroded condition [Kcor]:
 $= (2 + (\text{Inside Diameter} / (2 \cdot \text{Inside Head Depth}))^2) / 6$
 $= (2 + (2303.0 / (2 \cdot 576.5))^2) / 6$
 $= 0.998267$

Percent Elong. per UCS-79, VIII-1-01-57 $(75 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o)$ 4.647 %

MDMT Calculations in the Knuckle Portion:

Govrn. thk, $t_g = 21.0$, $t_r = 19.5$, $c = 1.5$ mm., $E^* = 1.0$
 Thickness Ratio $= t_r \cdot E^* / (t_g - c) = 1.$, Temp. Reduction = 0 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -6 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

MDMT Calculations in the Head Straight Flange:

Govrn. thk, $t_g = 25.0$, $t_r = 19.701$, $c = 1.5$ mm., $E^* = 1.0$
 Thickness Ratio $= t_r \cdot E^* / (t_g - c) = 0.838$, Temp. Reduction = 9 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -1 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -10 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	23.369 bars
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-33 °C
Governing Thk.	25.000 mm.	Basic MDMT	-1 °C
Stress/Thk. Ratio	0.838	Temperature Difference	9 °C
UCS-66(c) red.	No	Computed Min. Temp.	-10 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E^*	100 %

Note: Temperature values displayed above are for the location with the highest basic MDMT

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 28.750 mm.
 Head inner surface maximum deviation inside the specified shape, 0.625% of D: 14.375 mm.

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Hydrostatic Test Pressure Results:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	30.201 bars
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	29.900 bars
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	32.472 bars
UG100	= 1.10 * M.A.W.P. * Sa/S	25.555 bars
PED	= max(1.43*DP, 1.25*DP*ratio)	32.890 bars
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	32.890 bars
App 27-4	= M.A.W.P.	23.231 bars

UG-99(b), Test Pressure Calculation:
 = Test Factor • MAWP • Stress Ratio
 = 1.3 • 23.231 • 1.0
 = 30.201 bars

Horizontal Test performed per: UG-99b

Please note that Nozzle, Shell, Head, Flange, etc MAWPs are all considered when determining the hydrotest pressure for those test types that are based on the MAWP of the vessel.

Stresses on Elements due to Test Pressure (N./mm² & bars):

From To	Stress	Allowable	Ratio	Pressure
Left D end	166.9	234.0	0.713	30.43
Shell 1	160.9	234.0	0.688	30.43
Shell 2	160.9	234.0	0.688	30.43
Shell 3	160.9	234.0	0.688	30.43
Shell 4	160.9	234.0	0.688	30.43
Shell 5	160.9	234.0	0.688	30.43
Right D end	166.9	234.0	0.713	30.43

Stress ratios for Nozzle and Pad Materials (N./mm²):

Description	Pad/Nozzle	Ambient	Operating	Ratio
MH1_24"	Nozzle	138.00	138.00	1.000
MH1_24"	Pad	138.00	138.00	1.000
N1_8"	Nozzle	118.00	118.00	1.000
N1_8"	Pad	138.00	138.00	1.000
N2_2"	Nozzle	118.00	118.00	1.000
N6_4"	Nozzle	118.00	118.00	1.000
N6_4"	Pad	138.00	138.00	1.000
K8_1"	Nozzle	138.00	138.00	1.000
N10_1"	Nozzle	138.00	138.00	1.000
N7_20"	Nozzle	138.00	138.00	1.000
N7_20"	Pad	138.00	138.00	1.000
K3A_2"	Nozzle	138.00	138.00	1.000
K3B_2"	Nozzle	138.00	138.00	1.000
K1A_2"	Nozzle	138.00	138.00	1.000
K1B_2"	Nozzle	138.00	138.00	1.000
K2A_2"	Nozzle	138.00	138.00	1.000
K2B_2"	Nozzle	138.00	138.00	1.000
N4_3"	Nozzle	118.00	118.00	1.000
N4_3"	Pad	138.00	138.00	1.000
K9_12"	Nozzle	138.00	138.00	1.000
K9_12"	Pad	138.00	138.00	1.000
K7_1"	Nozzle	138.00	138.00	1.000
N3_6"	Nozzle	118.00	118.00	1.000
N3_6"	Pad	138.00	138.00	1.000
N8_8"	Nozzle	118.00	118.00	1.000
N8_8"	Pad	138.00	138.00	1.000
MH2_24"	Nozzle	138.00	138.00	1.000
MH2_24"	Pad	138.00	138.00	1.000

Minimum

1.000



Stress ratios for Pressurized Vessel Elements (N./mm²):

Description	Ambient	Operating	Ratio
Left D end	138.00	138.00	1.000
Shell 1	138.00	138.00	1.000
Shell 2	138.00	138.00	1.000
Shell 3	138.00	138.00	1.000
Shell 4	138.00	138.00	1.000
Shell 5	138.00	138.00	1.000
Right D end	138.00	138.00	1.000

Minimum

1.000

Hoop Stress in Nozzle Wall during Pressure Test (N./mm²):

Description	Ambient	Operating	Ratio
MH1_24"	117.76	179.40	0.656
N1_8"	28.78	153.40	0.188
N2_2"	10.79	153.40	0.070
N6_4"	22.00	153.40	0.143
K8_1"	4.54	179.40	0.025
N10_1"	4.54	179.40	0.025
N7_20"	79.11	179.40	0.441
K3A_2"	6.44	179.40	0.036
K3B_2"	6.44	179.40	0.036
K1A_2"	6.44	179.40	0.036
K1B_2"	6.44	179.40	0.036
K2A_2"	6.44	179.40	0.036
K2B_2"	6.44	179.40	0.036
N4_3"	19.07	216.00	0.088
K9_12"	48.20	179.40	0.269
K7_1"	4.54	179.40	0.025
N3_6"	25.45	153.40	0.166
N8_8"	28.78	153.40	0.188
MH2_24"	117.76	179.40	0.656

Elements Suitable for Test Pressure.

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External Pressure Calculation Results Summary:

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External Pressure Calculations:

From	To	Section Length mm.	Outside Diameter mm.	Corroded Thickness mm.	Factor A	Factor B N./mm ²
10	20	No Calc	2342	19.5	0.0011564	89.6426
20	30	11383.3	2344	20.5	0.00020475	20.4704
30	40	11383.3	2344	20.5	0.00020475	20.4704
40	50	11383.3	2344	20.5	0.00020475	20.4704
50	60	11383.3	2344	20.5	0.00020475	20.4704
60	70	11383.3	2344	20.5	0.00020475	20.4704
70	80	No Calc	2342	19.5	0.0011564	89.6426

External Pressure Calculations:

From	To	External Actual T. mm.	External Required T. mm.	External Design Pressure bars	External M.A.W.P. bars
10	20	21	No Calc	...	8.29267
20	30	22	No Calc	...	2.38691
30	40	22	No Calc	...	2.38691
40	50	22	No Calc	...	2.38691
50	60	22	No Calc	...	2.38691
60	70	22	No Calc	...	2.38691
70	80	21	No Calc	...	8.29267

Minimum

2

External Pressure Calculations:

From	To	Actual Length Bet. Stiffeners mm.	Allowable Length Bet. Stiffeners mm.	Ring Inertia Required cm ⁴	Ring Inertia Available cm ⁴
10	20	No Calc	No Calc	No Calc	No Calc
20	30	11383.3	No Calc	No Calc	No Calc
30	40	11383.3	No Calc	No Calc	No Calc
40	50	11383.3	No Calc	No Calc	No Calc
50	60	11383.3	No Calc	No Calc	No Calc
60	70	11383.3	No Calc	No Calc	No Calc
70	80	No Calc	No Calc	No Calc	No Calc

Elements Suitable for External Pressure.

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Element and Detail Weights:

From	To	Element Metal Wgt. kg.	Element ID Volume cm3	Corroded Metal Wgt. kg.	Corroded ID Volume cm3	Extra due Misc % kg.
10	20	1324.36	1800717	1244.9	1807500	92.705
20	30	2487.69	8311000	2319.56	8332696	174.138
30	40	2487.68	8311002	2319.57	8332696	174.137
40	50	3731.52	12466504	3479.35	12499045	261.206
50	60	2487.68	8311002	2319.57	8332696	174.137
60	70	2363.3	7895452	2203.59	7916061	165.431
70	80	1324.36	1800717	1244.9	1807500	92.705
Total		16206	48896396.00	15131	49028200.00	1134

Weight of Details:

From	Type	Weight of Detail kg.	X Offset, Dtl. Cent. mm.	Y Offset, Dtl. Cent. mm.	Z Offset Dtl. Cent. mm.	Description
10	Plat	4795.54	5500	1247.2	...	PLAT:[1 OF 1]
10	Liqd	1093.94	-191.667	...	...	Liquid: 10
10	Insl	23.7041	-262.5	...	...	Ins: 10
10	Nozl	754.413	-575	...	...	MH1_24"
10	Wght	360	-575	...	...	manway davit
20	Sadl	385.271	1400	1273	...	Fixed Saddle
20	Liqd	5048.95	1000	...	...	Liquid 20
20	Insl	42.5191	1000	...	...	Ins: 10
20	Nozl	73.5032	400	1248.42	...	N1_8"
20	Nozl	10.1207	800	1172.52	...	N2_2"
20	Nozl	20.5899	750	-1199.66	...	N6_4"
20	Wght	140	400	...	...	N1_Dip pipe
20	Wght	25	800	...	...	N2_Dip pipe
20	Wght	2	750	-1150	...	vortex N6
30	Liqd	5048.95	1000	...	...	Liquid 30
30	Insl	42.5191	1000	...	...	Ins: 10
30	Nozl	5.61363	1200	1162.7	...	K8_1"
30	Nozl	5.61363	1500	1162.7	...	N10_1"
30	Wght	325	1000	...	...	Baffle type A
40	Liqd	7573.42	1500	...	...	Liquid 40
40	Insl	63.7786	1500	...	...	Ins: 10
40	Nozl	590.608	1250	1404	...	N7_20"
40	Nozl	11.3175	2500	1175.4	...	K3A_2"
40	Nozl	11.3175	2500	-1175.4	...	K3B_2"
40	Nozl	11.3175	2000	952.756	688.346	K1A_2"
40	Nozl	11.3175	2000	-952.756	688.346	K1B_2"
40	Nozl	11.3175	2500	952.756	688.346	K2A_2"
40	Nozl	11.3175	2500	-952.756	688.346	K2B_2"
40	Wght	200	1500	...	...	Clip Wt
40	Wght	630	1700	...	...	Lifting Wt
40	Wght	2500	1250	3000	...	Exchanger weigh
40	Wght	325	2000	...	...	baffle B
40	Wght	700	1250	2000	...	Vessel internal
40	Wght	200	1300	2000	...	N7 ext int
50	Liqd	5048.95	1000	...	...	Liquid 50
50	Insl	42.5191	1000	...	...	Ins: 10
50	Nozl	19.002	1400	1194.45	...	N4_3"
60	Sadl	385.271	600	1273	...	New Saddle
60	Liqd	4796.5	950	...	...	Liquid 60
60	Insl	40.3931	950	...	...	Ins: 10
60	Nozl	139.863	1150	1302.4	...	K9_12"
60	Nozl	6.71732	1500	843.256	800.494	K7_1"
60	Nozl	38.3735	1700	1224.54	...	N3_6"
60	Nozl	73.5032	1370	-1248.43	...	N8_8"
60	Wght	90	1700	...	...	N3_Dip pipe
60	Wght	5	1370	-1150	...	vortex N8
70	Liqd	1093.94	241.667	...	...	Liquid 70



70 Insl	23.7041	312.5	...	...	Ins: 10
70 Nozl	799.324	625	...	...	MH2_24"
70 Wght	360	625	...	...	Manway davit

Total Weight of Each Detail Type:

Saddles	770.5
Platforms	4795.5
Liquid	29704.6
Insulation	279.1
Nozzles	2605.2
Weights	5862.0

Sum of the Detail Weights 44017.0 kg.

Weight Summation Results: (kg.)

	Fabricated	Shop Test	Shipping	Erected	Empty	Operating
Main Elements	17341.0	17341.0	17341.0	17341.0	17341.0	17341.0
Saddles	770.5	770.5	770.5	770.5	770.5	770.5
Nozzles	2605.2	2605.2	2605.2	2605.2	2605.2	2605.2
Wld Weights	4942.0	4942.0	4942.0	4942.0	4942.0	4942.0
Platforms	...	...	4795.5	4795.5	4795.5	4795.5
Insulation	...	...	279.1	279.1	279.1	279.1
Empty Weights	...	...	...	200.0	200.0	...
Test Weights	...	200.0	...	...	...	...
Ope Weights	...	...	...	...	...	920.0
Ope. Liquid	...	...	...	...	...	29704.6
Test Liquid	...	48866.6	...	...	...	...
Totals	25658.7	74725.3	30733.4	30933.4	30933.4	61358.0

Field Installation Options:

Miscellaneous Weight Percent: 7.0 %

Note that the above value for the miscellaneous weight percent has been applied to the shells/heads/flange/tubesheets/tubes etc. in the weight calculations for metallic components.

Weight Summary:

Fabricated Wt.	- Bare weight without removable internals	25658.7 kg.
Shop Test Wt.	- Fabricated weight + water (full)	74725.3 kg.
Shipping Wt.	- Fab. weight + removable intls.+ shipping app.	30733.4 kg.
Erected Wt.	- Fab. wt + or - loose items (trays,platforms etc.)	30933.4 kg.
Ope. Wt. no Liq	- Fab. weight + internals. + details + weights	30933.4 kg.
Operating Wt.	- Empty weight + operating liq. uncorroded	61358.0 kg.
Oper. Wt. + CA	- Corr wt. + operating liquid	60207.6 kg.

The Corroded Weight and thickness are used in the Horizontal Vessel Analysis (Ope Case) and Earthquake Load Calculations.

Surface Areas of Elements:

From	To	Outside Surface Area cm^2	Inside Surface Area cm^2
10	20	63554.5	60955.7
20	30	147278	144513
30	40	147278	144513
40	50	220917	216770
50	60	147278	144513
60	70	139914	137288
70	80	63554.5	60955.7

Total 929773.4 909508.6 cm2

PV Elite 28 Licensee: SL User (Modify in Config->Save Options)
FileName : 401-V-32001 - 7% Page 47 of 216
Element and Detail Weights: Step: 13 10:59am Jul 20,2026



HEXAGON

Total 1000.8 979.0 ft²

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Nozzle Flange MAWP Results: (bars & °C)

Nozzle Description	Flange Rating		Des. Temp	Class	Grade/Group	Equiv. Press	Max Pressure		
	Ope.	Ambient					UG-44	50%	DNV
MH1_24"	45.10	51.00	150	300	GR 1.1	...	...	...	...
N1_8"	45.10	51.00	150	300	GR 1.1	27.731	39.919	45.100	39.919
N2_2"	90.10	102.00	150	600	GR 1.1	72.402	62.749	90.101	62.749
N6_4"	45.10	51.00	150	300	GR 1.1	41.018	26.633	45.100	26.633
K8_1"	45.10	51.00	150	300	GR 1.1	...	...	...	...
N10_1"	45.10	51.00	150	300	GR 1.1	...	...	...	...
N7_20"	45.10	51.00	150	300	GR 1.1	...	...	...	...
K3A_2"	45.10	51.00	150	300	GR 1.1	...	...	...	...
K3B_2"	45.10	51.00	150	300	GR 1.1	...	...	...	...
K1A_2"	45.10	51.00	150	300	GR 1.1	...	...	...	...
K1B_2"	45.10	51.00	150	300	GR 1.1	...	...	...	...
K2A_2"	45.10	51.00	150	300	GR 1.1	...	...	...	...
K2B_2"	45.10	51.00	150	300	GR 1.1	...	...	...	...
N4_3"	45.10	51.00	150	300	GR 1.1	...	...	...	...
K9_12"	45.10	51.00	150	300	GR 1.1	20.955	45.100	45.100	45.100
K7_1"	45.10	51.00	150	300	GR 1.1	...	...	...	...
N3_6"	45.10	51.00	150	300	GR 1.1	32.066	35.584	45.100	35.584
N8_8"	45.10	51.00	150	300	GR 1.1	27.731	39.919	45.100	39.919
MH2_24"	45.10	51.00	150	300	GR 1.1	...	...	...	...
Min Rating	26.633	51.000, [for Core Elements]					...	...	...

Selected Method for Derating ASME B16.5 Flange MAWP: ASME UG-44(b)

The UG-44(b) Method is based on the paper PVP 2013-97814. PV Elite uses the maximum loads from each load category to determine ME and FE. In many cases, the computed maximum allowable pressure will be greater than the flange rating. In these cases, the minimum of the rating from the table and the UG-44(b) method will be used. SA-193 B8 Cl. 2 bolts or ones with higher allowable stresses at the specified bolt size shall be used.

The axial force FE must put the nozzle in tension (-P for 107/297; +Fr for PD 5500 AnnexG) to be considered.

Nozzle Description	Gasket Dimension G mm.	Fm Value	Resolved Moment ME Kg-m.	Axial Force FE Kg	MAWP PD bars	Maximum Pressure Pmax bars	Pass Fail
N1_8"	248.539	0.50	818	555	23.231	39.919	Pass
N2_2"	77.851	0.50	65	165	23.231	62.749	Pass
N6_4"	138.176	0.50	207	280	23.231	26.633	Pass
K9_12"	357.251	0.50	1839	830	23.231	45.100	Pass
N3_6"	196.088	0.50	464	410	23.231	35.584	Pass
N8_8"	248.539	0.50	818	555	23.231	39.919	Pass

Evaluating Flange on Nozzle [N18"]:

$$= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 \cdot [(PR - PD) + FM \cdot PR]$$

$$= 16(818) + 4 \cdot \text{abs}(555)248.539 \leq \pi \cdot 248.539^3 [(45.100 - 23.231) + .500 \cdot 45.100]$$

$$= 0.13640E+05 \leq 0.21848E+05 \text{ Check Passes}$$

Maximum Pressure for Flange on Nozzle N18" [Pmax]:

$$= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3))$$

$$= \min(45.100, 0.500 \cdot 45.100 + 45.100 - 13639.8 / (\pi \cdot 248.539^3))$$

$$= \min(45.100, 39.919)$$

$$= 39.919 \text{ bars}$$

Evaluating Flange on Nozzle [N22"]:

$$= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 \cdot [(PR - PD) + FM \cdot PR]$$

$$= 16(65) + 4 \cdot \text{abs}(165)77.851 \leq \pi \cdot 77.851^3 [(90.101 - 23.231) + .500 \cdot 90.101]$$

$$= 0.10945E+04 \leq 0.16918E+04 \text{ Check Passes}$$

Maximum Pressure for Flange on Nozzle N22" [Pmax]:

$$= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3))$$

$$= \min(90.101, 0.500 \cdot 90.101 + 90.101 - 1094.5 / (\pi \cdot 77.851^3))$$

$$= \min(90.101, 62.749)$$

$$= 62.749 \text{ bars}$$

Evaluating Flange on Nozzle [N64"]:

$$= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 [(PR - PD) + FM \cdot PR]$$

$$= 16(207) + 4 \cdot \text{abs}(280) 138.176 \leq \pi \cdot 138.176^3 [(45.100 - 23.231) + .500 \cdot 45.100]$$

$$= 0.34668E+04 \leq 0.37542E+04 \text{ Check Passes}$$

Maximum Pressure for Flange on Nozzle N64" [Pmax]:

$$= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3))$$

$$= \min(45.100, 0.500 \cdot 45.100 + 45.100 - 3466.8 / (\pi \cdot 138.176^3))$$

$$= \min(45.100, 26.633)$$

$$= 26.633 \text{ bars}$$

Evaluating Flange on Nozzle [K912"]:

$$= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 [(PR - PD) + FM \cdot PR]$$

$$= 16(1839) + 4 \cdot \text{abs}(830) 357.251 \leq \pi \cdot 357.251^3 [(45.100 - 23.231) + .500 \cdot 45.100]$$

$$= 0.30610E+05 \leq 0.64885E+05 \text{ Check Passes}$$

Maximum Pressure for Flange on Nozzle K912" [Pmax]:

$$= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3))$$

$$= \min(45.100, 0.500 \cdot 45.100 + 45.100 - 30610.0 / (\pi \cdot 357.251^3))$$

$$= \min(45.100, 46.696)$$

$$= 45.100 \text{ bars}$$

Evaluating Flange on Nozzle [N36"]:

$$= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 [(PR - PD) + FM \cdot PR]$$

$$= 16(464) + 4 \cdot \text{abs}(410) 196.088 \leq \pi \cdot 196.088^3 [(45.100 - 23.231) + .500 \cdot 45.100]$$

$$= 0.77457E+04 \leq 0.10729E+05 \text{ Check Passes}$$

Maximum Pressure for Flange on Nozzle N36" [Pmax]:

$$= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3))$$

$$= \min(45.100, 0.500 \cdot 45.100 + 45.100 - 7745.7 / (\pi \cdot 196.088^3))$$

$$= \min(45.100, 35.584)$$

$$= 35.584 \text{ bars}$$

Evaluating Flange on Nozzle [N88"]:

$$= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 [(PR - PD) + FM \cdot PR]$$

$$= 16(818) + 4 \cdot \text{abs}(555) 248.539 \leq \pi \cdot 248.539^3 [(45.100 - 23.231) + .500 \cdot 45.100]$$

$$= 0.13640E+05 \leq 0.21848E+05 \text{ Check Passes}$$

Maximum Pressure for Flange on Nozzle N88" [Pmax]:

$$= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3))$$

$$= \min(45.100, 0.500 \cdot 45.100 + 45.100 - 13639.8 / (\pi \cdot 248.539^3))$$

$$= \min(45.100, 39.919)$$

$$= 39.919 \text{ bars}$$

Flange Pressure Ratings are per ASME B16.5 2025 Metric Edition

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Wind Load Calculation Results:

Wind Load Calculations per India Std. IS 875 (PART 3):2015, Amd. 1&2 (2020):

Input Values:

Wind Design Code	IS 875 (PART 3):2015-2020
Basic Wind Speed for IS-875	180 Km/hr
Wind Zone Number	5
Base Elevation	2600 mm.
Percent Wind for Hydrotest	33.0
Risk Factor [k1]	1.0
Terrain Category	2
Equipment Class	Class A
Topography Factor [k3]	1.0
Importance Factor [k4]	1.15
Damping Factor (Beta) for Wind (Ope)	0.0100
Damping Factor (Beta) for Wind (Empty)	0.0000
Damping Factor (Beta) for Wind (Filled)	0.0000
Use Gust Response Factor (Dynamic Analysis)	No

Calculated Values:

Vessel Length to Diameter ratio: 4.696

Design Wind Speed [Vz]:

= Basic Wind Speed • k1 • k2,1 • k3 • k4 (for the 1st element)

= 50.0 • 1.0 • 1.0 • 1.0 • 1.15

= 57.496 meters/s

Design Wind Pressure per 4.5.2 [pd]:

= max(0.7 • pz, Kd • Ka • Kc • 0.6 • Vz²)

= max(0.7 • 1983.503, 1.0 • 0.88 • 1.0 • 0.6 • 57.5²)

= 1745.2 N/m²

Element Wind Load (operating condition):

= Wind Area • Cf • pd

= 1.4 • 0.79 • 1745.229

= 1995.0 N

From	Height mm.	k1	k2	k3	Vz m/sec	Cf
10	4030.00	1.0000	1.0000	1.0000	57.50	0.7899
20	4030.00	1.0000	1.0000	1.0000	57.50	0.7899
30	4030.00	1.0000	1.0000	1.0000	57.50	0.7899
40	4030.00	1.0000	1.0000	1.0000	57.50	0.7899
50	4030.00	1.0000	1.0000	1.0000	57.50	0.7899
60	4030.00	1.0000	1.0000	1.0000	57.50	0.7899
70	4030.00	1.0000	1.0000	1.0000	57.50	0.7899

Platform Load Calculations

ID	Wind Area cm2	Elevation mm.	Pressure Kgs/m2	Force Kgf	Cf
PLAT:[1 OF 1]	171089.00	4030.00	178.04	3046.06	1.20

Wind Loads on Masses/Equipment/Piping

ID	Wind Area cm2	Elevation mm.	Pressure Kgs/m2	Force Kgf
manway davit	0.00	4030.00	178.04	0.00
N1dip pipe	0.00	4030.00	178.04	0.00
N2dip pipe	0.00	4030.00	178.04	0.00
vortex N6	0.00	2600.00	178.04	0.00



Baffle type A	0.00	4030.00	178.04	0.00
Clip Wt	0.00	4030.00	178.04	0.00
Lifting Wt	0.00	4030.00	178.04	0.00
Exchanger wei	0.00	7030.00	178.26	0.00
baffle B	0.00	4030.00	178.04	0.00
Vessel intern	0.00	6030.00	178.19	0.00
N7 ext int	0.00	6030.00	178.19	0.00
N3Dip pipe	0.00	4030.00	178.04	0.00
vortex N8	0.00	2600.00	178.04	0.00
Manway davit	0.00	4030.00	178.04	0.00

Wind Load Calculation:

From	To	Wind Height mm.	Wind Diameter mm.	Wind Area cm^2	Wind Pressure Kgs/m^2	Element Wind Load Kgf
10	20	4030	2978.4	14471.9	178.039	3574.46
20	30	4030	2980.8	59616	178.039	921.889
30	40	4030	2980.8	59616	178.039	921.889
40	50	4030	2980.8	89424	178.039	1382.83
50	60	4030	2980.8	59616	178.039	921.889
60	70	4030	2980.8	56635.2	178.039	875.795
70	80	4030	2978.4	14471.9	178.039	223.791

The Wind Loads calculated and printed in the Wind Load calculation report have been factored by the input scalar/load reduction factor of: 1.100.

Be sure the wind speed is in accordance with the specified wind design code.

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Earthquake Load Calculation Results:

Seismic Analysis Results per IS 1893 (Part 4): 2024

Input Data:

Importance Factor as Entered by User I 2.0000
 Zone Number 2
 Soil Type Rocky or Hard Soil
 Response reduction factor (Force factor) R 2.0000
 Time Period of Vibration (program or user) 0.0303 Seconds
 Value of Sa/g from user Spectrum Data 0.0939
 Percent Damping 2.0 %
 Considering MCE for load determination No

Calculated results:

Horizontal Seismic Coefficient per 7.3.1 [Ah]:
 $= (Sa/g)_{ss} \cdot DampVal \cdot (I/R)$
 $= (0.0939 \cdot 1.0) \cdot 1.0$
 $= 0.094$

User Spectrum Data points specified for increments of Time Period

Time Period	Sa/g
0.010	0.070
0.050	0.117
0.100	0.175
0.130	0.175
0.200	0.175
0.270	0.175
0.300	0.175
0.400	0.175
0.500	0.140
0.550	0.127
0.600	0.117
0.670	0.105
0.800	0.087
1.000	0.070
1.200	0.058
1.400	0.050
1.600	0.044
1.800	0.039
2.000	0.035
2.400	0.029
0.000	0.000

Actual Time Period = 0.0303 secs
 Computed Sa/g = 0.0939

Note:

It was indicated to ignore the damping factor, assuming that the Site Spectrum specified is per the actual damping value of this vessel.
 The value of "DampVal" has been set to 1 (not used).

Earthquake Element Load, for the first Element:
 $= \text{Earthquake Weight} \cdot Ah \cdot \text{Scalar}$
 $= 6604.12 \cdot 0.0939 \cdot 1.0$
 $= 619.8 \text{ Kgf}$

Vertical load per 8.4:
 $= 2/3 \cdot Ah \cdot W \cdot \text{Scalar}$
 $= 2/3 \cdot 0.0939 \cdot 59437 \cdot 1.0$
 $= 3719.0 \text{ Kgf}$

Earthquake Load Calculation:

	Earthquake	Earthquake	Element
--	------------	------------	---------



From	To	Height mm.	Weight Kgf	Ope Load Kgf
10	20	1150	6604.12	619.837
20	30	1150	6604.12	619.837
20	30	1150	6604.12	619.837
30	40	1150	6604.12	619.837
40	50	1150	6604.12	619.837
50	60	1150	6604.12	619.837
60	0	1150	6604.12	619.837
60	70	1150	6604.12	619.837
70	80	1150	6604.12	619.837

The Earthquake Weight does not include the weight of the saddles.

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Shop/Field Installation Options :

Platform(s) installed in the Shop.
Insulation is installed in the Shop.

Note : The CG is computed from the first Element From Node

Center of Gravity of the Saddles	5550.000 mm.
Center of Gravity of the Platforms	5500.000 mm.
Center of Gravity of the Liquid	5500.000 mm.
Center of Gravity of the Insulation	5500.000 mm.
Center of Gravity of the Nozzles	5878.683 mm.
Center of Gravity of the Added Weights (Operating)	5251.697 mm.
Center of Gravity of the Added Weights (Empty)	5216.929 mm.
Center of Gravity of Bare Shell New and Cold	5499.999 mm.
Center of Gravity of Bare Shell Corroded	5500.002 mm.
Vessel CG in the Operating Condition	5492.851 mm.
Vessel CG in the Fabricated (Shop/Empty) Condition	5486.082 mm.
Vessel CG in the Test Condition	5494.605 mm.

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ASME VIII Division 2 Horizontal Vessel Analysis, Left Saddle:

Horizontal Vessel Stress Calculations : Operating Case

Note:

Wear Pad Width (230.00) is less than $1.56 \cdot \sqrt{r_m \cdot t}$
 and less than $2a$. The wear plate will be ignored.

Minimum Wear Plate Width to be considered in analysis [b1]:

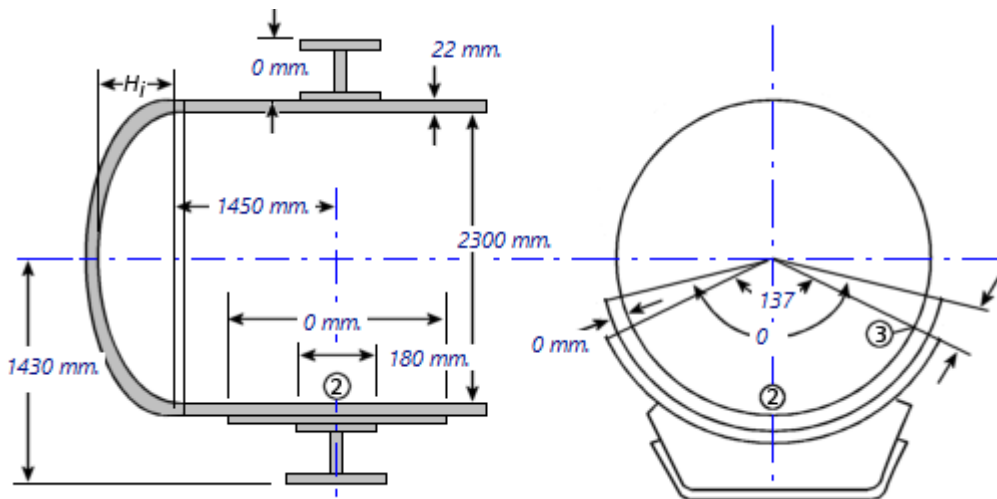
$$= \min(b + 1.56 \cdot \sqrt{R_m \cdot t}, 2a)$$

$$= \min(180.0 + 1.56 \cdot \sqrt{1161.75 \cdot 20.5}, 2 \cdot 1450.0)$$

$$= 420.7453 \text{ mm.}$$

Input and Calculated Values:

Vessel Mean Radius	Rm	1161.75	mm.
Shell Thickness used in this Case	t	22.000	mm.
Stiffened Vessel Length per 4.15.6	L	11000.00	mm.
Distance from Saddle to Vessel tangent	a1 or a	1450.00	mm.
Saddle Width	b1 or b	180.00	mm.
Saddle Bearing Angle	delta or theta	137.00	degrees
Inside or Mean Depth of Head	Hi or hm	586.25	mm.
Shell Allowable Stress used in Calculation		138.00	N./mm ²
Head Allowable Stress used in Calculation		138.00	N./mm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	1430.00	mm.
Coefficient of Friction	mu	0.00	



Results for Horizontal Vessel Analysis:

Saddle Force Q, Operating Case	40658.71	Kgf
Pressure used in Analysis	P	23.069 bars

Horizontal Vessel Analysis Results:		Actual N./mm ²	Allowable N./mm ²
Long. Stress at Top of Midspan		60.05	138.00
Long. Stress at Bottom of Midspan		70.69	138.00
Long. Stress at Top of Saddles		73.22	138.00
Long. Stress at Bottom of Saddles		60.93	138.00
Tangential Shear in Shell		10.76	110.40
Circ. Stress at Horn of Saddle		68.35	172.50

Circ. Compressive Stress in Shell	3.26	138.00
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Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:
 $= Ftr(Ft/Num\ of\ Saddles + Z\ Force\ Load) \cdot B / E$
 $= 3.0(8822.5/2 + 0) \cdot 1430.0/2185.0$
 $= 8661.0\ Kg$

Longitudinal Saddle Reaction Force [Fwl]:
 $= \max(Fl,wind; Sum\ of\ X\ Forces) \cdot B / Ls$
 $= \max(2027.72, 0) \cdot 1430.0/8200.0$
 $= 353.6\ Kg$

Saddle Reaction Force due to Earthquake Fl [Fsl]:
 $= \max(Fl,eq; Sum\ of\ X\ Forces) \cdot B / Ls$
 $= \max(5578.53, 0) \cdot 1430.0/8200.0$
 $= 972.8\ Kg$

Saddle Reaction Force due to Earthquake Ft [Fst]:
 $= Ftr(Ft/Num\ of\ Saddles + Z\ Force\ Load) \cdot B / E$
 $= 3.0(5579/2 + 0) \cdot 1430.0/2185.0$
 $= 5476.4\ Kg$

Load Combination Results for Q + Wind or Seismic [Q]:
 $= Saddle\ Load + \max(Fwl, Fwt, Fsl, Fst)$
 $= 31998 + \max(354, 8661, 973, 5476)$
 $= 40658.7\ Kg$

Longitudinal Wind Force [Fl]:
 $= WindScalar \cdot WindPress(Platform\ Area + (\pi/4(OD \cdot WindDiaMult)^2))$
 $= 1.1 \cdot 1745.935(3.375 + \pi/4(2.484 \cdot 1.2)^2)$
 $= 19884.830\ N$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	41043.99	Kgf
Transverse Shear Load	4411.27	Kgf
Longitudinal Shear Load	5578.53	Kgf

Note: Wear Plate is Welded to the Shell, $k = 0.1$

Saddle Dimension [E]:
 $= Baseplate\ Length$
 $= 2185.000\ mm.$

The Computed K values from Table 4.15.1:

K1 = 0.1356	K2 = 0.9345	K3 = 0.6284	K4 = 0.3370
K5 = 0.7052	K6 = 0.0399	K7 = 0.0399	K8 = 0.3208
K9 = 0.2427	K10 = 0.0455	K1* = 0.2397	

Note: Dimension a is greater than or equal to $Rm/2$.

Moment per Equation 4.15.1 [M1]:
 $= -Q \cdot a [1 - (1 - a/L + (Rm^2 - hm^2)/(2a \cdot L)) / (1 + (4hm)/3L)]$
 $= -40659 \cdot 1450.0 [1 - (1 - 1450.0/11000.0 + (1161.75^2 - 586.25^2) / (2 \cdot 1450.0 \cdot 11000.0)) / (1 + (4 \cdot 586.25) / (3 \cdot 11000.0))]$
 $= -9431.6\ Kg \cdot m.$

Moment per Equation 4.15.2 [M2]:
 $= Q \cdot L / 4 ((1 + 2(Rm^2 - hm^2)/(L^2)) / (1 + (4hm^2)/(3L)) - 4a/L)$
 $= 40659 \cdot 11000 / 4 ((1 + 2(1162^2 - 586^2)/(11000^2)) / (1 + (4 \cdot 586) / (3 \cdot 11000)) - 4 \cdot 1450 / 11000)$
 $= 47174.8\ Kg \cdot m.$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:
 $= P \cdot Rm / (2t) - M2 / (\pi \cdot Rm^2 t)$
 $= 23.069 \cdot 1161.75 / (2 \cdot 20.5) - 47174.8 / (\pi \cdot 1161.8^2 \cdot 20.5)$
 $= 60.05\ N./mm^2$



Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$= P \cdot R_m / (2t) + M_2 / (\pi \cdot R_m^2 \cdot t)$$

$$= 23.069 \cdot 1161.75 / (2 \cdot 20.5) + 47174.8 / (\pi \cdot 1161.8^2 \cdot 20.5)$$

$$= 70.69 \text{ N./mm}^2$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ_{*3}]:

$$= P \cdot R_m / (2t) - M_1 / (K_1 \cdot \pi \cdot R_m^2 \cdot t)$$

$$= 23.069 \cdot 1161.75 / (2 \cdot 20.5) - 9431.6 / (0.1356 \cdot \pi \cdot 1161.8^2 \cdot 20.5)$$

$$= 73.22 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ_{*4}]:

$$= P \cdot R_m / (2t) + M_1 / (K_1 \cdot \pi \cdot R_m^2 \cdot t)$$

$$= 23.069 \cdot 1161.75 / (2 \cdot 20.5) + 9431.6 / (0.2397 \cdot \pi \cdot 1161.8^2 \cdot 20.5)$$

$$= 60.93 \text{ N./mm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L-2a) / (L + (4 \cdot h_m / 3))$$

$$= 40659(11000.0 - 2 \cdot 1450.0) / (11000.0 + (4 \cdot 586.25 / 3))$$

$$= 27953.2 \text{ Kgf}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K_2 \cdot T / (R_m \cdot t)$$

$$= 0.9345 \cdot 27953.23 / (1161.75 \cdot 20.5)$$

$$= 10.76 \text{ N./mm}^2$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{(R_m \cdot t)}$$

$$= 0.78 \cdot \sqrt{(1161.75 \cdot 20.5)}$$

$$= 120.373 \text{ mm.}$$

Circumferential Stress in shell, no rings (4.15.21) [σ_6]:

$$= -K_5 \cdot Q \cdot k / (t(b + X_1 + X_2))$$

$$= -0.7052 \cdot 40659 \cdot 0.1 / (20.5(180.0 + 120.37 + 120.37))$$

$$= -3.26 \text{ N./mm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L \geq 8R_m$ (4.15.22) [σ_7]:

$$= -Q / (4 \cdot t(b + X_1 + X_2)) - 3 \cdot K_7 \cdot Q / (2 \cdot t^2)$$

$$= -40659 / (4 \cdot 20.5(180.0 + 120.373 + 120.373)) - 3 \cdot 0.0399 \cdot 40659 / (2 \cdot 20.5^2)$$

$$= -68.35 \text{ N./mm}^2$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(R_m \cdot t)}, 2a)$$

$$= \min(180.0 + 1.56 \cdot \sqrt{(1161.75 \cdot 20.5)}, 2 \cdot 1450.0)$$

$$= 420.75 \text{ mm.}$$

Distance between Saddle Supports [Ls]:

$$= 8200.0 \text{ mm.}$$

Free Un-Restrained Thermal Expansion between the Saddles [Exp]:

$$= \alpha \cdot L_s(\text{Design Temperature} - \text{Ambient Temperature})$$

$$= 0.000012 \cdot 8200.0(150.0 - 21.1)$$

$$= 13.133 \text{ mm.}$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	2185.0000	mm.
Baseplate Thickness	Bpthk	20.0000	mm.
Baseplate Width	Bpwid	200.0000	mm.
Number of Ribs (inc. outside ribs)	Nribs	3	
Rib Thickness	Ribtk	16.0000	mm.
Web Thickness	Webtk	16.0000	mm.
Web Location	Webloc	Center	
Saddle Design Temperature		21.1	°C
Saddle Yield Stress	Sy	232.0	N./mm ²
Height of Web at Center	Hw,c	306.0	mm.
Friction Coefficient	mu	0.000	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:



$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth}) t^3) / 12$$

$$= ((1.56 \cdot \sqrt{(1151.5 \cdot 20.5)} + 230.0) 20.5^3) / 12$$

$$= 33.720 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}} (C1 - Y)^2$$

$$= 33.72 + 96.285 (88.578 - 10.25)^2$$

$$= 5941.075 \text{ cm}^4$$

	B mm.	D mm.	Y mm.	A cm ²	AY mm. ³	I + AD ⁽²⁾ cm ⁴
Shell	469.7	20.5	10.2	96.3	98691.7	0.594E+04
Wearplate	230.0	12.0	26.5	27.6	73140.0	0.107E+04
Web	16.0	226.0	145.5	36.2	526128.0	0.271E+04
BasePlate	200.0	20.0	268.5	40.0	1074000.0	0.130E+05
Totals	...	...	...	200.0	1771959.6	0.227E+05

Distance to Centroid [C1]:

$$= AY / A$$

$$= 1771960 / 200.045$$

$$= 88.578 \text{ mm.}$$

Angle [β]:

$$= 180 - \text{Saddle Angle} / 2$$

$$= 180 - 137.0 / 2$$

$$= 111.5$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(111.5) - 0.5 \cdot \sin(111.5)^2) / (\pi - 1.946 + \sin(111.5) \cos(111.5))$$

$$= 0.2348$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.235 \cdot 40658.715$$

$$= 9547.2188 \text{ Kgf}$$

Tension Stress, St = (Fh/As) = 9.0235 N./mm2
 Allowed Stress, Sa = 0.6 * Yield Str = 139.1985 N./mm2

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\text{theta}/2) / (\text{theta}/2 \text{ in radians})$$

$$= 1430.0 - 1151.5 \cdot \sin(137.0/2) / 1.1956$$

$$= 533.864 \text{ mm.}$$

Bending Moment, M = Fh * d = 5097.0186 Kg-m.

Bending Stress, Sb = (M * C1 / I) = 19.5211 N./mm2
 Allowed Stress, Sa = 2/3 * Yield Str = 154.6650 N./mm2

Minimum Thickness of Baseplate per Moss:

$$= \sqrt{3(Q + \text{SaddleWt}) \text{BasePlateWidth} / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress})}$$

$$= \sqrt{3(40659 + 385)200.0 / (4 \cdot 2185.0 \cdot 154.665)}$$

$$= 13.366 \text{ mm.}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle} / 2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 137.0/2) (1150.0 + 22.0 + 12.0)$$

$$= 2203.229 \text{ mm.}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (\text{Nr ribs} - 1)$$

$$= 2203.2288 / (3 - 1)$$

$$= 1101.614 \text{ mm.}$$

Baseplate Pressure Area [Ap]:

$$= e \cdot \text{Bpwid} / 2$$

$$= 1101.6144 \cdot 200.0/2$$

$$= 1101.615 \text{ cm}^2$$

Bearing Pressure [Bp]:

$$= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$$

$$= 41043.988 / (2185.0 \cdot 200.0)$$

$$= 9.392 \text{ Kg/cm}^2$$

Axial Load [P]:

$$= A_p \cdot B_p$$

$$= 1101.6 \cdot 9.39$$

$$= 10346.603 \text{ Kg}$$

Area of the Rib and Web [Ar]:

$$= \text{Rib Area} + \text{Web Area}$$

$$= 26.24 + 88.129$$

$$= 114.369 \text{ cm}^2$$

Compressive Stress [Sc]:

$$= P / A_r$$

$$= 10346.6 / 114.3692$$

$$= 8.872 \text{ N./mm}^2$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	16.0	180.0	...	28.8	...	778.

Rib dimension [D]:

$$= \text{Saddle Width} - \text{Web Thickness}$$

$$= 180.0 - 16.0$$

$$= 164.000 \text{ mm.}$$

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0.0 / 114.369$$

$$= 0.000 \text{ mm.}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 180.0 / 2$$

$$= 90.000 \text{ mm.}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(777.6 / 114.369)}$$

$$= 26.075 \text{ mm.}$$

Length of Outer Rib, step 1, [Lv]:

$$= \text{Saddle Height} - \cos(\text{saddle bearing angle} / 2) (\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk}$$

$$= 1430.0 - \cos(137.0/2) (1150.0 + 22.0 + 12.0) - 20.0$$

$$= 976.063 \text{ mm.}$$

Length of Outer Rib, step 2, [Lw]:

$$= 2 \cdot \sin(\text{saddle bearing angle} / 2) (\text{radius} + \text{shlthk} + \text{wpdthk})$$

$$= 2 \cdot \sin(137.0/2) (1150.0 + 22.0 + 12.0)$$

$$= 2203.229 \text{ mm.}$$

Length of Outer Rib, step 3, [Bd]:

$$= \text{Baseplate length} - \text{clearance}$$

$$= 2185.0 - 50.8$$

$$= 2134.200 \text{ mm.}$$

Length of Outer Rib, step 4, [Dd]:

$$= (L_w - B_d) / 2$$

$$= (2203.229 - 2134.2) / 2$$

$$= 34.514 \text{ mm.}$$

Length of Outer Rib [L]:

$$= \sqrt{(L_v^2 + D_d^2)}$$

$$= \sqrt{(976.063^2 + 34.514^2)}$$

$$= 976.673 \text{ mm.}$$

Intermediate Term [Cc]:

$$= \sqrt{(2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress})}$$

$$= \sqrt{(2 \cdot \pi^2 \cdot 0.19994\text{E}+09/232.0)}$$

$$= 130.433$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 976.673/26.075$$

$$= 37.456$$

Bending Moment [Rm]:

$$= FI / (2 \cdot B_{plen}) \cdot e \cdot L / 2$$

$$= 5578.5 / (2 \cdot 2185.0) \cdot 1101.614 \cdot 976.67/2$$

$$= 686.745 \text{ Kg-m.}$$

Compressive Allowable, $KL/r < C_c$ (37.4563 < 130.4335) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (37.46)^2 / (2 \cdot 130.43^2)) 232 /$$

$$(5/3 + 3 \cdot (37.46) / (8 \cdot 130.43) - (37.46^3) / (8 \cdot 130.43^3))$$

$$= 125.6 \text{ N/mm}^2$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= S_c / S_{ca} + (R_m \cdot C_1 / I) / S_{ba}$$

$$= 8.87/125.57 + (686.75 \cdot 90.0/7775999) / 154.66$$

$$= 0.575$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	16.0	164.0	0.0	26.2	0.0	777.
Web	1101.6	16.0	0.0	176.3	0.0	37.6
Totals	...	...	...	202.5	...	815.

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0.0/202.498$$

$$= 0.000 \text{ mm.}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 180.0/2$$

$$= 90.000 \text{ mm.}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \text{Outside Radius} - B_{pthk}$$

$$= 1430.0 - \cos(1184.0/2) (20.0 + 0.0 + 0.0) - 0.0$$

$$= 226.000 \text{ mm.}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(814.7/202.498)}$$

$$= 20.057 \text{ mm.}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 226.0/20.057$$

$$= 11.268$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 5578.532 / (2 \cdot 2185.0)$$

$$= 1.277 \text{ Kg/mm.}$$



Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 1.277 \cdot 1101.614 \cdot 226.0$$

$$= 317.823 \text{ Kg-m.}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 317.823/90517.281$$

$$= 34.433 \text{ N./mm}^2$$

Compressive Allowable, $KL/r < C_c$ (11.2676 < 130.4335) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (11.27)^2 / (2 \cdot 130.43^2)) 232 / (5/3 + 3 \cdot (11.27) / (8 \cdot 130.43) - (11.27)^3 / (8 \cdot 130.43^3))$$

$$= 136 \text{ N./mm}^2$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= S_c / S_{ca} + (M_{base,c} \cdot C_{1/I}) / S_{ba}$$

$$= 10.06/136.04 + (317.82 \cdot 90.0/814.655) / 154.66$$

$$= 0.297$$

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	4
Total Number of Bolts in Tension/Baseplate	Nbt	2
Bolt Design Temperature		21.1 °C
Bolt Material Specification		SA-285 C
Bolt Allowable Stress	Stba	137.30 N./mm ²
Bolt Corrosion Allowance	Bca	3.0 mm.
Distance from Bolts to Edge	Edgedis	50.8 mm.
User Defined Area in a Single Bolt	UserBoltArea	5.4000 cm ²
BasePlate Allowable Stress	S	138.00 N./mm ²
Area Available in a Single Bolt	BlArea	3.2115 cm ²
Saddle Load QO (Weight)	QO	32382.9 Kg
Saddle Load QL (Wind/Seismic contribution)	QL	972.8 Kg
Maximum Transverse Force	Ft	4411.3 Kg
Maximum Longitudinal Force	Fl	5578.5 Kg
Saddle Bolted to Steel Foundation		No

Friction force due to thermal growth, [Ff]:

$$= 3154.4 \text{ Kg}$$

Shear Stress in a Single Bolt, Longitudinal Direction [τ_b ,l]:

$$= \max(F_f, F_l) / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 5578.5 / (3.211 \cdot 4)$$

$$= 42.6 \text{ N./mm}^2. \text{ Must be less than } 107.9 \text{ N./mm}^2.$$

Shear Stress in a Single Bolt, Transverse Direction [τ_b ,t]:

$$= F_t / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 4411.3 / (3.211 \cdot 4)$$

$$= 33.7 \text{ N./mm}^2. \text{ Must be less than } 107.9 \text{ N./mm}^2.$$

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:

$$= 0.0 \text{ (} Q_O > Q_L \text{ --> No Uplift in Longitudinal direction)}$$

Bolt Area due to Shear Load [Bltarears]:

$$= F_l / (\text{BoltShearAllowable} \cdot \text{Nbolts})$$

$$= 5578.53 / (107.87 \cdot 4.0)$$

$$= 1.2679 \text{ cm}^2$$

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:

$$= B \cdot F_t + \text{Sum of X Moments}$$

$$= 1430.0 \cdot 4411.27 + 0.0$$

$$= 6308.25 \text{ Kg-m.}$$



Eccentricity (e):
 = Rmom / QO
 = 6308.25/32382.95
 = 194.80 mm. < Bplen/6 --> No Uplift in Transverse direction

Bolt Area due to Transverse Load [Bltareart]:
 = 0 (No Uplift)

Required Area of a Single Bolt [Bltarear]:
 = max[Bltarearl, Bltarears, Bltareart]
 = max[0.0, 1.2679, 0.0]
 = 1.2679 cm²

MDMT Calculations for: Fixed Saddle, Saddle to Wearplate Junction, Curve: B

Govern. thk, tg = 12.0, c = 1.5 mm., E* = 1.0
 Thickness Ratio = tr * E*/(tg - c) = 0.962, Temp. Reduction = 2 °C
 Attachment governing, Attachment stress = Shell stress per Code

Min Metal Temp. w/o impact per UCS-66, Curve B -23 °C
 Min Metal Temp per UCS-66 and UCS-68(c), PWHt credit -40 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -42 °C

ASME VIII Division 2 Horizontal Vessel Analysis, Right Saddle:

Note:
 Wear Pad Width (230.00) is less than $1.56 \cdot \sqrt{rm \cdot t}$
 and less than 2a. The wear plate will be ignored.

Minimum Wear Plate Width to be considered in analysis [b1]:
 = $\min(b + 1.56 \cdot \sqrt{Rm \cdot t}, 2a)$
 = $\min(180.0 + 1.56 \cdot \sqrt{1161.75 \cdot 20.5}, 2 \cdot 1450.0)$
 = 420.7453 mm.

Input and Calculated Values:

Vessel Mean Radius	Rm	1161.75	mm.
Shell Thickness used in this Case	t	22.000	mm.
Stiffened Vessel Length per 4.15.6	L	11000.00	mm.
Distance from Saddle to Vessel tangent	a1 or a	1450.00	mm.
Saddle Width	b1 or b	180.00	mm.
Saddle Bearing Angle	delta or theta	137.00	degrees
Inside or Mean Depth of Head	Hi or hm	586.25	mm.
Shell Allowable Stress used in Calculation		138.00	N./mm ²
Head Allowable Stress used in Calculation		138.00	N./mm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	1430.00	mm.
Coefficient of Friction	mu	0.10	

Results for Horizontal Vessel Analysis:

Saddle Force Q, Operating Case		39819.49	Kgf
Pressure used in Analysis	P	23.069	bars

Horizontal Vessel Analysis Results:	Actual N./mm ²	Allowable N./mm ²
Long. Stress at Top of Midspan	60.16	138.00
Long. Stress at Bottom of Midspan	70.58	138.00
Long. Stress at Top of Saddles	73.05	138.00
Long. Stress at Bottom of Saddles	61.02	138.00
Tangential Shear in Shell	10.53	110.40
Circ. Stress at Horn of Saddle	66.94	172.50

Circ. Compressive Stress in Shell 3.19 138.00

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:
 $= Ftr(Ft/Num\ of\ Saddles + Z\ Force\ Load) \cdot B / E$
 $= 3.0(8822.5/2 + 0) \cdot 1430.0/2185.0$
 $= 8661.0\ Kg$

Longitudinal Saddle Reaction Force [Fwl]:
 $= \max(Fl,wind; Sum\ of\ X\ Forces) \cdot B / Ls$
 $= \max(2027.72, 0) \cdot 1430.0/8200.0$
 $= 353.6\ Kg$

Saddle Reaction Force due to Earthquake Fl [Fsl]:
 $= \max(Fl,eq; Sum\ of\ X\ Forces) \cdot B / Ls$
 $= \max(5578.53, 0) \cdot 1430.0/8200.0$
 $= 972.8\ Kg$

Saddle Reaction Force due to Earthquake Ft [Fst]:
 $= Ftr(Ft/Num\ of\ Saddles + Z\ Force\ Load) \cdot B / E$
 $= 3.0(5579/2 + 0) \cdot 1430.0/2185.0$
 $= 5476.4\ Kg$

Load Combination Results for Q + Wind or Seismic [Q]:
 $= Saddle\ Load + \max(Fwl, Fwt, Fsl, Fst)$
 $= 31158 + \max(354, 8661, 973, 5476)$
 $= 39819.5\ Kg$

Longitudinal Wind Force [FI]:
 $= WindScalar \cdot WindPress(Platform\ Area + (\pi/4(OD \cdot WindDiaMult)^2))$
 $= 1.1 \cdot 1745.935(3.375 + \pi/4(2.484 \cdot 1.2)^2)$
 $= 19884.834\ N$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	40204.76	Kgf
Transverse Shear Load	4411.27	Kgf
Longitudinal Shear Load	5578.53	Kgf

Note: Wear Plate is Welded to the Shell, $k = 0.1$

Saddle Dimension [E]:
 $= Baseplate\ Length$
 $= 2185.000\ mm.$

The Computed K values from Table 4.15.1:

K1 = 0.1356	K2 = 0.9345	K3 = 0.6284	K4 = 0.3370
K5 = 0.7052	K6 = 0.0399	K7 = 0.0399	K8 = 0.3208
K9 = 0.2427	K10 = 0.0455	K1* = 0.2397	

Note: Dimension a is greater than or equal to $Rm/2$.

Moment per Equation 4.15.1 [M1]:
 $= -Q \cdot a [1 - (1 - a/L + (Rm^2 - hm^2)/(2a \cdot L)) / (1 + (4hm)/3L)]$
 $= -39819 \cdot 1450.0 [1 - (1 - 1450.0/11000.0 + (1161.75^2 - 586.25^2) / (2 \cdot 1450.0 \cdot 11000.0)) / (1 + (4 \cdot 586.25) / (3 \cdot 11000.0))]$
 $= -9236.9\ Kg\cdot m.$

Moment per Equation 4.15.2 [M2]:
 $= Q \cdot L / 4 ((1 + 2(Rm^2 - hm^2)/(L^2)) / (1 + (4hm^2)/(3L)) - 4a/L)$
 $= 39819 \cdot 11000 / 4 ((1 + 2(1162^2 - 586^2)/(11000^2)) / (1 + (4 \cdot 586) / (3 \cdot 11000)) - 4 \cdot 1450 / 11000)$
 $= 46201.1\ Kg\cdot m.$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:
 $= P \cdot Rm / (2t) - M2 / (\pi \cdot Rm^2 t)$
 $= 23.069 \cdot 1161.75 / (2 \cdot 20.5) - 46201.1 / (\pi \cdot 1161.8^2 \cdot 20.5)$
 $= 60.16\ N./mm^2$



Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$= P \cdot R_m / (2t) + M_2 / (\pi \cdot R_m^2 \cdot t)$$

$$= 23.069 \cdot 1161.75 / (2 \cdot 20.5) + 46201.1 / (\pi \cdot 1161.8^2 \cdot 20.5)$$

$$= 70.58 \text{ N./mm}^2$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ_{*3}]:

$$= P \cdot R_m / (2t) - M_1 / (K_1 \cdot \pi \cdot R_m^2 \cdot t)$$

$$= 23.069 \cdot 1161.75 / (2 \cdot 20.5) - 9236.9 / (0.1356 \cdot \pi \cdot 1161.8^2 \cdot 20.5)$$

$$= 73.05 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ_{*4}]:

$$= P \cdot R_m / (2t) + M_1 / (K_1 \cdot \pi \cdot R_m^2 \cdot t)$$

$$= 23.069 \cdot 1161.75 / (2 \cdot 20.5) + 9236.9 / (0.2397 \cdot \pi \cdot 1161.8^2 \cdot 20.5)$$

$$= 61.02 \text{ N./mm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L-2a) / (L + (4 \cdot h_m / 3))$$

$$= 39819(11000.0 - 2 \cdot 1450.0) / (11000.0 + (4 \cdot 586.25 / 3))$$

$$= 27376.3 \text{ Kgf}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K_2 \cdot T / (R_m \cdot t)$$

$$= 0.9345 \cdot 27376.25 / (1161.75 \cdot 20.5)$$

$$= 10.53 \text{ N./mm}^2$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{(R_m \cdot t)}$$

$$= 0.78 \cdot \sqrt{(1161.75 \cdot 20.5)}$$

$$= 120.373 \text{ mm.}$$

Circumferential Stress in shell, no rings (4.15.21) [σ_6]:

$$= -K_5 \cdot Q \cdot k / (t(b + X_1 + X_2))$$

$$= -0.7052 \cdot 39819 \cdot 0.1 / (20.5(180.0 + 120.37 + 120.37))$$

$$= -3.19 \text{ N./mm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L \geq 8R_m$ (4.15.22) [σ_7]:

$$= -Q / (4 \cdot t(b + X_1 + X_2)) - 3 \cdot K_7 \cdot Q / (2 \cdot t^2)$$

$$= -39819 / (4 \cdot 20.5(180.0 + 120.373 + 120.373)) - 3 \cdot 0.0399 \cdot 39819 / (2 \cdot 20.5^2)$$

$$= -66.94 \text{ N./mm}^2$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(R_m \cdot t)}, 2a)$$

$$= \min(180.0 + 1.56 \cdot \sqrt{(1161.75 \cdot 20.5)}, 2 \cdot 1450.0)$$

$$= 420.75 \text{ mm.}$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	2185.0000	mm.
Baseplate Thickness	Bpthk	20.0000	mm.
Baseplate Width	Bpwid	200.0000	mm.
Number of Ribs (inc. outside ribs)	Nribs	3	
Rib Thickness	Ribtk	16.0000	mm.
Web Thickness	Webtk	16.0000	mm.
Web Location	Webloc	Center	
Saddle Design Temperature		21.1	°C
Saddle Yield Stress	Sy	232.0	N./mm ²
Height of Web at Center	Hw,c	306.0	mm.
Friction Coefficient	mu	0.100	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})^3) / 12$$

$$= ((1.56 \cdot \sqrt{(1151.5 \cdot 20.5)} + 230.0)^3) / 12$$

$$= 33.720 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}}(C_1 - Y)^2$$

$$= 33.72 + 96.285(88.578 - 10.25)^2$$



= 5941.070 cm⁴

	B mm.	D mm.	Y mm.	A cm ²	AY mm. ³	I + AD ⁽²⁾ cm ⁴
Shell	469.7	20.5	10.2	96.3	98691.7	0.594E+04
Wearplate	230.0	12.0	26.5	27.6	73140.0	0.107E+04
Web	16.0	226.0	145.5	36.2	526127.6	0.271E+04
BasePlate	200.0	20.0	268.5	40.0	1073999.5	0.130E+05
Totals	...	...	...	200.0	1771958.8	0.227E+05

Distance to Centroid [C1]:

$$= AY / A$$

$$= 1771959/200.045$$

$$= 88.578 \text{ mm.}$$

Angle [β]:

$$= 180 - \text{Saddle Angle}/2$$

$$= 180 - 137.0/2$$

$$= 111.5$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(111.5) - 0.5 \cdot \sin(111.5)^2) / (\pi - 1.946 + \sin(111.5) \cos(111.5))$$

$$= 0.2348$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.235 \cdot 39819.488$$

$$= 9350.1562 \text{ Kgf}$$

$$\text{Tension Stress, } St = (Fh/As) = 8.8373 \text{ N./mm}^2$$

$$\text{Allowed Stress, } Sa = 0.6 \cdot \text{Yield Str} = 139.1985 \text{ N./mm}^2$$

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians})$$

$$= 1430.0 - 1151.5 \cdot \sin(137.0/2) / 1.1956$$

$$= 533.864 \text{ mm.}$$

$$\text{Bending Moment, } M = Fh \cdot d = 4991.8110 \text{ Kg-m.}$$

$$\text{Bending Stress, } Sb = (M \cdot C1 / I) = 19.1182 \text{ N./mm}^2$$

$$\text{Allowed Stress, } Sa = 2/3 \cdot \text{Yield Str} = 154.6650 \text{ N./mm}^2$$

Minimum Thickness of Baseplate per Moss:

$$= \sqrt{(3(Q + \text{SaddleWt}) \cdot \text{BasePlateWidth}) / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress})}$$

$$= \sqrt{(3(39819 + 385)200.0) / (4 \cdot 2185.0 \cdot 154.665)}$$

$$= 13.229 \text{ mm.}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 137.0/2) (1150.0 + 22.0 + 12.0)$$

$$= 2203.229 \text{ mm.}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (\text{Nr ribs} - 1)$$

$$= 2203.229 / (3 - 1)$$

$$= 1101.615 \text{ mm.}$$

Baseplate Pressure Area [Ap]:

$$= e \cdot \text{Bpwid} / 2$$

$$= 1101.6145 \cdot 200.0/2$$

$$= 1101.615 \text{ cm}^2$$

Bearing Pressure [Bp]:

$$= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$$

$$= 40204.762 / (2185.0 \cdot 200.0)$$

$$= 9.200 \text{ Kgf/cm}^2$$

Axial Load [P]:
 = $A_p \cdot B_p$
 = $1101.6 \cdot 9.2$
 = 10135.046 Kgf

Area of the Rib and Web [Ar]:
 = Rib Area + Web Area
 = $26.24 + 88.129$
 = 114.369 cm²

Compressive Stress [Sc]:
 = P/Ar
 = $10135.0/114.3692$
 = 8.691 N./mm²

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	16.0	180.0	...	28.8	...	778.

Rib dimension [D]:
 = Saddle Width - Web Thickness
 = $180.0 - 16.0$
 = 164.000 mm.

Distance to Centroid from Datum [ytot]:
 = AY / A
 = $0.0/114.369$
 = 0.000 mm.

Distance to Centroid [C1]:
 = Saddle Width / 2
 = $180.0/2$
 = 90.000 mm.

Radius of Gyration [r]:
 = $\sqrt{\text{Total Inertia} / \text{Total Area}}$
 = $\sqrt{777.6/114.369}$
 = 26.075 mm.

Length of Outer Rib, step 1, [Lv]:
 = Saddle Height - $\cos(\text{saddle bearing angle} / 2) (\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk}$
 = $1430.0 - \cos(137.0/2) (1150.0 + 22.0 + 12.0) - 20.0$
 = 976.062 mm.

Length of Outer Rib, step 2, [Lw]:
 = $2 \cdot \sin(\text{saddle bearing angle} / 2) (\text{radius} + \text{shlthk} + \text{wpdthk})$
 = $2 \cdot \sin(137.0/2) (1150.0 + 22.0 + 12.0)$
 = 2203.229 mm.

Length of Outer Rib, step 3, [Bd]:
 = Baseplate length - clearance
 = $2185.0 - 50.8$
 = 2134.200 mm.

Length of Outer Rib, step 4, [Dd]:
 = $(Lw - Bd) / 2$
 = $(2203.229 - 2134.2) / 2$
 = 34.515 mm.

Length of Outer Rib [L]:
 = $\sqrt{Lv^2 + Dd^2}$
 = $\sqrt{976.062^2 + 34.515^2}$
 = 976.673 mm.

Intermediate Term [Cc]:

$$= \sqrt{(2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress})}$$

$$= \sqrt{(2 \cdot \pi^2 \cdot 0.19994\text{E}+09 / 232.0)}$$

$$= 130.433$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 976.673 / 26.075$$

$$= 37.456$$

Bending Moment [Rm]:

$$= FI / (2 \cdot B_{\text{plen}}) \cdot e \cdot L / 2$$

$$= 5578.5 / (2 \cdot 2185.0) \cdot 1101.615 \cdot 976.67 / 2$$

$$= 686.745 \text{ Kg-m.}$$

Compressive Allowable, $KL/r < C_c$ ($37.4563 < 130.4335$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (37.46)^2 / (2 \cdot 130.43^2)) 232 /$$

$$(5/3 + 3 \cdot (37.46) / (8 \cdot 130.43) - (37.46^3) / (8 \cdot 130.43^3))$$

$$= 125.6 \text{ N./mm}^2$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= S_c / S_{ca} + (R_m \cdot C_1 / I) / S_{ba}$$

$$= 8.69 / 125.57 + (686.75 \cdot 90.0 / 7775999) / 154.66$$

$$= 0.573$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	16.0	164.0	0.0	26.2	0.0	777.
Web	1101.6	16.0	0.0	176.3	0.0	37.6
Totals	...	...	...	202.5	...	815.

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0.0 / 202.498$$

$$= 0.000 \text{ mm.}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 180.0 / 2$$

$$= 90.000 \text{ mm.}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \text{Outside Radius} - B_{\text{pthk}}$$

$$= 1430.0 - \cos(1184.0/2) (20.0 + 0.0 + 0.0) - 0.0$$

$$= 226.000 \text{ mm.}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(814.7 / 202.498)}$$

$$= 20.057 \text{ mm.}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 226.0 / 20.057$$

$$= 11.268$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 5578.532 / (2 \cdot 2185.0)$$

$$= 1.277 \text{ Kg/mm.}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 1.277 \cdot 1101.615 \cdot 226.0$$

$$= 317.823 \text{ Kg-m.}$$

Bending Stress due to Transverse Force and Weight Load [$\sigma_{B, \text{base}, c}$]:



= Bending Moment / Section Modulus
 = 317.823/90517.281
 = 34.433 N./mm²

Compressive Allowable, $KL/r < C_c$ (11.2676 < 130.4335) per AISC E2-1 [Sca]:
 $= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$
 $= (1 - (11.27)^2 / (2 \cdot 130.43^2)) 232 / (5/3 + 3 \cdot (11.27) / (8 \cdot 130.43) - (11.27^3) / (8 \cdot 130.43^3))$
 = 136 N./mm²

AISC Unity Check of Inside Ribs (must be ≤ 1)

= $Sc/Sca + (M_{base,c} \cdot C1/I) / S_{ba}$
 = $9.86/136.04 + (317.82 \cdot 90.0/814.655) / 154.66$
 = 0.295

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	4
Total Number of Bolts in Tension/Baseplate	Nbt	2
Bolt Design Temperature		21.1 °C
Bolt Material Specification		SA-285 C
Bolt Allowable Stress	Stba	137.30 N./mm ²
Bolt Corrosion Allowance	Bca	3.0 mm.
Distance from Bolts to Edge	Edgedis	50.8 mm.
User Defined Area in a Single Bolt	UserBoltArea	5.4000 cm ²
BasePlate Allowable Stress	S	138.00 N./mm ²
Area Available in a Single Bolt	BltArea	3.2115 cm ²
Saddle Load QO (Weight)	QO	31543.7 Kg
Saddle Load QL (Wind/Seismic contribution)	QL	972.8 Kg
Maximum Transverse Force	Ft	4411.3 Kg
Maximum Longitudinal Force	Fl	5578.5 Kg
Saddle Bolted to Steel Foundation		No

Shear Stress in a Single Bolt, Longitudinal Direction [τ_{b,l}]:
 = $Fl / (\text{Bolt Area} \cdot \text{Number of Bolts})$
 = $5578.5 / (3.211 \cdot 4)$
 = 42.6 N./mm². Must be less than 107.9 N./mm².

Shear Stress in a Single Bolt, Transverse Direction [τ_{b,t}]:
 = $Ft / (\text{Bolt Area} \cdot \text{Number of Bolts})$
 = $4411.3 / (3.211 \cdot 4)$
 = 33.7 N./mm². Must be less than 107.9 N./mm².

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:
 = 0.0 (QO > QL --> No Uplift in Longitudinal direction)

Bolt Area due to Shear Load [Bltarears]:
 = $Fl / (\text{BoltShearAllowable} \cdot \text{Nbolts})$
 = $5578.53 / (107.87 \cdot 4.0)$
 = 1.2679 cm²

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:
 = $B \cdot Ft + \text{Sum of X Moments}$
 = $1430.0 \cdot 4411.27 + 0.0$
 = 6308.25 Kg-m.

Eccentricity (e):
 = $Rmom / QO$
 = $6308.25 / 31543.72$
 = 199.98 mm. < Bplen/6 --> No Uplift in Transverse direction

Bolt Area due to Transverse Load [Bltareart]:
 = 0 (No Uplift)

Required Area of a Single Bolt [Bltarear]:
 = max[Bltarearl, Bltarears, Bltareart]



= max[0.0, 1.2679, 0.0]
= 1.2679 cm²

MDMT Calculations for: New Saddle, Saddle to Wearplate Junction, Curve: B

Govrn. thk, tg = 12.0, c = 1.5 mm., E* = 1.0
Thickness Ratio = tr * E*/(tg - c) = 0.961, Temp. Reduction = 2 °C
Attachment governing, Attachment stress = Shell stress per Code

Min Metal Temp. w/o impact per UCS-66, Curve B	-23 °C
Min Metal Temp per UCS-66 and UCS-68(c),PWHT credit	-40 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-42 °C

Review notes about nozzle loadings and supports in the Warnings report.

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ASME VIII Division 2 Horizontal Vessel Analysis, Left Saddle:

Horizontal Vessel Stress Calculations : Test Case

Note:

Wear Pad Width (230.00) is less than $1.56 \cdot \sqrt{r_m \cdot t}$
 and less than $2a$. The wear plate will be ignored.

Minimum Wear Plate Width to be considered in analysis [b1]:

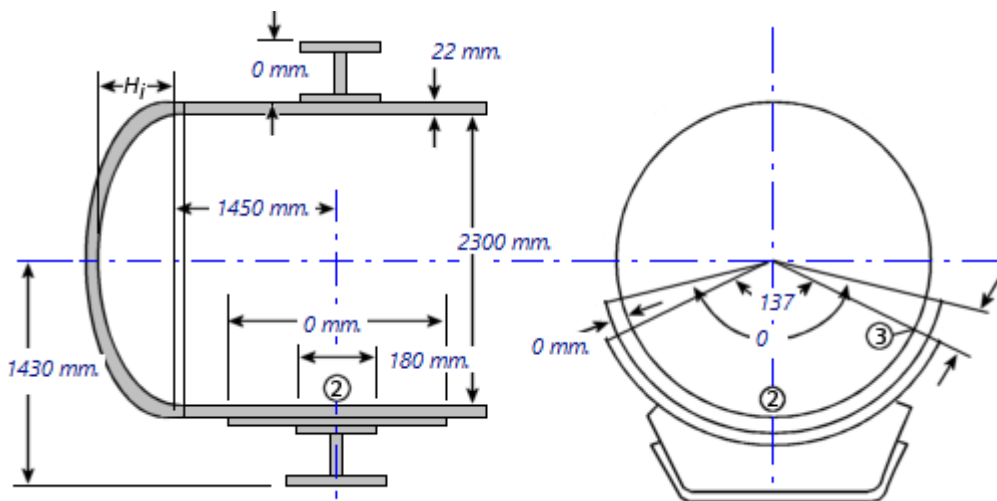
$$= \min(b + 1.56 \cdot \sqrt{R_m \cdot t}, 2a)$$

$$= \min(180.0 + 1.56 \cdot \sqrt{1161.0 \cdot 22.0}, 2 \cdot 1450.0)$$

$$= 429.3170 \text{ mm.}$$

Input and Calculated Values:

Vessel Mean Radius	Rm	1161.00	mm.
Shell Thickness used in this Case	t	22.000	mm.
Stiffened Vessel Length per 4.15.6	L	11000.00	mm.
Distance from Saddle to Vessel tangent	a1 or a	1450.00	mm.
Saddle Width	b1 or b	180.00	mm.
Saddle Bearing Angle	delta or theta	137.00	degrees
Inside or Mean Depth of Head	Hi or hm	585.50	mm.
Shell Allowable Stress used in Calculation		234.00	N./mm ²
Head Allowable Stress used in Calculation		234.00	N./mm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	1430.00	mm.
Coefficient of Friction	mu	0.00	



Results for Horizontal Vessel Analysis:

Saddle Force Q, Test Case, no Ext. Forces	41934.51	Kgf
Pressure used in Analysis	P	30.315 bars

Horizontal Vessel Analysis Results:		Actual N./mm ²	Allowable N./mm ²
Long. Stress at Top of Midspan		74.87	234.00
Long. Stress at Bottom of Midspan		85.12	234.00
Long. Stress at Top of Saddles		87.54	234.00
Long. Stress at Bottom of Saddles		75.72	234.00
Tangential Shear in Shell		10.35	138.58
Circ. Stress at Horn of Saddle		61.74	350.99

Circ. Compressive Stress in Shell	3.07	234.00
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Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:
 $= F_{tr}(F_t / \text{Num of Saddles} + Z \text{ Force Load}) \cdot B / E$
 $= 3.0(2911.4/2 + 0) \cdot 1430.0/2185.0$
 $= 2858.1 \text{ Kg}$

Longitudinal Saddle Reaction Force [Fwl]:
 $= \max(F_{l,wind}; \text{Sum of X Forces}) \cdot B / L_s$
 $= \max(669.15, 0) \cdot 1430.0/8200.0$
 $= 116.7 \text{ Kg}$

Load Combination Results for Q + Wind or Seismic [Q]:
 $= \text{Saddle Load} + \max(F_{wl}, F_{wt}, F_{sl}, F_{st})$
 $= 39076 + \max(117, 2858, 0, 0)$
 $= 41934.5 \text{ Kg}$

Longitudinal Wind Force [FI]:
 $= \text{WindScalar} \cdot \text{WindPress}(\text{Platform Area} + (\pi/4(\text{OD} \cdot \text{WindDiaMult})^2))$
 $= 1.1 \cdot 1745.935(3.375 + \pi/4(2.484 \cdot 1.2)^2)$
 $= 19884.830 \text{ N}$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	42319.79	Kgf
Transverse Shear Load	1455.72	Kgf
Longitudinal Shear Load	669.15	Kgf

Note: Wear Plate is Welded to the Shell, $k = 0.1$

Saddle Dimension [E]:
 $= \text{Baseplate Length}$
 $= 2185.000 \text{ mm}$

The Computed K values from Table 4.15.1:

K1 = 0.1356	K2 = 0.9345	K3 = 0.6284	K4 = 0.3370
K5 = 0.7052	K6 = 0.0399	K7 = 0.0399	K8 = 0.3208
K9 = 0.2427	K10 = 0.0455	K1* = 0.2397	

Note: Dimension a is greater than or equal to $R_m/2$.

Moment per Equation 4.15.1 [M1]:
 $= -Q \cdot a [1 - (1 - a/L + (R_m^2 - h_m^2)/(2a \cdot L))/(1 + (4h_m)/3L)]$
 $= -41935 \cdot 1450.0 [1 - (1 - 1450.0/11000.0 + (1161.0^2 - 585.5^2)/(2 \cdot 1450.0 \cdot 11000.0))/(1 + (4 \cdot 585.5)/(3 \cdot 11000.0))]$
 $= -9724.7 \text{ Kg-m}$

Moment per Equation 4.15.2 [M2]:
 $= Q \cdot L/4((1 + 2(R_m^2 - h_m^2)/(L^2))/(1 + (4h_m)/3L) - 4a/L)$
 $= 41935 \cdot 11000/4((1 + 2(1161^2 - 586^2)/(11000^2))/(1 + (4 \cdot 586)/(3 \cdot 11000)) - 4 \cdot 1450/11000)$
 $= 48662.8 \text{ Kg-m}$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:
 $= P \cdot R_m/(2t) - M_2/(\pi \cdot R_m^2 \cdot t)$
 $= 30.315 \cdot 1161.0/(2 \cdot 22.0) - 48662.8/(\pi \cdot 1161.0^2 \cdot 22.0)$
 $= 74.87 \text{ N/mm}^2$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:
 $= P \cdot R_m/(2t) + M_2/(\pi \cdot R_m^2 \cdot t)$
 $= 30.315 \cdot 1161.0/(2 \cdot 22.0) + 48662.8/(\pi \cdot 1161.0^2 \cdot 22.0)$
 $= 85.12 \text{ N/mm}^2$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ_3]:
 $= P \cdot R_m/(2t) - M_1/(K_1 \cdot \pi \cdot R_m^2 \cdot t)$
 $= 30.315 \cdot 1161.0/(2 \cdot 22.0) - 9724.7/(0.1356 \cdot \pi \cdot 1161.0^2 \cdot 22.0)$
 $= 87.54 \text{ N/mm}^2$



Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ_4]:

$$= P \cdot R_m / (2t) + M1 / (K1 \cdot \pi \cdot R_m^2 \cdot t)$$

$$= 30.315 \cdot 1161.0 / (2 \cdot 22.0) + 9724.7 / (0.2397 \cdot \pi \cdot 1161.0^2 \cdot 22.0)$$

$$= 75.72 \text{ N./mm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L-2a) / (L + (4 \cdot h_m / 3))$$

$$= 41935(11000.0 - 2 \cdot 1450.0) / (11000.0 + (4 \cdot 585.5 / 3))$$

$$= 28832.8 \text{ Kgf}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K2 \cdot T / (R_m \cdot t)$$

$$= 0.9345 \cdot 28832.8 / (1161.0 \cdot 22.0)$$

$$= 10.35 \text{ N./mm}^2$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{R_m \cdot t}$$

$$= 0.78 \cdot \sqrt{1161.0 \cdot 22.0}$$

$$= 124.659 \text{ mm.}$$

Circumferential Stress in shell, no rings (4.15.21) [σ_6]:

$$= -K5 \cdot Q \cdot k / (t(b + X1 + X2))$$

$$= -0.7052 \cdot 41935 \cdot 0.1 / (22.0(180.0 + 124.66 + 124.66))$$

$$= -3.07 \text{ N./mm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L \geq 8R_m$ (4.15.22) [σ_7]:

$$= -Q / (4 \cdot t(b + X1 + X2)) - 3 \cdot K7 \cdot Q / (2 \cdot t^2)$$

$$= -41935 / (4 \cdot 22.0(180.0 + 124.659 + 124.659)) - 3 \cdot 0.0399 \cdot 41935 / (2 \cdot 22.0^2)$$

$$= -61.74 \text{ N./mm}^2$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{R_m \cdot t}, 2a)$$

$$= \min(180.0 + 1.56 \cdot \sqrt{1161.0 \cdot 22.0}, 2 \cdot 1450.0)$$

$$= 429.32 \text{ mm.}$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	2185.0000	mm.
Baseplate Thickness	Bpthk	20.0000	mm.
Baseplate Width	Bpwid	200.0000	mm.
Number of Ribs (inc. outside ribs)	Nribs	3	
Rib Thickness	Ribtk	16.0000	mm.
Web Thickness	Webtk	16.0000	mm.
Web Location	Webloc	Center	
Saddle Yield Stress	Sy	260.0	N./mm2
Height of Web at Center	Hw,c	306.0	mm.
Friction Coefficient	mu	0.000	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{R \cdot t} + \text{WearPlateWidth})t^3) / 12$$

$$= ((1.56 \cdot \sqrt{1150.0 \cdot 22.0} + 230.0)22.0^3) / 12$$

$$= 42.426 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}}(C1 - Y)^2$$

$$= 42.426 + 105.189(86.363 - 11.0)^2$$

$$= 6016.668 \text{ cm}^4$$

	B mm.	D mm.	Y mm.	A cm ²	AY mm. ³	I + AD ⁽²⁾ cm ⁴
Shell	478.1	22.0	11.0	105.2	115708.2	0.602E+04
Wearplate	230.0	12.0	28.0	27.6	77280.0	943.
Web	16.0	226.0	147.0	36.2	531552.0	0.287E+04
BasePlate	200.0	20.0	270.0	40.0	1080000.0	0.135E+05
Totals	...	...	...	208.9	1804540.2	0.233E+05

Distance to Centroid [C1]:



$$= AY / A$$

$$= 1804540/208.949$$

$$= 86.363 \text{ mm.}$$

Angle [β]:

$$= 180 - \text{Saddle Angle}/2$$

$$= 180 - 137.0/2$$

$$= 111.5$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(111.5) - 0.5 \cdot \sin(111.5)^2) / (\pi - 1.946 + \sin(111.5) \cos(111.5))$$

$$= 0.2348$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.235 \cdot 41934.512$$

$$= 9846.7930 \text{ Kgf}$$

Tension Stress, $St = (Fh/As) = 9.3067 \text{ N./mm}^2$
 Allowed Stress, $Sa = 0.6 \cdot \text{Yield Str} = 155.9972 \text{ N./mm}^2$

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians})$$

$$= 1430.0 - 1150.0 \cdot \sin(137.0/2) / 1.1956$$

$$= 535.031 \text{ mm.}$$

Bending Moment, $M = Fh \cdot d = 5268.4473 \text{ Kg-m.}$

Bending Stress, $Sb = (M \cdot C1 / I) = 19.1246 \text{ N./mm}^2$
 Allowed Stress, $Sa = 2/3 \cdot \text{Yield Str} = 173.3302 \text{ N./mm}^2$

Minimum Thickness of Baseplate per Moss:

$$= \sqrt{3(Q + \text{SaddleWt}) \text{BasePlateWidth} / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress})}$$

$$= \sqrt{3(41935 + 385)200.0 / (4 \cdot 2185.0 \cdot 173.33)}$$

$$= 12.821 \text{ mm.}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 137.0/2) (1150.0 + 22.0 + 12.0)$$

$$= 2203.229 \text{ mm.}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (N_{\text{ribs}} - 1)$$

$$= 2203.2288 / (3 - 1)$$

$$= 1101.614 \text{ mm.}$$

Baseplate Pressure Area [Ap]:

$$= e \cdot \text{Bpwid} / 2$$

$$= 1101.6144 \cdot 200.0/2$$

$$= 1101.615 \text{ cm}^2$$

Bearing Pressure [Bp]:

$$= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$$

$$= 42319.789 / (2185.0 \cdot 200.0)$$

$$= 9.684 \text{ Kgf/cm}^2$$

Axial Load [P]:

$$= Ap \cdot Bp$$

$$= 1101.6 \cdot 9.68$$

$$= 10668.213 \text{ Kgf}$$

Area of the Rib and Web [Ar]:

$$= \text{Rib Area} + \text{Web Area}$$

$$= 26.24 + 88.129$$

$$= 114.369 \text{ cm}^2$$

Compressive Stress [Sc]:

$$= P/Ar$$

$$= 10668.2/114.3692$$

$$= 9.148 \text{ N./mm}^2$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	16.0	180.0	...	28.8	...	778.

Rib dimension [D]:

$$= \text{Saddle Width} - \text{Web Thickness}$$

$$= 180.0 - 16.0$$

$$= 164.000 \text{ mm.}$$

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0.0/114.369$$

$$= 0.000 \text{ mm.}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 180.0/2$$

$$= 90.000 \text{ mm.}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(777.6/114.369)}$$

$$= 26.075 \text{ mm.}$$

Length of Outer Rib, step 1, [Lv]:

$$= \text{Saddle Height} - \cos(\text{saddle bearing angle} / 2) (\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk}$$

$$= 1430.0 - \cos(137.0/2) (1150.0 + 22.0 + 12.0) - 20.0$$

$$= 976.063 \text{ mm.}$$

Length of Outer Rib, step 2, [Lw]:

$$= 2 \cdot \sin(\text{saddle bearing angle} / 2) (\text{radius} + \text{shlthk} + \text{wpdthk})$$

$$= 2 \cdot \sin(137.0/2) (1150.0 + 22.0 + 12.0)$$

$$= 2203.229 \text{ mm.}$$

Length of Outer Rib, step 3, [Bd]:

$$= \text{Baseplate length} - \text{clearance}$$

$$= 2185.0 - 50.8$$

$$= 2134.200 \text{ mm.}$$

Length of Outer Rib, step 4, [Dd]:

$$= (Lw - Bd) / 2$$

$$= (2203.229 - 2134.2) / 2$$

$$= 34.514 \text{ mm.}$$

Length of Outer Rib [L]:

$$= \sqrt{Lv^2 + Dd^2}$$

$$= \sqrt{976.063^2 + 34.514^2}$$

$$= 976.673 \text{ mm.}$$

Intermediate Term [Cc]:

$$= \sqrt{(2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress})}$$

$$= \sqrt{(2 \cdot \pi^2 \cdot 0.19994\text{E}+09/260.0)}$$

$$= 123.211$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 976.673/26.075$$

$$= 37.456$$

Bending Moment [Rm]:

$$= FI / (2 \cdot Bplen) \cdot e \cdot L / 2$$

$$= 669.1 / (2 \cdot 2185.0) \cdot 1101.614 \cdot 976.67/2$$

= 82.375 Kg-m.

Compressive Allowable, $KL/r < Cc$ (37.4563 < 123.2105) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot Cc^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$$

$$= (1 - (37.46)^2 / (2 \cdot 123.21^2)) 260 / (5/3 + 3 \cdot (37.46) / (8 \cdot 123.21) - (37.46^3) / (8 \cdot 123.21^3))$$

$$= 139.5 \text{ N./mm}^2$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= Sc/Sca + (Rm \cdot C1 / I) / Sba$$

$$= 9.15/139.54 + (82.38 \cdot 90.0/7775999) / 173.33$$

$$= 0.119$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	16.0	164.0	0.0	26.2	0.0	777.
Web	1101.6	16.0	0.0	176.3	0.0	37.6
Totals	...	...	...	202.5	...	815.

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0.0/202.498$$

$$= 0.000 \text{ mm.}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 180.0/2$$

$$= 90.000 \text{ mm.}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \text{Outside Radius} - Bpthk$$

$$= 1430.0 - \cos(1184.0/2) (20.0 + 0.0 + 0.0) - 0.0$$

$$= 226.000 \text{ mm.}$$

Radius of Gyration [r]:

$$= \sqrt{ (\text{Total Inertia} / \text{Total Area}) }$$

$$= \sqrt{ (814.7/202.498) }$$

$$= 20.057 \text{ mm.}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 226.0/20.057$$

$$= 11.268$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 669.147 / (2 \cdot 2185.0)$$

$$= 0.153 \text{ Kg/mm.}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 0.153 \cdot 1101.614 \cdot 226.0$$

$$= 38.123 \text{ Kg-m.}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 38.123/90517.281$$

$$= 4.130 \text{ N./mm}^2$$

Compressive Allowable, $KL/r < Cc$ (11.2676 < 123.2105) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot Cc^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$$

$$= (1 - (11.27)^2 / (2 \cdot 123.21^2)) 260 / (5/3 + 3 \cdot (11.27) / (8 \cdot 123.21) - (11.27^3) / (8 \cdot 123.21^3))$$

$$= 152.2 \text{ N./mm}^2$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= Sc/Sca + (Mbase,c \cdot C1 / I) / Sba$$



$$= 10.38/152.22 + (38.12 \cdot 90.0/814.655)/173.33$$

$$= 0.092$$

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	4
Total Number of Bolts in Tension/Baseplate	Nbt	2
Bolt Material Specification		SA-285 C
Bolt Allowable Stress	Stba	108.00 N./mm2
Bolt Corrosion Allowance	Bca	3.0 mm.
Distance from Bolts to Edge	Edgedis	50.8 mm.
User Defined Area in a Single Bolt	UserBoltArea	5.4000 cm2
BasePlate Allowable Stress	S	138.00 N./mm2
Area Available in a Single Bolt	BltArea	3.2115 cm2
Saddle Load Q0 (Weight)	Q0	39461.6 Kgf
Saddle Load QL (Wind/Seismic contribution)	QL	116.7 Kgf
Maximum Transverse Force	Ft	1455.7 Kgf
Maximum Longitudinal Force	Fl	669.1 Kgf
Saddle Bolted to Steel Foundation		No

Shear Stress in a Single Bolt, Longitudinal Direction [rb,l]:
 $= Fl / (\text{Bolt Area} \cdot \text{Number of Bolts})$
 $= 669.1 / (3.211 \cdot 4)$
 $= 5.1 \text{ N./mm}^2$. Must be less than 64.8 N./mm².

Shear Stress in a Single Bolt, Transverse Direction [rb,t]:
 $= Ft / (\text{Bolt Area} \cdot \text{Number of Bolts})$
 $= 1455.7 / (3.211 \cdot 4)$
 $= 11.1 \text{ N./mm}^2$. Must be less than 64.8 N./mm².

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:
 $= 0.0$ (Q0 > QL --> No Uplift in Longitudinal direction)

Bolt Area due to Shear Load [Bltarears]:
 $= Fl / (\text{BoltShearAllowable} \cdot \text{Nbolts})$
 $= 669.15 / (64.8 \cdot 4.0)$
 $= 0.2532 \text{ cm}^2$

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:
 $= B \cdot Ft + \text{Sum of X Moments}$
 $= 1430.0 \cdot 1455.72 + 0.0$
 $= 2081.72 \text{ Kg-m.}$

Eccentricity (e):
 $= Rmom / Q0$
 $= 2081.72 / 39461.64$
 $= 52.75 \text{ mm.}$ < Bplen/6 --> No Uplift in Transverse direction

Bolt Area due to Transverse Load [Bltareart]:
 $= 0$ (No Uplift)

Required Area of a Single Bolt [Bltarear]:
 $= \max[\text{Bltarearl}, \text{Bltarears}, \text{Bltareart}]$
 $= \max[0.0, 0.2532, 0.0]$
 $= 0.2532 \text{ cm}^2$

ASME VIII Division 2 Horizontal Vessel Analysis, Right Saddle:

Note:

Wear Pad Width (230.00) is less than $1.56 \cdot \sqrt{rm \cdot t}$
 and less than 2a. The wear plate will be ignored.

Minimum Wear Plate Width to be considered in analysis [b1]:
 $= \min(b + 1.56 \cdot \sqrt{Rm \cdot t}, 2a)$
 $= \min(180.0 + 1.56 \cdot \sqrt{1161.0 \cdot 22.0}, 2 \cdot 1450.0)$



= 429.3171 mm.

Input and Calculated Values:

Vessel Mean Radius	Rm	1161.00	mm.
Shell Thickness used in this Case	t	22.000	mm.
Stiffened Vessel Length per 4.15.6	L	11000.00	mm.
Distance from Saddle to Vessel tangent	a1 or a	1450.00	mm.
Saddle Width	b1 or b	180.00	mm.
Saddle Bearing Angle	delta or theta	137.00	degrees
Inside or Mean Depth of Head	Hi or hm	585.50	mm.
Shell Allowable Stress used in Calculation		234.00	N./mm ²
Head Allowable Stress used in Calculation		234.00	N./mm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	1430.00	mm.
Coefficient of Friction	mu	0.10	

Results for Horizontal Vessel Analysis:

Saddle Force Q, Test Case, no Ext. Forces		41069.18	Kgf
Pressure used in Analysis	P	30.315	bars

Horizontal Vessel Analysis Results:		Actual N./mm ²	Allowable N./mm ²
Long. Stress at Top of Midspan		74.98	234.00
Long. Stress at Bottom of Midspan		85.01	234.00
Long. Stress at Top of Saddles		87.39	234.00
Long. Stress at Bottom of Saddles		75.81	234.00
Tangential Shear in Shell		10.13	138.58
Circ. Stress at Horn of Saddle		60.47	350.99
Circ. Compressive Stress in Shell		3.01	234.00

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:
 $= F_{tr} (F_t / \text{Num of Saddles} + Z \text{ Force Load}) \cdot B / E$
 $= 3.0 (2911.4/2 + 0) \cdot 1430.0/2185.0$
 $= 2858.1 \text{ Kgf}$

Longitudinal Saddle Reaction Force [Fwl]:
 $= \max(F_{l,wind}; \text{Sum of X Forces}) \cdot B / L_s$
 $= \max(669.15, 0) \cdot 1430.0/8200.0$
 $= 116.7 \text{ Kgf}$

Load Combination Results for Q + Wind or Seismic [Q]:
 $= \text{Saddle Load} + \max(F_{wl}, F_{wt}, F_{sl}, F_{st})$
 $= 38211 + \max(117, 2858, 0, 0)$
 $= 41069.2 \text{ Kgf}$

Longitudinal Wind Force [FI]:
 $= \text{WindScalar} \cdot \text{WindPress} (\text{Platform Area} + (\pi/4 (\text{OD} \cdot \text{WindDiaMult})^2))$
 $= 1.1 \cdot 1745.935 (3.375 + \pi/4 (2.484 \cdot 1.2)^2)$
 $= 19884.834 \text{ N}$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	41454.46	Kgf
Transverse Shear Load	1455.72	Kgf
Longitudinal Shear Load	669.15	Kgf

Note: Wear Plate is Welded to the Shell, k = 0.1

Saddle Dimension [E]:

= Baseplate Length
 = 2185.000 mm.

The Computed K values from Table 4.15.1:

K1 = 0.1356 K2 = 0.9345 K3 = 0.6284 K4 = 0.3370
 K5 = 0.7052 K6 = 0.0399 K7 = 0.0399 K8 = 0.3208
 K9 = 0.2427 K10 = 0.0455 K1* = 0.2397

Note: Dimension a is greater than or equal to Rm/2.

Moment per Equation 4.15.1 [M1]:

$$= -Q \cdot a \left[1 - \left(1 - \frac{a}{L} + \frac{(Rm^2 - hm^2)}{(2a \cdot L)} \right) / \left(1 + \frac{(4hm)}{3L} \right) \right]$$

$$= -41069 \cdot 1450.0 \left[1 - \left(1 - \frac{1450.0}{11000.0} + \frac{(1161.0^2 - 585.5^2)}{(2 \cdot 1450.0 \cdot 11000.0)} \right) / \left(1 + \frac{(4 \cdot 585.5)}{(3 \cdot 11000.0)} \right) \right]$$

$$= -9524.0 \text{ Kg-m.}$$

Moment per Equation 4.15.2 [M2]:

$$= Q \cdot L / 4 \left(\frac{(1 + 2(Rm^2 - hm^2)/(L^2)) / (1 + (4hm^2)/(3L)) - 4a/L}{(3 \cdot 11000)} \right) - 4 \cdot 1450 / 11000$$

$$= 41069 \cdot 11000 / 4 \left(\frac{(1 + 2(1161^2 - 586^2)/(11000^2)) / (1 + (4 \cdot 586)/(3 \cdot 11000)) - 4 \cdot 1450 / 11000}{(3 \cdot 11000)} \right) - 4 \cdot 1450 / 11000$$

$$= 47658.7 \text{ Kg-m.}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$= P \cdot Rm / (2t) - M2 / (\pi \cdot Rm^2 \cdot t)$$

$$= 30.315 \cdot 1161.0 / (2 \cdot 22.0) - 47658.7 / (\pi \cdot 1161.0^2 \cdot 22.0)$$

$$= 74.98 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$= P \cdot Rm / (2t) + M2 / (\pi \cdot Rm^2 \cdot t)$$

$$= 30.315 \cdot 1161.0 / (2 \cdot 22.0) + 47658.7 / (\pi \cdot 1161.0^2 \cdot 22.0)$$

$$= 85.01 \text{ N./mm}^2$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ_{*3}]:

$$= P \cdot Rm / (2t) - M1 / (K1 \cdot \pi \cdot Rm^2 \cdot t)$$

$$= 30.315 \cdot 1161.0 / (2 \cdot 22.0) - 9524.0 / (0.1356 \cdot \pi \cdot 1161.0^2 \cdot 22.0)$$

$$= 87.39 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ_{*4}]:

$$= P \cdot Rm / (2t) + M1 / (K1* \cdot \pi \cdot Rm^2 \cdot t)$$

$$= 30.315 \cdot 1161.0 / (2 \cdot 22.0) + 9524.0 / (0.2397 \cdot \pi \cdot 1161.0^2 \cdot 22.0)$$

$$= 75.81 \text{ N./mm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L - 2a) / (L + (4 \cdot hm / 3))$$

$$= 41069(11000.0 - 2 \cdot 1450.0) / (11000.0 + (4 \cdot 585.5 / 3))$$

$$= 28237.8 \text{ Kg}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K2 \cdot T / (Rm \cdot t)$$

$$= 0.9345 \cdot 28237.83 / (1161.0 \cdot 22.0)$$

$$= 10.13 \text{ N./mm}^2$$

Decay Length (4.15.20) [x1,x2]:

$$= 0.78 \cdot \sqrt{(Rm \cdot t)}$$

$$= 0.78 \cdot \sqrt{(1161.0 \cdot 22.0)}$$

$$= 124.659 \text{ mm.}$$

Circumferential Stress in shell, no rings (4.15.21) [σ_6]:

$$= -K5 \cdot Q \cdot k / (t(b + X1 + X2))$$

$$= -0.7052 \cdot 41069 \cdot 0.1 / (22.0(180.0 + 124.66 + 124.66))$$

$$= -3.01 \text{ N./mm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L \geq 8Rm$ (4.15.22) [σ_7]:

$$= -Q / (4 \cdot t(b + X1 + X2)) - 3 \cdot K7 \cdot Q / (2 \cdot t^2)$$

$$= -41069 / (4 \cdot 22.0(180.0 + 124.659 + 124.659)) - 3 \cdot 0.0399 \cdot 41069 / (2 \cdot 22.0^2)$$

$$= -60.47 \text{ N./mm}^2$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(Rm \cdot t)}, 2a)$$



$$= \min(180.0 + 1.56 \cdot \sqrt{(1161.0 \cdot 22.0)}, 2 \cdot 1450.0)$$

$$= 429.32 \text{ mm.}$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	2185.0000	mm.
Baseplate Thickness	Bpthk	20.0000	mm.
Baseplate Width	Bpwid	200.0000	mm.
Number of Ribs (inc. outside ribs)	Nribs	3	
Rib Thickness	Ribtk	16.0000	mm.
Web Thickness	Webtk	16.0000	mm.
Web Location	Webloc	Center	
Saddle Yield Stress	Sy	260.0	N./mm2
Height of Web at Center	Hw,c	306.0	mm.
Friction Coefficient	mu	0.100	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth}) t^3) / 12$$

$$= ((1.56 \cdot \sqrt{(1150.0 \cdot 22.0)} + 230.0) 22.0^3) / 12$$

$$= 42.426 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}} (C1 - Y)^2$$

$$= 42.426 + 105.189 (86.363 - 11.0)^2$$

$$= 6016.663 \text{ cm}^4$$

	B mm.	D mm.	Y mm.	A cm ²	AY mm. ³	I + AD ² cm ⁴
Shell	478.1	22.0	11.0	105.2	115708.2	0.602E+04
Wearplate	230.0	12.0	28.0	27.6	77280.0	943.
Web	16.0	226.0	147.0	36.2	531551.6	0.287E+04
BasePlate	200.0	20.0	270.0	40.0	1079999.6	0.135E+05
Totals	...	...	...	208.9	1804539.5	0.233E+05

Distance to Centroid [C1]:

$$= AY / A$$

$$= 1804540 / 208.949$$

$$= 86.363 \text{ mm.}$$

Angle [β]:

$$= 180 - \text{Saddle Angle} / 2$$

$$= 180 - 137.0 / 2$$

$$= 111.5$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(111.5) - 0.5 \cdot \sin(111.5)^2) / (\pi - 1.946 + \sin(111.5) \cos(111.5))$$

$$= 0.2348$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.235 \cdot 41069.184$$

$$= 9643.6025 \text{ Kg}$$

Tension Stress, St = (Fh/As) = 9.1146 N./mm2

Allowed Stress, Sa = 0.6 * Yield Str = 155.9972 N./mm2

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians})$$

$$= 1430.0 - 1150.0 \cdot \sin(137.0/2) / 1.1956$$

$$= 535.031 \text{ mm.}$$

Bending Moment, M = Fh * d = 5159.7310 Kg-m.

Bending Stress, Sb = (M * C1 / I) = 18.7299 N./mm2

Allowed Stress, Sa = 2/3 * Yield Str = 173.3302 N./mm2

Minimum Thickness of Baseplate per Moss:

$$= \sqrt{\frac{3(Q + \text{SaddleWt}) \cdot \text{BasePlateWidth}}{4 \cdot \text{BasePlateLength} \cdot \text{AllStress}}}$$

$$= \sqrt{\frac{3(41069 + 385)200.0}{4 \cdot 2185.0 \cdot 173.33}}$$

$$= 12.689 \text{ mm.}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 137.0/2) (1150.0 + 22.0 + 12.0)$$

$$= 2203.229 \text{ mm.}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (\text{Nr ribs} - 1)$$

$$= 2203.229 / (3 - 1)$$

$$= 1101.615 \text{ mm.}$$

Baseplate Pressure Area [Ap]:

$$= e \cdot \text{Bpwid} / 2$$

$$= 1101.615 \cdot 200.0 / 2$$

$$= 1101.615 \text{ cm}^2$$

Bearing Pressure [Bp]:

$$= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$$

$$= 41454.461 / (2185.0 \cdot 200.0)$$

$$= 9.486 \text{ Kg/cm}^2$$

Axial Load [P]:

$$= \text{Ap} \cdot \text{Bp}$$

$$= 1101.6 \cdot 9.49$$

$$= 10450.077 \text{ Kg}$$

Area of the Rib and Web [Ar]:

$$= \text{Rib Area} + \text{Web Area}$$

$$= 26.24 + 88.129$$

$$= 114.369 \text{ cm}^2$$

Compressive Stress [Sc]:

$$= P / \text{Ar}$$

$$= 10450.1 / 114.3692$$

$$= 8.961 \text{ N./mm}^2$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	16.0	180.0	...	28.8	...	778.

Rib dimension [D]:

$$= \text{Saddle Width} - \text{Web Thickness}$$

$$= 180.0 - 16.0$$

$$= 164.000 \text{ mm.}$$

Distance to Centroid from Datum [ytot]:

$$= \text{AY} / \text{A}$$

$$= 0.0 / 114.369$$

$$= 0.000 \text{ mm.}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 180.0 / 2$$

$$= 90.000 \text{ mm.}$$

Radius of Gyration [r]:

$$= \sqrt{\text{Total Inertia} / \text{Total Area}}$$

$$= \sqrt{777.6 / 114.369}$$

$$= 26.075 \text{ mm.}$$



Length of Outer Rib, step 1, [Lv]:
 = Saddle Height - cos(saddle bearing angle / 2)(radius + shlthk + wpdthk) - bpthk
 = 1430.0 - cos(137.0/2)(1150.0 + 22.0 + 12.0) - 20.0
 = 976.062 mm.

Length of Outer Rib, step 2, [Lw]:
 = 2 • sin(saddle bearing angle / 2)(radius + shlthk + wpdthk)
 = 2 • sin(137.0/2)(1150.0 + 22.0 + 12.0)
 = 2203.229 mm.

Length of Outer Rib, step 3, [Bd]:
 = Baseplate length - clearance
 = 2185.0 - 50.8
 = 2134.200 mm.

Length of Outer Rib, step 4, [Dd]:
 = (Lw - Bd)/2
 = (2203.229 - 2134.2)/2
 = 34.515 mm.

Length of Outer Rib [L]:
 = $\sqrt{Lv^2 + Dd^2}$
 = $\sqrt{976.062^2 + 34.515^2}$
 = 976.673 mm.

Intermediate Term [Cc]:
 = $\sqrt{2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress}}$
 = $\sqrt{2 \cdot \pi^2 \cdot 0.19994\text{E}+09 / 260.0}$
 = 123.211

Slenderness ratio [KL/r]:
 = KL/r
 = 1 • 976.673/26.075
 = 37.456

Bending Moment [Rm]:
 = FI / (2 • Bplen) • e • L / 2
 = 669.1/ (2 • 2185.0) • 1101.615 • 976.67/2
 = 82.375 Kg-m.

Compressive Allowable, KL/r < Cc (37.4563 < 123.2105) per AISC E2-1 [Sca]:
 = $(1 - (KL/r)^2 / (2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$
 = $(1 - (37.46)^2 / (2 \cdot 123.21^2)) 260 / (5/3 + 3 \cdot (37.46) / (8 \cdot 123.21) - (37.46^3) / (8 \cdot 123.21^3))$
 = 139.5 N./mm²

AISC Unity Check of Outside Ribs (must be ≤ 1)
 = Sc/Sca + (Rm • C1 / I)/Sba
 = 8.96/139.54 + (82.38 • 90.0/7775999)/173.33
 = 0.118

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	16.0	164.0	0.0	26.2	0.0	777.
Web	1101.6	16.0	0.0	176.3	0.0	37.6
Totals	...	...	...	202.5	...	815.

Distance to Centroid from Datum [ytot]:
 = AY / A
 = 0.0/202.498
 = 0.000 mm.

Distance to Centroid [C1]:
 = Saddle Width / 2
 = 180.0/2
 = 90.000 mm.



Length of Inner Rib [L]:

$$= \text{Saddle Height} - \text{Outside Radius} - \text{Bpthk}$$

$$= 1430.0 - \cos(1184.0/2)(20.0 + 0.0 + 0.0) - 0.0$$

$$= 226.000 \text{ mm.}$$

Radius of Gyration [r]:

$$= \sqrt{\text{Total Inertia} / \text{Total Area}}$$

$$= \sqrt{814.7/202.498}$$

$$= 20.057 \text{ mm.}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 226.0/20.057$$

$$= 11.268$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 669.147 / (2 \cdot 2185.0)$$

$$= 0.153 \text{ Kg/mm.}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 0.153 \cdot 1101.615 \cdot 226.0$$

$$= 38.123 \text{ Kg-m.}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 38.123/90517.281$$

$$= 4.130 \text{ N./mm}^2$$

Compressive Allowable, $KL/r < C_c$ ($11.2676 < 123.2105$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (11.27)^2 / (2 \cdot 123.21^2)) 260 /$$

$$(5/3 + 3 \cdot (11.27) / (8 \cdot 123.21) - (11.27)^3 / (8 \cdot 123.21^3))$$

$$= 152.2 \text{ N./mm}^2$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= S_c/S_{ca} + (M_{base,c} \cdot C_{1/I})/S_{ba}$$

$$= 10.17/152.22 + (38.12 \cdot 90.0/814.655)/173.33$$

$$= 0.091$$

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	4
Total Number of Bolts in Tension/Baseplate	Nbt	2
Bolt Material Specification		SA-285 C
Bolt Allowable Stress	Stba	137.30 N./mm ²
Bolt Corrosion Allowance	Bca	3.0 mm.
Distance from Bolts to Edge	Edgedis	50.8 mm.
User Defined Area in a Single Bolt	UserBoltArea	5.4000 cm ²
BasePlate Allowable Stress	S	138.00 N./mm ²
Area Available in a Single Bolt	BltArea	3.2115 cm ²
Saddle Load Q0 (Weight)	Q0	38596.3 Kg
Saddle Load QL (Wind/Seismic contribution)	QL	116.7 Kg
Maximum Transverse Force	Ft	1455.7 Kg
Maximum Longitudinal Force	F1	669.1 Kg
Saddle Bolted to Steel Foundation	No	

Shear Stress in a Single Bolt, Longitudinal Direction [τ_b ,l]:

$$= F1 / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 669.1 / (3.211 \cdot 4)$$

$$= 5.1 \text{ N./mm}^2. \text{ Must be less than } 107.9 \text{ N./mm}^2.$$

Shear Stress in a Single Bolt, Transverse Direction [τ_b ,t]:

$$= Ft / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 1455.7 / (3.211 \cdot 4)$$

$$= 11.1 \text{ N./mm}^2. \text{ Must be less than } 107.9 \text{ N./mm}^2.$$

Bolt Area Calculation per Dennis R. Moss



Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:
= 0.0 (QO > QL --> No Uplift in Longitudinal direction)

Bolt Area due to Shear Load [Bltarears]:
= $F_l / (BoltShearAllowable \cdot Nbolts)$
= $669.15 / (107.87 \cdot 4.0)$
= 0.1521 cm²

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:
= $B \cdot Ft + \text{Sum of X Moments}$
= $1430.0 \cdot 1455.72 + 0.0$
= 2081.72 Kg-m.

Eccentricity (e):
= $Rmom / QO$
= $2081.72 / 38596.31$
= 53.93 mm. < Bplen/6 --> No Uplift in Transverse direction

Bolt Area due to Transverse Load [Bltareart]:
= 0 (No Uplift)

Required Area of a Single Bolt [Bltarear]:
= max[Bltarearl, Bltarears, Bltareart]
= max[0.0, 0.1521, 0.0]
= 0.1521 cm²

Review notes about nozzle loadings and supports in the Warnings report.

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Input, Nozzle Desc: MH124"

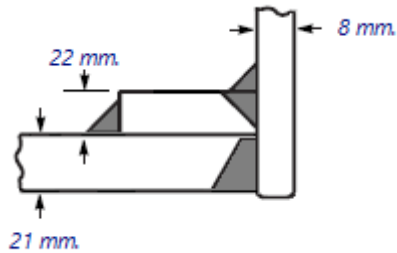
From: 10

Pressure for Reinforcement Calculations	P	23.069	bars
Temperature for Internal Pressure	Temp	150	°C
Parent Material		SA-516 70	+IT
Parent Allowable Stress at Temperature	Sv	138.00	N./mm2
Parent Allowable Stress At Ambient	Sva	138.00	N./mm2
Inside Diameter of Elliptical Head	D	2300.00	mm.
Aspect Ratio of Elliptical Head	Ar	2.00	
Head Finished (Minimum) Thickness	t	21.0000	mm.
Head Internal Corrosion Allowance	c	1.5000	mm.
Head External Corrosion Allowance	co	0.0000	mm.
Distance from Head Centerline	L1	0.0000	mm.
User Entered Minimum Design Metal Temperature		-33.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-516 70	+IT
Material UNS Number		K02700	
Material Specification/Type		Plate	
Allowable Stress at Temperature	Sn	138.00	N./mm2
Allowable Stress At Ambient	Sna	138.00	N./mm2
Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		24.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	8.0000	mm.
Flange Material		SA-350 LF2	+IT
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	1.5000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	158.3760	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	22.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Pad Material		SA-516 70	+IT
Pad Allowable Stress at Temperature	Sp	138.00	N./mm2
Pad Allowable Stress At Ambient	Spa	138.00	N./mm2
Diameter of Pad along vessel surface	Dp	1060.0000	mm.
Thickness of Pad	te	22.0000	mm.
Weld leg size between Pad and Shell	Wp	18.0000	mm.
Groove weld depth between Pad and Nozzle	Wgpn	22.0000	mm.
Reinforcing Pad Width		217.2000	mm.
This is a Manway, Coupling or Access Opening.			
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: MH124"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	24.000 in.
Actual Thickness Used in Calculation	0.315 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Elliptical Head, tr [Int. Press]
 $= P \cdot D \cdot K1 / (2 \cdot Sv \cdot E - 0.2 \cdot P)$ per Appendix 1-4(c)
 $= 23.07 \cdot 2303.0 \cdot 0.8989 / (2 \cdot 138.0 \cdot 1.0 - 0.2 \cdot 23.07)$
 $= 17.3328 \text{ mm.}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (Sn \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 23.07 \cdot 306.3 / (138 \cdot 1.0 - 0.6 \cdot 23.07)$
 $= 5.1724 \text{ mm.}$

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	1225.2000 mm.
Parallel to Vessel Wall, opening length	d	612.6000 mm.
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		38.2500 mm.

Weld Strength Reduction Factor [fr1]:
 $= \min(1, Sn/Sv)$
 $= \min(1, 138.0/138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, Sn/Sv)$
 $= \min(1, 138.0/138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr4]:
 $= \min(1, Sp/Sv)$
 $= \min(1, 138.0/138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(1.0, 1.0)$
 $= 1.000$

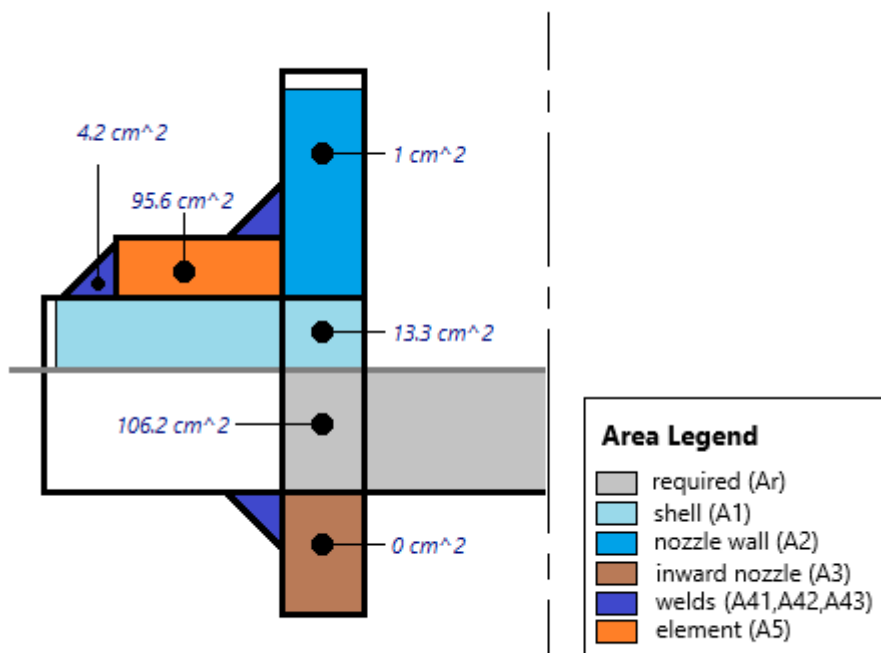


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (cm2)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	106.181	NA	NA
Area in Shell	A1	13.276	NA	NA
Area in Nozzle Wall	A2	1.016	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	4.240	NA	NA
Area in Element	A5	95.568	NA	NA
TOTAL AREA AVAILABLE	Atot	114.100	NA	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Insufficient.
 The area available with the given pad is Sufficient.

SELECTION OF POSSIBLE REINFORCING PADS: Diameter Thickness
 Based on given Pad Thickness: 1024.0029 22.0000 mm.
 Based on given Pad Diameter: 1060.0000 20.1769 mm.
 Based on the Estimated Diameter Limit: 1223.6125 14.6567 mm.

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$
 $= (612.6 \cdot 17.3328 \cdot 1.0 + 2 \cdot 6.5 \cdot 17.3328 \cdot 1.0 \cdot (1 - 1.0))$
 $= 106.181 \text{ cm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 612.6(1.0 \cdot 19.5 - 1.0 \cdot 17.333) - 2 \cdot 6.5$
 $(1.0 \cdot 19.5 - 1.0 \cdot 17.3328) \cdot (1 - 1.0)$
 $= 13.276 \text{ cm}^2$

Area Available in Nozzle Wall Projecting Outward [A2]:
 $= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2$
 $= (2 \cdot 38.25) \cdot (6.5 - 5.17) \cdot 1.0$



$$= 1.016 \text{ cm}^2$$

Area Available in Welds [A41 + A42 + A43]:
 $= W_o^2 \cdot fr3 + (W_i - can/0.707)^2 \cdot fr2 + W_p^2 \cdot fr4$
 $= 10.0^2 \cdot 1.0 + (0.0)^2 \cdot 1.0 + 18.0^2 \cdot 1.0$
 $= 4.240 \text{ cm}^2$

Area Available in Element [A5]:
 $= (\min(D_p, DL) - (\text{Nozzle OD})) \cdot (\min(t_p, Tlwp, te)) \cdot fr4$
 $= (1060.0 - 625.6) \cdot 22.0 \cdot 1.0$
 $= 95.568 \text{ cm}^2$

UG-45 Minimum Nozzle Neck Thickness Requirement:

Wall Thickness for Internal/External pressures $t_a = 6.6724 \text{ mm}$.

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= t_a$$

$$= 6.6724 \text{ mm}.$$

Available Nozzle Neck Thickness = 8.0000 mm. -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Nozzle Neck to Pad Weld for the Nozzle (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Nozzle Neck to Pad Weld for Reinf. pad (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Shell to Pad Weld at Pad OD for pad (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Gov. MDMT of the Nozzle	:	-44 °C
Gov. MDMT of the Reinforcement Pad	:	-39 °C
Gov. MDMT of the nozzle to shell joint welded assembly	:	-39 °C

Flange MDMT including Temperature reduction per UCS-66.1:

MDMT of ASME B16.5/47 flange per Matl. Specification	-18 °C
Flange MDMT with Temp reduction per UCS-66(i)(2)	-57 °C

Where the Stress Reduction Ratio per UCS-66(i)(2) is :

$$\text{Design Pressure/Ambient Rating} = 23.07/51.00 = 0.452$$

Weld Size Calculations, Description: MH124"

Intermediate Calc. for nozzle/shell Welds	T_{min}	6.5000 mm.
Intermediate Calc. for pad/shell Welds	T_{minPad}	19.0000 mm.

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$4.5500 = 0.7 \cdot t_{min}$	$7.0700 = 0.7 \cdot W_o \text{ mm.}$
Pad Weld	$9.5000 = 0.5 \cdot T_{minPad}$	$12.7260 = 0.7 \cdot W_p \text{ mm.}$

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

$$= \max(0, (A-A1+2 \cdot t_n \cdot f_{r1} \cdot (E1 \cdot t - t_r)) \cdot S_v)$$

$$= \max(0, (106.1806 - 13.2764 + 2 \cdot 6.5 \cdot 1.0 \cdot (1.0 \cdot 19.5 - 17.3328)) \cdot 138)$$

$$= 131126.89 \text{ Kgf}$$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$= (A2+A5+A4-(W_i \cdot \text{Can}/.707)^2 \cdot f_{r2}) \cdot S_v$$

$$= (1.0156 + 95.568 + 4.24 - 0.0 \cdot 1.0) \cdot 138$$

$$= 141874.16 \text{ Kgf}$$

Weld Load [W2]:

$$= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S_v$$

$$= (1.0156 + 0.0 + 1. + (2.535)) \cdot 138$$

$$= 6403.42 \text{ Kgf}$$

Weld Load [W3]:

$$= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S$$

$$= (1.0156 + 0.0 + 4.24 + 95.568 + (2.535)) \cdot 138$$

$$= 145441.28 \text{ Kgf}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$$

$$= (3.1416/2.0) \cdot 625.6 \cdot 10.0 \cdot 0.49 \cdot 138$$

$$= 67758. \text{ Kgf}$$

Shear, Pad Element Weld [Spew]:

$$= (\pi/2) \cdot D_P \cdot W_P \cdot 0.49 \cdot S_{EW}$$

$$= (3.1416/2.0) \cdot 1060.0 \cdot 18.0 \cdot 0.49 \cdot 138$$

$$= 206650. \text{ Kgf}$$

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - \text{Can}) \cdot 0.7 \cdot S_n$$

$$= (3.1416 \cdot 309.55) \cdot (8.0 - 1.5) \cdot 0.7 \cdot 138$$

$$= 62265. \text{ Kgf}$$

Tension, Pad Groove Weld [Tpgw]:

$$= (\pi/2) \cdot D_{lo} \cdot W_{gpn} \cdot 0.74 \cdot S_{eg}$$

$$= (3.1416/2) \cdot 625.6 \cdot 22.0 \cdot 0.74 \cdot 138$$

$$= 225124. \text{ Kgf}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot D_{lo} \cdot (T - \text{cas}) \cdot 0.74 \cdot S_{ng}$$

$$= (3.1416/2.0) \cdot 625.6 \cdot (21.0 - 1.5) \cdot 0.74 \cdot 138$$

$$= 199542. \text{ Kgf}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (S_{PEW} + S_{NW}) = (206650 + 62265) = 268915 \text{ Kgf}$$

Failure Path [Path 2-2]:

$$= (S_{onw} + T_{pgw} + T_{ngw} + S_{inw})$$

$$= (67758 + 225124 + 199542 + 0) = 492424 \text{ Kgf}$$

Failure Path [Path 3-3]:

$$= (S_{pew} + T_{ngw} + S_{inw})$$

$$= (206650 + 199542 + 0) = 406191 \text{ Kgf}$$

Summary of Failure Path Calculations:

Path 1-1 = 268914 Kgf, must exceed W = 131126 Kgf or W1 = 141874 Kgf
 Path 2-2 = 492423 Kgf, must exceed W = 131126 Kgf or W2 = 6403 Kgf



Path 3-3 = 406191 Kgf, must exceed W = 131126 Kgf or W3 = 145441 Kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 23 bars

Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 23.5767 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 202.9527 mm.

Percent Elongation Calculations:

% Elongation per Table UG-79-1 ($50 \cdot t_{nom} / R_f \cdot (1 - R_f / R_o)$) 1.295 %

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Input, Nozzle Desc: N18"

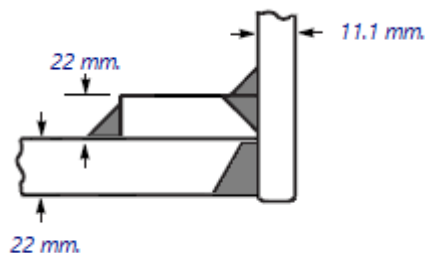
From: 20

Pressure for Reinforcement Calculations	P	23.000	bars
Temperature for Internal Pressure	Temp	150	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm2
Parent Allowable Stress At Ambient	Sva	138.00	N./mm2
Inside Diameter of Cylindrical Shell	D	2300.00	mm.
Shell Finished (Minimum) Thickness	t	22.0000	mm.
Shell Internal Corrosion Allowance	c	1.5000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		450.00	mm.
User Entered Minimum Design Metal Temperature		-33.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-333 6 +IT	
Material UNS Number		K03006	
Material Specification/Type	Smls. & wld. pipe		
Allowable Stress at Temperature	Sn	118.00	N./mm2
Allowable Stress At Ambient	Sna	118.00	N./mm2
Diameter Basis (for tr calc only)		Inside	
Layout Angle		90.00	deg
Diameter		8.0000	in.
Size and Thickness Basis		Minimum	
Nominal Thickness		80	
Flange Material		SA-350 LF2 +IT	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	1.5000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	418.2720	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	22.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Pad Material		SA-516 70 +IT	
Pad Allowable Stress at Temperature	Sp	138.00	N./mm2
Pad Allowable Stress At Ambient	Spa	138.00	N./mm2
Diameter of Pad along vessel surface	Dp	360.0000	mm.
Thickness of Pad	te	22.0000	mm.
Weld leg size between Pad and Shell	Wp	18.0000	mm.
Groove weld depth between Pad and Nozzle	Wgpn	22.0000	mm.
Reinforcing Pad Width		70.4625	mm.
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: N18"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	7.750 in.
Actual Thickness Used in Calculation	0.438 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 23.0 \cdot 1151.5 / (138 \cdot 1.0 - 0.6 \cdot 23.0)$
 $= 19.3870 \text{ mm.}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 23.0 \cdot 99.92 / (118 \cdot 1.0 - 0.6 \cdot 23.0)$
 $= 1.9709 \text{ mm.}$

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	399.7000 mm.
Parallel to Vessel Wall, opening length	d	199.8500 mm.
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		46.0312 mm.

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 118.0 / 138.0)$
 $= 0.855$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 118.0 / 138.0)$
 $= 0.855$

Weld Strength Reduction Factor [fr4]:
 $= \min(1, S_p / S_v)$
 $= \min(1, 138.0 / 138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(0.855, 1.0)$
 $= 0.855$

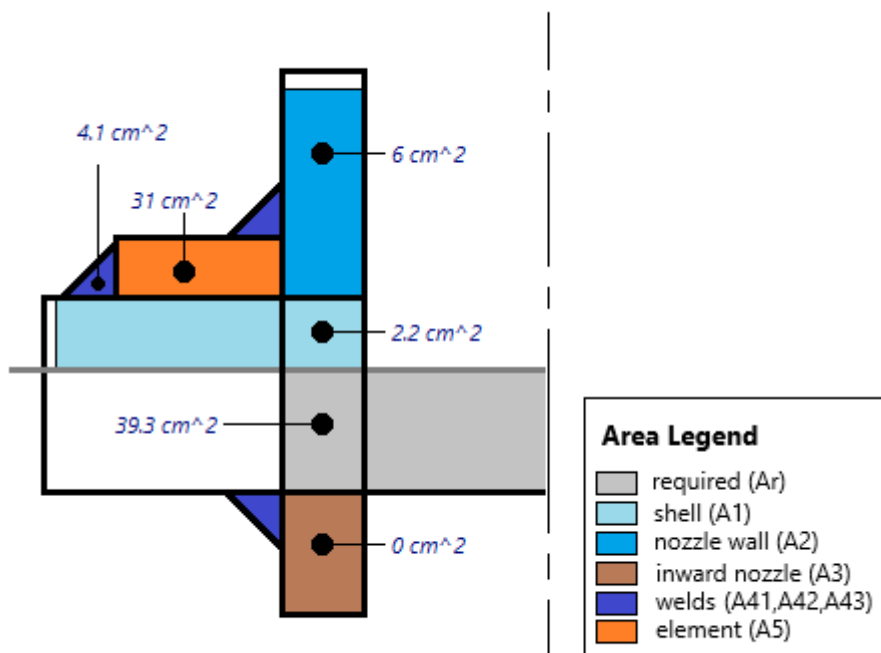


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (cm2)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	39.285	NA	NA
Area in Shell	A1	2.193	NA	NA
Area in Nozzle Wall	A2	6.016	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	4.095	NA	NA
Area in Element	A5	31.003	NA	NA
TOTAL AREA AVAILABLE	Atot	43.307	NA	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Insufficient.
 The area available with the given pad is Sufficient.

SELECTION OF POSSIBLE REINFORCING PADS: Diameter Thickness
 Based on given Pad Thickness: 341.7161 22.0000 mm.
 Based on given Pad Diameter: 360.0000 19.1457 mm.
 Based on the Estimated Diameter Limit: 398.1125 15.0701 mm.

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$
 $= (199.85 \cdot 19.387 \cdot 1.0 + 2 \cdot 9.6125 \cdot 19.387 \cdot 1.0 \cdot (1 - 0.86))$
 $= 39.285 \text{ cm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 199.85(1.0 \cdot 20.5 - 1.0 \cdot 19.387) - 2 \cdot 9.613$
 $(1.0 \cdot 20.5 - 1.0 \cdot 19.387) \cdot (1 - 0.855)$
 $= 2.193 \text{ cm}^2$

Area Available in Nozzle Wall Projecting Outward [A2]:
 $= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2$
 $= (2 \cdot 46.03) \cdot (9.61 - 1.97) \cdot 0.8551$



= 6.016 cm²

Area Available in Welds [A41 + A42 + A43]:
 = $W_o^2 \cdot fr3 + (W_i - can/0.707)^2 \cdot fr2 + W_p^2 \cdot fr4$
 = $10.0^2 \cdot 0.86 + (0.0)^2 \cdot 0.86 + 18.0^2 \cdot 1.0$
 = 4.095 cm²

Area Available in Element [A5]:
 = $(\min(D_p, DL) - (\text{Nozzle OD})) \cdot (\min(t_p, Tlwp, te)) \cdot fr4$
 = $(360.0 - 219.075) \cdot 22.0 \cdot 1.0$
 = 31.003 cm²

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures $t_a = 3.4709$ mm.
 Wall Thickness per UG16(b), $tr16b = 3.0000$ mm.
 Wall Thickness, shell/head, internal pressure $trb1 = 20.8870$ mm.
 Wall Thickness $tb1 = \max(trb1, tr16b) = 20.8870$ mm.
 Wall Thickness $tb2 = \max(trb2, tr16b) = 3.0000$ mm.
 Wall Thickness per table UG-45 $tb3 = 8.6600$ mm.

Determine Nozzle Thickness candidate [tb]:
 = $\min[tb3, \max(tb1, tb2)]$
 = $\min[8.66, \max(20.887, 3.0)]$
 = 8.6600 mm.

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 = $\max(t_a, t_b)$
 = $\max(3.4709, 8.66)$
 = 8.6600 mm.

Available Nozzle Neck Thickness = 11.1125 mm. -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	:	35.8,	Allowable	:	118.0 N./mm ²	Passed
Expansion	:	0.0,	Allowable	:	259.2 N./mm ²	Passed
Occasional	:	11.4,	Allowable	:	156.9 N./mm ²	Passed
Shear	:	17.1,	Allowable	:	82.6 N./mm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld (Impact tested) :

Note:
This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification	-46 °C
Calculated Minimum Design Metal Temperature	-104 °C

Nozzle Neck to Pad Weld for the Nozzle (Impact tested) :

Note:
This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification	-46 °C
Calculated Minimum Design Metal Temperature	-104 °C

Nozzle Neck to Pad Weld for Reinf. pad (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Shell to Pad Weld Junction at Pad OD, Curve: B



Govrn. thk, tg = 22.0, tr = 19.387, c = 1.5 mm., E* = 1.0
 Thickness Ratio = $tr \cdot E^*/(tg - c) = 0.946$, Temp. Reduction = 3 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-5 °C
Min Metal Temp per UCS-66 and UCS-68(c),PWHT credit	-21 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-24 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle-Shell/Head Weld (UCS-66(a)(1)(b)), Curve: B

Govrn. thk, tg = 11.113, tr = 1.971, c = 1.5 mm., E* = 1.0
 Thickness Ratio = $tr \cdot E^*/(tg - c) = 0.205$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-26 °C
Min Metal Temp per UCS-66 and UCS-68(c),PWHT credit	-42 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Gov. MDMT of the Nozzle	: -104 °C
Gov. MDMT of the Reinforcement Pad	: -29 °C
Gov. MDMT of the nozzle to shell joint welded assembly	: -29 °C

Flange MDMT including Temperature reduction per UCS-66.1:

MDMT of ASME B16.5/47 flange per Matl. Specification	-46 °C
Flange MDMT with Temp reduction per UCS-66(i)(2)	-85 °C

Where the Stress Reduction Ratio per UCS-66(i)(2) is :
 Design Pressure/Ambient Rating = $23.00/51.00 = 0.451$

Weld Size Calculations, Description: N18"

Intermediate Calc. for nozzle/shell Welds	Tmin	9.6125 mm.
Intermediate Calc. for pad/shell Welds	TminPad	19.0000 mm.

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = $0.7 \cdot W_o$ mm.
Pad Weld	9.5000 = $0.5 \cdot T_{minPad}$	12.7260 = $0.7 \cdot W_p$ mm.

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A-A1+2 \cdot t_n \cdot f_{r1} \cdot (E1 \cdot t - tr)) \cdot S_v)$
 $= \max(0, (39.2851 - 2.1933 + 2 \cdot 9.6125 \cdot 0.8551 \cdot (1.0 \cdot 20.5 - 19.387)) \cdot 138)$
 $= 52451.39 \text{ Kgf}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2+A5+A4-(W_i-Can/.707)^2 \cdot f_{r2}) \cdot S_v$
 $= (6.0156 + 31.0035 + 4.0951 - 0.0 \cdot 0.86) \cdot 138$
 $= 57854.14 \text{ Kgf}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S_v$
 $= (6.0156 + 0.0 + 0.8551 + (3.37)) \cdot 138$
 $= 14410.30 \text{ Kgf}$

Weld Load [W3]:
 $= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S$
 $= (6.0156 + 0.0 + 4.0951 + 31.0035 + (3.37)) \cdot 138$
 $= 62596.27 \text{ Kgf}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$
 $= (3.1416/2.0) \cdot 219.075 \cdot 10.0 \cdot 0.49 \cdot 118$
 $= 20289. \text{ Kgf}$



Shear, Pad Element Weld [Spew]:
 $= (\pi/2) \cdot DP \cdot WP \cdot 0.49 \cdot SEW$
 $= (3.1416/2.0) \cdot 360.0 \cdot 18.0 \cdot 0.49 \cdot 138$
 $= 70183. \text{ Kg}$

Shear, Nozzle Wall [Snw]:
 $= (\pi \cdot (Dlr + Dlo) / 4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn$
 $= (3.1416 \cdot 104.7313) \cdot (11.1125 - 1.5) \cdot 0.7 \cdot 118$
 $= 26639. \text{ Kg}$

Tension, Pad Groove Weld [Tpgw]:
 $= (\pi/2) \cdot Dlo \cdot Wgpn \cdot 0.74 \cdot Seg$
 $= (3.1416/2) \cdot 219.075 \cdot 22.0 \cdot 0.74 \cdot 138$
 $= 78835. \text{ Kg}$

Tension, Shell Groove Weld [Tngw]:
 $= (\pi/2) \cdot Dlo \cdot (T - cas) \cdot 0.74 \cdot Sng$
 $= (3.1416/2.0) \cdot 219.075 \cdot (22.0 - 1.5) \cdot 0.74 \cdot 138$
 $= 73458. \text{ Kg}$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:
 $= (SPEW + SNW) = (70183 + 26639) = 96822 \text{ Kg}$

Failure Path [Path 2-2]:
 $= (Sonw + Tpgw + Tngw + Sinw)$
 $= (20289 + 78835 + 73458 + 0) = 172582 \text{ Kg}$

Failure Path [Path 3-3]:
 $= (Spew + Tngw + Sinw)$
 $= (70183 + 73458 + 0) = 143642 \text{ Kg}$

Summary of Failure Path Calculations:

Path 1-1 = 96822 Kg, must exceed W = 52451 Kg or W1 = 57854 Kg
 Path 2-2 = 172582 Kg, must exceed W = 52451 Kg or W2 = 14410 Kg
 Path 3-3 = 143641 Kg, must exceed W = 52451 Kg or W3 = 62596 Kg

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 24 bars

The Drop for this Nozzle is : 5.2287 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 445.5007 mm.

Input Echo, WRC297 Item 1, Description: N18" :

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	1.5000	mm.
Shell External Corrosion Allowance	0.0000	mm.
Shell Diameter	2300.000	mm.
Shell Thickness	22.0000	mm.
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc	138.00 N./mm ²
Vessel Hot S.I. Allowable	Smh	138.00 N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter Basis for Nozzle	ID	
Nozzle Corrosion Allowance	1.5000	mm.
Nozzle Diameter	196.850	mm.
Nozzle Thickness	11.1125	mm.
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-333 6	
Nozzle Cold S.I. Allowable	SNmc	118.00 N./mm ²
Nozzle Hot S.I. Allowable	SNmh	118.00 N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter of Reinforcing Pad 360.000 mm.
 Thickness of Reinforcing Pad 22.0000 mm.

Note:

*External Forces and Moments in WRC 107/537 Convention:
 These loads are assumed to be SUSstained loads.*

Design Internal Pressure	Dp	23.00	bars
Radial Load	P	-555.00	Kgf
Circumferential Shear	Vc	820.00	Kgf
Longitudinal Shear	Vl	820.00	Kgf
Circumferential Moment	Mc	490.00	Kg-m.
Longitudinal Moment	Ml	655.00	Kg-m.
Torsional Moment	Mt	820.00	Kg-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(\text{Rmean}(t - ca))}$
 $= 219.075 + 2 \cdot 1.65 \cdot \sqrt{(1172.75(22.0 - 1.5))}$
 $= 730.749 \text{ mm.}$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	2345.500 mm.
Nozzle Outside Diameter	(d) =	219.075 mm.
Vessel Thickness used	(T) =	42.500 mm.
Nozzle Thickness used	(t) =	9.613 mm.
T / t	=	4.421
d / t	=	22.791

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 0.694$$

Nr/P	=	0.092
Mr/P	=	0.079
M0/P	=	0.138
N0/P	=	0.299
MrD/Mc	=	0.124
NrDL/Mc	=	0.122
M0d/Mc	=	0.268
N0DL/Mc	=	0.124
MrD/Ml	=	0.116
NrDL/Ml	=	0.115
M0D/Ml	=	0.238
N0DL/Ml	=	0.289

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	0	0	0	0
Outplane Bending (P)	2	-2	2	-2
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-4	-4	4	4
Outplane Bending (ML)	-23	23	23	-23
Normal Pressure Stress	61	63	61	63
Outplane Stress Summary	36	80	92	43



Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	0	0	0	0
Inplane Bending (P)	1	-1	1	-1
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-1	-1	1	1
Inplane Bending (ML)	-11	11	11	-11
Inplane Pressure Stress	30	30	30	30
Inplane Stress Summary	19	38	45	19

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	2	2	2	2
Shear Stress Summary	3	3	1	1

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	37	80	92	43

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Membrane (P)	0	0	0	0
Outplane Bending (P)	2	-2	2	-2
Outplane Membrane (Mc)	-1	-1	1	1
Outplane Bending (Mc)	-19	19	19	-19
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	30	30	30	30
Outplane Stress Summary	12	47	54	10

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Inplane Membrane (P)	0	0	0	0
Inplane Bending (P)	1	-1	1	-1
Inplane Membrane (Mc)	-1	-1	1	1
Inplane Bending (Mc)	-9	9	9	-9
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	61	63	61	63
Inplane Stress Summary	52	69	73	54

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0



Torsional Shear (Mt)	2	2	2	2
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Shear Stress Summary	1	1	3	3
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Two * Max Shear Stress	52	70	73	55

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Hoop Membrane (P)	0	0	0	0
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-4	-4	4	4
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	27	30	27	30
Hoop Stress Summary	23	26	32	35

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Axial Membrane (P)	0	0	0	0
Axial Bending (P)	11	-11	11	-11
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-20	-20	20	20
Axial Bending (ML)	-112	112	112	-112
Axial Pressure Stress	13	13	13	13
Axial Stress Summary	-106	94	158	-89

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress)	Au Outside	Al Inside	Bu Outside	Bl Inside
Shear due to (Vc)	2	2	-2	-2
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	11	11	11	11
Shear Stress Summary	13	13	8	8

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Outside	Al Inside	Bu Outside	Bl Inside
Two * Max Shear Stress	132	97	159	126

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Hoop Membrane (P)	0	0	0	0
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-1	-1	1	1
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	27	30	27	30



Hoop Stress Summary 26 29 29 32

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses Normal to pipe cross-section)	Left Outside	Left Inside	Right Outside	Right Inside
Axial Membrane (P)	0	0	0	0
Axial Bending (P)	11	-11	11	-11
Axial Membrane (Mc)	-15	-15	15	15
Axial Bending (Mc)	-89	89	89	-89
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	13	13	13	13
Axial Stress Summary	-78	77	131	-71

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Shear stress)	Outside	Inside	Outside	Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-2	-2	2	2
Shear due to Torsion	11	11	11	11
Shear Stress Summary	8	8	13	13

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stress Intensities)	Outside	Inside	Outside	Inside
Two * Max Shear Stress	106	78	132	107

Stress Computations at the Edge of the Reinforcing Pad:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(\text{Rmean}(t - ca))}$
 $= 219.075 + 2 \cdot 1.65 \cdot \sqrt{1161.75(22.0 - 1.5)}$
 $= 728.344 \text{ mm.}$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	2323.500 mm.
Nozzle Outside Diameter	(d) =	360.000 mm.
Vessel Thickness used	(T) =	20.500 mm.
Nozzle Thickness used	(t) =	80.075 mm.
T / t	=	0.256
d / t	=	10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 1.650$$

Note: Re-pad thickness is added to nozzle thickness to simulate the Re-pad.

Nr/P	=	0.078
Mr/P	=	0.166
M0/P	=	0.048
N0/P	=	0.092
MrD/Mc	=	0.296
NrDL/Mc	=	0.088
M0d/Mc	=	0.091
N0DL/Mc	=	0.068
MrD/Ml	=	0.131
NrDL/Ml	=	0.090
M0D/Ml	=	0.041
N0DL/Ml	=	0.113

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses Normal to	Top	Top	Bottom	Bottom



longitudinal plane)	Outside	Inside	Outside	Inside
Outplane Membrane (P)	1	1	1	1
Outplane Bending (P)	3	-3	3	-3
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-4	-4	4	4
Outplane Bending (ML)	-10	10	10	-10
Normal Pressure Stress	128	130	128	130
Outplane Stress Summary	117	133	148	122

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	1	1	1	1
Inplane Bending (P)	12	-12	12	-12
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-3	-3	3	3
Inplane Bending (ML)	-33	33	33	-33
Inplane Pressure Stress	64	64	64	64
Inplane Stress Summary	40	81	115	22

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	1	1	1	1
Shear Stress Summary	2	2	1	1

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	117	133	148	122

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Membrane (P)	1	1	1	1
Outplane Bending (P)	3	-3	3	-3
Outplane Membrane (Mc)	-2	-2	2	2
Outplane Bending (Mc)	-17	17	17	-17
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	64	64	64	64
Outplane Stress Summary	49	76	88	46

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Inplane Membrane (P)	1	1	1	1
Inplane Bending (P)	12	-12	12	-12
Inplane Membrane (Mc)	-2	-2	2	2
Inplane Bending (Mc)	-56	56	56	-56
Inplane Membrane (ML)	0	0	0	0



Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	128	130	128	130
Inplane Stress Summary	82	172	201	64

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu	Cl	Du	Dl
	Left Outside	Left Inside	Right Outside	Right Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Torsional Shear (Mt)	1	1	1	1
Shear Stress Summary	1	1	2	2

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu	Cl	Du	Dl
	Left Outside	Left Inside	Right Outside	Right Inside
Two * Max Shear Stress	82	172	201	65

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	61	63	61	63	61	63	61	63
Circ. Pl (SUS)	-4	-4	4	4	-1	-1	1	1
Circ. Q (SUS)	-21	21	25	-25	-8	8	10	-10
Long. Pm (SUS)	30	30	30	30	30	30	30	30
Long. Pl (SUS)	-1	-1	1	1	-1	-1	1	1
Long. Q (SUS)	-10	10	12	-12	-17	17	21	-21
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	2	2	2	2	2	2	2	2
Pm (SUS)	61.0	63.0	61.0	63.0	61.0	63.0	61.0	63.0
Pm+Pl (SUS)	57.0	59.0	65.0	67.0	60.0	62.0	62.0	64.0
Pm+Pl+Q (Total)	36.2	80.1	90.1	42.2	52.1	70.2	72.2	54.1

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	63.00	138.00	Passed
Pm+Pl (SUS)	67.00	207.00	Passed
Pm+Pl+Q (TOTAL)	90.08	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Reinforcing Pad Edge (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	128	130	128	130	128	130	128	130



Circ. P1 (SUS)	-3	-3	5	5	-1	-1	3	3
Circ. Q (SUS)	-7	7	13	-13	-44	44	68	-68
Long. Pm (SUS)	64	64	64	64	64	64	64	64
Long. P1 (SUS)	-2	-2	4	4	-1	-1	3	3
Long. Q (SUS)	-21	21	45	-45	-14	14	20	-20
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear P1 (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	1	1	1	1	1	1	1	1
Pm (SUS)	128.0	130.0	128.0	130.0	128.0	130.0	128.0	130.0
Pm+P1 (SUS)	125.0	127.0	133.0	135.0	127.0	129.0	131.0	133.0
Pm+P1+Q (Total)	118.0	134.0	146.0	122.0	83.0	173.0	199.0	65.1

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	130.00	138.00	Passed
Pm+P1 (SUS)	135.00	207.00	Passed
Pm+P1+Q (TOTAL)	199.01	413.99	Passed

Because only sustained loads were specified, the Pm+P1+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	27	30	27	30	27	30	27	30
Circ. P1 (SUS)	-4	-4	4	4	-1	-1	1	1
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	13	13	13	13	13	13	13	13
Long. P1 (SUS)	-20	-20	20	20	-15	-15	15	15
Long. Q (SUS)	-101	101	123	-123	-78	78	100	-100
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear P1 (SUS)	2	2	-2	-2	-2	-2	2	2
Shear Q (SUS)	11	11	11	11	11	11	11	11
Pm (SUS)	27.0	30.0	27.0	30.0	27.0	30.0	27.0	30.0
Pm+P1 (SUS)	30.3	33.2	34.2	35.6	28.3	31.3	30.0	32.0
Pm+P1+Q (Total)	133.6	96.4	156.6	125.3	107.5	77.7	129.7	106.2

Nozzle Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	30.00	118.00	Passed
Pm+P1 (SUS)	35.56	177.00	Passed
Pm+P1+Q (TOTAL)	156.64	354.00	Passed

Because only sustained loads were specified, the Pm+P1+Q allowable was 3 * Smh.



Input, Nozzle Desc: N22"

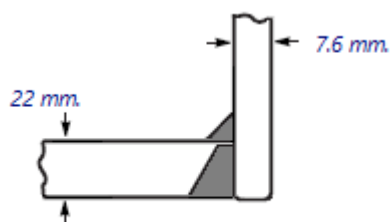
From: 20

Pressure for Reinforcement Calculations	P	23.000	bars
Temperature for Internal Pressure	Temp	150	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm2
Parent Allowable Stress At Ambient	Sva	138.00	N./mm2
Inside Diameter of Cylindrical Shell	D	2300.00	mm.
Shell Finished (Minimum) Thickness	t	22.0000	mm.
Shell Internal Corrosion Allowance	c	1.5000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		850.00	mm.
User Entered Minimum Design Metal Temperature		-33.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-333 6 +IT	
Material UNS Number		K03006	
Material Specification/Type	Smls. & wld. pipe		
Allowable Stress at Temperature	Sn	118.00	N./mm2
Allowable Stress At Ambient	Sna	118.00	N./mm2
Diameter Basis (for tr calc only)		Inside	
Layout Angle		90.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Minimum	
Nominal Thickness		160	
Flange Material		SA-350 LF2 +IT	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	1.5000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	454.8480	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	22.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Flange Class		600	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: N22"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	1.773	in.
Actual Thickness Used in Calculation	0.301	in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 23.0 \cdot 1151.5 / (138 \cdot 1.0 - 0.6 \cdot 23.0)$
 $= 19.3870$ mm.

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 23.0 \cdot 24.02 / (118 \cdot 1.0 - 0.6 \cdot 23.0)$
 $= 0.4737$ mm.

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	101.3250	mm.
Parallel to Vessel Wall	Rn+tn+tr	50.6625	mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	15.3635	mm.

*Taking a UG-36(c)(3)(a) exemption for nozzle: N22".
 This calculation is valid for nozzles that meet all the requirements of
 paragraph UG-36. Please check the Code carefully, especially for nozzles
 that are not isolated or do not meet Code spacing requirements. To force
 the computation of areas for small nozzles go to Tools->Configuration
 and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 1.9737	mm.
Wall Thickness per UG16(b),	tr16b	= 3.0000	mm.
Wall Thickness, shell/head, internal pressure	trb1	= 20.8870	mm.
Wall Thickness	tb1 = max(trb1, tr16b)	= 20.8870	mm.
Wall Thickness	tb2 = max(trb2, tr16b)	= 3.0000	mm.
Wall Thickness per table UG-45	tb3	= 4.9200	mm.

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[4.92, \max(20.887, 3.0)]$
 $= 4.9200$ mm.

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(1.9737, 4.92)$
 $= 4.9200$ mm.

Available Nozzle Neck Thickness = 7.6454 mm. -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	:	52.0,	Allowable	:	118.0 N./mm2	Passed
Expansion	:	0.0,	Allowable	:	243.0 N./mm2	Passed
Occasional	:	4.0,	Allowable	:	156.9 N./mm2	Passed
Shear	:	36.4,	Allowable	:	82.6 N./mm2	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld (Impact tested) :

*Note:
 This Material was specified as being an Impact Tested (Low Temperature) Material.*

Impact Test Temperature provided per Specification	-46 °C
Calculated Minimum Design Metal Temperature	-104 °C

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), Curve: B

Govrn. thk, tg = 7.645, tr = 0.474, c = 1.5 mm., E* = 1.0



Thickness Ratio = $tr * E / (tg - c) = 0.077$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -29 °C
 Min Metal Temp per UCS-66 and UCS-68(c),PWHT credit -46 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:

MDMT of ASME B16.5/47 flange per Matl. Specification -46 °C
 Flange MDMT with Temp reduction per UCS-66(i)(2) -104 °C

Where the Stress Reduction Ratio per UCS-66(i)(2) is :
 Design Pressure/Ambient Rating = $23.00/102.00 = 0.225$

Weld Size Calculations, Description: N22"

Intermediate Calc. for nozzle/shell Welds Tmin 6.1454 mm.

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$4.3018 = 0.7 * t_{min}$	$7.0700 = 0.7 * W_o$ mm.

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 0.3956 mm.
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 477.2436 mm.

Input Echo, WRC297 Item 1, Description: N22" :

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	1.5000	mm.
Shell External Corrosion Allowance	0.0000	mm.
Shell Diameter	2300.000	mm.
Shell Thickness	22.0000	mm.
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc	138.00 N./mm ²
Vessel Hot S.I. Allowable	Smh	138.00 N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter Basis for Nozzle	ID	
Nozzle Corrosion Allowance	1.5000	mm.
Nozzle Diameter	45.034	mm.
Nozzle Thickness	7.6454	mm.
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-333 6	
Nozzle Cold S.I. Allowable	SNmc	118.00 N./mm ²
Nozzle Hot S.I. Allowable	SNmh	118.00 N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Note:

External Forces and Moments in WRC 107/537 Convention:
 These loads are assumed to be SUSstained loads.

Design Internal Pressure	Dp	23.00	bars
Radial Load	P	-165.00	Kgf
Circumferential Shear	Vc	260.00	Kgf
Longitudinal Shear	Vl	260.00	Kgf
Circumferential Moment	Mc	35.00	Kg-m.
Longitudinal Moment	Ml	55.00	Kg-m.
Torsional Moment	Mt	65.00	Kg-m.

Include Axial Pressure Thrust No



Include Pressure Stress Indices per Div. 2 No
 Local Loads applied at end of Nozzle/Attachment No

*Warning - The ratio, Dn/Tv, (2.742) is < 5
 check the Limitations of WRC bulletin 297.*
*Warning - The ratio, Dn/Tn (7.890) is < 10
 check the Limitations of WRC bulletin 297.*

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(\text{Rmean}(t - ca))}$
 $= 60.325 + 2 \cdot 1.65 \cdot \sqrt{(1161.75(22.0 - 1.5))}$
 $= 569.594 \text{ mm.}$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter (D) = 2323.500 mm.
 Nozzle Outside Diameter (d) = 60.325 mm.
 Vessel Thickness used (T) = 20.500 mm.
 Nozzle Thickness used (t) = 6.145 mm.
 T / t = 3.336
 d / t = 10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 0.276$$

Nr/P = 0.229
 Mr/P = 0.194
 M0/P = 0.217
 N0/P = 0.211
 MrD/Mc = 0.178
 NrDL/Mc = 0.181
 M0d/Mc = 0.234
 N0DL/Mc = -0.125
 MrD/Ml = 0.175
 NrDL/Ml = 0.184
 M0D/Ml = 0.229
 N0DL/Ml = -0.100

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	0	0	0	0
Outplane Bending (P)	4	-4	4	-4
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	2	2	-2	-2
Outplane Bending (ML)	-29	29	29	-29
Normal Pressure Stress	128	130	128	130
Outplane Stress Summary	106	157	160	94

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	0	0	0	0
Inplane Bending (P)	4	-4	4	-4
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-3	-3	3	3
Inplane Bending (ML)	-22	22	22	-22
Inplane Pressure Stress	64	64	64	64
Inplane Stress Summary	43	78	95	41

Vessel Stresses (N./mm²):



LONGITUDINAL PLANE			Au	Al	Bu	B1
(Shear stress normal to longitudinal plane)			Top Outside	Top Inside	Bottom Outside	Bottom Inside
Outplane Shear (Vc)			1	1	-1	-1
Outplane Shear (Vl)			0	0	0	0
Outplane Shear (Mt)			5	5	5	5
Shear Stress Summary			6	6	4	4

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE			Au	Al	Bu	B1
(Stress Intensities)			Top Outside	Top Inside	Bottom Outside	Bottom Inside
Two * Max Shear Stress			107	158	161	95

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE			Cu	Cl	Du	D1
(Stresses Normal to circumferential plane)			Left Outside	Left Inside	Right Outside	Right Inside
Outplane Membrane (P)			0	0	0	0
Outplane Bending (P)			4	-4	4	-4
Outplane Membrane (Mc)			1	1	-1	-1
Outplane Bending (Mc)			-19	19	19	-19
Outplane Membrane (ML)			0	0	0	0
Outplane Bending (ML)			0	0	0	0
Normal Pressure Stress			64	64	64	64
Outplane Stress Summary			52	80	87	39

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE			Cu	Cl	Du	D1
(Stresses parallel to circumferential plane)			Left Outside	Left Inside	Right Outside	Right Inside
Inplane Membrane (P)			0	0	0	0
Inplane Bending (P)			4	-4	4	-4
Inplane Membrane (Mc)			-2	-2	2	2
Inplane Bending (Mc)			-14	14	14	-14
Inplane Membrane (ML)			0	0	0	0
Inplane Bending (ML)			0	0	0	0
Inplane Pressure Stress			128	130	128	130
Inplane Stress Summary			116	138	150	114

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE			Cu	Cl	Du	D1
(Shear stress normal to circumferential plane)			Left Outside	Left Inside	Right Outside	Right Inside
Outplane Shear (Vc)			0	0	0	0
Outplane Shear (Vl)			-1	-1	1	1
Torsional Shear (Mt)			5	5	5	5
Shear Stress Summary			4	4	6	6

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE			Cu	Cl	Du	D1
(Stress Intensities)			Left Outside	Left Inside	Right Outside	Right Inside
Two * Max Shear Stress			116	139	151	115

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE			Au	Al	Bu	B1
(Stresses in the hoop direction)			Top Outside	Top Inside	Bottom Outside	Bottom Inside



Hoop Membrane (P)	0	0	0	0
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	2	2	-2	-2
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	9	12	9	12
Hoop Stress Summary	12	15	8	11

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Axial Membrane (P)	1	1	1	1
Axial Bending (P)	20	-20	20	-20
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-41	-41	41	41
Axial Bending (ML)	-118	118	118	-118
Axial Pressure Stress	4	4	4	4
Axial Stress Summary	-133	62	187	-90

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress)	Au Outside	Al Inside	Bu Outside	Bl Inside
Shear due to (Vc)	4	4	-4	-4
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	18	18	18	18
Shear Stress Summary	22	22	13	13

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Outside	Al Inside	Bu Outside	Bl Inside
Two * Max Shear Stress	153	71	188	105

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Hoop Membrane (P)	0	0	0	0
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	1	1	-1	-1
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	9	12	9	12
Hoop Stress Summary	12	14	8	11

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Axial Membrane (P)	1	1	1	1
Axial Bending (P)	20	-20	20	-20
Axial Membrane (Mc)	-26	-26	26	26
Axial Bending (Mc)	-79	79	79	-79
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	4	4	4	4
Axial Stress Summary	-79	38	132	-66



Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-4	-4	4	4
Shear due to Torsion	18	18	18	18
Shear Stress Summary	13	13	22	22

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Two * Max Shear Stress	95	44	136	89

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	128	130	128	130	128	130	128	130
Circ. Pl (SUS)	2	2	-2	-2	-2	-2	2	2
Circ. Q (SUS)	-25	25	33	-33	-10	10	18	-18
Long. Pm (SUS)	64	64	64	64	64	64	64	64
Long. Pl (SUS)	-3	-3	3	3	1	1	-1	-1
Long. Q (SUS)	-18	18	26	-26	-15	15	23	-23
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	5	5	5	5	5	5	5	5
Pm (SUS)	128.0	130.0	128.0	130.0	128.0	130.0	128.0	130.0
Pm+Pl (SUS)	130.0	132.0	126.0	128.0	126.0	128.0	130.0	132.0
Pm+Pl+Q (Total)	105.6	157.5	159.2	95.3	116.2	138.3	148.6	114.5

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	130.00	138.00	Passed
Pm+Pl (SUS)	132.01	207.00	Passed
Pm+Pl+Q (TOTAL)	159.24	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	9	12	9	12	9	12	9	12
Circ. Pl (SUS)	2	2	-2	-2	1	1	-1	-1
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	4	4	4	4	4	4	4	4
Long. Pl (SUS)	-40	-40	42	42	-25	-25	27	27
Long. Q (SUS)	-98	98	138	-138	-59	59	99	-99



Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	4	4	-4	-4	-4	-4	4	4
Shear Q (SUS)	18	18	18	18	18	18	18	18
Pm (SUS)	9.0	12.0	9.0	12.0	9.0	12.0	9.0	12.0
Pm+Pl (SUS)	47.7	50.6	46.4	46.4	32.0	34.9	31.7	31.8
Pm+Pl+Q (Total)	151.5	70.6	185.1	105.8	94.3	44.3	133.8	90.4

Nozzle Stress Summation Comparison (N./mm2):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	12.00	118.00	Passed
Pm+Pl (SUS)	50.64	177.00	Passed
Pm+Pl+Q (TOTAL)	185.10	354.00	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

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Input, Nozzle Desc: N64"

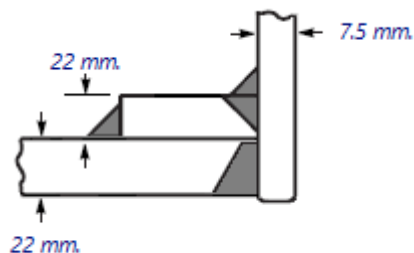
From: 20

Pressure for Reinforcement Calculations	P	23.137	bars
Temperature for Internal Pressure	Temp	150	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm2
Parent Allowable Stress At Ambient	Sva	138.00	N./mm2
Inside Diameter of Cylindrical Shell	D	2300.00	mm.
Shell Finished (Minimum) Thickness	t	22.0000	mm.
Shell Internal Corrosion Allowance	c	1.5000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		800.00	mm.
User Entered Minimum Design Metal Temperature		-33.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-333 6 +IT	
Material UNS Number		K03006	
Material Specification/Type	Smls. & wld. pipe		
Allowable Stress at Temperature	Sn	118.00	N./mm2
Allowable Stress At Ambient	Sna	118.00	N./mm2
Diameter Basis (for tr calc only)		Inside	
Layout Angle		270.00	deg
Diameter		4.0000	in.
Size and Thickness Basis		Minimum	
Nominal Thickness		80	
Flange Material		SA-350 LF2 +IT	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	1.5000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	443.6720	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	22.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Pad Material		SA-516 70 +IT	
Pad Allowable Stress at Temperature	Sp	138.00	N./mm2
Pad Allowable Stress At Ambient	Spa	138.00	N./mm2
Diameter of Pad along vessel surface	Dp	203.2000	mm.
Thickness of Pad	te	22.0000	mm.
Weld leg size between Pad and Shell	Wp	18.0000	mm.
Groove weld depth between Pad and Nozzle	Wgpn	22.0000	mm.
Reinforcing Pad Width		44.4500	mm.
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: N64"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	3.910 in.
Actual Thickness Used in Calculation	0.295 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 23.14 \cdot 1151.5 / (138 \cdot 1.0 - 0.6 \cdot 23.14)$
 $= 19.5037 \text{ mm.}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 23.14 \cdot 51.16 / (118 \cdot 1.0 - 0.6 \cdot 23.14)$
 $= 1.0151 \text{ mm.}$

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	204.6407 mm.
Parallel to Vessel Wall, opening length	d	102.3204 mm.
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		36.9746 mm.

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 118.0 / 138.0)$
 $= 0.855$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 118.0 / 138.0)$
 $= 0.855$

Weld Strength Reduction Factor [fr4]:
 $= \min(1, S_p / S_v)$
 $= \min(1, 138.0 / 138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(0.855, 1.0)$
 $= 0.855$

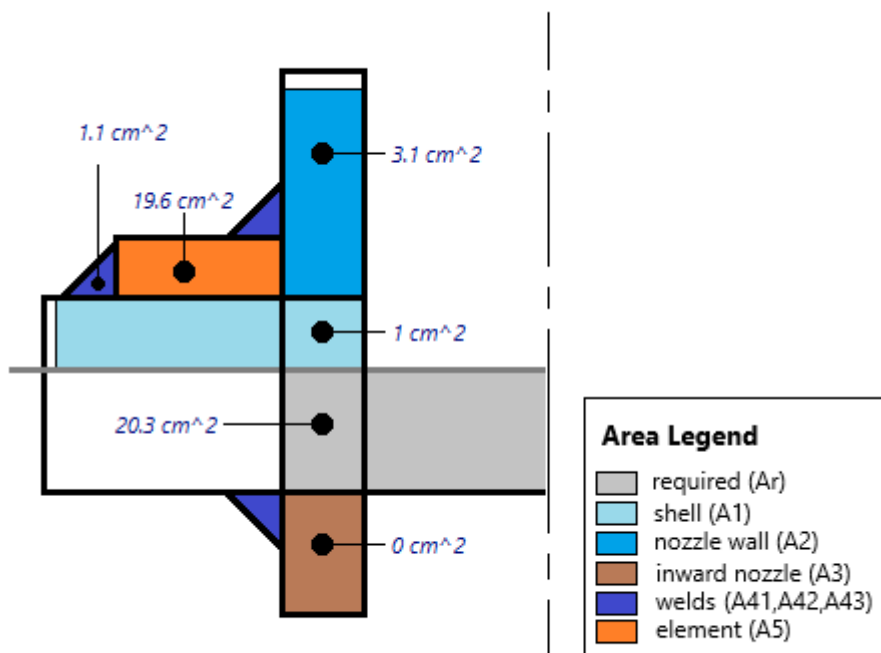


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (cm2)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	20.295	NA	NA
Area in Shell	A1	1.002	NA	NA
Area in Nozzle Wall	A2	3.146	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	1.109	NA	NA
Area in Element	A5	19.558	NA	NA
TOTAL AREA AVAILABLE	Atot	24.815	NA	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Insufficient.
 The area available with the given pad is Sufficient.

SELECTION OF POSSIBLE REINFORCING PADS: Diameter Thickness
 Based on given Pad Thickness: 182.6539 22.0000 mm.
 Based on given Pad Diameter: 203.2000 16.9155 mm.
 Based on the Estimated Diameter Limit: 203.0532 16.9434 mm.

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$
 $= (102.3204 \cdot 19.5037 \cdot 1.0 + 2 \cdot 5.9898 \cdot 19.5037 \cdot 1.0 \cdot (1 - 0.86))$
 $= 20.295 \text{ cm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 102.32(1.0 \cdot 20.5 - 1.0 \cdot 19.504) - 2 \cdot 5.99$
 $(1.0 \cdot 20.5 - 1.0 \cdot 19.5037) \cdot (1 - 0.855)$
 $= 1.002 \text{ cm}^2$

Area Available in Nozzle Wall Projecting Outward [A2]:
 $= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2$
 $= (2 \cdot 36.97) \cdot (5.99 - 1.02) \cdot 0.8551$



= 3.146 cm²

Area Available in Welds [A41 + A42 + A43]:
 = (Wo² - Ar Lost)•Fr3+((Wi-can/0.707)² - Ar Lost)•fr2 + Trapfr4
 = (1.) • 0.86 + (0.0) • 0.86 + 25.4137² • 1.0
 = 1.109 cm²

Area Available in Element [A5]:
 = (min(Dp,DL)-(Nozzle OD))•(min(tp,Tlwp,te)) • fr4
 = (203.2 - 114.3) • 22.0 • 1.0
 = 19.558 cm²

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures ta = 2.5151 mm.
 Wall Thickness per UG16(b), tr16b = 3.0000 mm.
 Wall Thickness, shell/head, internal pressure trb1 = 21.0037 mm.
 Wall Thickness tb1 = max(trb1, tr16b) = 21.0037 mm.
 Wall Thickness tb2 = max(trb2, tr16b) = 3.0000 mm.
 Wall Thickness per table UG-45 tb3 = 6.7578 mm.

Determine Nozzle Thickness candidate [tb]:
 = min[tb3, max(tb1,tb2)]
 = min[6.758, max(21.0037, 3.0)]
 = 6.7578 mm.

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 = max(ta, tb)
 = max(2.5151, 6.7578)
 = 6.7578 mm.

Available Nozzle Neck Thickness = 7.4898 mm. -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 46.7,	Allowable	: 118.0 N./mm ²	Passed
Expansion	: 0.0,	Allowable	: 248.3 N./mm ²	Passed
Occasional	: 9.3,	Allowable	: 156.9 N./mm ²	Passed
Shear	: 26.3,	Allowable	: 82.6 N./mm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld (Impact tested) :

Note:
This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification	-46 °C
Calculated Minimum Design Metal Temperature	-104 °C

Nozzle Neck to Pad Weld for the Nozzle (Impact tested) :

Note:
This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification	-46 °C
Calculated Minimum Design Metal Temperature	-104 °C

Nozzle Neck to Pad Weld for Reinf. pad (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Shell to Pad Weld Junction at Pad OD, Curve: B



Govrn. thk, tg = 22.0, tr = 19.504, c = 1.5 mm., E* = 1.0
 Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.951$, Temp. Reduction = 3 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-5 °C
Min Metal Temp per UCS-66 and UCS-68(c),PWHT credit	-21 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-24 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle-Shell/Head Weld (UCS-66(a)(b)), Curve: B

Govrn. thk, tg = 7.49, tr = 1.015, c = 1.5 mm., E* = 1.0
 Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.169$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-29 °C
Min Metal Temp per UCS-66 and UCS-68(c),PWHT credit	-46 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Gov. MDMT of the Nozzle	: -104 °C
Gov. MDMT of the Reinforcement Pad	: -29 °C
Gov. MDMT of the nozzle to shell joint welded assembly	: -29 °C

Flange MDMT including Temperature reduction per UCS-66.1:

MDMT of ASME B16.5/47 flange per Matl. Specification	-46 °C
Flange MDMT with Temp reduction per UCS-66(i)(2)	-85 °C

Where the Stress Reduction Ratio per UCS-66(i)(2) is :
 Design Pressure/Ambient Rating = $23.14/51.00 = 0.454$

Weld Size Calculations, Description: N64"

Intermediate Calc. for nozzle/shell Welds	Tmin	5.9898 mm.
Intermediate Calc. for pad/shell Welds	TminPad	19.0000 mm.

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$4.1929 = 0.7 \cdot t_{min}$	$7.0700 = 0.7 \cdot W_o$ mm.
Pad Weld	$9.5000 = 0.5 \cdot T_{minPad}$	$12.7260 = 0.7 \cdot W_p$ mm.

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A-A1+2 \cdot t_n \cdot f_{r1} \cdot (E1 \cdot t - tr)) \cdot S_v)$
 $= \max(0, (20.2948 - 1.0021 + 2 \cdot 5.9898 \cdot 0.8551 \cdot (1.0 \cdot 20.5 - 19.5037)) \cdot 138)$
 $= 27291.50 \text{ Kgf}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2+A5+A4-(W_i-Can/.707)^2 \cdot f_{r2}) \cdot S_v$
 $= (3.1456 + 19.558 + 1.1092 - 0.0 \cdot 0.86) \cdot 138$
 $= 33508.45 \text{ Kgf}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S_v$
 $= (3.1456 + 0.0 + 0.8551 + (2.1)) \cdot 138$
 $= 8584.62 \text{ Kgf}$

Weld Load [W3]:
 $= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S$
 $= (3.1456 + 0.0 + 1.1092 + 19.558 + (2.1)) \cdot 138$
 $= 36463.41 \text{ Kgf}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$
 $= (3.1416/2.0) \cdot 114.3 \cdot 10.0 \cdot 0.49 \cdot 118$
 $= 10586. \text{ Kgf}$

Shear, Pad Element Weld [Spew]:
 $= (\pi/2) \cdot DP \cdot WP \cdot 0.49 \cdot SEW$
 $= (3.1416/2.0) \cdot 203.2 \cdot 18.0 \cdot 0.49 \cdot 138$
 $= 39615. \text{ Kg}$

Shear, Nozzle Wall [Snw]:
 $= (\pi \cdot (Dlr + Dlo) / 4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn$
 $= (3.1416 \cdot 54.1551) \cdot (7.4898 - 1.5) \cdot 0.7 \cdot 118$
 $= 8583. \text{ Kg}$

Tension, Pad Groove Weld [Tpgw]:
 $= (\pi/2) \cdot Dlo \cdot Wgpn \cdot 0.74 \cdot Seg$
 $= (3.1416/2) \cdot 114.3 \cdot 22.0 \cdot 0.74 \cdot 138$
 $= 41131. \text{ Kg}$

Tension, Shell Groove Weld [Tngw]:
 $= (\pi/2) \cdot Dlo \cdot (T - cas) \cdot 0.74 \cdot Sng$
 $= (3.1416/2.0) \cdot 114.3 \cdot (22.0 - 1.5) \cdot 0.74 \cdot 138$
 $= 38326. \text{ Kg}$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:
 $= (SPEW + SNW) = (39615 + 8583) = 48198 \text{ Kg}$

Failure Path [Path 2-2]:
 $= (Sonw + Tpgw + Tngw + Sinw)$
 $= (10586 + 41131 + 38326 + 0) = 90043 \text{ Kg}$

Failure Path [Path 3-3]:
 $= (Spew + Tngw + Sinw)$
 $= (39615 + 38326 + 0) = 77941 \text{ Kg}$

Summary of Failure Path Calculations:

Path 1-1 = 48197	Kgf, must exceed W = 27291 Kg	or W1 = 33508 Kg
Path 2-2 = 90042	Kgf, must exceed W = 27291 Kg	or W2 = 8584 Kg
Path 3-3 = 77940	Kgf, must exceed W = 27291 Kg	or W3 = 36463 Kg

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 24 bars

Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 1.4209 mm.
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 467.0930 mm.

Input Echo, WRC297 Item 1, Description: N64" :

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	1.5000	mm.
Shell External Corrosion Allowance	0.0000	mm.
Shell Diameter	2300.000	mm.
Shell Thickness	22.0000	mm.
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc 138.00	N./mm ²
Vessel Hot S.I. Allowable	Smh 138.00	N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter Basis for Nozzle	ID	
Nozzle Corrosion Allowance	1.5000	mm.
Nozzle Diameter	99.320	mm.
Nozzle Thickness	7.4898	mm.
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-333 6	



Nozzle Cold S.I. Allowable	SNmc	118.00	N./mm2
Nozzle Hot S.I. Allowable	SNmh	118.00	N./mm2

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter of Reinforcing Pad	203.200	mm.
Thickness of Reinforcing Pad	22.0000	mm.

Note:

*External Forces and Moments in WRC 107/537 Convention:
 These loads are assumed to be SUSained loads.*

Design Internal Pressure	Dp	23.14	bars
Radial Load	P	-280.00	Kgf
Circumferential Shear	Vc	410.00	Kgf
Longitudinal Shear	Vl	410.00	Kgf
Circumferential Moment	Mc	125.00	Kg-m.
Longitudinal Moment	Ml	165.00	Kg-m.
Torsional Moment	Mt	205.00	Kg-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{\text{mean}}(t - c_a))}$$

$$= 114.3 + 2 \cdot 1.65 \cdot \sqrt{(1172.75(22.0 - 1.5))}$$

$$= 625.974 \text{ mm.}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	2345.500	mm.
Nozzle Outside Diameter	(d) =	114.300	mm.
Vessel Thickness used	(T) =	42.500	mm.
Nozzle Thickness used	(t) =	5.990	mm.
T / t	=	7.095	
d / t	=	19.082	

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 0.362$$

Nr/P	=	0.114
Mr/P	=	0.074
M0/P	=	0.274
N0/P	=	0.359
MrD/Mc	=	0.094
NrDL/Mc	=	0.106
M0d/Mc	=	0.316
N0DL/Mc	=	-0.140
MrD/Ml	=	0.095
NrDL/Ml	=	0.107
M0D/Ml	=	0.305
N0DL/Ml	=	0.084

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	0	0	0	0
Outplane Bending (P)	2	-2	2	-2
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (Ml)	0	0	0	0
Outplane Bending (Ml)	-14	14	14	-14
Normal Pressure Stress	61	63	61	63



Outplane Stress Summary 49 75 79 48

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE
 (Stresses parallel to longitudinal plane)

	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
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Inplane Membrane (P)	0	0	0	0
Inplane Bending (P)	0	0	0	0
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	-4	4	4	-4
Inplane Pressure Stress	30	30	30	30

Inplane Stress Summary 26 33 36 26

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE
 (Shear stress normal to longitudinal plane)

	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
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Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	2	2	2	2

Shear Stress Summary 2 2 1 1

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE
 (Stress Intensities)

	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
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Two * Max Shear Stress 49 75 79 48

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE
 (Stresses Normal to circumferential plane)

	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
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Outplane Membrane (P)	0	0	0	0
Outplane Bending (P)	2	-2	2	-2
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	-11	11	11	-11
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	30	30	30	30

Outplane Stress Summary 23 40 44 16

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE
 (Stresses parallel to circumferential plane)

	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
--	-----------------------	----------------------	------------------------	-----------------------

Inplane Membrane (P)	0	0	0	0
Inplane Bending (P)	0	0	0	0
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	-3	3	3	-3
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	61	63	61	63

Inplane Stress Summary 58 66 66 60

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE
 (Shear stress normal to circumferential plane)

	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
--	-----------------------	----------------------	------------------------	-----------------------



Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Torsional Shear (Mt)	2	2	2	2
Shear Stress Summary	1	1	2	2

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Two * Max Shear Stress	58	66	66	60

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Hoop Membrane (P)	0	0	0	0
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	22	25	22	25
Hoop Stress Summary	22	25	23	26

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Axial Membrane (P)	1	1	1	1
Axial Bending (P)	7	-7	7	-7
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-30	-30	30	30
Axial Bending (ML)	-97	97	97	-97
Axial Pressure Stress	11	11	11	11
Axial Stress Summary	-108	71	149	-62

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress)	Au Outside	Al Inside	Bu Outside	Bl Inside
Shear due to (Vc)	3	3	-3	-3
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	16	16	16	16
Shear Stress Summary	20	20	12	12

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Outside	Al Inside	Bu Outside	Bl Inside
Two * Max Shear Stress	136	78	150	92

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Hoop Membrane (P)	0	0	0	0
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0



Hoop Pressure Stress	22	25	22	25
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Hoop Stress Summary	23	26	22	24
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Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
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Axial Membrane (P)	1	1	1	1
Axial Bending (P)	7	-7	7	-7
Axial Membrane (Mc)	-23	-23	23	23
Axial Bending (Mc)	-73	73	73	-73
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	11	11	11	11

Axial Stress Summary	-76	54	117	-45
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Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu Outside	Cl Inside	Du Outside	Dl Inside
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Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-3	-3	3	3
Shear due to Torsion	16	16	16	16

Shear Stress Summary	12	12	20	20
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Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities	Cu Outside	Cl Inside	Du Outside	Dl Inside
--	---------------	--------------	---------------	--------------

Two * Max Shear Stress	103	59	121	81
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Stress Computations at the Edge of the Reinforcing Pad:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - ca)}$$

$$= 114.3 + 2 \cdot 1.65 \cdot \sqrt{1161.75(22.0 - 1.5)}$$

$$= 623.569 \text{ mm.}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	2323.500 mm.
Nozzle Outside Diameter	(d) =	203.200 mm.
Vessel Thickness used	(T) =	20.500 mm.
Nozzle Thickness used	(t) =	50.440 mm.
T / t	=	0.406
d / t	=	10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 0.931$$

Note: Re-pad thickness is added to nozzle thickness to simulate the Re-pad.

Nr/P	=	0.109
Mr/P	=	0.235
M0/P	=	0.069
N0/P	=	0.133
MrD/Mc	=	0.309
NrDL/Mc	=	0.082
M0d/Mc	=	0.096
N0DL/Mc	=	0.037
MrD/Ml	=	0.204
NrDL/Ml	=	0.090
M0D/Ml	=	0.065
N0DL/Ml	=	0.111

Vessel Stresses (N./mm²):



LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	0	0	0	0
Outplane Bending (P)	2	-2	2	-2
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-2	-2	2	2
Outplane Bending (ML)	-7	7	7	-7
Normal Pressure Stress	128	131	128	131
Outplane Stress Summary	122	134	141	123

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	0	0	0	0
Inplane Bending (P)	9	-9	9	-9
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-1	-1	1	1
Inplane Bending (ML)	-23	23	23	-23
Inplane Pressure Stress	64	64	64	64
Inplane Stress Summary	49	77	99	34

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	1	1	1	1
Shear Stress Summary	2	2	0	0

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	122	134	141	123

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Membrane (P)	0	0	0	0
Outplane Bending (P)	2	-2	2	-2
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	-8	8	8	-8
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	64	64	64	64
Outplane Stress Summary	59	70	76	54

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Inplane Membrane (P)	0	0	0	0
Inplane Bending (P)	9	-9	9	-9
Inplane Membrane (Mc)	-1	-1	1	1



Inplane Bending (Mc)	-26	26	26	-26
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	128	131	128	131

Inplane Stress Summary	110	148	166	97
------------------------	-----	-----	-----	----

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu	Cl	Du	Dl
	Left Outside	Left Inside	Right Outside	Right Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Torsional Shear (Mt)	1	1	1	1

Shear Stress Summary	0	0	2	2
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu	Cl	Du	Dl
	Left Outside	Left Inside	Right Outside	Right Inside
Two * Max Shear Stress	110	148	166	97

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	61	63	61	63	61	63	61	63
Circ. Pl (SUS)	0	0	0	0	0	0	0	0
Circ. Q (SUS)	-12	12	16	-16	-3	3	3	-3
Long. Pm (SUS)	30	30	30	30	30	30	30	30
Long. Pl (SUS)	0	0	0	0	0	0	0	0
Long. Q (SUS)	-4	4	4	-4	-9	9	13	-13
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	2	2	2	2	2	2	2	2
Pm (SUS)	61.0	63.0	61.0	63.0	61.0	63.0	61.0	63.0
Pm+Pl (SUS)	61.0	63.0	61.0	63.0	61.0	63.0	61.0	63.0
Pm+Pl+Q (Total)	49.2	75.1	77.1	47.2	58.1	66.1	64.2	60.1

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	63.00	138.00	Passed
Pm+Pl (SUS)	63.00	207.00	Passed
Pm+Pl+Q (TOTAL)	77.09	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Reinforcing Pad Edge (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl



Circ. Pm (SUS)	128	131	128	131	128	131	128	131
Circ. Pl (SUS)	-2	-2	2	2	-1	-1	1	1
Circ. Q (SUS)	-5	5	9	-9	-17	17	35	-35
Long. Pm (SUS)	64	64	64	64	64	64	64	64
Long. Pl (SUS)	-1	-1	1	1	0	0	0	0
Long. Q (SUS)	-14	14	32	-32	-6	6	10	-10
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	1	1	1	1	1	1	1	1
Pm (SUS)	128.0	131.0	128.0	131.0	128.0	131.0	128.0	131.0
Pm+Pl (SUS)	126.0	129.0	130.0	133.0	127.0	130.0	129.0	132.0
Pm+Pl+Q (Total)	121.0	134.0	139.0	124.0	110.0	147.0	164.0	97.0

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	131.00	138.00	Passed
Pm+Pl (SUS)	133.00	207.00	Passed
Pm+Pl+Q (TOTAL)	164.01	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	22	25	22	25	22	25	22	25
Circ. Pl (SUS)	0	0	0	0	0	0	0	0
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	11	11	11	11	11	11	11	11
Long. Pl (SUS)	-29	-29	31	31	-22	-22	24	24
Long. Q (SUS)	-90	90	104	-104	-66	66	80	-80
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	3	3	-3	-3	-3	-3	3	3
Shear Q (SUS)	16	16	16	16	16	16	16	16
Pm (SUS)	22.0	25.0	22.0	25.0	22.0	25.0	22.0	25.0
Pm+Pl (SUS)	40.4	43.4	42.4	42.5	33.5	36.5	35.7	35.8
Pm+Pl+Q (Total)	135.4	78.7	147.3	90.8	102.4	59.8	118.7	79.6

Nozzle Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	25.00	118.00	Passed
Pm+Pl (SUS)	43.42	177.00	Passed
Pm+Pl+Q (TOTAL)	147.35	354.00	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.



Input, Nozzle Desc: K81"

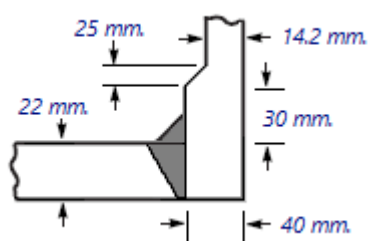
From: 30

Pressure for Reinforcement Calculations	P	23.000	bars
Temperature for Internal Pressure	Temp	150	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm2
Parent Allowable Stress At Ambient	Sva	138.00	N./mm2
Inside Diameter of Cylindrical Shell	D	2300.00	mm.
Shell Finished (Minimum) Thickness	t	22.0000	mm.
Shell Internal Corrosion Allowance	c	1.5000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		3250.00	mm.
User Entered Minimum Design Metal Temperature		-33.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-350 LF2 +IT	
Material UNS Number		K03011	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	N./mm2
Allowable Stress At Ambient	Sna	138.00	N./mm2
Diameter Basis (for tr calc only)		Inside	
Layout Angle		90.00	deg
Diameter		1.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	14.2240	mm.
Flange Material		SA-350 LF2 +IT	
Flange Type		N/A	
Hub Height of Integral Nozzle	h	30.0000	mm.
Height of Beveled Transition	L'	25.0000	mm.
Hub Thickness of Integral Nozzle (tn or x+tp)		40.0000	mm.
Style of FVC Nozzle		Long Weld Neck	
Corrosion Allowance	can	1.5000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	512.2520	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	22.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation 1.000 in.
 Actual Thickness Used in Calculation 0.560 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 23.0 \cdot 1151.5 / (138 \cdot 1.0 - 0.6 \cdot 23.0)$
 $= 19.3870$ mm.

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]
 $= R \cdot (\exp([P / (S \cdot E)] - 1))$ per Appendix 1-2 (a)(1)
 $= 14.2 \cdot (\exp([23.0 / (138 \cdot 1.0)] - 1))$
 $= 0.2387$ mm.

Intermediate Hub Nozzle Calculations:

The Diameter Limit is less than the Hub OD !

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
 $= (\text{Hub Thickness} - \text{Neck Thickness}) / \tan(30)$
 $= 20.5 / 0.5773$
 $= 35.5071$ mm.

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	94.8480 mm.
Parallel to Vessel Wall	Rn+tn+t	47.4240 mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	51.2500 mm.

*Taking a UG-36(c)(3)(a) exemption for nozzle: K81".
 This calculation is valid for nozzles that meet all the requirements of
 paragraph UG-36. Please check the Code carefully, especially for nozzles
 that are not isolated or do not meet Code spacing requirements. To force
 the computation of areas for small nozzles go to Tools->Configuration
 and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 1.7387 mm.
Wall Thickness per UG16(b),	tr16b = 3.0000 mm.
Wall Thickness, shell/head, internal pressure	trb1 = 20.8870 mm.
Wall Thickness	tb1 = max(trb1, tr16b) = 20.8870 mm.
Wall Thickness	tb2 = max(trb2, tr16b) = 3.0000 mm.
Wall Thickness per table UG-45	tb3 = 4.9200 mm.

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[4.92, \max(20.887, 3.0)]$
 $= 4.9200$ mm.

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(1.7387, 4.92)$
 $= 4.9200$ mm.

Available Nozzle Neck Thickness = 14.2240 mm. -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)(1)(b)), Curve: B

Govrn. thk, tg = 22.0, tr = 19.387, c = 1.5 mm., E* = 1.0
 Thickness Ratio = $tr \cdot E* / (tg - c) = 0.946$, Temp. Reduction = 3 °C



Min Metal Temp. w/o impact per UCS-66, Curve B -5 °C
Min Metal Temp per UCS-66 and UCS-68(c),PWHT credit -21 °C
Min Metal Temp. at Required thickness (UCS 66.1) -24 °C
Min Metal Temp. w/o impact per UG-20(f) -29 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -29 °C

Weld Size Calculations, Description: K81"

Intermediate Calc. for nozzle/shell Welds Tmin 19.0000 mm.

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm.

[Skipping the nozzle attachment weld strength calculations. Per UW-15\(b\)\(2\) the nozzles exempted by UG-36\(c\)\(3\)\(a\) \(small nozzles\) do not require a weld strength check.](#)

The Drop for this Nozzle is : 0.9782 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 535.2303 mm.

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Input, Nozzle Desc: N101"

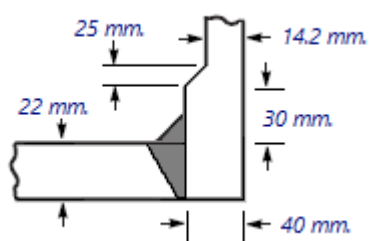
From: 30

Pressure for Reinforcement Calculations	P	23.000	bars
Temperature for Internal Pressure	Temp	150	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm2
Parent Allowable Stress At Ambient	Sva	138.00	N./mm2
Inside Diameter of Cylindrical Shell	D	2300.00	mm.
Shell Finished (Minimum) Thickness	t	22.0000	mm.
Shell Internal Corrosion Allowance	c	1.5000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		3550.00	mm.
User Entered Minimum Design Metal Temperature		-33.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-350 LF2 +IT	
Material UNS Number		K03011	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	N./mm2
Allowable Stress At Ambient	Sna	138.00	N./mm2
Diameter Basis (for tr calc only)		Inside	
Layout Angle		90.00	deg
Diameter		1.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	14.2240	mm.
Flange Material		SA-350 LF2 +IT	
Flange Type		N/A	
Hub Height of Integral Nozzle	h	30.0000	mm.
Height of Beveled Transition	L'	25.0000	mm.
Hub Thickness of Integral Nozzle (tn or x+tp)		40.0000	mm.
Style of FVC Nozzle		Long Weld Neck	
Corrosion Allowance	can	1.5000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	512.2520	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	22.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation 1.000 in.
 Actual Thickness Used in Calculation 0.560 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 23.0 \cdot 1151.5 / (138 \cdot 1.0 - 0.6 \cdot 23.0)$
 $= 19.3870$ mm.

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]
 $= R \cdot (\exp([P / (S \cdot E)] - 1))$ per Appendix 1-2 (a)(1)
 $= 14.2 \cdot (\exp([23.0 / (138 \cdot 1.0)] - 1))$
 $= 0.2387$ mm.

Intermediate Hub Nozzle Calculations:

The Diameter Limit is less than the Hub OD !

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
 $= (\text{Hub Thickness} - \text{Neck Thickness}) / \tan(30)$
 $= 20.5 / 0.5773$
 $= 35.5071$ mm.

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	94.8480 mm.
Parallel to Vessel Wall	Rn+tn+t	47.4240 mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	51.2500 mm.

*Taking a UG-36(c)(3)(a) exemption for nozzle: N101".
 This calculation is valid for nozzles that meet all the requirements of
 paragraph UG-36. Please check the Code carefully, especially for nozzles
 that are not isolated or do not meet Code spacing requirements. To force
 the computation of areas for small nozzles go to Tools->Configuration
 and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 1.7387 mm.
Wall Thickness per UG16(b),	tr16b = 3.0000 mm.
Wall Thickness, shell/head, internal pressure	trb1 = 20.8870 mm.
Wall Thickness	tb1 = max(trb1, tr16b) = 20.8870 mm.
Wall Thickness	tb2 = max(trb2, tr16b) = 3.0000 mm.
Wall Thickness per table UG-45	tb3 = 4.9200 mm.

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[4.92, \max(20.887, 3.0)]$
 $= 4.9200$ mm.

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(1.7387, 4.92)$
 $= 4.9200$ mm.

Available Nozzle Neck Thickness = 14.2240 mm. -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)(1)(b)), Curve: B

Govrn. thk, tg = 22.0, tr = 19.387, c = 1.5 mm., E* = 1.0
 Thickness Ratio = $tr \cdot E* / (tg - c) = 0.946$, Temp. Reduction = 3 °C



Min Metal Temp. w/o impact per UCS-66, Curve B -5 °C
Min Metal Temp per UCS-66 and UCS-68(c),PWHT credit -21 °C
Min Metal Temp. at Required thickness (UCS 66.1) -24 °C
Min Metal Temp. w/o impact per UG-20(f) -29 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -29 °C

Weld Size Calculations, Description: N101"

Intermediate Calc. for nozzle/shell Welds Tmin 19.0000 mm.

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm.

[Skipping the nozzle attachment weld strength calculations. Per UW-15\(b\)\(2\) the nozzles exempted by UG-36\(c\)\(3\)\(a\) \(small nozzles\) do not require a weld strength check.](#)

The Drop for this Nozzle is : 0.9782 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 535.2303 mm.

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Input, Nozzle Desc: N720"

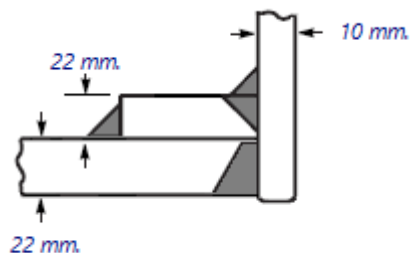
From: 40

Pressure for Reinforcement Calculations	P	23.000	bars
Temperature for Internal Pressure	Temp	150	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm2
Parent Allowable Stress At Ambient	Sva	138.00	N./mm2
Inside Diameter of Cylindrical Shell	D	2300.00	mm.
Shell Finished (Minimum) Thickness	t	22.0000	mm.
Shell Internal Corrosion Allowance	c	1.5000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		5300.00	mm.
User Entered Minimum Design Metal Temperature		-33.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-516 70 +IT	
Material UNS Number		K02700	
Material Specification/Type		Plate	
Allowable Stress at Temperature	Sn	138.00	N./mm2
Allowable Stress At Ambient	Sna	138.00	N./mm2
Diameter Basis (for tr calc only)		Inside	
Layout Angle		90.00	deg
Diameter		20.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	10.0000	mm.
Flange Material		SA-350 LF2 +IT	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	1.5000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	367.4720	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	22.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Pad Material		SA-516 70 +IT	
Pad Allowable Stress at Temperature	Sp	138.00	N./mm2
Pad Allowable Stress At Ambient	Spa	138.00	N./mm2
Diameter of Pad along vessel surface	Dp	979.9999	mm.
Thickness of Pad	te	22.0000	mm.
Weld leg size between Pad and Shell	Wp	18.0000	mm.
Groove weld depth between Pad and Nozzle	Wgpn	22.0000	mm.
Reinforcing Pad Width		226.0000	mm.
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: N720"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	20.000 in.
Actual Thickness Used in Calculation	0.394 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 23.0 \cdot 1151.5 / (138 \cdot 1.0 - 0.6 \cdot 23.0)$
 $= 19.3870$ mm.

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 23.0 \cdot 255.5 / (138 \cdot 1.0 - 0.6 \cdot 23.0)$
 $= 4.3016$ mm.

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	1022.0000	mm.
Parallel to Vessel Wall, opening length	d	511.0000	mm.
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		43.2500	mm.

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 138.0 / 138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 138.0 / 138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr4]:
 $= \min(1, S_p / S_v)$
 $= \min(1, 138.0 / 138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(1.0, 1.0)$
 $= 1.000$

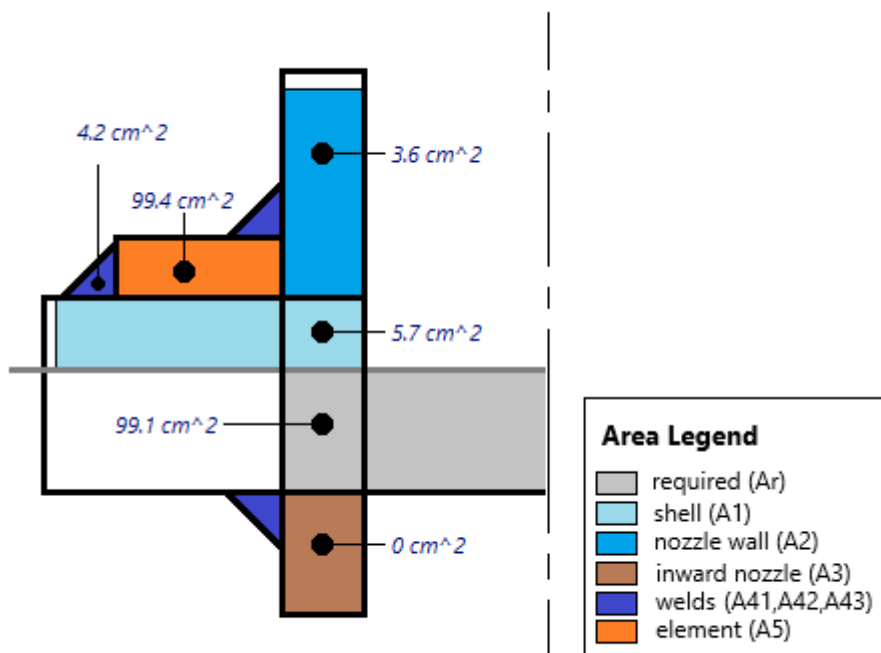


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (cm2)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	99.068	NA	NA
Area in Shell	A1	5.687	NA	NA
Area in Nozzle Wall	A2	3.632	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	4.240	NA	NA
Area in Element	A5	99.440	NA	NA
TOTAL AREA AVAILABLE	Atot	112.999	NA	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Insufficient.
 The area available with the given pad is Sufficient.

SELECTION OF POSSIBLE REINFORCING PADS: Diameter Thickness
 Based on given Pad Thickness: 916.6758 22.0000 mm.
 Based on given Pad Diameter: 979.9999 18.9179 mm.
 Based on the Estimated Diameter Limit: 1020.4125 17.3653 mm.

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$
 $= (511.0 \cdot 19.387 \cdot 1.0 + 2 \cdot 8.5 \cdot 19.387 \cdot 1.0 \cdot (1 - 1.0))$
 $= 99.068 \text{ cm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 511.0(1.0 \cdot 20.5 - 1.0 \cdot 19.387) - 2 \cdot 8.5$
 $(1.0 \cdot 20.5 - 1.0 \cdot 19.387) \cdot (1 - 1.0)$
 $= 5.687 \text{ cm}^2$

Area Available in Nozzle Wall Projecting Outward [A2]:
 $= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2$
 $= (2 \cdot 43.25) \cdot (8.5 - 4.3) \cdot 1.0$



= 3.632 cm²

Area Available in Welds [A41 + A42 + A43]:
 = $W_o^2 \cdot fr_3 + (W_i - can/0.707)^2 \cdot fr_2 + W_p^2 \cdot fr_4$
 = $10.0^2 \cdot 1.0 + (0.0)^2 \cdot 1.0 + 18.0^2 \cdot 1.0$
 = 4.240 cm²

Area Available in Element [A5]:
 = $(\min(D_p, DL) - (\text{Nozzle OD})) \cdot (\min(t_p, Tlwp, te)) \cdot fr_4$
 = $(980.0 - 528.0) \cdot 22.0 \cdot 1.0$
 = 99.440 cm²

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 5.8016 mm.
Wall Thickness per UG16(b),	tr16b = 3.0000 mm.
Wall Thickness, shell/head, internal pressure	trb1 = 20.8870 mm.
Wall Thickness	tb1 = max(trb1, tr16b) = 20.8870 mm.
Wall Thickness	tb2 = max(trb2, tr16b) = 3.0000 mm.
Wall Thickness per table UG-45	tb3 = 9.8312 mm.

Determine Nozzle Thickness candidate [tb]:
 = min[tb3, max(tb1, tb2)]
 = min[9.831, max(20.887, 3.0)]
 = 9.8312 mm.

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 = max(ta, tb)
 = max(5.8016, 9.8312)
 = 9.8312 mm.

Available Nozzle Neck Thickness = 10.0000 mm. -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Nozzle Neck to Pad Weld for the Nozzle (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Nozzle Neck to Pad Weld for Reinf. pad (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Shell to Pad Weld Junction at Pad OD, Curve: B

Govrn. thk, tg = 22.0, tr = 19.387, c = 1.5 mm., E* = 1.0
 Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.946$, Temp. Reduction = 3 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-5 °C
Min Metal Temp per UCS-66 and UCS-68(c),PWHT credit	-21 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-24 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), Curve: B

Govrn. thk, tg = 10.0, tr = 4.302, c = 1.5 mm., E* = 1.0
 Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.506$, Temp. Reduction = 32 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-29 °C
Min Metal Temp per UCS-66 and UCS-68(c),PWHT credit	-46 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-48 °C

Gov. MDMT of the Nozzle	:	-48 °C
Gov. MDMT of the Reinforcement Pad	:	-29 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -29 °C

Flange MDMT including Temperature reduction per UCS-66.1:

MDMT of ASME B16.5/47 flange per Matl. Specification -46 °C
 Flange MDMT with Temp reduction per UCS-66(i)(2) -85 °C

Where the Stress Reduction Ratio per UCS-66(i)(2) is :
 Design Pressure/Ambient Rating = 23.00/51.00 = 0.451

Weld Size Calculations, Description: N720"

Intermediate Calc. for nozzle/shell Welds Tmin 8.5000 mm.
 Intermediate Calc. for pad/shell Welds TminPad 19.0000 mm.

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	5.9500 = 0.7 * tmin.	7.0700 = 0.7 * Wo mm.
Pad Weld	9.5000 = 0.5 * TminPad	12.7260 = 0.7 * Wp mm.

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A-A1+2 \cdot tn \cdot fr1 \cdot (E1 \cdot t-tr)) \cdot Sv)$
 $= \max(0, (99.0676 - 5.6874 + 2 \cdot 8.5 \cdot 1.0 \cdot (1.0 \cdot 20.5 - 19.387)) \cdot 138)$
 $= 131666.98 \text{ Kgf}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2+A5+A4-(Wi-Can/.707)^2 \cdot fr2) \cdot Sv$
 $= (3.6316 + 99.44 + 4.24 - 0.0 \cdot 1.0) \cdot 138$
 $= 151004.28 \text{ Kgf}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot tn \cdot t \cdot fr1)) \cdot Sv$
 $= (3.6316 + 0.0 + 1. + (3.485)) \cdot 138$
 $= 11421.35 \text{ Kgf}$

Weld Load [W3]:
 $= (A2+A3+A4+A5+(2 \cdot tn \cdot t \cdot fr1)) \cdot S$
 $= (3.6316 + 0.0 + 4.24 + 99.44 + (3.485)) \cdot 138$
 $= 155908.23 \text{ Kgf}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot Dlo \cdot Wo \cdot 0.49 \cdot Snw$
 $= (3.1416/2.0) \cdot 528.0 \cdot 10.0 \cdot 0.49 \cdot 138$
 $= 57187. \text{ Kgf}$

Shear, Pad Element Weld [Spew]:
 $= (\pi/2) \cdot DP \cdot WP \cdot 0.49 \cdot SEW$
 $= (3.1416/2.0) \cdot 980.0 \cdot 18.0 \cdot 0.49 \cdot 138$
 $= 191054. \text{ Kgf}$

Shear, Nozzle Wall [Snw]:
 $= (\pi \cdot (Dlr + Dlo) / 4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn$
 $= (3.1416 \cdot 259.75) \cdot (10.0 - 1.5) \cdot 0.7 \cdot 138$
 $= 68324. \text{ Kgf}$

Tension, Pad Groove Weld [Tpgw]:
 $= (\pi/2) \cdot Dlo \cdot Wgpn \cdot 0.74 \cdot Seg$
 $= (3.1416/2) \cdot 528.0 \cdot 22.0 \cdot 0.74 \cdot 138$
 $= 190002. \text{ Kgf}$

Tension, Shell Groove Weld [Tngw]:
 $= (\pi/2) \cdot Dlo \cdot (T - cas) \cdot 0.74 \cdot Sng$
 $= (3.1416/2.0) \cdot 528.0 \cdot (22.0 - 1.5) \cdot 0.74 \cdot 138$



= 177047. Kgf

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (\text{SPEW} + \text{SNW}) = (191054 + 68324) = 259378 \text{ Kgf}$$

Failure Path [Path 2-2]:

$$= (\text{Sonw} + \text{Tpgw} + \text{Tngw} + \text{Sinw}) \\ = (57187 + 190002 + 177047 + 0) = 424237 \text{ Kgf}$$

Failure Path [Path 3-3]:

$$= (\text{Spew} + \text{Tngw} + \text{Sinw}) \\ = (191054 + 177047 + 0) = 368102 \text{ Kgf}$$

Summary of Failure Path Calculations:

Path 1-1 = 259378 Kgf, must exceed W = 131666 Kgf or W1 = 151004 Kgf

Path 2-2 = 424237 Kgf, must exceed W = 131666 Kgf or W2 = 11421 Kgf

Path 3-3 = 368101 Kgf, must exceed W = 131666 Kgf or W3 = 155908 Kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 24 bars

Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 30.7128 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 420.1848 mm.

Percent Elongation Calculations:

% Elongation per Table UG-79-1 ($50 \cdot t_{nom} / R_f \cdot (1 - R_f / R_o)$) 1.931 %

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Input, Nozzle Desc: K3A2"

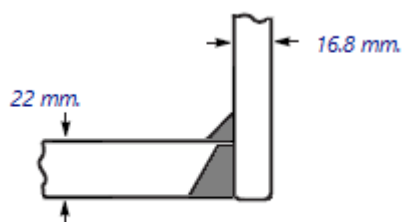
From: 40

Pressure for Reinforcement Calculations	P	23.000	bars
Temperature for Internal Pressure	Temp	150	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	2300.00	mm.
Shell Finished (Minimum) Thickness	t	22.0000	mm.
Shell Internal Corrosion Allowance	c	1.5000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		6550.00	mm.
User Entered Minimum Design Metal Temperature		-33.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-350 LF2 +IT	
Material UNS Number		K03011	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	N./mm ²
Allowable Stress At Ambient	Sna	138.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		90.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	16.7640	mm.
Flange Material		SA-105 +IT	
Flange Type		Long Weld Neck	
Corrosion Allowance	can	1.5000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	507.4260	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	22.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: K3A2"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	2.000	in.
Actual Thickness Used in Calculation	0.660	in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 23.0 \cdot 1151.5 / (138 \cdot 1.0 - 0.6 \cdot 23.0)$
 $= 19.3870$ mm.

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]
 $= R \cdot (\exp([P / (S_n \cdot E)] - 1))$ per Appendix 1-2 (a)(1)
 $= 26.9 \cdot (\exp([23.0 / (138.0 \cdot 1.0)] - 1))$
 $= 0.4521$ mm.

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	Dl	125.3280	mm.
Parallel to Vessel Wall	Rn+tn+tr	62.6640	mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	38.1600	mm.

*Taking a UG-36(c)(3)(a) exemption for nozzle: K3A2".
 This calculation is valid for nozzles that meet all the requirements of
 paragraph UG-36. Please check the Code carefully, especially for nozzles
 that are not isolated or do not meet Code spacing requirements. To force
 the computation of areas for small nozzles go to Tools->Configuration
 and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 1.9521	mm.
Wall Thickness per UG16(b),	tr16b	= 3.0000	mm.
Wall Thickness, shell/head, internal pressure	trb1	= 20.8870	mm.
Wall Thickness	tb1 = max(trb1, tr16b)	= 20.8870	mm.
Wall Thickness	tb2 = max(trb2, tr16b)	= 3.0000	mm.
Wall Thickness per table UG-45	tb3	= 6.3000	mm.

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[6.3, \max(20.887, 3.0)]$
 $= 6.3000$ mm.

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(1.9521, 6.3)$
 $= 6.3000$ mm.

Available Nozzle Neck Thickness = 16.7640 mm. -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)(1)(b)), Curve: B

Govrn. thk, tg = 16.764, tr = 0.452, c = 1.5 mm., $E^* = 1.0$
 Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.03$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-13 °C
Min Metal Temp per UCS-66 and UCS-68(c), PWHT credit	-29 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-48 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = $23.00 / 51.00 = 0.451$

Weld Size Calculations, Description: K3A2"

Intermediate Calc. for nozzle/shell welds	Tmin	15.2640	mm.
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Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm.

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 0.7732 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 530.1992 mm.

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Input, Nozzle Desc: K3B2"

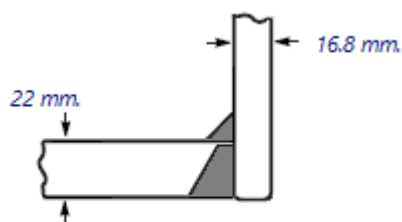
From: 40

Pressure for Reinforcement Calculations	P	23.137	bars
Temperature for Internal Pressure	Temp	150	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm2
Parent Allowable Stress At Ambient	Sva	138.00	N./mm2
Inside Diameter of Cylindrical Shell	D	2300.00	mm.
Shell Finished (Minimum) Thickness	t	22.0000	mm.
Shell Internal Corrosion Allowance	c	1.5000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		6550.00	mm.
User Entered Minimum Design Metal Temperature		-33.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-350 LF2 +IT	
Material UNS Number		K03011	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	N./mm2
Allowable Stress At Ambient	Sna	138.00	N./mm2
Diameter Basis (for tr calc only)		Inside	
Layout Angle		270.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	16.7640	mm.
Flange Material		SA-105 +IT	
Flange Type		Long Weld Neck	
Corrosion Allowance	can	1.5000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	206.3750	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	22.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: K3B2"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	2.000	in.
Actual Thickness Used in Calculation	0.660	in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 23.14 \cdot 1151.5 / (138 \cdot 1.0 - 0.6 \cdot 23.14)$
 $= 19.5037$ mm.

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]
 $= R \cdot (\exp([P / (S_n \cdot E)] - 1))$ per Appendix 1-2 (a)(1)
 $= 26.9 \cdot (\exp([23.14 / (138.0 \cdot 1.0)] - 1))$
 $= 0.4548$ mm.

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	125.3280	mm.
Parallel to Vessel Wall	Rn+tn+tr	62.6640	mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	38.1600	mm.

*Taking a UG-36(c)(3)(a) exemption for nozzle: K3E2".
 This calculation is valid for nozzles that meet all the requirements of
 paragraph UG-36. Please check the Code carefully, especially for nozzles
 that are not isolated or do not meet Code spacing requirements. To force
 the computation of areas for small nozzles go to Tools->Configuration
 and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 1.9548	mm.
Wall Thickness per UG16(b),	tr16b	= 3.0000	mm.
Wall Thickness, shell/head, internal pressure	trb1	= 21.0037	mm.
Wall Thickness	tb1 = max(trb1, tr16b)	= 21.0037	mm.
Wall Thickness	tb2 = max(trb2, tr16b)	= 3.0000	mm.
Wall Thickness per table UG-45	tb3	= 6.3000	mm.

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[6.3, \max(21.0037, 3.0)]$
 $= 6.3000$ mm.

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(1.9548, 6.3)$
 $= 6.3000$ mm.

Available Nozzle Neck Thickness = 16.7640 mm. -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)(1)(b)), Curve: B

Govrn. thk, tg = 16.764, tr = 0.455, c = 1.5 mm., $E^* = 1.0$
 Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.03$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-13 °C
Min Metal Temp per UCS-66 and UCS-68(c), PWHT credit	-29 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-48 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = $23.14 / 51.00 = 0.454$

Weld Size Calculations, Description: K3B2"

Intermediate Calc. for nozzle/shell welds	Tmin	15.2640	mm.
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Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm.

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 0.7732 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 229.1482 mm.

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Input, Nozzle Desc: K1A2"

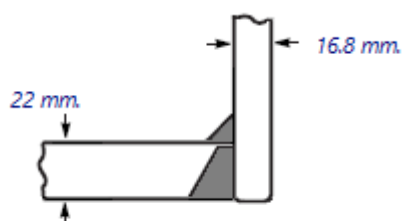
From: 40

Pressure for Reinforcement Calculations	P	23.013	bars
Temperature for Internal Pressure	Temp	150	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm2
Parent Allowable Stress At Ambient	Sva	138.00	N./mm2
Inside Diameter of Cylindrical Shell	D	2300.00	mm.
Shell Finished (Minimum) Thickness	t	22.0000	mm.
Shell Internal Corrosion Allowance	c	1.5000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Cylinder/Cone Centerline	L1	950.0001	mm.
Distance from Bottom/Left Tangent		6050.00	mm.
User Entered Minimum Design Metal Temperature		-33.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-350 LF2 +IT	
Material UNS Number		K03011	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	N./mm2
Allowable Stress At Ambient	Sna	138.00	N./mm2
Diameter Basis (for tr calc only)		Inside	
Layout Angle		54.15	deg
Diameter		2.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	16.7640	mm.
Flange Material		SA-105 +IT	
Flange Type		Long Weld Neck	
Corrosion Allowance	can	1.5000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	206.3750	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	22.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



Note : Checking Nozzle 90 degrees to the Longitudinal axis.

Reinforcement CALCULATION, Description: K1A2"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation 2.000 in.
 Actual Thickness Used in Calculation 0.660 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 23.01 \cdot 1151.5 / (138 \cdot 1.0 - 0.6 \cdot 23.01)$
 $= 19.3981$ mm.

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]
 $= R \cdot (\exp([P / (S_n \cdot E)] - 1))$ per Appendix 1-2 (a)(1)
 $= 26.9 \cdot (\exp([23.01 / (138.0 \cdot 1.0)] - 1))$
 $= 0.4524$ mm.

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit) D1 187.4754 mm.
 Parallel to Vessel Wall, opening length d 93.7377 mm.
 Normal to Vessel Wall (Thickness Limit), no pad Tlnp 38.1600 mm.

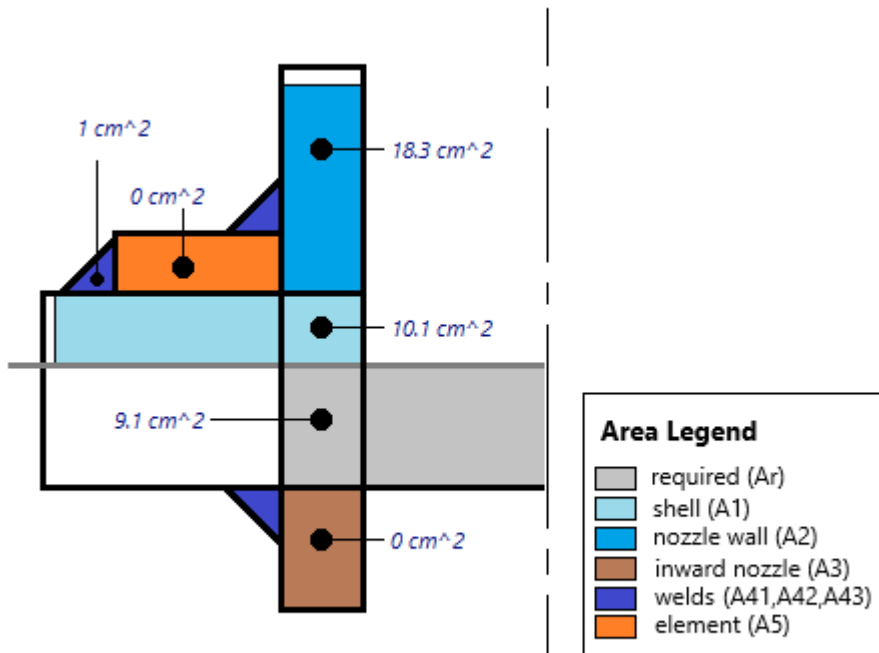


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (cm2)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	9.092	NA	NA
Area in Shell	A1	10.125	NA	NA
Area in Nozzle Wall	A2	18.347	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	1.000	NA	NA
Area in Element	A5	0.000	NA	NA
TOTAL AREA AVAILABLE	Atot	29.471	NA	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 35.03 degs.

The area available without a pad is Sufficient.

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \cdot UG-37(c)$
 $= (93.7377 \cdot 19.3981 \cdot 0.5 + 2 \cdot 15.264 \cdot 19.3981 \cdot 0.5 \cdot (1 - 1.0))$
 $= 9.092 \text{ cm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 93.738(1.0 \cdot 20.5 - 0.5 \cdot 19.398) - 2 \cdot 15.264$
 $(1.0 \cdot 20.5 - 0.5 \cdot 19.3981) \cdot (1 - 1.0)$
 $= 10.125 \text{ cm}^2$

Area Available in Nozzle Projecting Outward [A2]:
 $= (2 \cdot tlnp)(tn - trn)fr2/\sin(\alpha3)$
 $= (2 \cdot 38.16)(15.26 - 0.45)1.0/\sin(38.0)$
 $= 18.347 \text{ cm}^2$

See ASME VIII-1 2011(a) Appendix L, L-7.7.7(b) for more information.

Area Available in Inward Weld + Outward Weld [A41 + A43]:
 $= Wo^2 \cdot fr2 + (Wi - can/0.707)^2 \cdot fr2$
 $= 10.0^2 \cdot 1.0 + (0.0)^2 \cdot 1.0$
 $= 1.000 \text{ cm}^2$

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)(1)(b)), Curve: B

Govern. thk, $t_g = 16.764$, $t_r = 0.452$, $c = 1.5 \text{ mm.}$, $E* = 1.0$
 Thickness Ratio $= tr \cdot E*/(t_g - c) = 0.03$, Temp. Reduction $= 78 \text{ }^\circ\text{C}$

Min Metal Temp. w/o impact per UCS-66, Curve B	-13 $^\circ\text{C}$
Min Metal Temp per UCS-66 and UCS-68(c), PWHt credit	-29 $^\circ\text{C}$
Min Metal Temp. at Required thickness (UCS 66.1)	-104 $^\circ\text{C}$

Gov. MDMT of the nozzle to shell joint welded assembly : -104 $^\circ\text{C}$

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 $^\circ\text{C}$
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-48 $^\circ\text{C}$

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating $= 23.01/51.00 = 0.451$

Weld Size Calculations, Description: K1A2"

Intermediate Calc. for nozzle/shell Welds $T_{min} = 15.2640 \text{ mm.}$

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$6.0000 = \text{Min per Code}$	$7.0700 = 0.7 \cdot Wo \text{ mm.}$

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A - A1 + 2 \cdot tn \cdot fr1 \cdot (E1 \cdot t - tr)) \cdot Sv)$
 $= \max(0, (9.0916 - 10.1246 + 2 \cdot 15.264 \cdot 1.0 \cdot$
 $(1.0 \cdot 20.5 - 9.699)) \cdot 138)$
 $= 3186.36 \text{ Kg}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2 + A5 + A4 - (Wi - Can/0.707)^2 \cdot fr2) \cdot Sv$
 $= (18.3465 + 0.0 + 1.0 - 0.0 \cdot 1.0) \cdot 138$
 $= 27223.60 \text{ Kg}$

Weld Load [W2]:



$$= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot fr1)) \cdot Sv$$

$$= (18.3465 + 0.0 + 1. + (6.2582)) \cdot 138$$

$$= 36029.93 \text{ Kgf}$$

Weld Load [W3]:

$$= (A2 + A3 + A4 + A5 + (2 \cdot t_n \cdot t \cdot fr1)) \cdot S$$

$$= (18.3465 + 0.0 + 1. + 0.0 + (6.2582)) \cdot 138$$

$$= 36029.93 \text{ Kgf}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$= (\pi/2) \cdot Dlo \cdot Wo \cdot 0.49 \cdot Snw$$

$$= (3.1416/2.0) \cdot 146.9277 \cdot 10.0 \cdot 0.49 \cdot 138$$

$$= 15913. \text{ Kgf}$$

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (Dlr + Dlo) / 4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn$$

$$= (3.1416 \cdot 60.1664) \cdot (16.764 - 1.5) \cdot 0.7 \cdot 138$$

$$= 28420. \text{ Kgf}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot Dlo \cdot (T - cas) \cdot 0.74 \cdot Sng$$

$$= (3.1416/2.0) \cdot 146.9277 \cdot (22.0 - 1.5) \cdot 0.74 \cdot 138$$

$$= 49267. \text{ Kgf}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SONW + SNW) = (15913 + 28420) = 44333 \text{ Kgf}$$

Failure Path [Path 2-2]:

$$= (Sonw + Tpgw + Tngw + Sinw)$$

$$= (15913 + 0 + 49267 + 0) = 65181 \text{ Kgf}$$

Failure Path [Path 3-3]:

$$= (Sonw + Tngw + Sinw)$$

$$= (15913 + 49267 + 0) = 65181 \text{ Kgf}$$

Summary of Failure Path Calculations:

Path 1-1 = 44333	Kgf, must exceed W = 3186 Kgf	or W1 = 27223 Kgf
Path 2-2 = 65180	Kgf, must exceed W = 3186 Kgf	or W2 = 36029 Kgf
Path 3-3 = 65180	Kgf, must exceed W = 3186 Kgf	or W3 = 36029 Kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 24 bars

Note: The MAWP of this junction was limited by the parent Shell/Head.

Note : Checking Nozzle in plane parallel to the vessel axis.

Reinforcement CALCULATION, Description: K1A2"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	2.000 in.
Actual Thickness Used in Calculation	0.660 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]

$$= P \cdot R / (Sv \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= 23.01 \cdot 1151.5 / (138 \cdot 1.0 - 0.6 \cdot 23.01)$$

$$= 19.3981 \text{ mm.}$$

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]

$$= R \cdot (\exp([P / (Sn \cdot E)] - 1)) \text{ per Appendix 1-2 (a)(1)}$$

$$= 26.9 \cdot (\exp([23.01 / (138 \cdot 1.0)] - 1))$$



= 0.4524 mm.

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	125.3280 mm.
Parallel to Vessel Wall	Rn+tn+t	62.6640 mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	38.1600 mm.

*Taking a UG-36(c)(3)(a) exemption for nozzle: K1A2".
 This calculation is valid for nozzles that meet all the requirements of
 paragraph UG-36. Please check the Code carefully, especially for nozzles
 that are not isolated or do not meet Code spacing requirements. To force
 the computation of areas for small nozzles go to Tools->Configuration
 and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 1.9524 mm.
Wall Thickness per UG16(b),	tr16b = 3.0000 mm.
Wall Thickness, shell/head, internal pressure	trb1 = 20.8981 mm.
Wall Thickness	tb1 = max(trb1, tr16b) = 20.8981 mm.
Wall Thickness	tb2 = max(trb2, tr16b) = 3.0000 mm.
Wall Thickness per table UG-45	tb3 = 6.3000 mm.

Determine Nozzle Thickness candidate [tb]:

= min[tb3, max(tb1,tb2)]
 = min[6.3, max(20.8981, 3.0)]
 = 6.3000 mm.

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

= max(ta, tb)
 = max(1.9524, 6.3)
 = 6.3000 mm.

Available Nozzle Neck Thickness = 16.7640 mm. -> OK

Weld Size Calculations, Description: K1A2"

Intermediate Calc. for nozzle/shell Welds	Tmin	15.2640 mm.
---	------	-------------

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm.

*Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles
 exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.*

The Drop for this Nozzle is : 66.6013 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 311.2578 mm.

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Input, Nozzle Desc: K1B2"

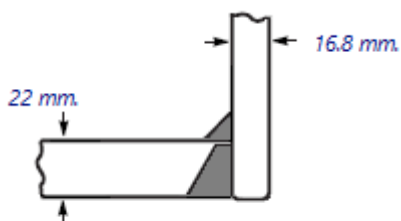
From: 40

Pressure for Reinforcement Calculations	P	23.124	bars
Temperature for Internal Pressure	Temp	150	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm2
Parent Allowable Stress At Ambient	Sva	138.00	N./mm2
Inside Diameter of Cylindrical Shell	D	2300.00	mm.
Shell Finished (Minimum) Thickness	t	22.0000	mm.
Shell Internal Corrosion Allowance	c	1.5000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Cylinder/Cone Centerline	L1	950.0001	mm.
Distance from Bottom/Left Tangent		6050.00	mm.
User Entered Minimum Design Metal Temperature		-33.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-350 LF2 +IT	
Material UNS Number		K03011	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	N./mm2
Allowable Stress At Ambient	Sna	138.00	N./mm2
Diameter Basis (for tr calc only)		Inside	
Layout Angle		-54.15	deg
Diameter		2.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	16.7640	mm.
Flange Material		SA-105 +IT	
Flange Type		Long Weld Neck	
Corrosion Allowance	can	1.5000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	206.3750	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	22.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



Note : Checking Nozzle 90 degrees to the Longitudinal axis.

Reinforcement CALCULATION, Description: K1B2"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation 2.000 in.
 Actual Thickness Used in Calculation 0.660 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 23.12 \cdot 1151.5 / (138 \cdot 1.0 - 0.6 \cdot 23.12)$
 $= 19.4927$ mm.

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]
 $= R \cdot (\exp([P / (S_n \cdot E)] - 1))$ per Appendix 1-2 (a)(1)
 $= 26.9 \cdot (\exp([23.12 / (138.0 \cdot 1.0)] - 1))$
 $= 0.4546$ mm.

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit) D1 187.4598 mm.
 Parallel to Vessel Wall, opening length d 93.7299 mm.
 Normal to Vessel Wall (Thickness Limit), no pad Tlnp 38.1600 mm.

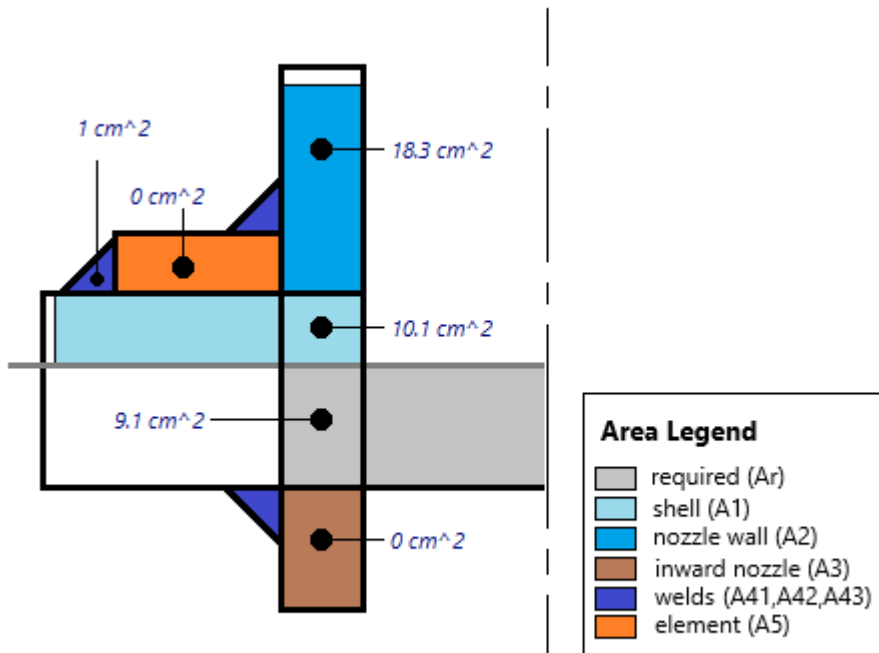


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (cm2)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	9.135	NA	NA
Area in Shell	A1	10.079	NA	NA
Area in Nozzle Wall	A2	18.344	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	1.000	NA	NA
Area in Element	A5	0.000	NA	NA
TOTAL AREA AVAILABLE	Atot	29.423	NA	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 35.03 degs.

The area available without a pad is Sufficient.

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \cdot UG-37(c)$
 $= (93.7299 \cdot 19.4927 \cdot 0.5 + 2 \cdot 15.264 \cdot 19.4927 \cdot 0.5 \cdot (1 - 1.0))$
 $= 9.135 \text{ cm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 93.73(1.0 \cdot 20.5 - 0.5 \cdot 19.493) - 2 \cdot 15.264$
 $(1.0 \cdot 20.5 - 0.5 \cdot 19.4927) \cdot (1 - 1.0)$
 $= 10.079 \text{ cm}^2$

Area Available in Nozzle Projecting Outward [A2]:
 $= (2 \cdot tnp)(tn - trn)fr2/\sin(\alpha3)$
 $= (2 \cdot 38.16)(15.26 - 0.45)1.0/\sin(38.0)$
 $= 18.344 \text{ cm}^2$

See ASME VIII-1 2011(a) Appendix L, L-7.7.7(b) for more information.

Area Available in Inward Weld + Outward Weld [A41 + A43]:
 $= Wo^2 \cdot fr2 + (Wi - can/0.707)^2 \cdot fr2$
 $= 10.0^2 \cdot 1.0 + (0.0)^2 \cdot 1.0$
 $= 1.000 \text{ cm}^2$

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)(1)(b)), Curve: B

Govern. thk, $t_g = 16.764$, $t_r = 0.455$, $c = 1.5 \text{ mm.}$, $E* = 1.0$
 Thickness Ratio $= tr \cdot E*/(t_g - c) = 0.03$, Temp. Reduction $= 78 \text{ }^\circ\text{C}$

Min Metal Temp. w/o impact per UCS-66, Curve B	-13 $^\circ\text{C}$
Min Metal Temp per UCS-66 and UCS-68(c), PWHT credit	-29 $^\circ\text{C}$
Min Metal Temp. at Required thickness (UCS 66.1)	-104 $^\circ\text{C}$

Gov. MDMT of the nozzle to shell joint welded assembly : -104 $^\circ\text{C}$

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 $^\circ\text{C}$
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-48 $^\circ\text{C}$

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating $= 23.12/51.00 = 0.453$

Weld Size Calculations, Description: K1B2"

Intermediate Calc. for nozzle/shell Welds $T_{min} = 15.2640 \text{ mm.}$

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$6.0000 = \text{Min per Code}$	$7.0700 = 0.7 \cdot Wo \text{ mm.}$

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A - A1 + 2 \cdot tn \cdot fr1 \cdot (E1 \cdot t - tr)) \cdot Sv)$
 $= \max(0, (9.1352 - 10.0794 + 2 \cdot 15.264 \cdot 1.0 \cdot$
 $(1.0 \cdot 20.5 - 9.7463)) \cdot 138)$
 $= 3290.91 \text{ Kg}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2 + A5 + A4 - (Wi - Can/0.707)^2 \cdot fr2) \cdot Sv$
 $= (18.3438 + 0.0 + 1.0 - 0.0 \cdot 1.0) \cdot 138$
 $= 27219.76 \text{ Kg}$

Weld Load [W2]:

$$= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot fr1)) \cdot S_v$$

$$= (18.3438 + 0.0 + 1. + (6.2582)) \cdot 138$$

$$= 36026.09 \text{ Kgf}$$

Weld Load [W3]:

$$= (A2 + A3 + A4 + A5 + (2 \cdot t_n \cdot t \cdot fr1)) \cdot S$$

$$= (18.3438 + 0.0 + 1. + 0.0 + (6.2582)) \cdot 138$$

$$= 36026.09 \text{ Kgf}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$$

$$= (3.1416/2.0) \cdot 146.9155 \cdot 10.0 \cdot 0.49 \cdot 138$$

$$= 15912. \text{ Kgf}$$

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - Can) \cdot 0.7 \cdot S_n$$

$$= (3.1416 \cdot 60.1613) \cdot (16.764 - 1.5) \cdot 0.7 \cdot 138$$

$$= 28417. \text{ Kgf}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot D_{lo} \cdot (T - cas) \cdot 0.74 \cdot S_{ng}$$

$$= (3.1416/2.0) \cdot 146.9155 \cdot (22.0 - 1.5) \cdot 0.74 \cdot 138$$

$$= 49263. \text{ Kgf}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SONW + SNW) = (15912 + 28417) = 44329 \text{ Kgf}$$

Failure Path [Path 2-2]:

$$= (Sonw + Tpgw + Tngw + Sinw)$$

$$= (15912 + 0 + 49263 + 0) = 65175 \text{ Kgf}$$

Failure Path [Path 3-3]:

$$= (Sonw + Tngw + Sinw)$$

$$= (15912 + 49263 + 0) = 65175 \text{ Kgf}$$

Summary of Failure Path Calculations:

Path 1-1 = 44329	Kgf, must exceed W = 3290 Kgf	or W1 = 27219 Kgf
Path 2-2 = 65175	Kgf, must exceed W = 3290 Kgf	or W2 = 36026 Kgf
Path 3-3 = 65175	Kgf, must exceed W = 3290 Kgf	or W3 = 36026 Kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 24 bars

[Note: The MAWP of this junction was limited by the parent Shell/Head.](#)

Note : Checking Nozzle in plane parallel to the vessel axis.

Reinforcement CALCULATION, Description: K1B2"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	2.000 in.
Actual Thickness Used in Calculation	0.660 in.

[Nozzle input data check completed without errors.](#)

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]

$$= P \cdot R / (S_v \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= 23.12 \cdot 1151.5 / (138 \cdot 1.0 - 0.6 \cdot 23.12)$$

$$= 19.4927 \text{ mm.}$$

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]

$$= R \cdot (\exp([P / (S_n \cdot E)]) - 1) \text{ per Appendix 1-2 (a)(1)}$$

$$= 26.9 \cdot (\exp([23.12 / (138.0 \cdot 1.0)] - 1))$$



= 0.4546 mm.

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	125.3280 mm.
Parallel to Vessel Wall	Rn+tn+t	62.6640 mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	38.1600 mm.

*Taking a UG-36(c)(3)(a) exemption for nozzle: K1B2".
 This calculation is valid for nozzles that meet all the requirements of
 paragraph UG-36. Please check the Code carefully, especially for nozzles
 that are not isolated or do not meet Code spacing requirements. To force
 the computation of areas for small nozzles go to Tools->Configuration
 and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 1.9546 mm.
Wall Thickness per UG16(b),	tr16b = 3.0000 mm.
Wall Thickness, shell/head, internal pressure	trb1 = 20.9927 mm.
Wall Thickness	tb1 = max(trb1, tr16b) = 20.9927 mm.
Wall Thickness	tb2 = max(trb2, tr16b) = 3.0000 mm.
Wall Thickness per table UG-45	tb3 = 6.3000 mm.

Determine Nozzle Thickness candidate [tb]:

= min[tb3, max(tb1,tb2)]
 = min[6.3, max(20.9927, 3.0)]
 = 6.3000 mm.

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

= max(ta, tb)
 = max(1.9546, 6.3)
 = 6.3000 mm.

Available Nozzle Neck Thickness = 16.7640 mm. -> OK

Weld Size Calculations, Description: K1B2"

Intermediate Calc. for nozzle/shell Welds Tmin 15.2640 mm.

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm.

*Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles
 exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.*

The Drop for this Nozzle is : 66.6013 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 311.2578 mm.

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Input, Nozzle Desc: K2A2"

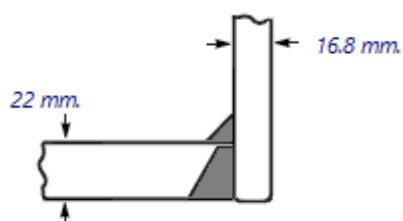
From: 40

Pressure for Reinforcement Calculations	P	23.013	bars
Temperature for Internal Pressure	Temp	150	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm2
Parent Allowable Stress At Ambient	Sva	138.00	N./mm2
Inside Diameter of Cylindrical Shell	D	2300.00	mm.
Shell Finished (Minimum) Thickness	t	22.0000	mm.
Shell Internal Corrosion Allowance	c	1.5000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Cylinder/Cone Centerline	L1	950.0001	mm.
Distance from Bottom/Left Tangent		6550.00	mm.
User Entered Minimum Design Metal Temperature		-33.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-350 LF2 +IT	
Material UNS Number		K03011	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	N./mm2
Allowable Stress At Ambient	Sna	138.00	N./mm2
Diameter Basis (for tr calc only)		Inside	
Layout Angle		54.15	deg
Diameter		2.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	16.7640	mm.
Flange Material		SA-105 +IT	
Flange Type		Long Weld Neck	
Corrosion Allowance	can	1.5000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	206.3750	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	22.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



Note : Checking Nozzle 90 degrees to the Longitudinal axis.

Reinforcement CALCULATION, Description: K2A2"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation 2.000 in.
 Actual Thickness Used in Calculation 0.660 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 23.01 \cdot 1151.5 / (138 \cdot 1.0 - 0.6 \cdot 23.01)$
 $= 19.3981$ mm.

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]
 $= R \cdot (\exp([P / (S_n \cdot E)] - 1))$ per Appendix 1-2 (a)(1)
 $= 26.9 \cdot (\exp([23.01 / (138.0 \cdot 1.0)] - 1))$
 $= 0.4524$ mm.

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit) D1 187.4754 mm.
 Parallel to Vessel Wall, opening length d 93.7377 mm.
 Normal to Vessel Wall (Thickness Limit), no pad Tlnp 38.1600 mm.

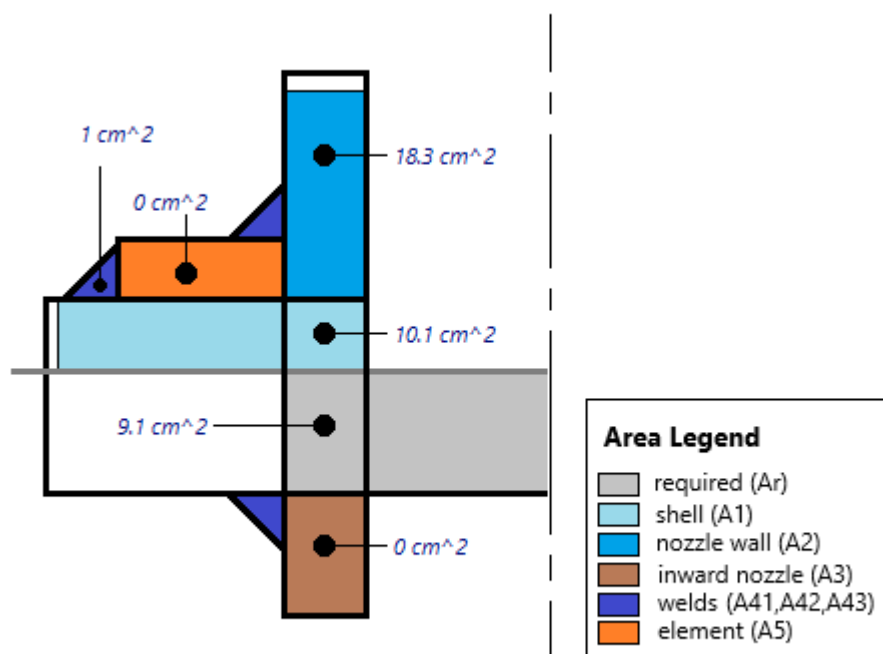


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	9.092	NA	NA
Area in Shell	A1	10.125	NA	NA
Area in Nozzle Wall	A2	18.347	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	1.000	NA	NA
Area in Element	A5	0.000	NA	NA
TOTAL AREA AVAILABLE	Atot	29.471	NA	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 35.03 degs.

The area available without a pad is Sufficient.

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \cdot UG-37(c)$
 $= (93.7377 \cdot 19.3981 \cdot 0.5 + 2 \cdot 15.264 \cdot 19.3981 \cdot 0.5 \cdot (1 - 1.0))$
 $= 9.092 \text{ cm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 93.738(1.0 \cdot 20.5 - 0.5 \cdot 19.398) - 2 \cdot 15.264$
 $(1.0 \cdot 20.5 - 0.5 \cdot 19.3981) \cdot (1 - 1.0)$
 $= 10.125 \text{ cm}^2$

Area Available in Nozzle Projecting Outward [A2]:
 $= (2 \cdot tnp)(tn - trn)fr2/\sin(\alpha3)$
 $= (2 \cdot 38.16)(15.26 - 0.45)1.0/\sin(38.0)$
 $= 18.347 \text{ cm}^2$

See ASME VIII-1 2011(a) Appendix L, L-7.7.7(b) for more information.

Area Available in Inward Weld + Outward Weld [A41 + A43]:
 $= Wo^2 \cdot fr2 + (Wi - can/0.707)^2 \cdot fr2$
 $= 10.0^2 \cdot 1.0 + (0.0)^2 \cdot 1.0$
 $= 1.000 \text{ cm}^2$

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)(1)(b)), Curve: B

Govern. thk, $t_g = 16.764$, $t_r = 0.452$, $c = 1.5 \text{ mm.}$, $E* = 1.0$
 Thickness Ratio $= tr \cdot E*/(t_g - c) = 0.03$, Temp. Reduction $= 78 \text{ }^\circ\text{C}$

Min Metal Temp. w/o impact per UCS-66, Curve B	-13 $^\circ\text{C}$
Min Metal Temp per UCS-66 and UCS-68(c), PWHt credit	-29 $^\circ\text{C}$
Min Metal Temp. at Required thickness (UCS 66.1)	-104 $^\circ\text{C}$

Gov. MDMT of the nozzle to shell joint welded assembly : -104 $^\circ\text{C}$

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 $^\circ\text{C}$
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-48 $^\circ\text{C}$

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating $= 23.01/51.00 = 0.451$

Weld Size Calculations, Description: K2A2"

Intermediate Calc. for nozzle/shell Welds Tmin 15.2640 mm.

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$6.0000 = \text{Min per Code}$	$7.0700 = 0.7 \cdot Wo \text{ mm.}$

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A - A1 + 2 \cdot tn \cdot fr1 \cdot (E1 \cdot t - tr)) \cdot Sv)$
 $= \max(0, (9.0916 - 10.1246 + 2 \cdot 15.264 \cdot 1.0 \cdot$
 $(1.0 \cdot 20.5 - 9.699)) \cdot 138)$
 $= 3186.36 \text{ Kg}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2 + A5 + A4 - (Wi - Can/0.707)^2 \cdot fr2) \cdot Sv$
 $= (18.3465 + 0.0 + 1.0 - 0.0 \cdot 1.0) \cdot 138$
 $= 27223.60 \text{ Kg}$

Weld Load [W2]:



$$= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot fr1)) \cdot Sv$$

$$= (18.3465 + 0.0 + 1. + (6.2582)) \cdot 138$$

$$= 36029.93 \text{ Kgf}$$

Weld Load [W3]:

$$= (A2 + A3 + A4 + A5 + (2 \cdot t_n \cdot t \cdot fr1)) \cdot S$$

$$= (18.3465 + 0.0 + 1. + 0.0 + (6.2582)) \cdot 138$$

$$= 36029.93 \text{ Kgf}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$= (\pi/2) \cdot Dlo \cdot Wo \cdot 0.49 \cdot Snw$$

$$= (3.1416/2.0) \cdot 146.9277 \cdot 10.0 \cdot 0.49 \cdot 138$$

$$= 15913. \text{ Kgf}$$

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (Dlr + Dlo) / 4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn$$

$$= (3.1416 \cdot 60.1664) \cdot (16.764 - 1.5) \cdot 0.7 \cdot 138$$

$$= 28420. \text{ Kgf}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot Dlo \cdot (T - cas) \cdot 0.74 \cdot Sng$$

$$= (3.1416/2.0) \cdot 146.9277 \cdot (22.0 - 1.5) \cdot 0.74 \cdot 138$$

$$= 49267. \text{ Kgf}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SONW + SNW) = (15913 + 28420) = 44333 \text{ Kgf}$$

Failure Path [Path 2-2]:

$$= (Sonw + Tpgw + Tngw + Sinw)$$

$$= (15913 + 0 + 49267 + 0) = 65181 \text{ Kgf}$$

Failure Path [Path 3-3]:

$$= (Sonw + Tngw + Sinw)$$

$$= (15913 + 49267 + 0) = 65181 \text{ Kgf}$$

Summary of Failure Path Calculations:

Path 1-1 = 44333	Kgf, must exceed W = 3186 Kgf	or W1 = 27223 Kgf
Path 2-2 = 65180	Kgf, must exceed W = 3186 Kgf	or W2 = 36029 Kgf
Path 3-3 = 65180	Kgf, must exceed W = 3186 Kgf	or W3 = 36029 Kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 24 bars

Note: The MAWP of this junction was limited by the parent Shell/Head.

Note : Checking Nozzle in plane parallel to the vessel axis.

Reinforcement CALCULATION, Description: K2A2"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	2.000 in.
Actual Thickness Used in Calculation	0.660 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]

$$= P \cdot R / (Sv \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= 23.01 \cdot 1151.5 / (138 \cdot 1.0 - 0.6 \cdot 23.01)$$

$$= 19.3981 \text{ mm.}$$

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]

$$= R \cdot (\exp([P / (Sn \cdot E)] - 1)) \text{ per Appendix 1-2 (a)(1)}$$

$$= 26.9 \cdot (\exp([23.01 / (138 \cdot 1.0)] - 1))$$

= 0.4524 mm.

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	125.3280 mm.
Parallel to Vessel Wall	Rn+tn+t	62.6640 mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	38.1600 mm.

*Taking a UG-36(c)(3)(a) exemption for nozzle: K2A2".
 This calculation is valid for nozzles that meet all the requirements of
 paragraph UG-36. Please check the Code carefully, especially for nozzles
 that are not isolated or do not meet Code spacing requirements. To force
 the computation of areas for small nozzles go to Tools->Configuration
 and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 1.9524 mm.
Wall Thickness per UG16(b),	tr16b = 3.0000 mm.
Wall Thickness, shell/head, internal pressure	trb1 = 20.8981 mm.
Wall Thickness	tb1 = max(trb1, tr16b) = 20.8981 mm.
Wall Thickness	tb2 = max(trb2, tr16b) = 3.0000 mm.
Wall Thickness per table UG-45	tb3 = 6.3000 mm.

Determine Nozzle Thickness candidate [tb]:

= min[tb3, max(tb1,tb2)]
 = min[6.3, max(20.8981, 3.0)]
 = 6.3000 mm.

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

= max(ta, tb)
 = max(1.9524, 6.3)
 = 6.3000 mm.

Available Nozzle Neck Thickness = 16.7640 mm. -> OK

Weld Size Calculations, Description: K2A2"

Intermediate Calc. for nozzle/shell Welds Tmin 15.2640 mm.

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm.

*Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles
 exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.*

The Drop for this Nozzle is : 66.6013 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 311.2578 mm.

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Input, Nozzle Desc: K2B2"

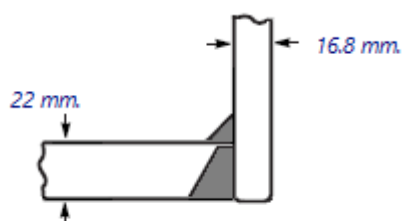
From: 40

Pressure for Reinforcement Calculations	P	23.124	bars
Temperature for Internal Pressure	Temp	150	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm2
Parent Allowable Stress At Ambient	Sva	138.00	N./mm2
Inside Diameter of Cylindrical Shell	D	2300.00	mm.
Shell Finished (Minimum) Thickness	t	22.0000	mm.
Shell Internal Corrosion Allowance	c	1.5000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Cylinder/Cone Centerline	L1	950.0001	mm.
Distance from Bottom/Left Tangent		6550.00	mm.
User Entered Minimum Design Metal Temperature		-33.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-350 LF2 +IT	
Material UNS Number		K03011	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	N./mm2
Allowable Stress At Ambient	Sna	138.00	N./mm2
Diameter Basis (for tr calc only)		Inside	
Layout Angle		-54.15	deg
Diameter		2.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	16.7640	mm.
Flange Material		SA-105 +IT	
Flange Type		Long Weld Neck	
Corrosion Allowance	can	1.5000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	206.3750	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	22.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



Note : Checking Nozzle 90 degrees to the Longitudinal axis.

Reinforcement CALCULATION, Description: K2B2"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation 2.000 in.
 Actual Thickness Used in Calculation 0.660 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 23.12 \cdot 1151.5 / (138 \cdot 1.0 - 0.6 \cdot 23.12)$
 $= 19.4927$ mm.

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]
 $= R \cdot (\exp([P / (S_n \cdot E)] - 1))$ per Appendix 1-2 (a)(1)
 $= 26.9 \cdot (\exp([23.12 / (138.0 \cdot 1.0)] - 1))$
 $= 0.4546$ mm.

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit) D1 187.4598 mm.
 Parallel to Vessel Wall, opening length d 93.7299 mm.
 Normal to Vessel Wall (Thickness Limit), no pad Tlnp 38.1600 mm.

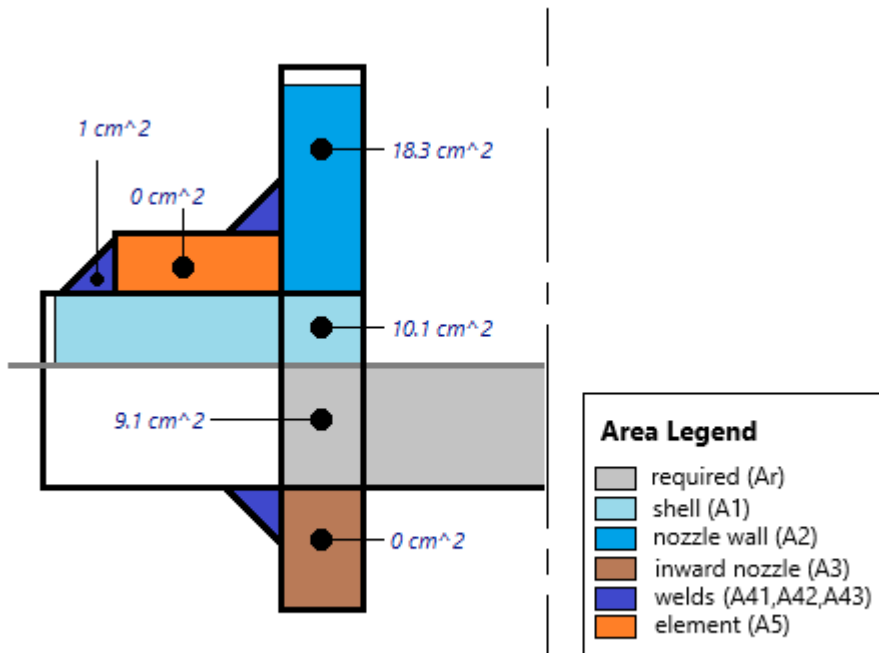


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (cm2)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	9.135	NA	NA
Area in Shell	A1	10.079	NA	NA
Area in Nozzle Wall	A2	18.344	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	1.000	NA	NA
Area in Element	A5	0.000	NA	NA
TOTAL AREA AVAILABLE	Atot	29.423	NA	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 35.03 degs.

The area available without a pad is Sufficient.

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \cdot UG-37(c)$
 $= (93.7299 \cdot 19.4927 \cdot 0.5 + 2 \cdot 15.264 \cdot 19.4927 \cdot 0.5 \cdot (1 - 1.0))$
 $= 9.135 \text{ cm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 93.73(1.0 \cdot 20.5 - 0.5 \cdot 19.493) - 2 \cdot 15.264$
 $(1.0 \cdot 20.5 - 0.5 \cdot 19.4927) \cdot (1 - 1.0)$
 $= 10.079 \text{ cm}^2$

Area Available in Nozzle Projecting Outward [A2]:
 $= (2 \cdot tnp)(tn - trn)fr2/\sin(\alpha3)$
 $= (2 \cdot 38.16)(15.26 - 0.45)1.0/\sin(38.0)$
 $= 18.344 \text{ cm}^2$

See ASME VIII-1 2011(a) Appendix L, L-7.7.7(b) for more information.

Area Available in Inward Weld + Outward Weld [A41 + A43]:
 $= Wo^2 \cdot fr2 + (Wi - can/0.707)^2 \cdot fr2$
 $= 10.0^2 \cdot 1.0 + (0.0)^2 \cdot 1.0$
 $= 1.000 \text{ cm}^2$

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)(1)(b)), Curve: B

Govrn. thk, $t_g = 16.764$, $t_r = 0.455$, $c = 1.5 \text{ mm.}$, $E* = 1.0$
 Thickness Ratio $= tr \cdot E*/(t_g - c) = 0.03$, Temp. Reduction $= 78 \text{ }^\circ\text{C}$

Min Metal Temp. w/o impact per UCS-66, Curve B	-13 $^\circ\text{C}$
Min Metal Temp per UCS-66 and UCS-68(c), PWHt credit	-29 $^\circ\text{C}$
Min Metal Temp. at Required thickness (UCS 66.1)	-104 $^\circ\text{C}$

Gov. MDMT of the nozzle to shell joint welded assembly : -104 $^\circ\text{C}$

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 $^\circ\text{C}$
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-48 $^\circ\text{C}$

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating $= 23.12/51.00 = 0.453$

Weld Size Calculations, Description: K2B2"

Intermediate Calc. for nozzle/shell Welds $T_{min} = 15.2640 \text{ mm.}$

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$6.0000 = \text{Min per Code}$	$7.0700 = 0.7 \cdot Wo \text{ mm.}$

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A - A1 + 2 \cdot tn \cdot fr1 \cdot (E1 \cdot t - tr)) \cdot Sv)$
 $= \max(0, (9.1352 - 10.0794 + 2 \cdot 15.264 \cdot 1.0 \cdot$
 $(1.0 \cdot 20.5 - 9.7463)) \cdot 138)$
 $= 3290.91 \text{ Kg}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2 + A5 + A4 - (Wi - Can/0.707)^2 \cdot fr2) \cdot Sv$
 $= (18.3438 + 0.0 + 1.0 - 0.0 \cdot 1.0) \cdot 138$
 $= 27219.76 \text{ Kg}$

Weld Load [W2]:



$$= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot fr1)) \cdot Sv$$

$$= (18.3438 + 0.0 + 1. + (6.2582)) \cdot 138$$

$$= 36026.09 \text{ Kgf}$$

Weld Load [W3]:

$$= (A2 + A3 + A4 + A5 + (2 \cdot t_n \cdot t \cdot fr1)) \cdot S$$

$$= (18.3438 + 0.0 + 1. + 0.0 + (6.2582)) \cdot 138$$

$$= 36026.09 \text{ Kgf}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$= (\pi/2) \cdot Dlo \cdot Wo \cdot 0.49 \cdot Snw$$

$$= (3.1416/2.0) \cdot 146.9155 \cdot 10.0 \cdot 0.49 \cdot 138$$

$$= 15912. \text{ Kgf}$$

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (Dlr + Dlo) / 4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn$$

$$= (3.1416 \cdot 60.1613) \cdot (16.764 - 1.5) \cdot 0.7 \cdot 138$$

$$= 28417. \text{ Kgf}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot Dlo \cdot (T - cas) \cdot 0.74 \cdot Sng$$

$$= (3.1416/2.0) \cdot 146.9155 \cdot (22.0 - 1.5) \cdot 0.74 \cdot 138$$

$$= 49263. \text{ Kgf}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SONW + SNW) = (15912 + 28417) = 44329 \text{ Kgf}$$

Failure Path [Path 2-2]:

$$= (Sonw + Tpgw + Tngw + Sinw)$$

$$= (15912 + 0 + 49263 + 0) = 65175 \text{ Kgf}$$

Failure Path [Path 3-3]:

$$= (Sonw + Tngw + Sinw)$$

$$= (15912 + 49263 + 0) = 65175 \text{ Kgf}$$

Summary of Failure Path Calculations:

Path 1-1 = 44329	Kgf, must exceed W = 3290 Kgf	or W1 = 27219 Kgf
Path 2-2 = 65175	Kgf, must exceed W = 3290 Kgf	or W2 = 36026 Kgf
Path 3-3 = 65175	Kgf, must exceed W = 3290 Kgf	or W3 = 36026 Kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 24 bars

[Note: The MAWP of this junction was limited by the parent Shell/Head.](#)

[Note : Checking Nozzle in plane parallel to the vessel axis.](#)

Reinforcement CALCULATION, Description: K2B2"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	2.000 in.
Actual Thickness Used in Calculation	0.660 in.

[Nozzle input data check completed without errors.](#)

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]

$$= P \cdot R / (Sv \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= 23.12 \cdot 1151.5 / (138 \cdot 1.0 - 0.6 \cdot 23.12)$$

$$= 19.4927 \text{ mm.}$$

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]

$$= R \cdot (\exp([P / (Sn \cdot E)] - 1)) \text{ per Appendix 1-2 (a)(1)}$$

$$= 26.9 \cdot (\exp([23.12 / (138.0 \cdot 1.0)] - 1))$$

= 0.4546 mm.

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	125.3280 mm.
Parallel to Vessel Wall	Rn+tn+t	62.6640 mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	38.1600 mm.

*Taking a UG-36(c)(3)(a) exemption for nozzle: K2B2".
 This calculation is valid for nozzles that meet all the requirements of
 paragraph UG-36. Please check the Code carefully, especially for nozzles
 that are not isolated or do not meet Code spacing requirements. To force
 the computation of areas for small nozzles go to Tools->Configuration
 and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 1.9546 mm.
Wall Thickness per UG16(b),	tr16b = 3.0000 mm.
Wall Thickness, shell/head, internal pressure	trb1 = 20.9927 mm.
Wall Thickness	tb1 = max(trb1, tr16b) = 20.9927 mm.
Wall Thickness	tb2 = max(trb2, tr16b) = 3.0000 mm.
Wall Thickness per table UG-45	tb3 = 6.3000 mm.

Determine Nozzle Thickness candidate [tb]:

= min[tb3, max(tb1,tb2)]
 = min[6.3, max(20.9927, 3.0)]
 = 6.3000 mm.

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

= max(ta, tb)
 = max(1.9546, 6.3)
 = 6.3000 mm.

Available Nozzle Neck Thickness = 16.7640 mm. -> OK

Weld Size Calculations, Description: K2B2"

Intermediate Calc. for nozzle/shell Welds	Tmin	15.2640 mm.
---	------	-------------

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm.

*Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles
 exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.*

The Drop for this Nozzle is : 66.6013 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 311.2578 mm.

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Input, Nozzle Desc: N43"

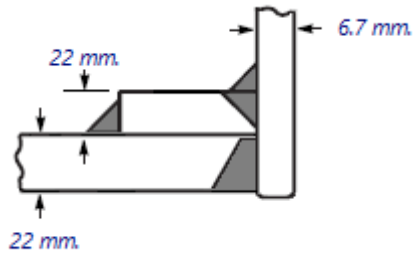
From: 50

Pressure for Reinforcement Calculations	P	23.000	bars
Temperature for Internal Pressure	Temp	150	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm2
Parent Allowable Stress At Ambient	Sva	138.00	N./mm2
Inside Diameter of Cylindrical Shell	D	2300.00	mm.
Shell Finished (Minimum) Thickness	t	22.0000	mm.
Shell Internal Corrosion Allowance	c	1.5000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		8450.00	mm.
User Entered Minimum Design Metal Temperature		-33.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-333 6 +IT	
Material UNS Number		K03006	
Material Specification/Type	Smls. & wld. pipe		
Allowable Stress at Temperature	Sn	118.00	N./mm2
Allowable Stress At Ambient	Sna	118.00	N./mm2
Diameter Basis (for tr calc only)		Outside	
Layout Angle		90.00	deg
Diameter		3.0000	in.
Size and Thickness Basis		Minimum	
Nominal Thickness		80	
Flange Material		SA-350 LF2 +IT	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	1.5000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	399.4760	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	22.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Pad Material		SA-516 70 +IT	
Pad Allowable Stress at Temperature	Sp	138.00	N./mm2
Pad Allowable Stress At Ambient	Spa	138.00	N./mm2
Diameter of Pad along vessel surface	Dp	152.4000	mm.
Thickness of Pad	te	22.0000	mm.
Weld leg size between Pad and Shell	Wp	18.0000	mm.
Groove weld depth between Pad and Nozzle	Wgpn	22.0000	mm.
Reinforcing Pad Width		31.7500	mm.
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: N43"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	3.500 in.
Actual Thickness Used in Calculation	0.263 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 23.0 \cdot 1151.5 / (138 \cdot 1.0 - 0.6 \cdot 23.0)$
 $= 19.3870 \text{ mm.}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= 23.0 \cdot 44.45 / (118 \cdot 1.0 + 0.4 \cdot 23.0)$
 $= 0.8598 \text{ mm.}$

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	157.1300 mm.
Parallel to Vessel Wall, opening length	d	78.5650 mm.
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		34.9188 mm.

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 118.0 / 138.0)$
 $= 0.855$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 118.0 / 138.0)$
 $= 0.855$

Weld Strength Reduction Factor [fr4]:
 $= \min(1, S_p / S_v)$
 $= \min(1, 138.0 / 138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(0.855, 1.0)$
 $= 0.855$

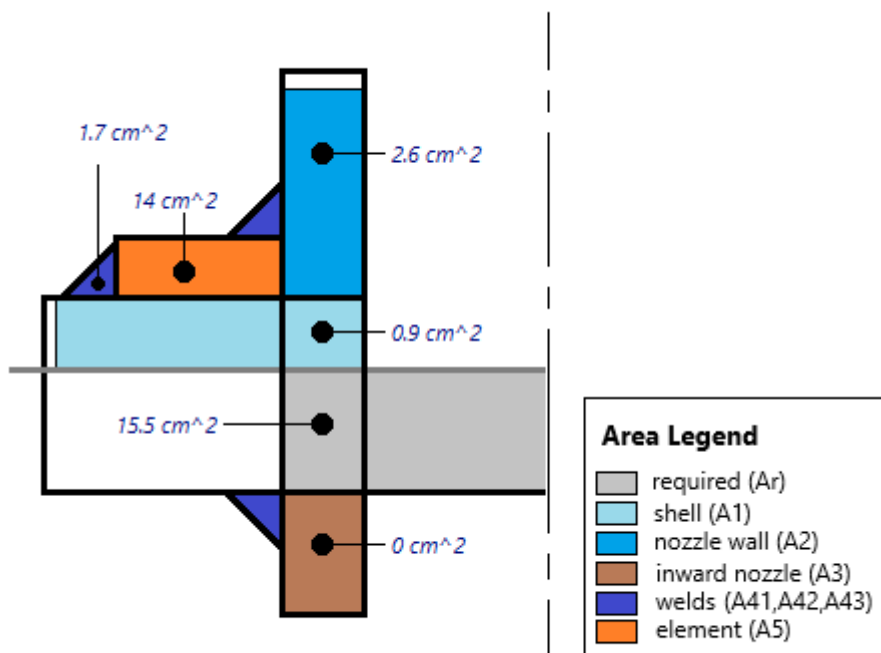


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (cm2)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	15.522	NA	NA
Area in Shell	A1	0.858	NA	NA
Area in Nozzle Wall	A2	2.572	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	1.651	NA	NA
Area in Element	A5	13.970	NA	NA
TOTAL AREA AVAILABLE	Atot	19.051	NA	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Insufficient.

The area available with the given pad is Sufficient.

SELECTION OF POSSIBLE REINFORCING PADS: Diameter Thickness
 Based on given Pad Thickness: 136.3595 22.0000 mm.
 Based on given Pad Diameter: 152.4000 16.4427 mm.
 Based on the Estimated Diameter Limit: 155.5425 15.6673 mm.

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$
 $= (78.565 \cdot 19.387 \cdot 1.0 + 2 \cdot 5.1675 \cdot 19.387 \cdot 1.0 \cdot (1 - 0.86))$
 $= 15.522 \text{ cm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 78.565(1.0 \cdot 20.5 - 1.0 \cdot 19.387) - 2 \cdot 5.168$
 $(1.0 \cdot 20.5 - 1.0 \cdot 19.387) \cdot (1 - 0.855)$
 $= 0.858 \text{ cm}^2$

Area Available in Nozzle Wall Projecting Outward [A2]:
 $= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2$
 $= (2 \cdot 34.92) \cdot (5.17 - 0.86) \cdot 0.8551$



$$= 2.572 \text{ cm}^2$$

Area Available in Welds [A41 + A42 + A43]:

$$= (W_o^2 - A_r \text{ Lost}) \cdot Fr_3 + ((W_i - c_{an}/0.707)^2 - A_r \text{ Lost}) \cdot fr_2 + Trapfr_4$$

$$= (1.) \cdot 0.86 + (0.0) \cdot 0.86 + 79.5465^2 \cdot 1.0$$

$$= 1.651 \text{ cm}^2$$

Area Available in Element [A5]:

$$= (\min(D_p, D_L) - (\text{Nozzle OD})) \cdot (\min(t_p, t_{lp}, t_e)) \cdot fr_4$$

$$= (152.4 - 88.9) \cdot 22.0 \cdot 1.0$$

$$= 13.970 \text{ cm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	$t_a = 2.3598 \text{ mm.}$
Wall Thickness per UG16(b),	$tr_{16b} = 3.0000 \text{ mm.}$
Wall Thickness, shell/head, internal pressure	$trb_1 = 20.8870 \text{ mm.}$
Wall Thickness	$tb_1 = \max(trb_1, tr_{16b}) = 20.8870 \text{ mm.}$
Wall Thickness	$tb_2 = \max(trb_2, tr_{16b}) = 3.0000 \text{ mm.}$
Wall Thickness per table UG-45	$tb_3 = 6.3000 \text{ mm.}$

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb_3, \max(tb_1, tb_2)]$$

$$= \min[6.3, \max(20.887, 3.0)]$$

$$= 6.3000 \text{ mm.}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(t_a, t_b)$$

$$= \max(2.3598, 6.3)$$

$$= 6.3000 \text{ mm.}$$

Available Nozzle Neck Thickness = 6.6675 mm. -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld (Impact tested) :

Note:
This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification	-46 °C
Calculated Minimum Design Metal Temperature	-104 °C

Nozzle Neck to Pad Weld for the Nozzle (Impact tested) :

Note:
This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification	-46 °C
Calculated Minimum Design Metal Temperature	-104 °C

Nozzle Neck to Pad Weld for Reinf. pad (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Shell to Pad Weld Junction at Pad OD, Curve: B

Govrn. thk, $t_g = 22.0$, $t_r = 19.387$, $c = 1.5 \text{ mm.}$, $E^* = 1.0$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.946$, Temp. Reduction = 3 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-5 °C
Min Metal Temp per UCS-66 and UCS-68(c), PWHt credit	-21 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-24 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), Curve: B

Govrn. thk, $t_g = 6.668$, $t_r = 0.86$, $c = 1.5 \text{ mm.}$, $E^* = 1.0$

Thickness Ratio = $tr \cdot E / (tg - c) = 0.166$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -29 °C
 Min Metal Temp per UCS-66 and UCS-68(c), PWHT credit -46 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Gov. MDMT of the Nozzle : -104 °C
 Gov. MDMT of the Reinforcement Pad : -29 °C
 Gov. MDMT of the nozzle to shell joint welded assembly : -29 °C

Flange MDMT including Temperature reduction per UCS-66.1:

MDMT of ASME B16.5/47 flange per Matl. Specification -46 °C
 Flange MDMT with Temp reduction per UCS-66(i)(2) -85 °C

Where the Stress Reduction Ratio per UCS-66(i)(2) is :
 Design Pressure/Ambient Rating = $23.00/51.00 = 0.451$

Weld Size Calculations, Description: N43"

Intermediate Calc. for nozzle/shell Welds T_{min} 5.1675 mm.
 Intermediate Calc. for pad/shell Welds T_{minPad} 19.0000 mm.

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$3.6173 = 0.7 \cdot t_{min}$	$7.0700 = 0.7 \cdot W_o$ mm.
Pad Weld	$9.5000 = 0.5 \cdot T_{minPad}$	$12.7260 = 0.7 \cdot W_p$ mm.

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A - A1 + 2 \cdot t_n \cdot fr1 \cdot (E1 \cdot t - tr)) \cdot Sv)$
 $= \max(0, (15.5218 - 0.8577 + 2 \cdot 5.1675 \cdot 0.8551 \cdot (1.0 \cdot 20.5 - 19.387)) \cdot 138)$
 $= 20773.02 \text{ Kgf}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2 + A5 + A4 - (W_i - Can / .707)^2 \cdot fr2) \cdot Sv$
 $= (2.5724 + 13.97 + 1.6505 - 0.0 \cdot 0.86) \cdot 138$
 $= 25600.34 \text{ Kgf}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot fr1)) \cdot Sv$
 $= (2.5724 + 0.0 + 0.8551 + (1.8116)) \cdot 138$
 $= 7372.25 \text{ Kgf}$

Weld Load [W3]:
 $= (A2 + A3 + A4 + A5 + (2 \cdot t_n \cdot t \cdot fr1)) \cdot S$
 $= (2.5724 + 0.0 + 1.6505 + 13.97 + (1.8116)) \cdot 138$
 $= 28149.58 \text{ Kgf}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$
 $= (3.1416/2.0) \cdot 88.9 \cdot 10.0 \cdot 0.49 \cdot 118$
 $= 8233. \text{ Kgf}$

Shear, Pad Element Weld [Spew]:
 $= (\pi/2) \cdot DP \cdot WP \cdot 0.49 \cdot SEW$
 $= (3.1416/2.0) \cdot 152.4 \cdot 18.0 \cdot 0.49 \cdot 138$
 $= 29711. \text{ Kgf}$

Shear, Nozzle Wall [Snw]:
 $= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - Can) \cdot 0.7 \cdot S_n$
 $= (3.1416 \cdot 41.8662) \cdot (6.6675 - 1.5) \cdot 0.7 \cdot 118$
 $= 5725. \text{ Kgf}$



Tension, Pad Groove Weld [Tpgw]:
 $= (\pi/2) \cdot Dlo \cdot Wgpn \cdot 0.74 \cdot Seg$
 $= (3.1416/2) \cdot 88.9 \cdot 22.0 \cdot 0.74 \cdot 138$
 $= 31991. \text{ Kg}$

Tension, Shell Groove Weld [Tngw]:
 $= (\pi/2) \cdot Dlo \cdot (T - cas) \cdot 0.74 \cdot Sng$
 $= (3.1416/2.0) \cdot 88.9 \cdot (22.0 - 1.5) \cdot 0.74 \cdot 138$
 $= 29809. \text{ Kg}$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:
 $= (\text{SPEW} + \text{SNW}) = (29711 + 5725) = 35435 \text{ Kg}$

Failure Path [Path 2-2]:
 $= (\text{Sonw} + \text{Tpgw} + \text{Tngw} + \text{Sinw})$
 $= (8233 + 31991 + 29809 + 0) = 70033 \text{ Kg}$

Failure Path [Path 3-3]:
 $= (\text{Spew} + \text{Tngw} + \text{Sinw})$
 $= (29711 + 29809 + 0) = 59520 \text{ Kg}$

Summary of Failure Path Calculations:

Path 1-1 = 35435	Kgf, must exceed W = 20773 Kg	or W1 = 25600 Kg
Path 2-2 = 70033	Kgf, must exceed W = 20773 Kg	or W2 = 7372 Kg
Path 3-3 = 59520	Kgf, must exceed W = 20773 Kg	or W3 = 28149 Kg

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 24 bars

Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 0.8593 mm.
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 422.3354 mm.

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Input, Nozzle Desc: K912"

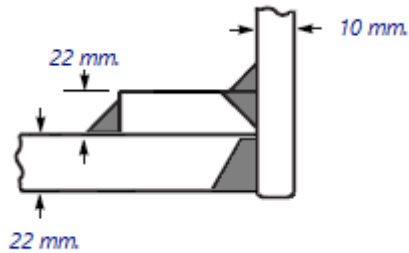
From: 60

Pressure for Reinforcement Calculations	P	23.000	bars
Temperature for Internal Pressure	Temp	150	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm2
Parent Allowable Stress At Ambient	Sva	138.00	N./mm2
Inside Diameter of Cylindrical Shell	D	2300.00	mm.
Shell Finished (Minimum) Thickness	t	22.0000	mm.
Shell Internal Corrosion Allowance	c	1.5000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		10200.00	mm.
User Entered Minimum Design Metal Temperature		-33.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-516 70 +IT	
Material UNS Number		K02700	
Material Specification/Type		Plate	
Allowable Stress at Temperature	Sn	138.00	N./mm2
Allowable Stress At Ambient	Sna	138.00	N./mm2
Diameter Basis (for tr calc only)		Inside	
Layout Angle		90.00	deg
Diameter		12.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	10.0000	mm.
Flange Material		SA-350 LF2 +IT	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	1.5000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	399.4760	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	22.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Pad Material		SA-516 70 +IT	
Pad Allowable Stress at Temperature	Sp	138.00	N./mm2
Pad Allowable Stress At Ambient	Spa	138.00	N./mm2
Diameter of Pad along vessel surface	Dp	580.0000	mm.
Thickness of Pad	te	22.0000	mm.
Weld leg size between Pad and Shell	Wp	18.0000	mm.
Groove weld depth between Pad and Nozzle	Wgpn	22.0000	mm.
Reinforcing Pad Width		127.6000	mm.
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: K912"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	12.000 in.
Actual Thickness Used in Calculation	0.394 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 23.0 \cdot 1151.5 / (138 \cdot 1.0 - 0.6 \cdot 23.0)$
 $= 19.3870$ mm.

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 23.0 \cdot 153.9 / (138 \cdot 1.0 - 0.6 \cdot 23.0)$
 $= 2.5911$ mm.

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	615.6000 mm.
Parallel to Vessel Wall, opening length	d	307.8000 mm.
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		43.2500 mm.

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 138.0 / 138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 138.0 / 138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr4]:
 $= \min(1, S_p / S_v)$
 $= \min(1, 138.0 / 138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(1.0, 1.0)$
 $= 1.000$

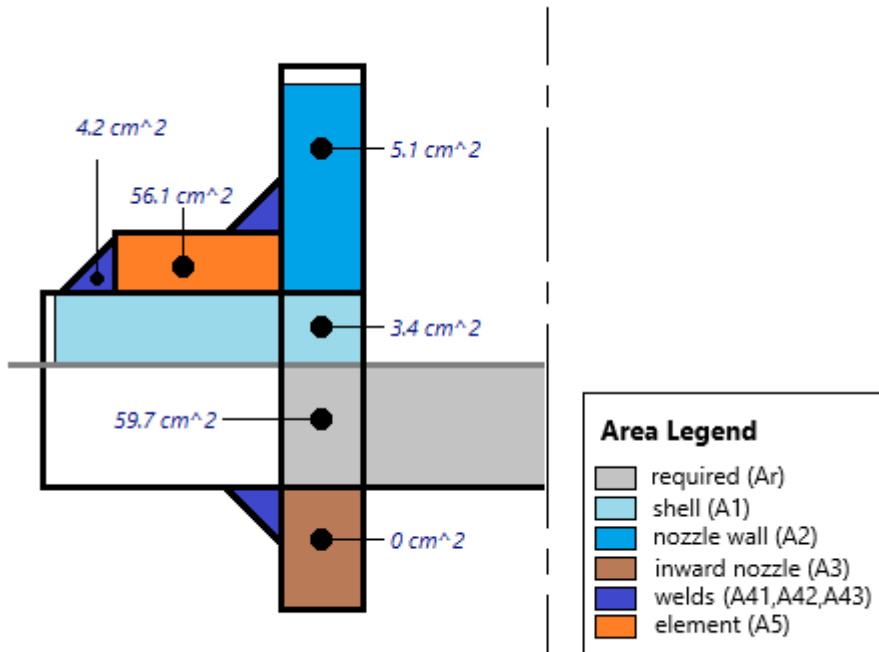


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (cm2)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	59.673	NA	NA
Area in Shell	A1	3.426	NA	NA
Area in Nozzle Wall	A2	5.111	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	4.240	NA	NA
Area in Element	A5	56.144	NA	NA
TOTAL AREA AVAILABLE	Atot	68.921	NA	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Insufficient.
 The area available with the given pad is Sufficient.

SELECTION OF POSSIBLE REINFORCING PADS: Diameter Thickness
 Based on given Pad Thickness: 537.9666 22.0000 mm.
 Based on given Pad Diameter: 580.0000 18.3764 mm.
 Based on the Estimated Diameter Limit: 614.0125 16.2153 mm.

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$
 $= (307.8 \cdot 19.387 \cdot 1.0 + 2 \cdot 8.5 \cdot 19.387 \cdot 1.0 \cdot (1 - 1.0))$
 $= 59.673 \text{ cm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 307.8(1.0 \cdot 20.5 - 1.0 \cdot 19.387) - 2 \cdot 8.5$
 $(1.0 \cdot 20.5 - 1.0 \cdot 19.387) \cdot (1 - 1.0)$
 $= 3.426 \text{ cm}^2$

Area Available in Nozzle Wall Projecting Outward [A2]:
 $= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2$
 $= (2 \cdot 43.25) \cdot (8.5 - 2.59) \cdot 1.0$



= 5.111 cm²

Area Available in Welds [A41 + A42 + A43]:
 = (Wo² - Ar Lost)•Fr3+((Wi-can/0.707)² - Ar Lost)•fr2 + Trapfr4
 = (1.) • 1.0 + (0.0) • 1.0 + 323.96² • 1.0
 = 4.240 cm²

Area Available in Element [A5]:
 = (min(Dp,DL)-(Nozzle OD))•(min(tp,Tlwp,te)) • fr4
 = (580.0 - 324.8) • 22.0 • 1.0
 = 56.144 cm²

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures ta = 4.0911 mm.
 Wall Thickness per UG16(b), tr16b = 3.0000 mm.
 Wall Thickness, shell/head, internal pressure trb1 = 20.8870 mm.
 Wall Thickness tb1 = max(trb1, tr16b) = 20.8870 mm.
 Wall Thickness tb2 = max(trb2, tr16b) = 3.0000 mm.
 Wall Thickness per table UG-45 tb3 = 9.8312 mm.

Determine Nozzle Thickness candidate [tb]:
 = min[tb3, max(tb1,tb2)]
 = min[9.831, max(20.887, 3.0)]
 = 9.8312 mm.

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 = max(ta, tb)
 = max(4.0911, 9.8312)
 = 9.8312 mm.

Available Nozzle Neck Thickness = 10.0000 mm. -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	:	47.0,	Allowable	:	138.0 N./mm ²	Passed
Expansion	:	0.0,	Allowable	:	298.0 N./mm ²	Passed
Occasional	:	20.3,	Allowable	:	183.5 N./mm ²	Passed
Shear	:	18.4,	Allowable	:	96.6 N./mm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Nozzle Neck to Pad Weld for the Nozzle (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Nozzle Neck to Pad Weld for Reinf. pad (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Shell to Pad Weld Junction at Pad OD, Curve: B

Govrn. thk, tg = 22.0, tr = 19.387, c = 1.5 mm., E* = 1.0
 Thickness Ratio = tr * E*/(tg - c) = 0.946, Temp. Reduction = 3 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-5 °C
Min Metal Temp per UCS-66 and UCS-68(c),PWHT credit	-21 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-24 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle-Shell/Head Weld (UCS-66(a)(1)(b)), Curve: B

Govrn. thk, tg = 10.0, tr = 2.591, c = 1.5 mm., E* = 1.0
 Thickness Ratio = $tr \cdot E^*/(tg - c) = 0.305$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -29 °C
 Min Metal Temp per UCS-66 and UCS-68(c),PWHT credit -46 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Gov. MDMT of the Nozzle : -104 °C
 Gov. MDMT of the Reinforcement Pad : -29 °C
 Gov. MDMT of the nozzle to shell joint welded assembly : -29 °C

Flange MDMT including Temperature reduction per UCS-66.1:

MDMT of ASME B16.5/47 flange per Matl. Specification -46 °C
 Flange MDMT with Temp reduction per UCS-66(i)(2) -85 °C

Where the Stress Reduction Ratio per UCS-66(i)(2) is :
 Design Pressure/Ambient Rating = $23.00/51.00 = 0.451$

Weld Size Calculations, Description: K912"

Intermediate Calc. for nozzle/shell Welds Tmin 8.5000 mm.
 Intermediate Calc. for pad/shell Welds TminPad 19.0000 mm.

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$5.9500 = 0.7 \cdot t_{min}$	$7.0700 = 0.7 \cdot W_o$ mm.
Pad Weld	$9.5000 = 0.5 \cdot T_{minPad}$	$12.7260 = 0.7 \cdot W_p$ mm.

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A - A1 + 2 \cdot t_n \cdot f_{r1} \cdot (E1 \cdot t - tr)) \cdot S_v)$
 $= \max(0, (59.6732 - 3.4258 + 2 \cdot 8.5 \cdot 1.0 \cdot (1.0 \cdot 20.5 - 19.387)) \cdot 138)$
 $= 79415.26 \text{ Kgf}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2 + A5 + A4 - (W_i - Can / .707)^2 \cdot f_{r2}) \cdot S_v$
 $= (5.1112 + 56.144 + 4.2396 - 0.0 \cdot 1.0) \cdot 138$
 $= 92161.52 \text{ Kgf}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S_v$
 $= (5.1112 + 0.0 + 1. + (3.485)) \cdot 138$
 $= 13503.40 \text{ Kgf}$

Weld Load [W3]:
 $= (A2 + A3 + A4 + A5 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S$
 $= (5.1112 + 0.0 + 4.2396 + 56.144 + (3.485)) \cdot 138$
 $= 97065.46 \text{ Kgf}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$
 $= (3.1416/2.0) \cdot 324.8 \cdot 10.0 \cdot 0.49 \cdot 138$
 $= 35179. \text{ Kgf}$

Shear, Pad Element Weld [Spew]:
 $= (\pi/2) \cdot DP \cdot WP \cdot 0.49 \cdot S_{EW}$
 $= (3.1416/2.0) \cdot 580.0 \cdot 18.0 \cdot 0.49 \cdot 138$
 $= 113073. \text{ Kgf}$

Shear, Nozzle Wall [Snw]:
 $= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - Can) \cdot 0.7 \cdot S_n$



$$= (3.1416 \cdot 158.15) \cdot (10.0 - 1.5) \cdot 0.7 \cdot 138$$

$$= 41599. \text{ Kgf}$$

Tension, Pad Groove Weld [Tpgw]:

$$= (\pi/2) \cdot Dlo \cdot Wgpn \cdot 0.74 \cdot Seg$$

$$= (3.1416/2) \cdot 324.8 \cdot 22.0 \cdot 0.74 \cdot 138$$

$$= 116880. \text{ Kgf}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot Dlo \cdot (T - cas) \cdot 0.74 \cdot Sng$$

$$= (3.1416/2.0) \cdot 324.8 \cdot (22.0 - 1.5) \cdot 0.74 \cdot 138$$

$$= 108911. \text{ Kgf}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (\text{SPEW} + \text{SNW}) = (113073 + 41599) = 154672 \text{ Kgf}$$

Failure Path [Path 2-2]:

$$= (\text{Sonw} + \text{Tpgw} + \text{Tngw} + \text{Sinw})$$

$$= (35179 + 116880 + 108911 + 0) = 260970 \text{ Kgf}$$

Failure Path [Path 3-3]:

$$= (\text{Spew} + \text{Tngw} + \text{Sinw})$$

$$= (113073 + 108911 + 0) = 221984 \text{ Kgf}$$

Summary of Failure Path Calculations:

Path 1-1 = 154672 Kgf, must exceed W = 79415 Kgf or W1 = 92161 Kgf
 Path 2-2 = 260970 Kgf, must exceed W = 79415 Kgf or W2 = 13503 Kgf
 Path 3-3 = 221984 Kgf, must exceed W = 79415 Kgf or W3 = 97065 Kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 24 bars

Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 11.5246 mm.
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 433.0006 mm.

Percent Elongation Calculations:

% Elongation per Table UG-79-1 ($50 \cdot t_{nom} / R_f \cdot (1 - R_f / R_o)$) 3.177 %

Input Echo, WRC297 Item 1, Description: K912" :

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	1.5000	mm.
Shell External Corrosion Allowance	0.0000	mm.
Shell Diameter	2300.000	mm.
Shell Thickness	22.0000	mm.
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc	138.00 N./mm ²
Vessel Hot S.I. Allowable	Smh	138.00 N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter Basis for Nozzle	ID	
Nozzle Corrosion Allowance	1.5000	mm.
Nozzle Diameter	304.800	mm.
Nozzle Thickness	10.0000	mm.
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-516 70	
Nozzle Cold S.I. Allowable	SNmc	138.00 N./mm ²
Nozzle Hot S.I. Allowable	SNmh	138.00 N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter of Reinforcing Pad 580.000 mm.
 Thickness of Reinforcing Pad 22.0000 mm.

Note:

*External Forces and Moments in WRC 107/537 Convention:
 These loads are assumed to be SUSTAINED loads.*

Design Internal Pressure	Dp	23.00	bars
Radial Load	P	-830.00	Kgf
Circumferential Shear	Vc	1230.00	Kgf
Longitudinal Shear	Vl	1230.00	Kgf
Circumferential Moment	Mc	1105.00	Kg-m.
Longitudinal Moment	Ml	1469.99	Kg-m.
Torsional Moment	Mt	1840.01	Kg-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(\text{Rmean}(t - ca))}$
 $= 324.8 + 2 \cdot 1.65 \cdot \sqrt{(1172.75(22.0 - 1.5))}$
 $= 836.474 \text{ mm.}$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	2345.500 mm.
Nozzle Outside Diameter	(d) =	324.800 mm.
Vessel Thickness used	(T) =	42.500 mm.
Nozzle Thickness used	(t) =	8.500 mm.
T / t	=	5.000
d / t	=	38.212

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 1.029$$

Nr/P	=	0.055
Mr/P	=	0.044
M0/P	=	0.107
N0/P	=	0.310
MrD/Mc	=	0.086
NrDL/Mc	=	0.092
M0d/Mc	=	0.264
N0DL/Mc	=	0.255
MrD/Ml	=	0.076
NrDL/Ml	=	0.081
M0D/Ml	=	0.196
N0DL/Ml	=	0.488

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	1	1	1	1
Outplane Bending (P)	2	-2	2	-2
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (Ml)	-11	-11	11	11
Outplane Bending (Ml)	-28	28	28	-28
Normal Pressure Stress	61	63	61	63
Outplane Stress Summary	24	78	106	45

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses parallel to	Au Top	Al Top	Bu Bottom	Bl Bottom
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longitudinal plane)	Outside	Inside	Outside	Inside
Inplane Membrane (P)	0	0	0	0
Inplane Bending (P)	1	-1	1	-1
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-1	-1	1	1
Inplane Bending (ML)	-11	11	11	-11
Inplane Pressure Stress	30	30	30	30
Inplane Stress Summary	18	38	45	20

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Shear stress normal to longitudinal plane)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	2	2	2	2
Shear Stress Summary	3	3	2	2

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stress Intensities)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Two * Max Shear Stress	25	79	106	45

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses Normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside
Outplane Membrane (P)	1	1	1	1
Outplane Bending (P)	2	-2	2	-2
Outplane Membrane (Mc)	-4	-4	4	4
Outplane Bending (Mc)	-29	29	29	-29
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	30	30	30	30
Outplane Stress Summary	0	53	68	4

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses parallel to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside
Inplane Membrane (P)	0	0	0	0
Inplane Bending (P)	1	-1	1	-1
Inplane Membrane (Mc)	-1	-1	1	1
Inplane Bending (Mc)	-9	9	9	-9
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	61	63	61	63
Inplane Stress Summary	51	70	73	54

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Shear stress normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Torsional Shear (Mt)	2	2	2	2
Shear Stress Summary	2	2	3	3



Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Two * Max Shear Stress	51	70	75	54

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Hoop Membrane (P)	1	1	1	1
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-11	-11	11	11
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	48	51	48	51
Hoop Stress Summary	38	40	62	64

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Axial Membrane (P)	0	0	0	0
Axial Bending (P)	11	-11	11	-11
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-22	-22	22	22
Axial Bending (ML)	-130	130	130	-130
Axial Pressure Stress	24	24	24	24
Axial Stress Summary	-115	122	188	-93

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress)	Au Outside	Al Inside	Bu Outside	Bl Inside
Shear due to (Vc)	2	2	-2	-2
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	12	12	12	12
Shear Stress Summary	15	15	10	10

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Outside	Al Inside	Bu Outside	Bl Inside
Two * Max Shear Stress	157	125	189	159

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Hoop Membrane (P)	1	1	1	1
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-4	-4	4	4
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	48	51	48	51
Hoop Stress Summary	45	48	54	57

Nozzle Stresses (N./mm²):



CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Axial Membrane (P)	0	0	0	0
Axial Bending (P)	11	-11	11	-11
Axial Membrane (Mc)	-16	-16	16	16
Axial Bending (Mc)	-111	111	111	-111
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	24	24	24	24
Axial Stress Summary	-91	108	164	-80

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-2	-2	2	2
Shear due to Torsion	12	12	12	12
Shear Stress Summary	10	10	15	15

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Two * Max Shear Stress	138	110	166	141

Stress Computations at the Edge of the Reinforcing Pad:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(\text{Rmean}(t - ca))}$
 $= 324.8 + 2 \cdot 1.65 \cdot \sqrt{(1161.75(22.0 - 1.5))}$
 $= 834.069 \text{ mm.}$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	2323.500 mm.
Nozzle Outside Diameter	(d) =	580.000 mm.
Vessel Thickness used	(T) =	20.500 mm.
Nozzle Thickness used	(t) =	136.100 mm.
T / t	=	0.151
d / t	=	10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 2.658$$

Note: Re-pad thickness is added to nozzle thickness to simulate the Re-pad.

Nr/P	=	0.065
Mr/P	=	0.118
M0/P	=	0.035
N0/P	=	0.056
MrD/Mc	=	0.276
NrDL/Mc	=	0.101
M0d/Mc	=	0.083
N0DL/Mc	=	0.074
MrD/Ml	=	0.090
NrDL/Ml	=	0.083
M0D/Ml	=	0.026
N0DL/Ml	=	0.073

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	1	1	1	1



Outplane Bending (P)	4	-4	4	-4
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-4	-4	4	4
Outplane Bending (ML)	-9	9	9	-9
Normal Pressure Stress	128	130	128	130

Outplane Stress Summary	119	132	146	122
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Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses parallel to longitudinal plane)	Top Outside	Top Inside	Bottom Outside	Bottom Inside

Inplane Membrane (P)	1	1	1	1
Inplane Bending (P)	13	-13	13	-13
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-4	-4	4	4
Inplane Bending (ML)	-31	31	31	-31
Inplane Pressure Stress	64	64	64	64

Inplane Stress Summary	42	78	115	24
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Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Shear stress normal to longitudinal plane)	Top Outside	Top Inside	Bottom Outside	Bottom Inside

Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	1	1	1	1

Shear Stress Summary	2	2	1	1
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Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stress Intensities)	Top Outside	Top Inside	Bottom Outside	Bottom Inside

Two * Max Shear Stress	119	132	146	122
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses Normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside

Outplane Membrane (P)	1	1	1	1
Outplane Bending (P)	4	-4	4	-4
Outplane Membrane (Mc)	-3	-3	3	3
Outplane Bending (Mc)	-22	22	22	-22
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	64	64	64	64

Outplane Stress Summary	43	79	94	42
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses parallel to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside

Inplane Membrane (P)	1	1	1	1
Inplane Bending (P)	13	-13	13	-13
Inplane Membrane (Mc)	-4	-4	4	4
Inplane Bending (Mc)	-73	73	73	-73
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	128	130	128	130



Inplane Stress Summary 64 187 221 48

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE		Cu	Cl	Du	Dl
(Shear stress normal to		Left	Left	Right	Right
circumferential plane)		Outside	Inside	Outside	Inside
Outplane Shear (Vc)		0	0	0	0
Outplane Shear (Vl)		0	0	0	0
Torsional Shear (Mt)		1	1	1	1

Shear Stress Summary 1 1 2 2

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE		Cu	Cl	Du	Dl
(Stress Intensities		Left	Left	Right	Right
		Outside	Inside	Outside	Inside
Two * Max Shear Stress		64	187	221	49

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	61	63	61	63	61	63	61	63
Circ. Pl (SUS)	-10	-10	12	12	-1	-1	1	1
Circ. Q (SUS)	-26	26	30	-30	-8	8	10	-10
Long. Pm (SUS)	30	30	30	30	30	30	30	30
Long. Pl (SUS)	-1	-1	1	1	-3	-3	5	5
Long. Q (SUS)	-10	10	12	-12	-27	27	31	-31
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	2	2	2	2	2	2	2	2
Pm (SUS)	61.0	63.0	61.0	63.0	61.0	63.0	61.0	63.0
Pm+Pl (SUS)	51.0	53.0	73.0	75.0	60.0	62.0	62.0	64.0
Pm+Pl+Q (Total)	25.6	79.1	103.1	45.2	52.2	70.2	72.6	54.1

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	63.00	138.00	Passed
Pm+Pl (SUS)	75.00	207.00	Passed
Pm+Pl+Q (TOTAL)	103.07	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Reinforcing Pad Edge (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	128	130	128	130	128	130	128	130
Circ. Pl (SUS)	-3	-3	5	5	-3	-3	5	5
Circ. Q (SUS)	-5	5	13	-13	-60	60	86	-86



Long. Pm (SUS)	64	64	64	64	64	64	64	64
Long. Pl (SUS)	-3	-3	5	5	-2	-2	4	4
Long. Q (SUS)	-18	18	44	-44	-18	18	26	-26
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	1	1	1	1	1	1	1	1
Pm (SUS)	128.0	130.0	128.0	130.0	128.0	130.0	128.0	130.0
Pm+Pl (SUS)	125.0	127.0	133.0	135.0	125.0	127.0	133.0	135.0
Pm+Pl+Q (Total)	120.0	132.0	146.0	122.0	65.0	187.0	219.0	49.1

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	130.00	138.00	Passed
Pm+Pl (SUS)	135.00	207.00	Passed
Pm+Pl+Q (TOTAL)	219.01	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	48	51	48	51	48	51	48	51
Circ. Pl (SUS)	-10	-10	12	12	-3	-3	5	5
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	24	24	24	24	24	24	24	24
Long. Pl (SUS)	-22	-22	22	22	-16	-16	16	16
Long. Q (SUS)	-119	119	141	-141	-100	100	122	-122
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	2	2	-2	-2	-2	-2	2	2
Shear Q (SUS)	12	12	12	12	12	12	12	12
Pm (SUS)	48.0	51.0	48.0	51.0	48.0	51.0	48.0	51.0
Pm+Pl (SUS)	38.1	41.1	60.3	63.2	45.1	48.1	53.3	56.2
Pm+Pl+Q (Total)	157.5	123.4	187.8	159.3	138.5	109.6	163.8	140.8

Nozzle Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	51.00	138.00	Passed
Pm+Pl (SUS)	63.23	207.00	Passed
Pm+Pl+Q (TOTAL)	187.78	414.00	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.



Input, Nozzle Desc: K71"

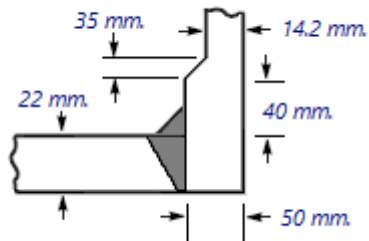
From: 60

Pressure for Reinforcement Calculations	P	23.019	bars
Temperature for Internal Pressure	Temp	150	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm2
Parent Allowable Stress At Ambient	Sva	138.00	N./mm2
Inside Diameter of Cylindrical Shell	D	2300.00	mm.
Shell Finished (Minimum) Thickness	t	22.0000	mm.
Shell Internal Corrosion Allowance	c	1.5000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Cylinder/Cone Centerline	L1	850.0001	mm.
Distance from Bottom/Left Tangent		10550.00	mm.
User Entered Minimum Design Metal Temperature		-33.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-350 LF2 +IT	
Material UNS Number		K03011	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	N./mm2
Allowable Stress At Ambient	Sna	138.00	N./mm2
Diameter Basis (for tr calc only)		Inside	
Layout Angle		46.49	deg
Diameter		1.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	14.2240	mm.
Flange Material		SA-350 LF2 +IT	
Flange Type		N/A	
Hub Height of Integral Nozzle	h	40.0000	mm.
Height of Beveled Transition	L'	35.0000	mm.
Hub Thickness of Integral Nozzle (tn or x+tp)		50.0000	mm.
Style of FVC Nozzle		Long Weld Neck	
Corrosion Allowance	can	1.5000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	211.1370	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	22.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



Note : Checking Nozzle 90 degrees to the Longitudinal axis.

Reinforcement CALCULATION, Description: K71"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	1.000 in.
Actual Thickness Used in Calculation	0.560 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 23.02 \cdot 1151.5 / (138 \cdot 1.0 - 0.6 \cdot 23.02)$
 $= 19.4030 \text{ mm.}$

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]
 $= R \cdot (\exp([P / (S_n \cdot E)] - 1))$ per Appendix 1-2 (a)(1)
 $= 14.2 (\exp([23.02 / (138.0 \cdot 1.0)] - 1))$
 $= 0.2389 \text{ mm.}$

Intermediate Hub Nozzle Calculations:

The Diameter Limit is less than the Hub OD !

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
 $= (\text{Hub Thickness} - \text{Neck Thickness}) / \tan(30)$
 $= 27.147 / 0.5773$
 $= 47.0207 \text{ mm.}$

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	Dl	108.1428 mm.
Parallel to Vessel Wall	Rn+tn+t	54.0714 mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	51.2500 mm.

*Taking a UG-36(c)(3)(a) exemption for nozzle: K71".
 This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.*

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)(b)), Curve: B

Govrn. thk, tg = 22.0, tr = 19.403, c = 1.5 mm., E* = 1.0
 Thickness Ratio = tr * E* / (tg - c) = 0.946, Temp. Reduction = 3 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-5 °C
Min Metal Temp per UCS-66 and UCS-68(c), PWHt credit	-21 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-24 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -29 °C

Weld Size Calculations, Description: K71"

Intermediate Calc. for nozzle/shell Welds Tmin 19.0000 mm.

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm.

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

Note : Checking Nozzle in plane parallel to the vessel axis.

Reinforcement CALCULATION, Description: K71"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	1.000 in.
Actual Thickness Used in Calculation	0.560 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 23.02 \cdot 1151.5 / (138 \cdot 1.0 - 0.6 \cdot 23.02)$
 $= 19.4030$ mm.

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]
 $= R \cdot (\exp([P / (S_n \cdot E)] - 1))$ per Appendix 1-2 (a)(1)
 $= 14.2 \cdot (\exp([23.02 / (138 \cdot 1.0)] - 1))$
 $= 0.2389$ mm.

Intermediate Hub Nozzle Calculations:

The Diameter Limit is less than the Hub OD !

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
 $= (\text{Hub Thickness} - \text{Neck Thickness}) / \tan(30)$
 $= 20.5 / 0.5773$
 $= 35.5071$ mm.

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	Dl	94.8480 mm.
Parallel to Vessel Wall	Rn+tn+t	47.4240 mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	51.2500 mm.

*Taking a UG-36(c)(3)(a) exemption for nozzle: K71".
 This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 1.7389 mm.
Wall Thickness per UG16(b),	tr16b = 3.0000 mm.
Wall Thickness, shell/head, internal pressure	trb1 = 20.9030 mm.
Wall Thickness	tb1 = max(trb1, tr16b) = 20.9030 mm.
Wall Thickness	tb2 = max(trb2, tr16b) = 3.0000 mm.
Wall Thickness per table UG-45	tb3 = 4.9200 mm.

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[4.92, \max(20.903, 3.0)]$



= 4.9200 mm.

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

= max(ta, tb)

= max(1.7389, 4.92)

= 4.9200 mm.

Available Nozzle Neck Thickness = 14.2240 mm. -> OK

Weld Size Calculations, Description: K7_1"

Intermediate Calc. for nozzle/shell Welds Tmin 19.0000 mm.

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm.

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 55.4791 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 298.9172 mm.

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Input, Nozzle Desc: N36"

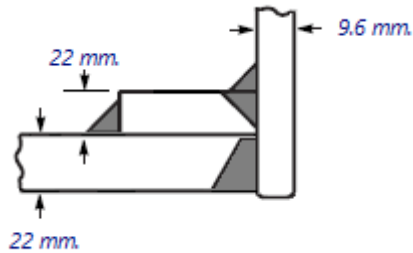
From: 60

Pressure for Reinforcement Calculations	P	23.000	bars
Temperature for Internal Pressure	Temp	150	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm2
Parent Allowable Stress At Ambient	Sva	138.00	N./mm2
Inside Diameter of Cylindrical Shell	D	2300.00	mm.
Shell Finished (Minimum) Thickness	t	22.0000	mm.
Shell Internal Corrosion Allowance	c	1.5000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		10750.00	mm.
User Entered Minimum Design Metal Temperature		-33.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-333 6 +IT	
Material UNS Number		K03006	
Material Specification/Type	Smls. & wld. pipe		
Allowable Stress at Temperature	Sn	118.00	N./mm2
Allowable Stress At Ambient	Sna	118.00	N./mm2
Diameter Basis (for tr calc only)		Inside	
Layout Angle		90.00	deg
Diameter		6.0000	in.
Size and Thickness Basis		Minimum	
Nominal Thickness		80	
Flange Material		SA-350 LF2 +IT	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	1.5000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	430.9720	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	22.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Pad Material		SA-516 70 +IT	
Pad Allowable Stress at Temperature	Sp	138.00	N./mm2
Pad Allowable Stress At Ambient	Spa	138.00	N./mm2
Diameter of Pad along vessel surface	Dp	280.0000	mm.
Thickness of Pad	te	22.0000	mm.
Weld leg size between Pad and Shell	Wp	18.0000	mm.
Groove weld depth between Pad and Nozzle	Wgpn	22.0000	mm.
Reinforcing Pad Width		55.8625	mm.
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: N36"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	5.869 in.
Actual Thickness Used in Calculation	0.378 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 23.0 \cdot 1151.5 / (138 \cdot 1.0 - 0.6 \cdot 23.0)$
 $= 19.3870$ mm.

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 23.0 \cdot 76.04 / (118 \cdot 1.0 - 0.6 \cdot 23.0)$
 $= 1.4997$ mm.

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	304.1452 mm.
Parallel to Vessel Wall, opening length	d	152.0726 mm.
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		42.2530 mm.

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 118.0 / 138.0)$
 $= 0.855$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 118.0 / 138.0)$
 $= 0.855$

Weld Strength Reduction Factor [fr4]:
 $= \min(1, S_p / S_v)$
 $= \min(1, 138.0 / 138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(0.855, 1.0)$
 $= 0.855$

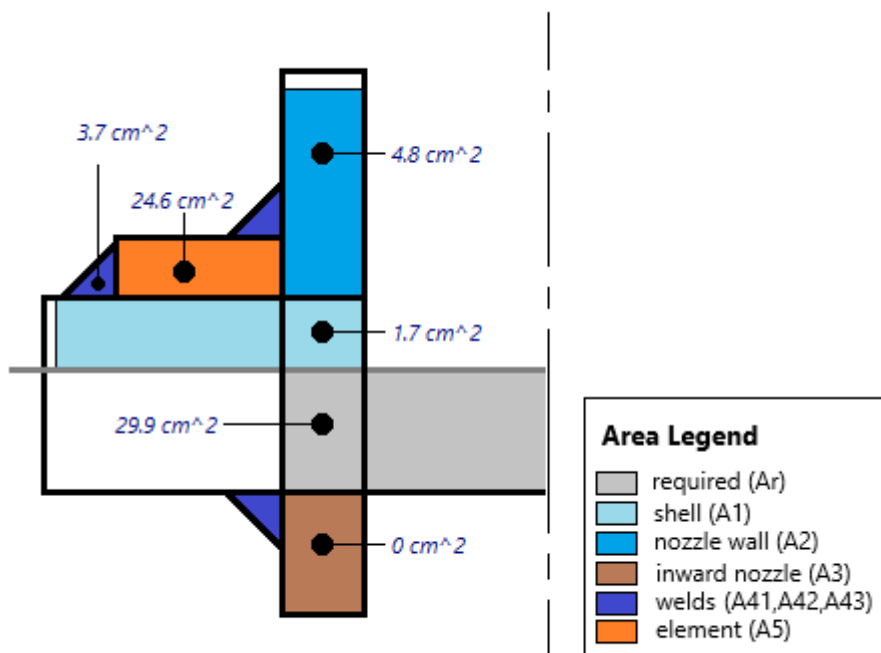


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (cm2)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	29.938	NA	NA
Area in Shell	A1	1.666	NA	NA
Area in Nozzle Wall	A2	4.770	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	3.744	NA	NA
Area in Element	A5	24.580	NA	NA
TOTAL AREA AVAILABLE	Atot	34.760	NA	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Insufficient.
 The area available with the given pad is Sufficient.

SELECTION OF POSSIBLE REINFORCING PADS: Diameter Thickness
 Based on given Pad Thickness: 258.0801 22.0000 mm.
 Based on given Pad Diameter: 280.0000 17.6837 mm.
 Based on the Estimated Diameter Limit: 302.5577 14.7131 mm.

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$
 $= (152.0726 \cdot 19.387 \cdot 1.0 + 2 \cdot 8.1012 \cdot 19.387 \cdot 1.0 \cdot (1 - 0.86))$
 $= 29.938 \text{ cm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 152.073(1.0 \cdot 20.5 - 1.0 \cdot 19.387) - 2 \cdot 8.101$
 $(1.0 \cdot 20.5 - 1.0 \cdot 19.387) \cdot (1 - 0.855)$
 $= 1.666 \text{ cm}^2$

Area Available in Nozzle Wall Projecting Outward [A2]:
 $= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2$
 $= (2 \cdot 42.25) \cdot (8.1 - 1.5) \cdot 0.8551$



= 4.770 cm²

Area Available in Welds [A41 + A42 + A43]:
 = (Wo² - Ar Lost)•Fr3+((Wi-can/0.707)² - Ar Lost)•fr2 + Trapfr4
 = (1.) • 0.86 + (0.0) • 0.86 + 288.8657² • 1.0
 = 3.744 cm²

Area Available in Element [A5]:
 = (min(Dp,DL)-(Nozzle OD))•(min(tp,Tlwp,te)) • fr4
 = (280.0 - 168.275) • 22.0 • 1.0
 = 24.580 cm²

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures ta = 2.9997 mm.
 Wall Thickness per UG16(b), tr16b = 3.0000 mm.
 Wall Thickness, shell/head, internal pressure trb1 = 20.8870 mm.
 Wall Thickness tb1 = max(trb1, tr16b) = 20.8870 mm.
 Wall Thickness tb2 = max(trb2, tr16b) = 3.0000 mm.
 Wall Thickness per table UG-45 tb3 = 7.7200 mm.

Determine Nozzle Thickness candidate [tb]:
 = min[tb3, max(tb1,tb2)]
 = min[7.72, max(20.887, 3.0)]
 = 7.7200 mm.

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 = max(ta, tb)
 = max(3.0, 7.72)
 = 7.7200 mm.

Available Nozzle Neck Thickness = 9.6012 mm. -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	:	38.5,	Allowable	:	118.0 N./mm ²	Passed
Expansion	:	0.0,	Allowable	:	256.5 N./mm ²	Passed
Occasional	:	10.2,	Allowable	:	156.9 N./mm ²	Passed
Shear	:	19.7,	Allowable	:	82.6 N./mm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld (Impact tested) :

Note:
This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification	-46 °C
Calculated Minimum Design Metal Temperature	-104 °C

Nozzle Neck to Pad Weld for the Nozzle (Impact tested) :

Note:
This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification	-46 °C
Calculated Minimum Design Metal Temperature	-104 °C

Nozzle Neck to Pad Weld for Reinf. pad (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Shell to Pad Weld Junction at Pad OD, Curve: B



Govrn. thk, tg = 22.0, tr = 19.387, c = 1.5 mm., E* = 1.0
 Thickness Ratio = $tr \cdot E^*/(tg - c) = 0.946$, Temp. Reduction = 3 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-5 °C
Min Metal Temp per UCS-66 and UCS-68(c),PWHT credit	-21 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-24 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle-Shell/Head Weld (UCS-66(a)(b)), Curve: B

Govrn. thk, tg = 9.601, tr = 1.5, c = 1.5 mm., E* = 1.0
 Thickness Ratio = $tr \cdot E^*/(tg - c) = 0.185$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-29 °C
Min Metal Temp per UCS-66 and UCS-68(c),PWHT credit	-46 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Gov. MDMT of the Nozzle	: -104 °C
Gov. MDMT of the Reinforcement Pad	: -29 °C
Gov. MDMT of the nozzle to shell joint welded assembly	: -29 °C

Flange MDMT including Temperature reduction per UCS-66.1:

MDMT of ASME B16.5/47 flange per Matl. Specification	-46 °C
Flange MDMT with Temp reduction per UCS-66(i)(2)	-85 °C

Where the Stress Reduction Ratio per UCS-66(i)(2) is :
 Design Pressure/Ambient Rating = $23.00/51.00 = 0.451$

Weld Size Calculations, Description: N36"

Intermediate Calc. for nozzle/shell Welds	Tmin	8.1012 mm.
Intermediate Calc. for pad/shell Welds	TminPad	19.0000 mm.

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$5.6708 = 0.7 \cdot t_{min}$	$7.0700 = 0.7 \cdot W_o$ mm.
Pad Weld	$9.5000 = 0.5 \cdot T_{minPad}$	$12.7260 = 0.7 \cdot W_p$ mm.

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A-A1+2 \cdot t_n \cdot f_{r1} \cdot (E1 \cdot t - tr)) \cdot S_v)$
 $= \max(0, (29.9375 - 1.6664 + 2 \cdot 8.1012 \cdot 0.8551 \cdot (1.0 \cdot 20.5 - 19.387)) \cdot 138)$
 $= 39998.88 \text{ Kgf}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2+A5+A4-(W_i-Can/.707)^2 \cdot f_{r2}) \cdot S_v$
 $= (4.7703 + 24.5795 + 3.7437 - 0.0 \cdot 0.86) \cdot 138$
 $= 46567.76 \text{ Kgf}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S_v$
 $= (4.7703 + 0.0 + 0.8551 + (2.8402)) \cdot 138$
 $= 11912.30 \text{ Kgf}$

Weld Load [W3]:
 $= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S$
 $= (4.7703 + 0.0 + 3.7437 + 24.5795 + (2.8402)) \cdot 138$
 $= 50564.32 \text{ Kgf}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$
 $= (3.1416/2.0) \cdot 168.275 \cdot 10.0 \cdot 0.49 \cdot 118$
 $= 15584. \text{ Kgf}$

Shear, Pad Element Weld [Spew]:
 $= (\pi/2) \cdot DP \cdot WP \cdot 0.49 \cdot SEW$
 $= (3.1416/2.0) \cdot 280.0 \cdot 18.0 \cdot 0.49 \cdot 138$
 $= 54587. \text{ Kg}$

Shear, Nozzle Wall [Snw]:
 $= (\pi \cdot (Dlr + Dlo) / 4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn$
 $= (3.1416 \cdot 80.0869) \cdot (9.6012 - 1.5) \cdot 0.7 \cdot 118$
 $= 17168. \text{ Kg}$

Tension, Pad Groove Weld [Tpgw]:
 $= (\pi/2) \cdot Dlo \cdot Wgpn \cdot 0.74 \cdot Seg$
 $= (3.1416/2) \cdot 168.275 \cdot 22.0 \cdot 0.74 \cdot 138$
 $= 60554. \text{ Kg}$

Tension, Shell Groove Weld [Tngw]:
 $= (\pi/2) \cdot Dlo \cdot (T - cas) \cdot 0.74 \cdot Sng$
 $= (3.1416/2.0) \cdot 168.275 \cdot (22.0 - 1.5) \cdot 0.74 \cdot 138$
 $= 56424. \text{ Kg}$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:
 $= (SPEW + SNW) = (54587 + 17168) = 71755 \text{ Kg}$

Failure Path [Path 2-2]:
 $= (Sonw + Tpgw + Tngw + Sinw)$
 $= (15584 + 60554 + 56424 + 0) = 132563 \text{ Kg}$

Failure Path [Path 3-3]:
 $= (Spew + Tngw + Sinw)$
 $= (54587 + 56424 + 0) = 111011 \text{ Kg}$

Summary of Failure Path Calculations:

Path 1-1 = 71754	Kgf, must exceed W = 39998 Kg	or W1 = 46567 Kg
Path 2-2 = 132563	Kgf, must exceed W = 39998 Kg	or W2 = 11912 Kg
Path 3-3 = 111011	Kgf, must exceed W = 39998 Kg	or W3 = 50564 Kg

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 24 bars

Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 3.0820 mm.
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 456.0540 mm.

Input Echo, WRC297 Item 1, Description: N36" :

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	1.5000	mm.
Shell External Corrosion Allowance	0.0000	mm.
Shell Diameter	2300.000	mm.
Shell Thickness	22.0000	mm.
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc	138.00 N./mm ²
Vessel Hot S.I. Allowable	Smh	138.00 N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter Basis for Nozzle	ID	
Nozzle Corrosion Allowance	1.5000	mm.
Nozzle Diameter	149.073	mm.
Nozzle Thickness	9.6012	mm.
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-333 6	



Nozzle Cold S.I. Allowable	SNmc	118.00	N./mm ²
Nozzle Hot S.I. Allowable	SNmh	118.00	N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter of Reinforcing Pad	280.000	mm.
Thickness of Reinforcing Pad	22.0000	mm.

Note:

*External Forces and Moments in WRC 107/537 Convention:
 These loads are assumed to be SUSained loads.*

Design Internal Pressure	Dp	23.00	bars
Radial Load	P	-410.00	Kgf
Circumferential Shear	Vc	615.00	Kgf
Longitudinal Shear	Vl	615.00	Kgf
Circumferential Moment	Mc	280.00	Kg-m.
Longitudinal Moment	Ml	370.00	Kg-m.
Torsional Moment	Mt	460.00	Kg-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(\text{Rmean}(t - ca))}$$

$$= 168.275 + 2 \cdot 1.65 \cdot \sqrt{(1172.75(22.0 - 1.5))}$$

$$= 679.949 \text{ mm.}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	2345.500	mm.
Nozzle Outside Diameter	(d) =	168.275	mm.
Vessel Thickness used	(T) =	42.500	mm.
Nozzle Thickness used	(t) =	8.101	mm.
T / t	=	5.246	
d / t	=	20.772	

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 0.533$$

Nr/P	=	0.099
Mr/P	=	0.069
M0/P	=	0.187
N0/P	=	0.337
MrD/Mc	=	0.101
NrDL/Mc	=	0.118
M0d/Mc	=	0.301
N0DL/Mc	=	0.120
MrD/Ml	=	0.101
NrDL/Ml	=	0.116
M0D/Ml	=	0.284
N0DL/Ml	=	0.211

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	0	0	0	0
Outplane Bending (P)	2	-2	2	-2
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-2	-2	2	2
Outplane Bending (ML)	-20	20	20	-20
Normal Pressure Stress	61	63	61	63



Outplane Stress Summary 41 79 87 43

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE
 (Stresses parallel to longitudinal plane)

	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
--	----------------------	---------------------	-------------------------	------------------------

Inplane Membrane (P)	0	0	0	0
Inplane Bending (P)	0	0	0	0
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-1	-1	1	1
Inplane Bending (ML)	-7	7	7	-7
Inplane Pressure Stress	30	30	30	30

Inplane Stress Summary 23 35 40 24

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE
 (Shear stress normal to longitudinal plane)

	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
--	----------------------	---------------------	-------------------------	------------------------

Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	2	2	2	2

Shear Stress Summary 2 2 1 1

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE
 (Stress Intensities)

	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
--	----------------------	---------------------	-------------------------	------------------------

Two * Max Shear Stress 42 79 87 44

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE
 (Stresses Normal to circumferential plane)

	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
--	-----------------------	----------------------	------------------------	-----------------------

Outplane Membrane (P)	0	0	0	0
Outplane Bending (P)	2	-2	2	-2
Outplane Membrane (Mc)	-1	-1	1	1
Outplane Bending (Mc)	-16	16	16	-16
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	30	30	30	30

Outplane Stress Summary 16 44 51 13

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE
 (Stresses parallel to circumferential plane)

	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
--	-----------------------	----------------------	------------------------	-----------------------

Inplane Membrane (P)	0	0	0	0
Inplane Bending (P)	0	0	0	0
Inplane Membrane (Mc)	-1	-1	1	1
Inplane Bending (Mc)	-5	5	5	-5
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	61	63	61	63

Inplane Stress Summary 55 67 68 58

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE
 (Shear stress normal to circumferential plane)

	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
--	-----------------------	----------------------	------------------------	-----------------------



Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Torsional Shear (Mt)	2	2	2	2

Shear Stress Summary	1	1	2	2
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
---	-----------------------	----------------------	------------------------	-----------------------

Two * Max Shear Stress	55	67	69	58
------------------------	----	----	----	----

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
---	----------------------	---------------------	-------------------------	------------------------

Hoop Membrane (P)	0	0	0	0
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-2	-2	2	2
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	24	27	24	27

Hoop Stress Summary	22	25	27	30
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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
--	----------------------	---------------------	-------------------------	------------------------

Axial Membrane (P)	0	0	0	0
Axial Bending (P)	7	-7	7	-7
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-23	-23	23	23
Axial Bending (ML)	-84	84	84	-84
Axial Pressure Stress	12	12	12	12

Axial Stress Summary	-87	67	128	-55
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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress)	Au Outside	Al Inside	Bu Outside	Bl Inside
--------------------------------------	---------------	--------------	---------------	--------------

Shear due to (Vc)	2	2	-2	-2
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	12	12	12	12

Shear Stress Summary	15	15	9	9
----------------------	----	----	---	---

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Outside	Al Inside	Bu Outside	Bl Inside
--	---------------	--------------	---------------	--------------

Two * Max Shear Stress	114	72	129	88
------------------------	-----	----	-----	----

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
--	-----------------------	----------------------	------------------------	-----------------------

Hoop Membrane (P)	0	0	0	0
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-1	-1	1	1
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0



Hoop Pressure Stress	24	27	24	27
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Hoop Stress Summary	24	27	26	29
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Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
---	-----------------------	----------------------	------------------------	-----------------------

Axial Membrane (P)	0	0	0	0
Axial Bending (P)	7	-7	7	-7
Axial Membrane (Mc)	-17	-17	17	17
Axial Bending (Mc)	-63	63	63	-63
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	12	12	12	12

Axial Stress Summary	-60	51	101	-39
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Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu Outside	Cl Inside	Du Outside	Dl Inside
---	---------------	--------------	---------------	--------------

Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-2	-2	2	2
Shear due to Torsion	12	12	12	12

Shear Stress Summary	9	9	15	15
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Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities	Cu Outside	Cl Inside	Du Outside	Dl Inside
--	---------------	--------------	---------------	--------------

Two * Max Shear Stress	86	54	104	75
------------------------	----	----	-----	----

Stress Computations at the Edge of the Reinforcing Pad:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - ca)}$$

$$= 168.275 + 2 \cdot 1.65 \cdot \sqrt{1161.75(22.0 - 1.5)}$$

$$= 677.544 \text{ mm.}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	2323.500 mm.
Nozzle Outside Diameter	(d) =	280.000 mm.
Vessel Thickness used	(T) =	20.500 mm.
Nozzle Thickness used	(t) =	63.964 mm.
T / t	=	0.320
d / t	=	10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 1.283$$

Note: Re-pad thickness is added to nozzle thickness to simulate the Re-pad.

Nr/P	=	0.092
Mr/P	=	0.194
M0/P	=	0.056
N0/P	=	0.110
MrD/Mc	=	0.305
NrDL/Mc	=	0.084
M0d/Mc	=	0.094
N0DL/Mc	=	0.055
MrD/Ml	=	0.163
NrDL/Ml	=	0.091
M0D/Ml	=	0.051
N0DL/Ml	=	0.120

Vessel Stresses (N./mm²):



LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	1	1	1	1
Outplane Bending (P)	3	-3	3	-3
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-3	-3	3	3
Outplane Bending (ML)	-9	9	9	-9
Normal Pressure Stress	128	130	128	130
Outplane Stress Summary	119	133	145	122

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	0	0	0	0
Inplane Bending (P)	11	-11	11	-11
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-2	-2	2	2
Inplane Bending (ML)	-30	30	30	-30
Inplane Pressure Stress	64	64	64	64
Inplane Stress Summary	43	81	108	26

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	1	1	1	1
Shear Stress Summary	2	2	1	1

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	119	134	145	122

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Membrane (P)	1	1	1	1
Outplane Bending (P)	3	-3	3	-3
Outplane Membrane (Mc)	-1	-1	1	1
Outplane Bending (Mc)	-13	13	13	-13
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	64	64	64	64
Outplane Stress Summary	53	73	82	50

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Inplane Membrane (P)	0	0	0	0
Inplane Bending (P)	11	-11	11	-11
Inplane Membrane (Mc)	-1	-1	1	1



Inplane Bending (Mc)	-42	42	42	-42
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	128	130	128	130

Inplane Stress Summary	95	160	184	79
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Shear stress normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Torsional Shear (Mt)	1	1	1	1

Shear Stress Summary	1	1	2	2
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stress Intensities)	Left Outside	Left Inside	Right Outside	Right Inside
Two * Max Shear Stress	95	160	184	79

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	61	63	61	63	61	63	61	63
Circ. Pl (SUS)	-2	-2	2	2	-1	-1	1	1
Circ. Q (SUS)	-18	18	22	-22	-5	5	5	-5
Long. Pm (SUS)	30	30	30	30	30	30	30	30
Long. Pl (SUS)	-1	-1	1	1	-1	-1	1	1
Long. Q (SUS)	-7	7	7	-7	-14	14	18	-18
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	2	2	2	2	2	2	2	2
Pm (SUS)	61.0	63.0	61.0	63.0	61.0	63.0	61.0	63.0
Pm+Pl (SUS)	59.0	61.0	63.0	65.0	60.0	62.0	62.0	64.0
Pm+Pl+Q (Total)	41.2	79.1	85.1	43.2	55.1	67.2	67.2	59.1

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	63.00	138.00	Passed
Pm+Pl (SUS)	65.00	207.00	Passed
Pm+Pl+Q (TOTAL)	85.08	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Reinforcing Pad Edge (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl



Circ. Pm (SUS)	128	130	128	130	128	130	128	130
Circ. Pl (SUS)	-2	-2	4	4	-1	-1	1	1
Circ. Q (SUS)	-6	6	12	-12	-31	31	53	-53
Long. Pm (SUS)	64	64	64	64	64	64	64	64
Long. Pl (SUS)	-2	-2	2	2	0	0	2	2
Long. Q (SUS)	-19	19	41	-41	-10	10	16	-16
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	1	1	1	1	1	1	1	1
Pm (SUS)	128.0	130.0	128.0	130.0	128.0	130.0	128.0	130.0
Pm+Pl (SUS)	126.0	128.0	132.0	134.0	127.0	129.0	129.0	131.0
Pm+Pl+Q (Total)	120.0	134.0	144.0	122.0	96.0	160.0	182.0	78.0

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	130.00	138.00	Passed
Pm+Pl (SUS)	134.00	207.00	Passed
Pm+Pl+Q (TOTAL)	182.01	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	24	27	24	27	24	27	24	27
Circ. Pl (SUS)	-2	-2	2	2	-1	-1	1	1
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	12	12	12	12	12	12	12	12
Long. Pl (SUS)	-23	-23	23	23	-17	-17	17	17
Long. Q (SUS)	-77	77	91	-91	-56	56	70	-70
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	2	2	-2	-2	-2	-2	2	2
Shear Q (SUS)	12	12	12	12	12	12	12	12
Pm (SUS)	24.0	27.0	24.0	27.0	24.0	27.0	24.0	27.0
Pm+Pl (SUS)	33.2	36.2	35.4	35.6	28.3	31.3	29.8	30.6
Pm+Pl+Q (Total)	113.5	70.3	127.0	87.3	86.3	54.5	101.6	74.5

Nozzle Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	27.00	118.00	Passed
Pm+Pl (SUS)	36.22	177.00	Passed
Pm+Pl+Q (TOTAL)	126.99	354.00	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.



Input, Nozzle Desc: N88"

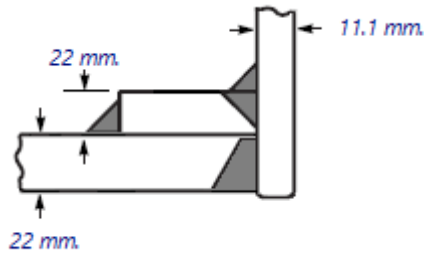
From: 60

Pressure for Reinforcement Calculations	P	23.137	bars
Temperature for Internal Pressure	Temp	150	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm2
Parent Allowable Stress At Ambient	Sva	138.00	N./mm2
Inside Diameter of Cylindrical Shell	D	2300.00	mm.
Shell Finished (Minimum) Thickness	t	22.0000	mm.
Shell Internal Corrosion Allowance	c	1.5000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		10420.00	mm.
User Entered Minimum Design Metal Temperature		-33.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-333 6 +IT	
Material UNS Number		K03006	
Material Specification/Type	Smls. & wld. pipe		
Allowable Stress at Temperature	Sn	118.00	N./mm2
Allowable Stress At Ambient	Sna	118.00	N./mm2
Diameter Basis (for tr calc only)		Inside	
Layout Angle		270.00	deg
Diameter		8.0000	in.
Size and Thickness Basis		Minimum	
Nominal Thickness		80	
Flange Material		SA-350 LF2 +IT	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	1.5000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	418.2720	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	22.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Pad Material		SA-516 70 +IT	
Pad Allowable Stress at Temperature	Sp	138.00	N./mm2
Pad Allowable Stress At Ambient	Spa	138.00	N./mm2
Diameter of Pad along vessel surface	Dp	360.0000	mm.
Thickness of Pad	te	22.0000	mm.
Weld leg size between Pad and Shell	Wp	18.0000	mm.
Groove weld depth between Pad and Nozzle	Wgpn	22.0000	mm.
Reinforcing Pad Width		70.4625	mm.
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: N88"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	7.750 in.
Actual Thickness Used in Calculation	0.438 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 23.14 \cdot 1151.5 / (138 \cdot 1.0 - 0.6 \cdot 23.14)$
 $= 19.5037 \text{ mm.}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 23.14 \cdot 99.92 / (118 \cdot 1.0 - 0.6 \cdot 23.14)$
 $= 1.9827 \text{ mm.}$

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	399.7000 mm.
Parallel to Vessel Wall, opening length	d	199.8500 mm.
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		46.0312 mm.

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 118.0 / 138.0)$
 $= 0.855$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 118.0 / 138.0)$
 $= 0.855$

Weld Strength Reduction Factor [fr4]:
 $= \min(1, S_p / S_v)$
 $= \min(1, 138.0 / 138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(0.855, 1.0)$
 $= 0.855$

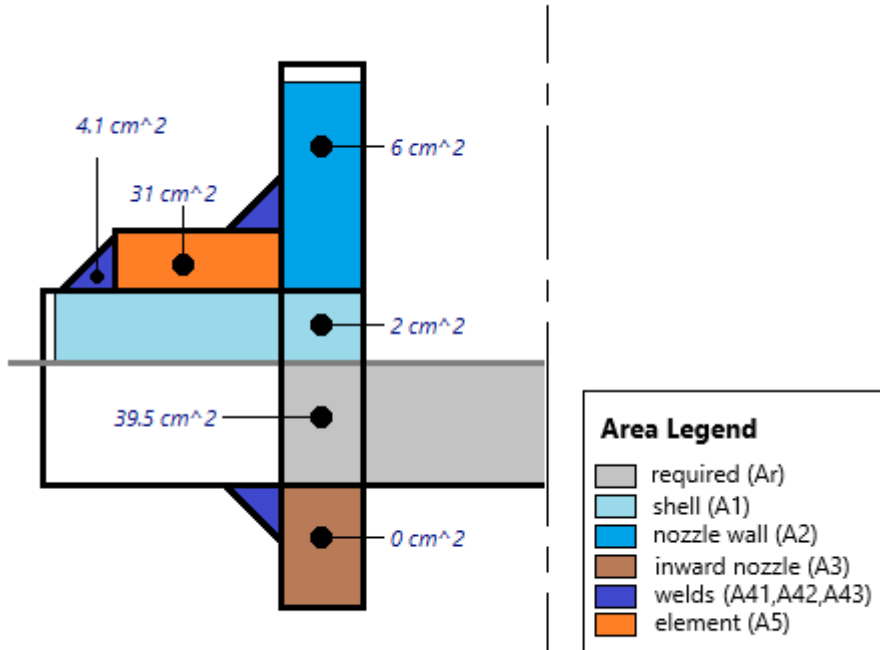


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (cm2)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	39.522	NA	NA
Area in Shell	A1	1.963	NA	NA
Area in Nozzle Wall	A2	6.006	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	4.095	NA	NA
Area in Element	A5	31.003	NA	NA
TOTAL AREA AVAILABLE	Atot	43.068	NA	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Insufficient.
 The area available with the given pad is Sufficient.

SELECTION OF POSSIBLE REINFORCING PADS: Diameter Thickness
 Based on given Pad Thickness: 343.8787 22.0000 mm.
 Based on given Pad Diameter: 360.0000 19.4833 mm.
 Based on the Estimated Diameter Limit: 398.1125 15.3358 mm.

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$
 $= (199.85 \cdot 19.5037 \cdot 1.0 + 2 \cdot 9.6125 \cdot 19.5037 \cdot 1.0 \cdot (1 - 0.86))$
 $= 39.522 \text{ cm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 199.85(1.0 \cdot 20.5 - 1.0 \cdot 19.504) - 2 \cdot 9.613$
 $(1.0 \cdot 20.5 - 1.0 \cdot 19.5037) \cdot (1 - 0.855)$
 $= 1.963 \text{ cm}^2$

Area Available in Nozzle Wall Projecting Outward [A2]:
 $= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2$
 $= (2 \cdot 46.03) \cdot (9.61 - 1.98) \cdot 0.8551$



= 6.006 cm²

Area Available in Welds [A41 + A42 + A43]:
 = $W_o^2 \cdot fr3 + (W_i - can/0.707)^2 \cdot fr2 + W_p^2 \cdot fr4$
 = $10.0^2 \cdot 0.86 + (0.0)^2 \cdot 0.86 + 18.0^2 \cdot 1.0$
 = 4.095 cm²

Area Available in Element [A5]:
 = $(\min(D_p, DL) - (\text{Nozzle OD})) \cdot (\min(t_p, Tlwp, te)) \cdot fr4$
 = $(360.0 - 219.075) \cdot 22.0 \cdot 1.0$
 = 31.003 cm²

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures $ta = 3.4827$ mm.
 Wall Thickness per UG16(b), $tr16b = 3.0000$ mm.
 Wall Thickness, shell/head, internal pressure $trb1 = 21.0037$ mm.
 Wall Thickness $tb1 = \max(trb1, tr16b) = 21.0037$ mm.
 Wall Thickness $tb2 = \max(trb2, tr16b) = 3.0000$ mm.
 Wall Thickness per table UG-45 $tb3 = 8.6600$ mm.

Determine Nozzle Thickness candidate [tb]:
 = $\min[tb3, \max(tb1, tb2)]$
 = $\min[8.66, \max(21.0037, 3.0)]$
 = 8.6600 mm.

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 = $\max(ta, tb)$
 = $\max(3.4827, 8.66)$
 = 8.6600 mm.

Available Nozzle Neck Thickness = 11.1125 mm. -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	:	35.9,	Allowable	:	118.0 N./mm ²	Passed
Expansion	:	0.0,	Allowable	:	259.1 N./mm ²	Passed
Occasional	:	11.5,	Allowable	:	156.9 N./mm ²	Passed
Shear	:	17.1,	Allowable	:	82.6 N./mm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld (Impact tested) :

Note:
This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification	-46 °C
Calculated Minimum Design Metal Temperature	-104 °C

Nozzle Neck to Pad Weld for the Nozzle (Impact tested) :

Note:
This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification	-46 °C
Calculated Minimum Design Metal Temperature	-104 °C

Nozzle Neck to Pad Weld for Reinf. pad (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Shell to Pad Weld Junction at Pad OD, Curve: B



Govrn. thk, tg = 22.0, tr = 19.504, c = 1.5 mm., E* = 1.0
 Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.951$, Temp. Reduction = 3 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-5 °C
Min Metal Temp per UCS-66 and UCS-68(c),PWHT credit	-21 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-24 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle-Shell/Head Weld (UCS-66(a)(b)), Curve: B

Govrn. thk, tg = 11.113, tr = 1.983, c = 1.5 mm., E* = 1.0
 Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.206$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-26 °C
Min Metal Temp per UCS-66 and UCS-68(c),PWHT credit	-42 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Gov. MDMT of the Nozzle	: -104 °C
Gov. MDMT of the Reinforcement Pad	: -29 °C
Gov. MDMT of the nozzle to shell joint welded assembly	: -29 °C

Flange MDMT including Temperature reduction per UCS-66.1:

MDMT of ASME B16.5/47 flange per Matl. Specification	-46 °C
Flange MDMT with Temp reduction per UCS-66(i)(2)	-85 °C

Where the Stress Reduction Ratio per UCS-66(i)(2) is :
 Design Pressure/Ambient Rating = $23.14/51.00 = 0.454$

Weld Size Calculations, Description: N88"

Intermediate Calc. for nozzle/shell Welds	Tmin	9.6125 mm.
Intermediate Calc. for pad/shell Welds	TminPad	19.0000 mm.

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$6.0000 = \text{Min per Code}$	$7.0700 = 0.7 \cdot W_o \text{ mm.}$
Pad Weld	$9.5000 = 0.5 \cdot T_{minPad}$	$12.7260 = 0.7 \cdot W_p \text{ mm.}$

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A-A1+2 \cdot t_n \cdot f_{r1} \cdot (E1 \cdot t - tr)) \cdot S_v)$
 $= \max(0, (39.5215 - 1.9633 + 2 \cdot 9.6125 \cdot 0.8551 \cdot (1.0 \cdot 20.5 - 19.5037)) \cdot 138)$
 $= 53080.71 \text{ Kgf}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2+A5+A4-(W_i-Can/.707)^2 \cdot f_{r2}) \cdot S_v$
 $= (6.0063 + 31.0035 + 4.0951 - 0.0 \cdot 0.86) \cdot 138$
 $= 57840.97 \text{ Kgf}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S_v$
 $= (6.0063 + 0.0 + 0.8551 + (3.37)) \cdot 138$
 $= 14397.13 \text{ Kgf}$

Weld Load [W3]:
 $= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S$
 $= (6.0063 + 0.0 + 4.0951 + 31.0035 + (3.37)) \cdot 138$
 $= 62583.10 \text{ Kgf}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$
 $= (3.1416/2.0) \cdot 219.075 \cdot 10.0 \cdot 0.49 \cdot 118$
 $= 20289. \text{ Kgf}$



Shear, Pad Element Weld [Spew]:
 $= (\pi/2) \cdot DP \cdot WP \cdot 0.49 \cdot SEW$
 $= (3.1416/2.0) \cdot 360.0 \cdot 18.0 \cdot 0.49 \cdot 138$
 $= 70183. \text{ Kg}$

Shear, Nozzle Wall [Snw]:
 $= (\pi \cdot (Dlr + Dlo) / 4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn$
 $= (3.1416 \cdot 104.7313) \cdot (11.1125 - 1.5) \cdot 0.7 \cdot 118$
 $= 26639. \text{ Kg}$

Tension, Pad Groove Weld [Tpgw]:
 $= (\pi/2) \cdot Dlo \cdot Wgpn \cdot 0.74 \cdot Seg$
 $= (3.1416/2) \cdot 219.075 \cdot 22.0 \cdot 0.74 \cdot 138$
 $= 78835. \text{ Kg}$

Tension, Shell Groove Weld [Tngw]:
 $= (\pi/2) \cdot Dlo \cdot (T - cas) \cdot 0.74 \cdot Sng$
 $= (3.1416/2.0) \cdot 219.075 \cdot (22.0 - 1.5) \cdot 0.74 \cdot 138$
 $= 73458. \text{ Kg}$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:
 $= (SPEW + SNW) = (70183 + 26639) = 96822 \text{ Kg}$

Failure Path [Path 2-2]:
 $= (Sonw + Tpgw + Tngw + Sinw)$
 $= (20289 + 78835 + 73458 + 0) = 172582 \text{ Kg}$

Failure Path [Path 3-3]:
 $= (Spew + Tngw + Sinw)$
 $= (70183 + 73458 + 0) = 143642 \text{ Kg}$

Summary of Failure Path Calculations:

Path 1-1 = 96822 Kg, must exceed W = 53080 Kg or W1 = 57840 Kg
 Path 2-2 = 172582 Kg, must exceed W = 53080 Kg or W2 = 14397 Kg
 Path 3-3 = 143641 Kg, must exceed W = 53080 Kg or W3 = 62583 Kg

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 24 bars

The Drop for this Nozzle is : 5.2286 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 445.5005 mm.

Input Echo, WRC297 Item 1, Description: N88" :

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	1.5000	mm.
Shell External Corrosion Allowance	0.0000	mm.
Shell Diameter	2300.000	mm.
Shell Thickness	22.0000	mm.
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc	138.00 N./mm ²
Vessel Hot S.I. Allowable	Smh	138.00 N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter Basis for Nozzle	ID	
Nozzle Corrosion Allowance	1.5000	mm.
Nozzle Diameter	196.850	mm.
Nozzle Thickness	11.1125	mm.
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-333 6	
Nozzle Cold S.I. Allowable	SNmc	118.00 N./mm ²
Nozzle Hot S.I. Allowable	SNmh	118.00 N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter of Reinforcing Pad 360.000 mm.
 Thickness of Reinforcing Pad 22.0000 mm.

Note:

*External Forces and Moments in WRC 107/537 Convention:
 These loads are assumed to be SUSTained loads.*

Design Internal Pressure	Dp	23.14	bars
Radial Load	P	-555.00	Kgf
Circumferential Shear	Vc	820.00	Kgf
Longitudinal Shear	Vl	820.00	Kgf
Circumferential Moment	Mc	490.00	Kg-m.
Longitudinal Moment	Ml	655.00	Kg-m.
Torsional Moment	Mt	820.00	Kg-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(\text{Rmean}(t - ca))}$
 $= 219.075 + 2 \cdot 1.65 \cdot \sqrt{(1172.75(22.0 - 1.5))}$
 $= 730.749 \text{ mm.}$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	2345.500 mm.
Nozzle Outside Diameter	(d) =	219.075 mm.
Vessel Thickness used	(T) =	42.500 mm.
Nozzle Thickness used	(t) =	9.613 mm.
T / t	=	4.421
d / t	=	22.791

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 0.694$$

Nr/P	=	0.092
Mr/P	=	0.079
M0/P	=	0.138
N0/P	=	0.299
MrD/Mc	=	0.124
NrDL/Mc	=	0.122
M0d/Mc	=	0.268
N0DL/Mc	=	0.124
MrD/Ml	=	0.116
NrDL/Ml	=	0.115
M0D/Ml	=	0.238
N0DL/Ml	=	0.289

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	0	0	0	0
Outplane Bending (P)	2	-2	2	-2
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-4	-4	4	4
Outplane Bending (ML)	-23	23	23	-23
Normal Pressure Stress	61	63	61	63
Outplane Stress Summary	37	80	92	43



Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	0	0	0	0
Inplane Bending (P)	1	-1	1	-1
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-1	-1	1	1
Inplane Bending (ML)	-11	11	11	-11
Inplane Pressure Stress	30	30	30	30
Inplane Stress Summary	19	39	45	20

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	2	2	2	2
Shear Stress Summary	3	3	1	1

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	37	80	92	43

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Membrane (P)	0	0	0	0
Outplane Bending (P)	2	-2	2	-2
Outplane Membrane (Mc)	-1	-1	1	1
Outplane Bending (Mc)	-19	19	19	-19
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	30	30	30	30
Outplane Stress Summary	13	47	55	11

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Inplane Membrane (P)	0	0	0	0
Inplane Bending (P)	1	-1	1	-1
Inplane Membrane (Mc)	-1	-1	1	1
Inplane Bending (Mc)	-9	9	9	-9
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	61	63	61	63
Inplane Stress Summary	52	70	73	55

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0



Torsional Shear (Mt)	2	2	2	2
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Shear Stress Summary	1	1	3	3
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Two * Max Shear Stress	52	70	74	55

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Hoop Membrane (P)	0	0	0	0
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-4	-4	4	4
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	27	30	27	30
Hoop Stress Summary	23	26	33	35

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Axial Membrane (P)	0	0	0	0
Axial Bending (P)	11	-11	11	-11
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-20	-20	20	20
Axial Bending (ML)	-112	112	112	-112
Axial Pressure Stress	13	13	13	13
Axial Stress Summary	-106	94	159	-89

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress)	Au Outside	Al Inside	Bu Outside	Bl Inside
Shear due to (Vc)	2	2	-2	-2
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	11	11	11	11
Shear Stress Summary	13	13	8	8

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Outside	Al Inside	Bu Outside	Bl Inside
Two * Max Shear Stress	132	97	159	126

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Hoop Membrane (P)	0	0	0	0
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-1	-1	1	1
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	27	30	27	30



Hoop Stress Summary 26 29 29 32

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses Normal to pipe cross-section)	Left Outside	Left Inside	Right Outside	Right Inside
Axial Membrane (P)	0	0	0	0
Axial Bending (P)	11	-11	11	-11
Axial Membrane (Mc)	-15	-15	15	15
Axial Bending (Mc)	-89	89	89	-89
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	13	13	13	13
Axial Stress Summary	-78	77	131	-71

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Shear stress)	Outside	Inside	Outside	Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-2	-2	2	2
Shear due to Torsion	11	11	11	11
Shear Stress Summary	8	8	13	13

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stress Intensities)	Outside	Inside	Outside	Inside
Two * Max Shear Stress	106	78	132	107

Stress Computations at the Edge of the Reinforcing Pad:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(\text{Rmean}(t - ca))}$
 $= 219.075 + 2 \cdot 1.65 \cdot \sqrt{1161.75(22.0 - 1.5)}$
 $= 728.344 \text{ mm.}$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter (D) = 2323.500 mm.
 Nozzle Outside Diameter (d) = 360.000 mm.
 Vessel Thickness used (T) = 20.500 mm.
 Nozzle Thickness used (t) = 80.075 mm.
 T / t = 0.256
 d / t = 10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 1.650$$

Note: Re-pad thickness is added to nozzle thickness to simulate the Re-pad.

Nr/P = 0.078
 Mr/P = 0.166
 M0/P = 0.048
 N0/P = 0.092
 MrD/Mc = 0.296
 NrDL/Mc = 0.088
 M0d/Mc = 0.091
 N0DL/Mc = 0.068
 MrD/Ml = 0.131
 NrDL/Ml = 0.090
 M0D/Ml = 0.041
 N0DL/Ml = 0.113

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses Normal to	Top	Top	Bottom	Bottom



longitudinal plane)	Outside	Inside	Outside	Inside
Outplane Membrane (P)	1	1	1	1
Outplane Bending (P)	3	-3	3	-3
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-4	-4	4	4
Outplane Bending (ML)	-10	10	10	-10
Normal Pressure Stress	128	131	128	131
Outplane Stress Summary	118	134	148	122

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses parallel to longitudinal plane)	Top	Top	Bottom	Bottom
	Outside	Inside	Outside	Inside
Inplane Membrane (P)	1	1	1	1
Inplane Bending (P)	12	-12	12	-12
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-3	-3	3	3
Inplane Bending (ML)	-33	33	33	-33
Inplane Pressure Stress	64	64	64	64
Inplane Stress Summary	41	82	115	22

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Shear stress normal to longitudinal plane)	Top	Top	Bottom	Bottom
	Outside	Inside	Outside	Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	1	1	1	1
Shear Stress Summary	2	2	1	1

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stress Intensities)	Top	Top	Bottom	Bottom
	Outside	Inside	Outside	Inside
Two * Max Shear Stress	118	134	149	122

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses Normal to circumferential plane)	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Outplane Membrane (P)	1	1	1	1
Outplane Bending (P)	3	-3	3	-3
Outplane Membrane (Mc)	-2	-2	2	2
Outplane Bending (Mc)	-17	17	17	-17
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	64	64	64	64
Outplane Stress Summary	49	77	88	46

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses parallel to circumferential plane)	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Inplane Membrane (P)	1	1	1	1
Inplane Bending (P)	12	-12	12	-12
Inplane Membrane (Mc)	-2	-2	2	2
Inplane Bending (Mc)	-56	56	56	-56
Inplane Membrane (ML)	0	0	0	0



Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	128	131	128	131
Inplane Stress Summary	83	172	201	65

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Torsional Shear (Mt)	1	1	1	1
Shear Stress Summary	1	1	2	2

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Two * Max Shear Stress	83	172	201	66

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	61	63	61	63	61	63	61	63
Circ. Pl (SUS)	-4	-4	4	4	-1	-1	1	1
Circ. Q (SUS)	-21	21	25	-25	-8	8	10	-10
Long. Pm (SUS)	30	30	30	30	30	30	30	30
Long. Pl (SUS)	-1	-1	1	1	-1	-1	1	1
Long. Q (SUS)	-10	10	12	-12	-17	17	21	-21
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	2	2	2	2	2	2	2	2
Pm (SUS)	61.0	63.0	61.0	63.0	61.0	63.0	61.0	63.0
Pm+Pl (SUS)	57.0	59.0	65.0	67.0	60.0	62.0	62.0	64.0
Pm+Pl+Q (Total)	36.2	80.1	90.1	42.2	52.1	70.2	72.2	54.1

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	63.00	138.00	Passed
Pm+Pl (SUS)	67.00	207.00	Passed
Pm+Pl+Q (TOTAL)	90.08	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Reinforcing Pad Edge (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	128	131	128	131	128	131	128	131



Circ. P1 (SUS)	-3	-3	5	5	-1	-1	3	3
Circ. Q (SUS)	-7	7	13	-13	-44	44	68	-68
Long. Pm (SUS)	64	64	64	64	64	64	64	64
Long. P1 (SUS)	-2	-2	4	4	-1	-1	3	3
Long. Q (SUS)	-21	21	45	-45	-14	14	20	-20
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear P1 (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	1	1	1	1	1	1	1	1
Pm (SUS)	128.0	131.0	128.0	131.0	128.0	131.0	128.0	131.0
Pm+P1 (SUS)	125.0	128.0	133.0	136.0	127.0	130.0	131.0	134.0
Pm+P1+Q (Total)	118.0	135.0	146.0	123.0	83.0	174.0	199.0	66.1

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	131.00	138.00	Passed
Pm+P1 (SUS)	136.00	207.00	Passed
Pm+P1+Q (TOTAL)	199.01	413.99	Passed

Because only sustained loads were specified, the Pm+P1+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	27	30	27	30	27	30	27	30
Circ. P1 (SUS)	-4	-4	4	4	-1	-1	1	1
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	13	13	13	13	13	13	13	13
Long. P1 (SUS)	-20	-20	20	20	-15	-15	15	15
Long. Q (SUS)	-101	101	123	-123	-78	78	100	-100
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear P1 (SUS)	2	2	-2	-2	-2	-2	2	2
Shear Q (SUS)	11	11	11	11	11	11	11	11
Pm (SUS)	27.0	30.0	27.0	30.0	27.0	30.0	27.0	30.0
Pm+P1 (SUS)	30.3	33.2	34.2	35.6	28.3	31.3	30.0	32.0
Pm+P1+Q (Total)	133.6	96.4	156.6	125.3	107.5	77.7	129.7	106.2

Nozzle Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	30.00	118.00	Passed
Pm+P1 (SUS)	35.56	177.00	Passed
Pm+P1+Q (TOTAL)	156.64	354.00	Passed

Because only sustained loads were specified, the Pm+P1+Q allowable was 3 * Smh.



Input, Nozzle Desc: MH224"

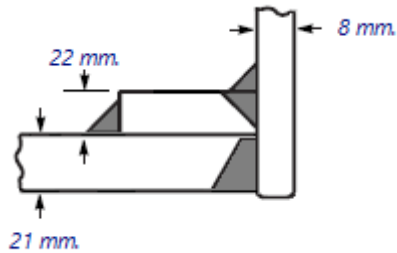
From: 70

Pressure for Reinforcement Calculations	P	23.069	bars
Temperature for Internal Pressure	Temp	150	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm2
Parent Allowable Stress At Ambient	Sva	138.00	N./mm2
Inside Diameter of Elliptical Head	D	2300.00	mm.
Aspect Ratio of Elliptical Head	Ar	2.00	
Head Finished (Minimum) Thickness	t	21.0000	mm.
Head Internal Corrosion Allowance	c	1.5000	mm.
Head External Corrosion Allowance	co	0.0000	mm.
Distance from Head Centerline	L1	0.0000	mm.
User Entered Minimum Design Metal Temperature		-33.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-516 70 +IT	
Material UNS Number		K02700	
Material Specification/Type		Plate	
Allowable Stress at Temperature	Sn	138.00	N./mm2
Allowable Stress At Ambient	Sna	138.00	N./mm2
Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		24.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	8.0000	mm.
Flange Material		SA-350 LF2 +IT	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	1.5000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	158.3760	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	22.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Pad Material		SA-516 70 +IT	
Pad Allowable Stress at Temperature	Sp	138.00	N./mm2
Pad Allowable Stress At Ambient	Spa	138.00	N./mm2
Diameter of Pad along vessel surface	Dp	1060.0000	mm.
Thickness of Pad	te	22.0000	mm.
Weld leg size between Pad and Shell	Wp	18.0000	mm.
Groove weld depth between Pad and Nozzle	Wgpn	22.0000	mm.
Reinforcing Pad Width		217.2000	mm.
This is a Manway, Coupling or Access Opening.			
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: MH224"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	24.000 in.
Actual Thickness Used in Calculation	0.315 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Elliptical Head, tr [Int. Press]
 $= P \cdot D \cdot K1 / (2 \cdot Sv \cdot E - 0.2 \cdot P)$ per Appendix 1-4(c)
 $= 23.07 \cdot 2303.0 \cdot 0.8989 / (2 \cdot 138.0 \cdot 1.0 - 0.2 \cdot 23.07)$
 $= 17.3327$ mm.

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (Sn \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 23.07 \cdot 306.3 / (138 \cdot 1.0 - 0.6 \cdot 23.07)$
 $= 5.1724$ mm.

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	1225.2000	mm.
Parallel to Vessel Wall, opening length	d	612.6000	mm.
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		38.2500	mm.

Weld Strength Reduction Factor [fr1]:
 $= \min(1, Sn/Sv)$
 $= \min(1, 138.0/138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, Sn/Sv)$
 $= \min(1, 138.0/138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr4]:
 $= \min(1, Sp/Sv)$
 $= \min(1, 138.0/138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(1.0, 1.0)$
 $= 1.000$

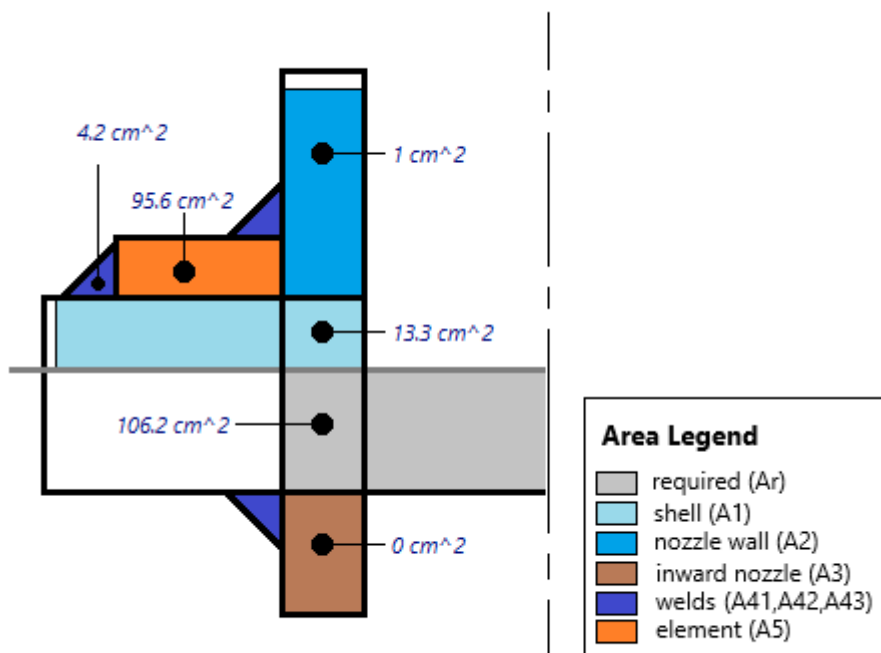


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (cm2)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	106.180	NA	NA
Area in Shell	A1	13.277	NA	NA
Area in Nozzle Wall	A2	1.016	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	4.240	NA	NA
Area in Element	A5	95.568	NA	NA
TOTAL AREA AVAILABLE	Atot	114.100	NA	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Insufficient.
 The area available with the given pad is Sufficient.

SELECTION OF POSSIBLE REINFORCING PADS: Diameter Thickness
 Based on given Pad Thickness: 1023.9994 22.0000 mm.
 Based on given Pad Diameter: 1060.0000 20.1768 mm.
 Based on the Estimated Diameter Limit: 1223.6125 14.6565 mm.

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1))$ UG-37(c)
 $= (612.6 \cdot 17.3327 \cdot 1.0 + 2 \cdot 6.5 \cdot 17.3327 \cdot 1.0 \cdot (1 - 1.0))$
 $= 106.180 \text{ cm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 612.6(1.0 \cdot 19.5 - 1.0 \cdot 17.333) - 2 \cdot 6.5$
 $(1.0 \cdot 19.5 - 1.0 \cdot 17.3327) \cdot (1 - 1.0)$
 $= 13.277 \text{ cm}^2$

Area Available in Nozzle Wall Projecting Outward [A2]:
 $= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2$
 $= (2 \cdot 38.25) \cdot (6.5 - 5.17) \cdot 1.0$

= 1.016 cm²

Area Available in Welds [A41 + A42 + A43]:
 = $W_o^2 \cdot fr_3 + (W_i - can/0.707)^2 \cdot fr_2 + W_p^2 \cdot fr_4$
 = $10.0^2 \cdot 1.0 + (0.0)^2 \cdot 1.0 + 18.0^2 \cdot 1.0$
 = 4.240 cm²

Area Available in Element [A5]:
 = $(\min(D_p, DL) - (\text{Nozzle OD})) \cdot (\min(t_p, Tlwp, te)) \cdot fr_4$
 = $(1060.0 - 625.6) \cdot 22.0 \cdot 1.0$
 = 95.568 cm²

UG-45 Minimum Nozzle Neck Thickness Requirement:

Wall Thickness for Internal/External pressures $t_a = 6.6724$ mm.

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 = t_a
 = 6.6724 mm.

Available Nozzle Neck Thickness = 8.0000 mm. -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Nozzle Neck to Pad Weld for the Nozzle (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Nozzle Neck to Pad Weld for Reinf. pad (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Shell to Pad Weld Junction at Pad OD, Curve: B

Govern. thk, $t_g = 21.0$, $t_r = 17.333$, $c = 1.5$ mm., $E^* = 1.0$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.889$, Temp. Reduction = 6 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-6 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-12 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle-Shell/Head Weld (UCS-66(a)(1)(b)), Curve: B

Govern. thk, $t_g = 8.0$, $t_r = 5.172$, $c = 1.5$ mm., $E^* = 1.0$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.796$, Temp. Reduction = 11 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-29 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-40 °C

Gov. MDMT of the Nozzle	:	-40 °C
Gov. MDMT of the Reinforcement Pad	:	-29 °C
Gov. MDMT of the nozzle to shell joint welded assembly	:	-29 °C

Flange MDMT including Temperature reduction per UCS-66.1:

MDMT of ASME B16.5/47 flange per Matl. Specification	-46 °C
Flange MDMT with Temp reduction per UCS-66(i)(2)	-85 °C

Where the Stress Reduction Ratio per UCS-66(i)(2) is :
 Design Pressure/Ambient Rating = $23.07/51.00 = 0.452$

Weld Size Calculations, Description: MH224"

Intermediate Calc. for nozzle/shell welds $T_{min} = 6.5000$ mm.

Intermediate Calc. for pad/shell Welds TminPad 19.0000 mm.

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$4.5500 = 0.7 * t_{min}$	$7.0700 = 0.7 * W_o$ mm.
Pad Weld	$9.5000 = 0.5 * T_{minPad}$	$12.7260 = 0.7 * W_p$ mm.

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A-A1+2 \cdot t_n \cdot f_{r1} \cdot (E1 \cdot t - t_r)) \cdot S_v)$
 $= \max(0, (106.1803 - 13.2768 + 2 \cdot 6.5 \cdot 1.0 \cdot (1.0 \cdot 19.5 - 17.3327)) \cdot 138)$
 $= 131126.27 \text{ Kgf}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2+A5+A4-(W_i-Can/.707)^2 \cdot f_{r2}) \cdot S_v$
 $= (1.0156 + 95.568 + 4.24 - 0.0 \cdot 1.0) \cdot 138$
 $= 141874.69 \text{ Kgf}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S_v$
 $= (1.0156 + 0.0 + 1. + (2.535)) \cdot 138$
 $= 6403.44 \text{ Kgf}$

Weld Load [W3]:
 $= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S$
 $= (1.0156 + 0.0 + 4.24 + 95.568 + (2.535)) \cdot 138$
 $= 145441.81 \text{ Kgf}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$
 $= (3.1416/2.0) \cdot 625.6 \cdot 10.0 \cdot 0.49 \cdot 138$
 $= 67758. \text{ Kgf}$

Shear, Pad Element Weld [Spew]:
 $= (\pi/2) \cdot D_P \cdot W_P \cdot 0.49 \cdot S_{EW}$
 $= (3.1416/2.0) \cdot 1060.0 \cdot 18.0 \cdot 0.49 \cdot 138$
 $= 206651. \text{ Kgf}$

Shear, Nozzle Wall [Snw]:
 $= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - Can) \cdot 0.7 \cdot S_n$
 $= (3.1416 \cdot 309.55) \cdot (8.0 - 1.5) \cdot 0.7 \cdot 138$
 $= 62265. \text{ Kgf}$

Tension, Pad Groove Weld [Tpgw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_{gpn} \cdot 0.74 \cdot S_{eg}$
 $= (3.1416/2) \cdot 625.6 \cdot 22.0 \cdot 0.74 \cdot 138$
 $= 225124. \text{ Kgf}$

Tension, Shell Groove Weld [Tngw]:
 $= (\pi/2) \cdot D_{lo} \cdot (T - cas) \cdot 0.74 \cdot S_{ng}$
 $= (3.1416/2.0) \cdot 625.6 \cdot (21.0 - 1.5) \cdot 0.74 \cdot 138$
 $= 199542. \text{ Kgf}$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:
 $= (S_{PEW} + S_{NW}) = (206651 + 62265) = 268916 \text{ Kgf}$

Failure Path [Path 2-2]:
 $= (S_{onw} + T_{pgw} + T_{ngw} + S_{inw})$
 $= (67758 + 225124 + 199542 + 0) = 492424 \text{ Kgf}$



Failure Path [Path 3-3]:
= (Spew + Tngw + Sinw)
= (206651 + 199542 + 0) = 406192 Kgf

Summary of Failure Path Calculations:

Path 1-1 = 268915 Kgf, must exceed W = 131126 Kgf or W1 = 141874 Kgf
Path 2-2 = 492423 Kgf, must exceed W = 131126 Kgf or W2 = 6403 Kgf
Path 3-3 = 406192 Kgf, must exceed W = 131126 Kgf or W3 = 145441 Kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 23 bars

Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 23.5767 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 202.9527 mm.

Percent Elongation Calculations:

% Elongation per Table UG-79-1 ($50 \cdot t_{nom} / R_f \cdot (1 - R_f / R_o)$) 1.295 %

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